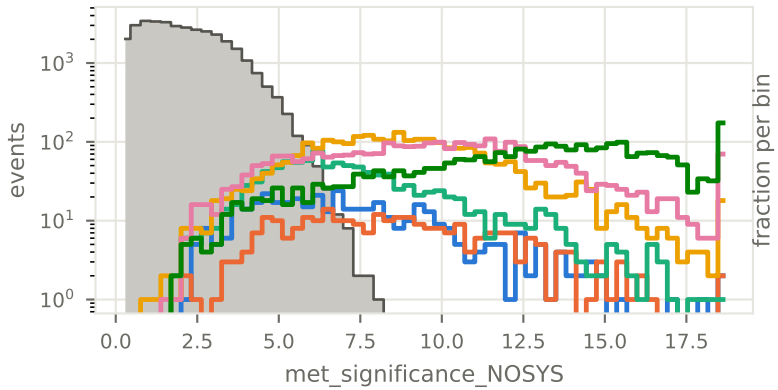
met_significance_NOSYS

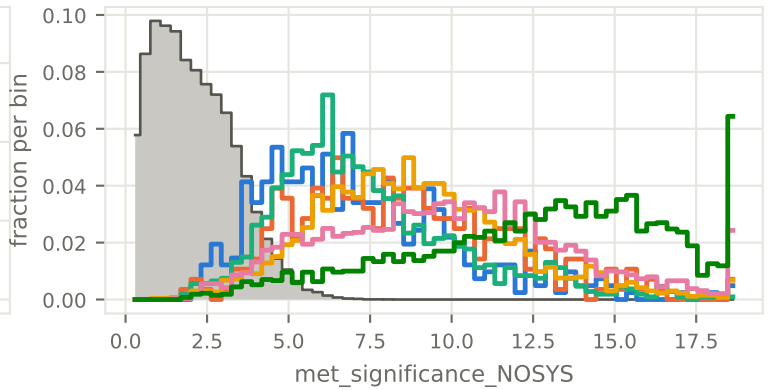
region: ENDCAP · rank 1 / 98

D_equal 0.876 (noise 0.022) D_pooled 0.896 min 0.810 (mH125_mS15) max 0.939 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

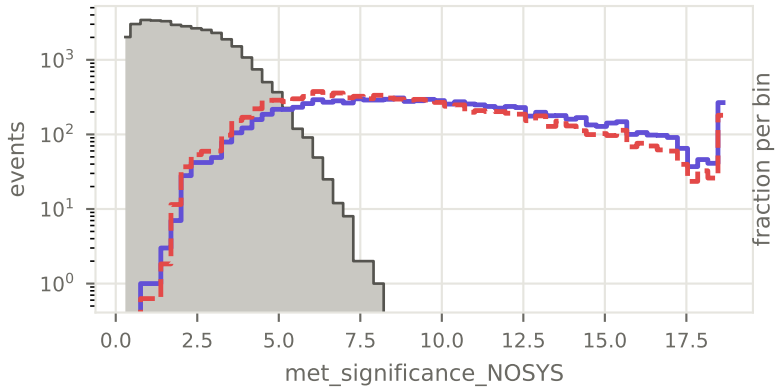


1b · each mass point vs background · normalized

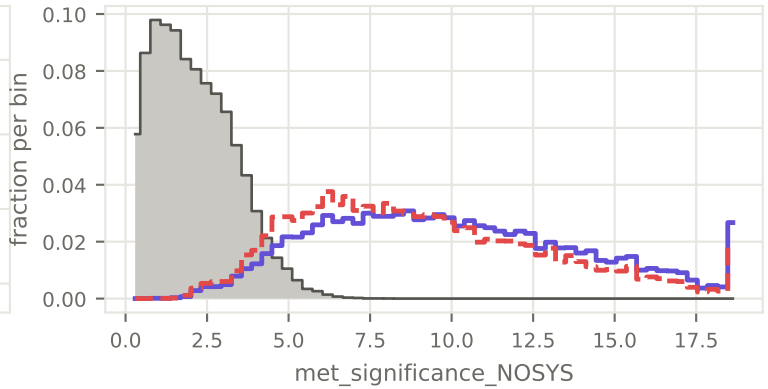


1 · each mass point
background n 34,970
mH125_mS15 D 0.810 n 411
mH125_mS55 D 0.887 n 281
mH200_mS50 D 0.859 n 1,071
mH400_mS100 D 0.906 n 2,648
mH600_mS150 D 0.895 n 2,882
mH1000_mS275 D 0.939 n 2,703

2a · combined signal vs background · raw

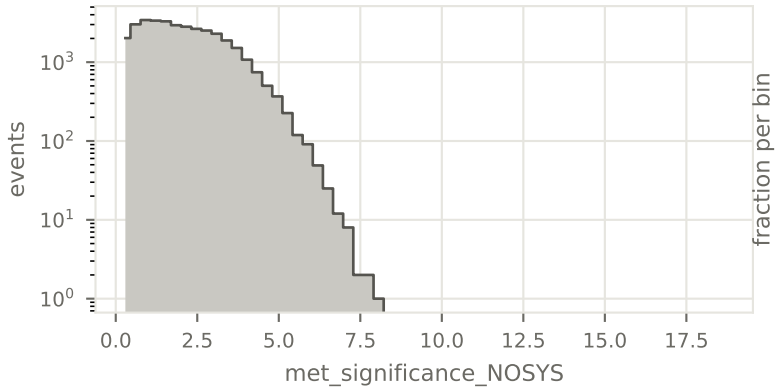


2b · combined signal vs background · normalized

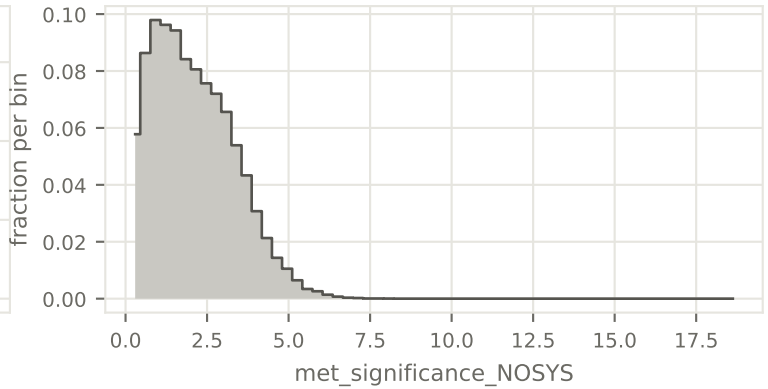


2 · combined signal
background n 34,970
combined, pooled D 0.896 n 9,996
equal weight per mass D 0.876

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

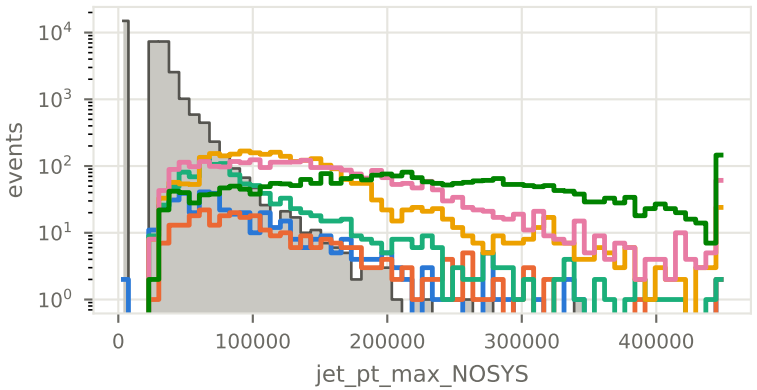
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

jet_pt_max_NOSYS

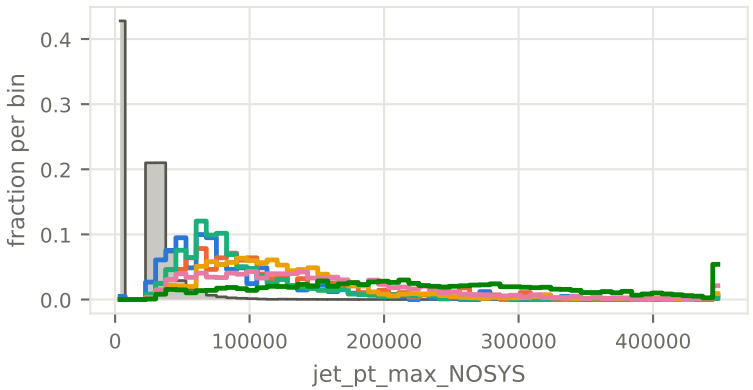
region: ENDCAP · rank 2 / 98

D_equal 0.854 (noise 0.022) D_pooled 0.874 min 0.785 (mH125_mS15) max 0.920 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

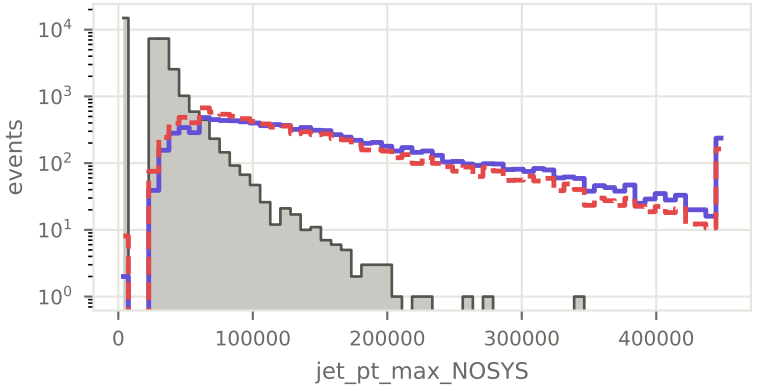


1b · each mass point vs background · normalized

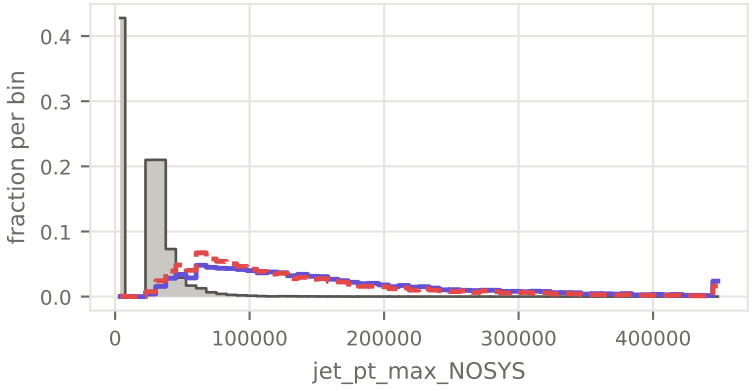


1 · each mass point
background n 34,970
mH125_mS15 D 0.785 n 411
mH125_mS55 D 0.866 n 281
mH200_mS50 D 0.851 n 1,071
mH400_mS100 D 0.894 n 2,648
mH600_mS150 D 0.874 n 2,882
mH1000_mS275 D 0.920 n 2,703

2a · combined signal vs background · raw

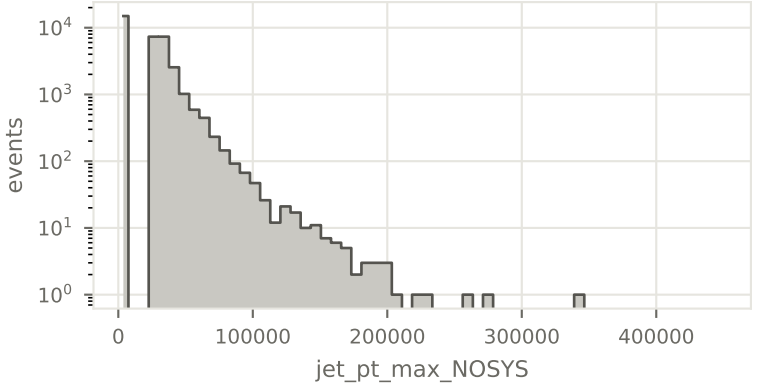


2b · combined signal vs background · normalized

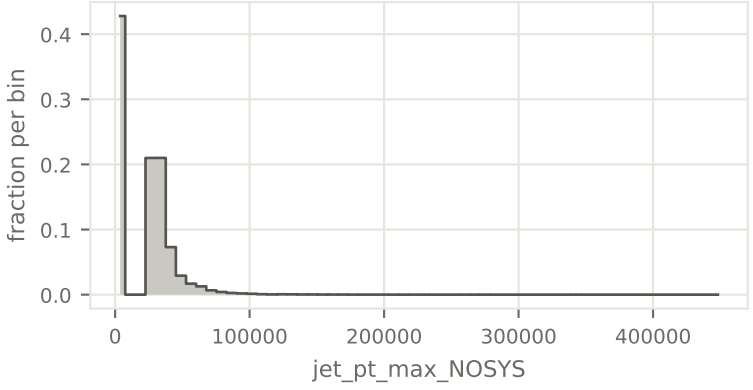


2 · combined signal
background n 34,970
combined, pooled D 0.874 n 9,996
equal weight per mass D 0.854

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

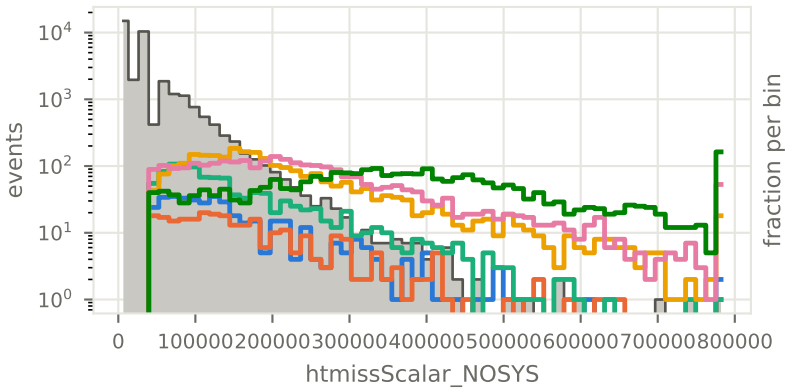
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

htmissScalar_NOSYS

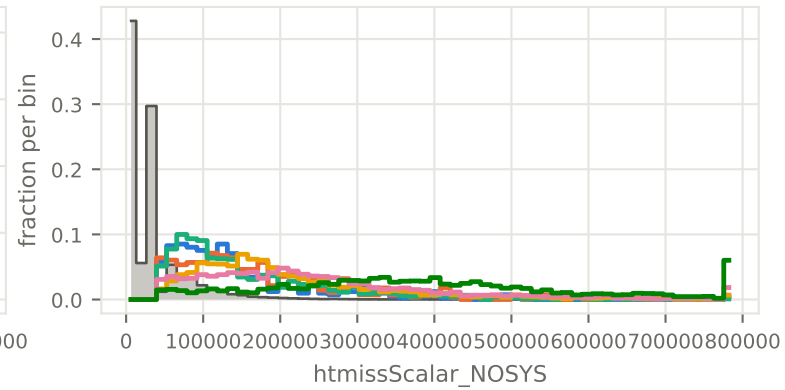
region: ENDCAP · rank 3 / 98

D_equal 0.789 (noise 0.022) D_pooled 0.789 min 0.789 (mH125_mS15) max 0.868 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

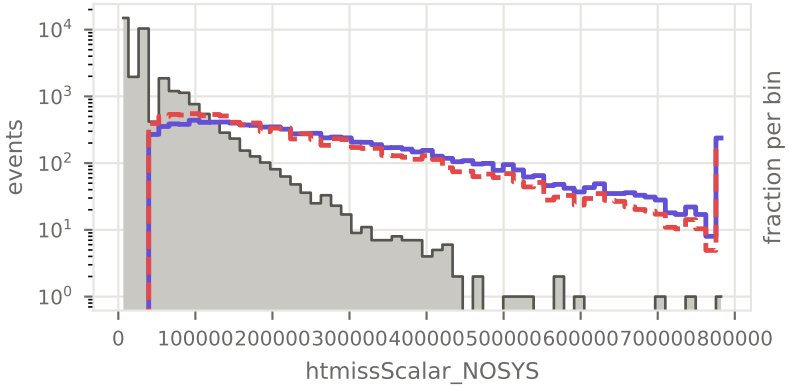


1b · each mass point vs background · normalized

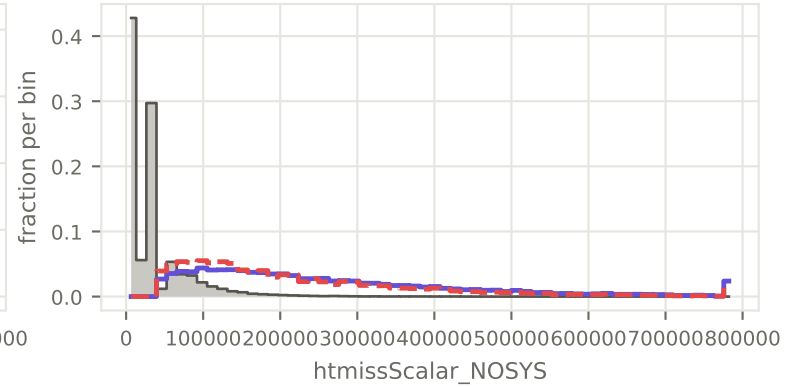


1 · each mass point
background n 34,970
mH125_mS15 D 0.789 n 411
mH125_mS55 D 0.789 n 281
mH200_mS50 D 0.789 n 1,071
mH400_mS100 D 0.803 n 2,648
mH600_mS150 D 0.789 n 2,882
mH1000_mS275 D 0.868 n 2,703

2a · combined signal vs background · raw

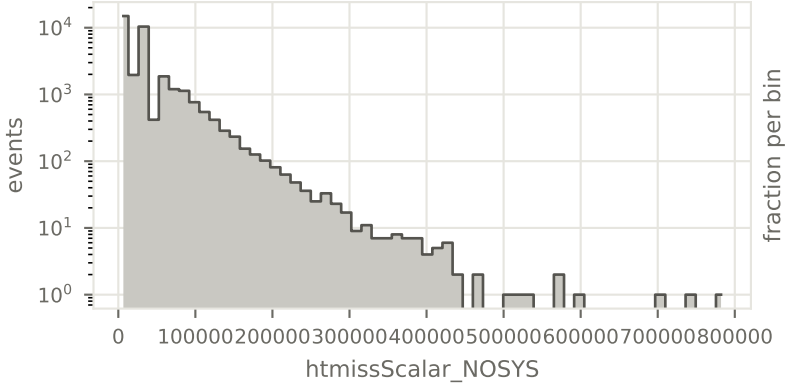


2b · combined signal vs background · normalized

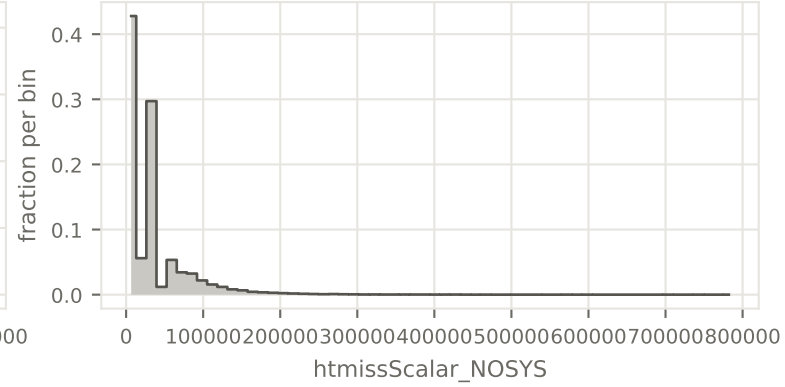


2 · combined signal
background n 34,970
combined, pooled D 0.789 n 9,996
equal weight per mass D 0.789

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

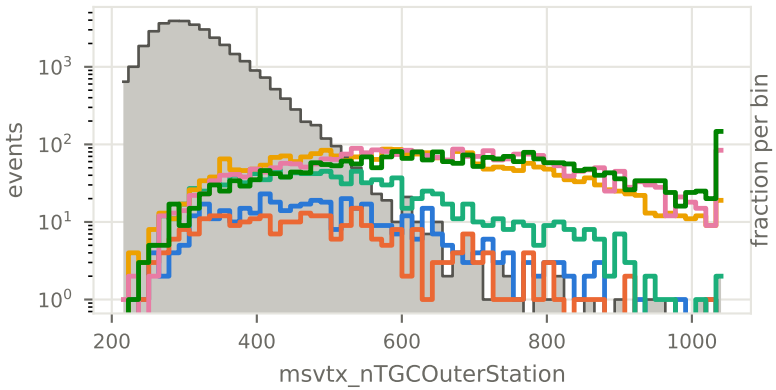
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCOuterStation

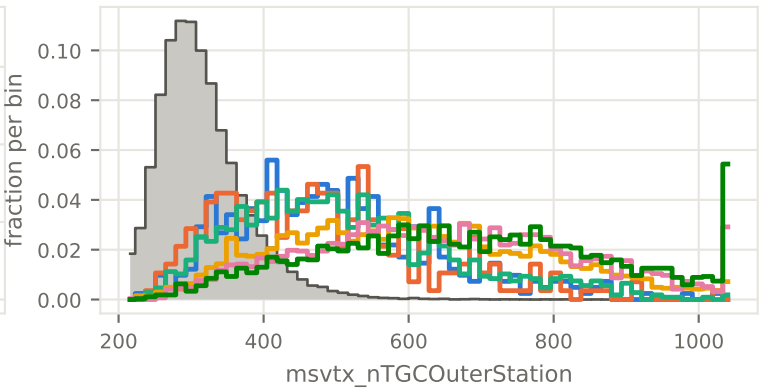
region: ENDCAP · rank 4 / 98

D_equal 0.725 (noise 0.022) D_pooled 0.785 min 0.612 (mH125_mS55) max 0.839 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

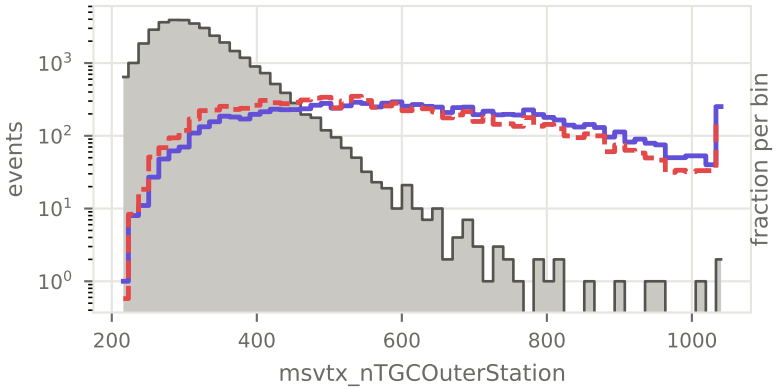


1b · each mass point vs background · normalized

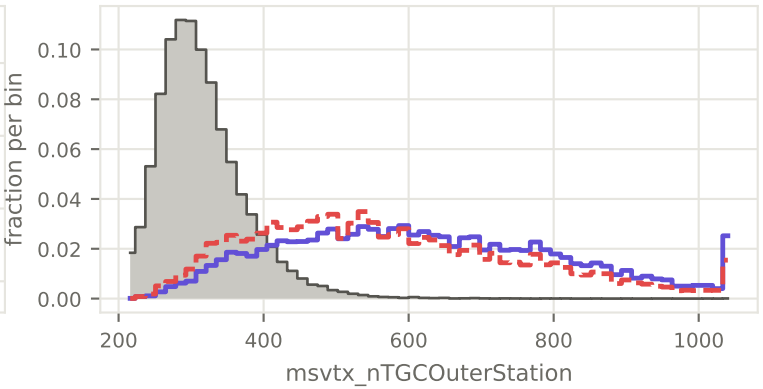


1 · each mass point
background n 34,970
mH125_mS15 D 0.666 n 411
mH125_mS55 D 0.612 n 281
mH200_mS50 D 0.678 n 1,071
mH400_mS100 D 0.785 n 2,648
mH600_mS150 D 0.826 n 2,882
mH1000_mS275 D 0.839 n 2,703

2a · combined signal vs background · raw

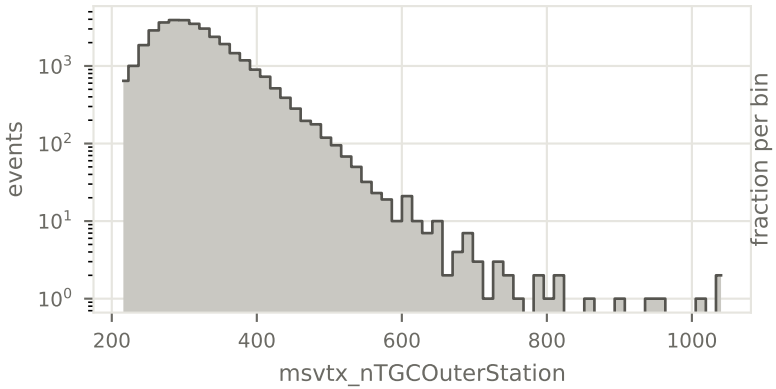


2b · combined signal vs background · normalized

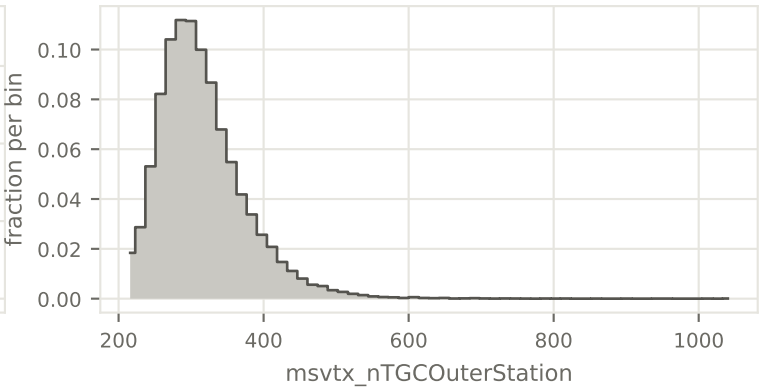


2 · combined signal
background n 34,970
combined, pooled D 0.785 n 9,996
equal weight per mass D 0.725

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

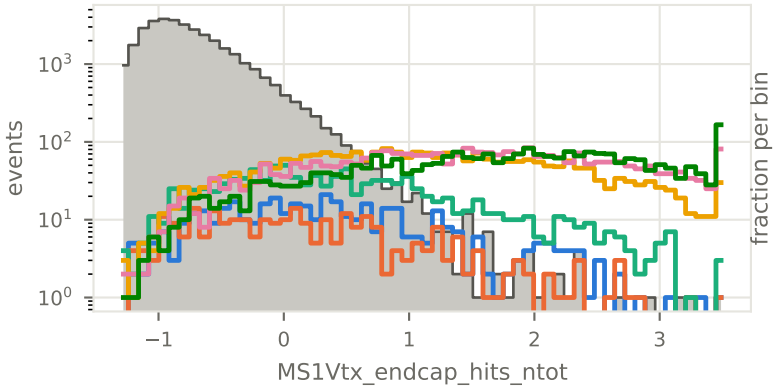
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_endcap_hits_ntot

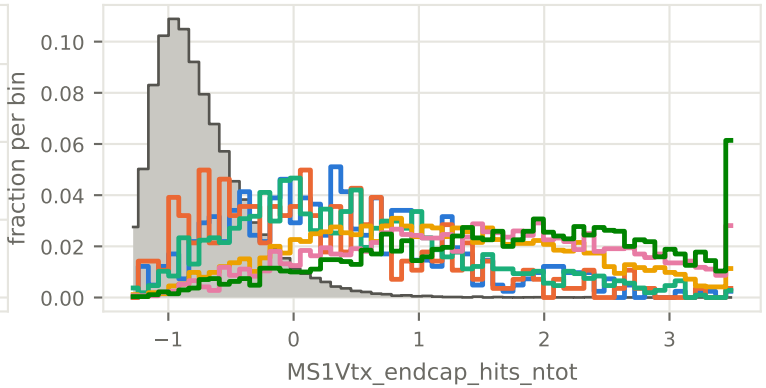
region: ENDCAP · rank 5 / 98 · in: NN2 endcap

D_equal 0.718 (noise 0.022) D_pooled 0.794 min 0.553 (mH125_mS55) max 0.864 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

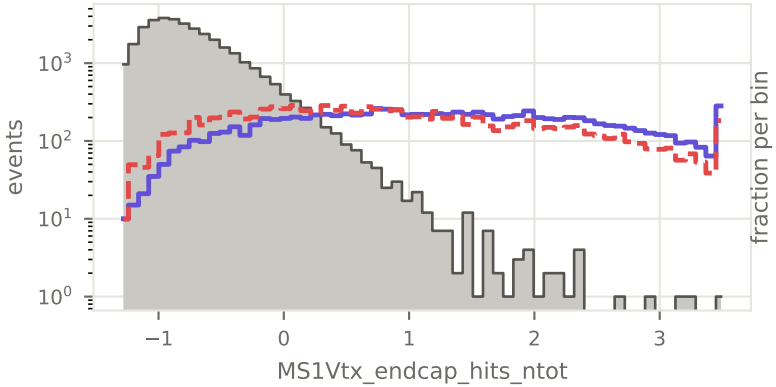


1b · each mass point vs background · normalized

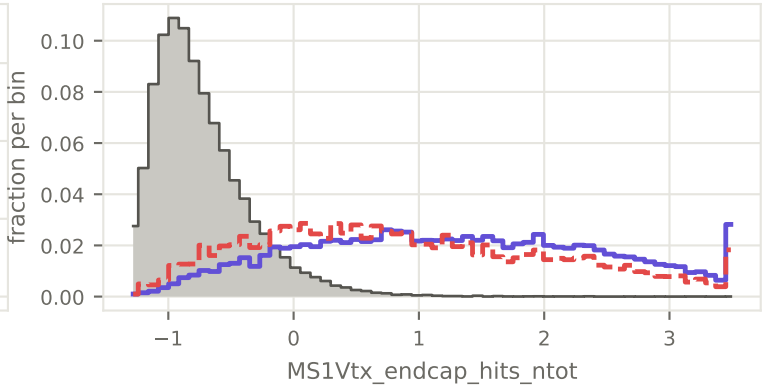


1 · each mass point
background n 34,970
mH125_mS15 D 0.635 n 411
mH125_mS55 D 0.553 n 281
mH200_mS50 D 0.668 n 1,071
mH400_mS100 D 0.803 n 2,648
mH600_mS150 D 0.839 n 2,882
mH1000_mS275 D 0.864 n 2,703

2a · combined signal vs background · raw

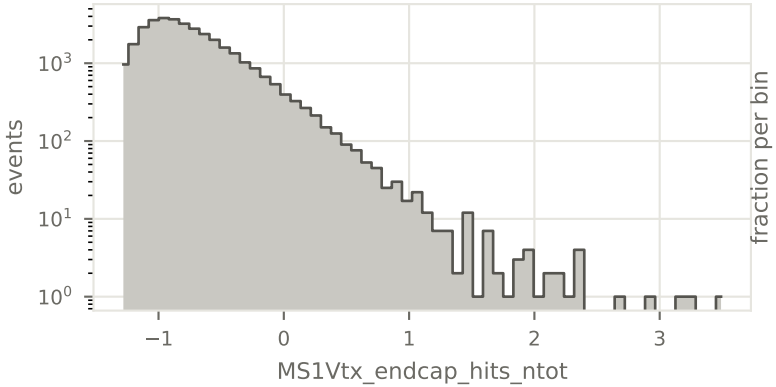


2b · combined signal vs background · normalized

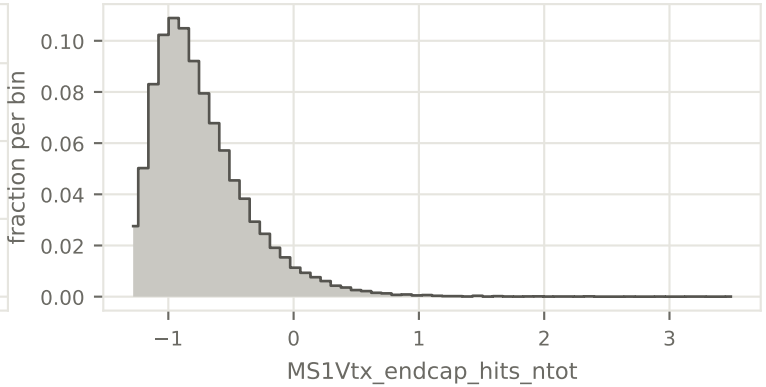


2 · combined signal
background n 34,970
combined, pooled D 0.794 n 9,996
equal weight per mass D 0.718

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

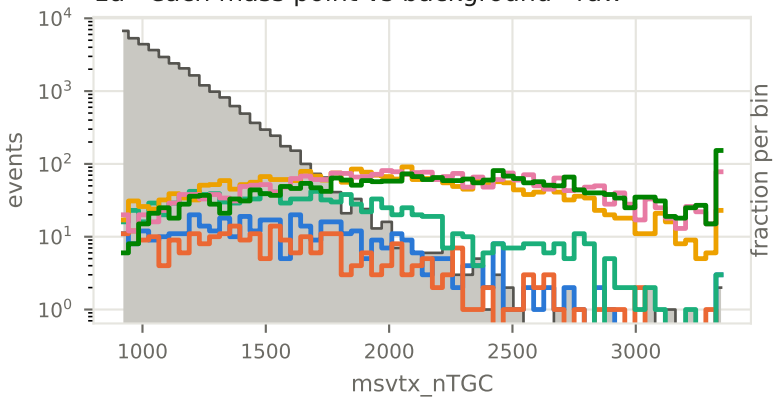
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTGC

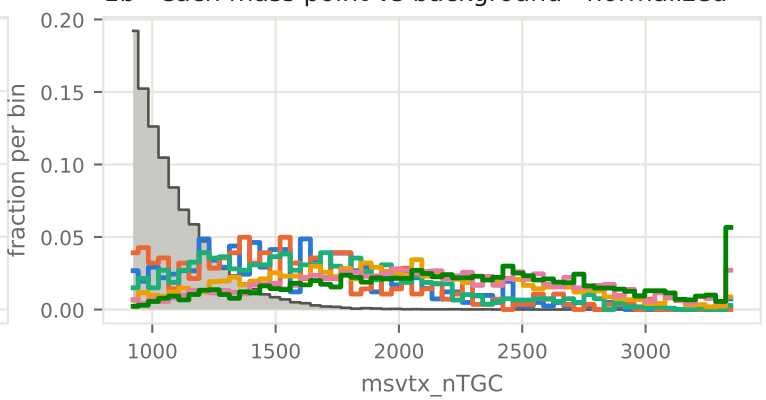
region: ENDCAP · rank 6 / 98

D_equal 0.701 (noise 0.022) D_pooled 0.773 min 0.590 (mH125_mS55) max 0.840 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

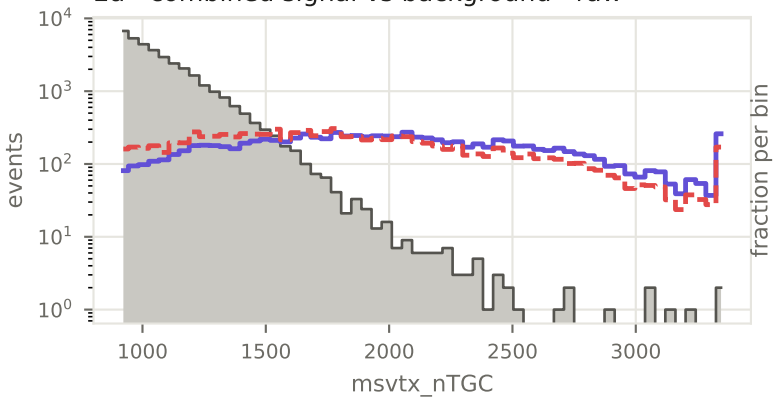


1b · each mass point vs background · normalized

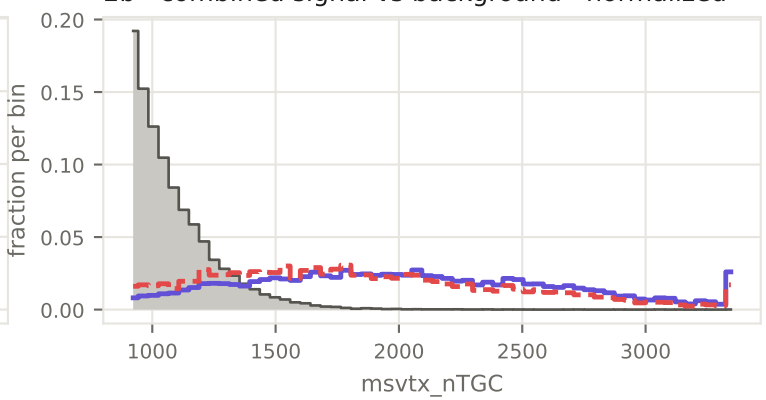


1 · each mass point
background n 34,970
mH125_mS15 D 0.625 n 411
mH125_mS55 D 0.590 n 281
mH200_mS50 D 0.640 n 1,071
mH400_mS100 D 0.774 n 2,648
mH600_mS150 D 0.823 n 2,882
mH1000_mS275 D 0.840 n 2,703

2a · combined signal vs background · raw

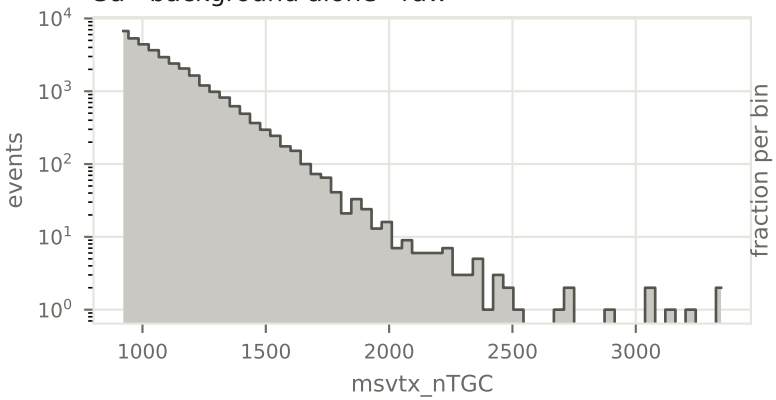


2b · combined signal vs background · normalized

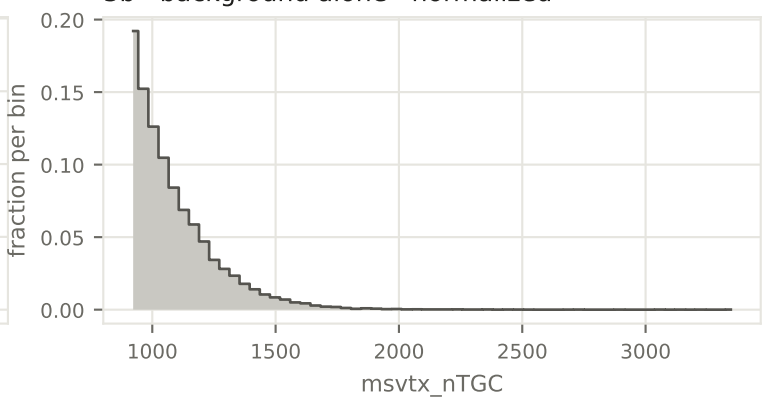


2 · combined signal
background n 34,970
combined, pooled D 0.773 n 9,996
equal weight per mass D 0.701

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

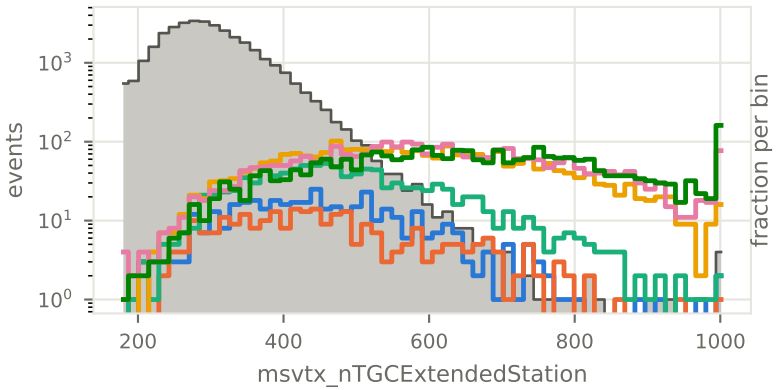
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCExtendedStation

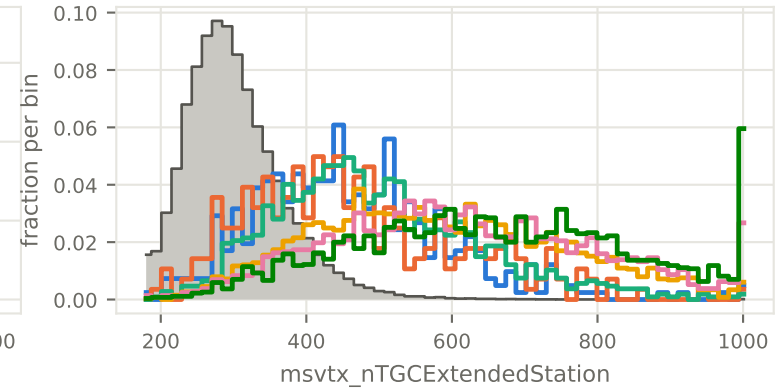
region: ENDCAP · rank 7 / 98

D_equal 0.700 (noise 0.022) D_pooled 0.766 min 0.585 (mH125_mS55) max 0.828 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

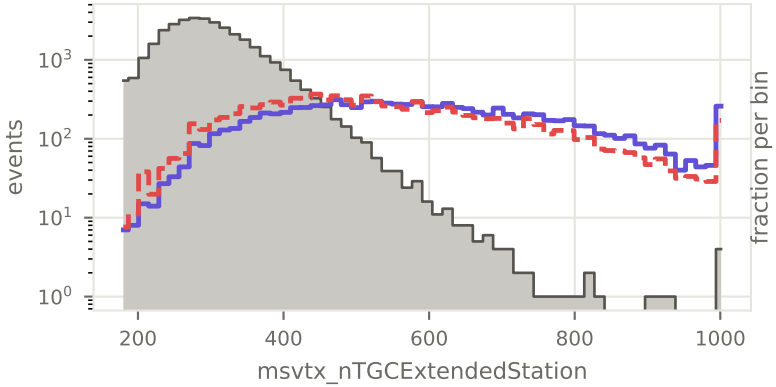


1b · each mass point vs background · normalized

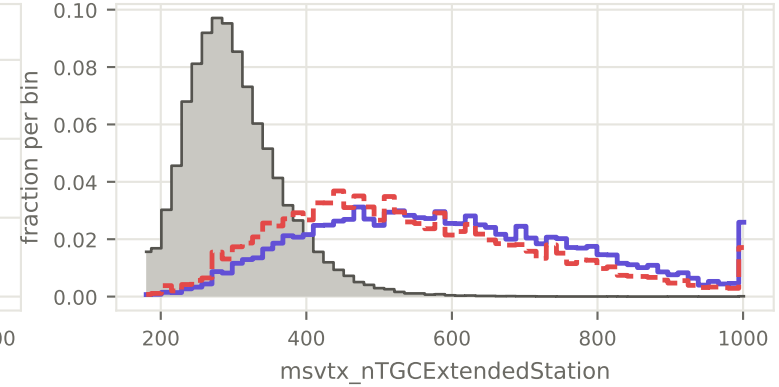


1 · each mass point
background n 34,970
mH125_mS15 D 0.599 n 411
mH125_mS55 D 0.585 n 281
mH200_mS50 D 0.681 n 1,071
mH400_mS100 D 0.774 n 2,648
mH600_mS150 D 0.799 n 2,882
mH1000_mS275 D 0.828 n 2,703

2a · combined signal vs background · raw

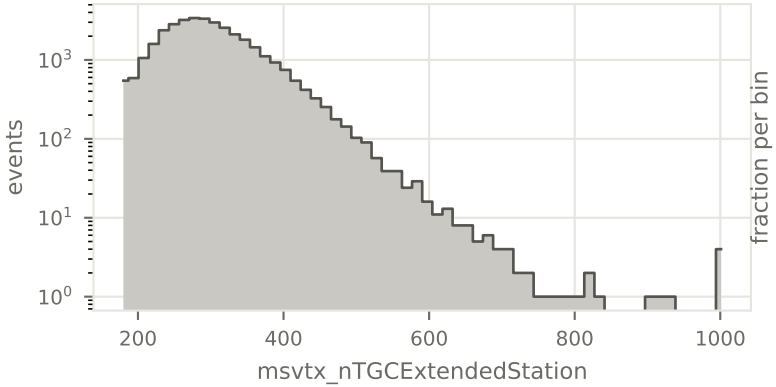


2b · combined signal vs background · normalized

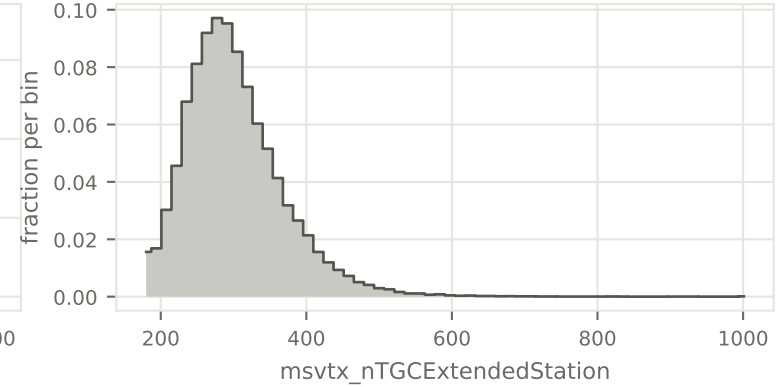


2 · combined signal
background n 34,970
combined, pooled D 0.766 n 9,996
equal weight per mass D 0.700

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

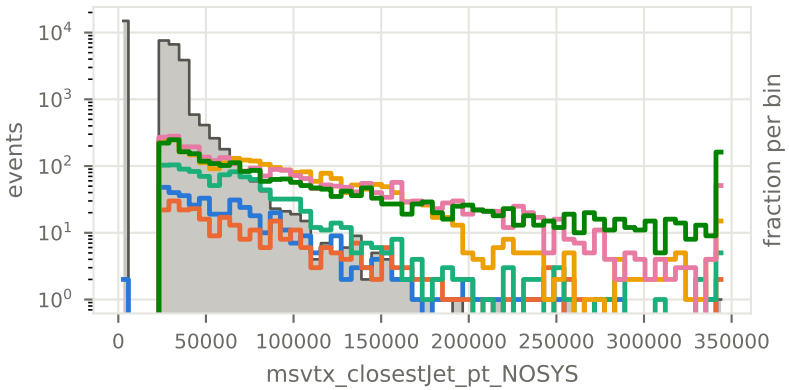
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_closestJet_pt_NOSYS

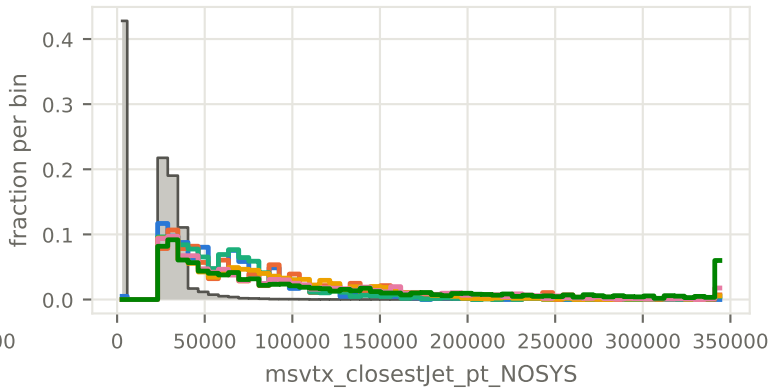
region: ENDCAP · rank 8 / 98

D_equal 0.687 (noise 0.022) D_pooled 0.697 min 0.647 (mH125_mS15) max 0.715 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

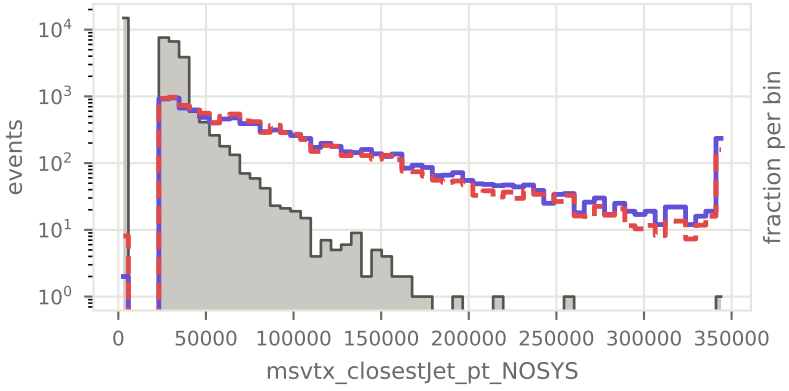


1b · each mass point vs background · normalized

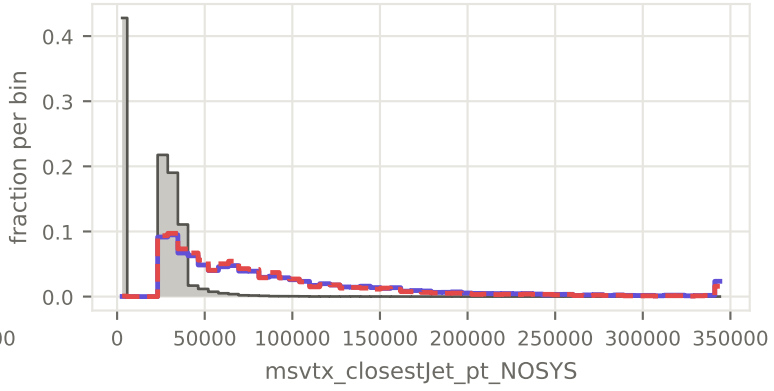


1 · each mass point
background n 34,970
mH125_mS15 D 0.647 n 411
mH125_mS55 D 0.691 n 281
mH200_mS50 D 0.681 n 1,071
mH400_mS100 D 0.699 n 2,648
mH600_mS150 D 0.693 n 2,882
mH1000_mS275 D 0.715 n 2,703

2a · combined signal vs background · raw

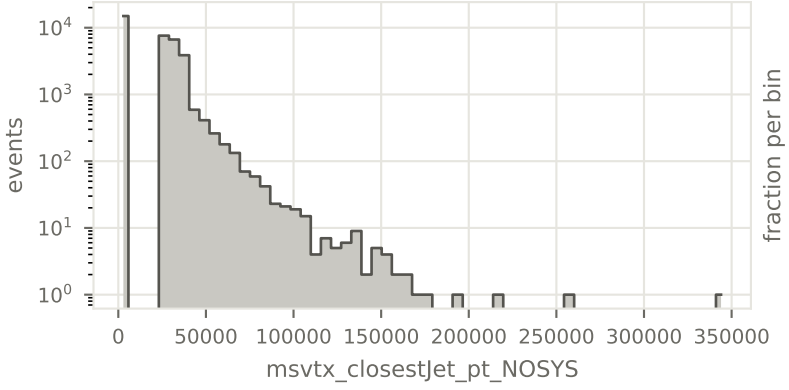


2b · combined signal vs background · normalized

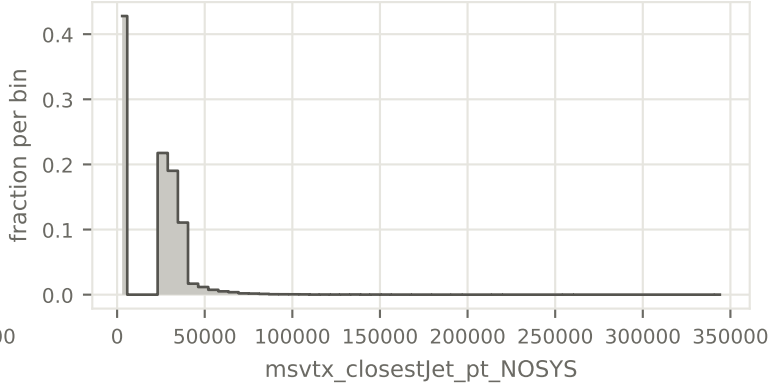


2 · combined signal
background n 34,970
combined, pooled D 0.697 n 9,996
equal weight per mass D 0.687

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

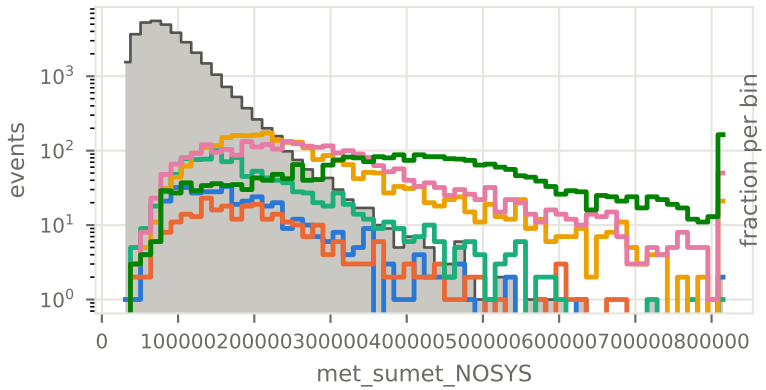
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

met_sumet_NOSYS

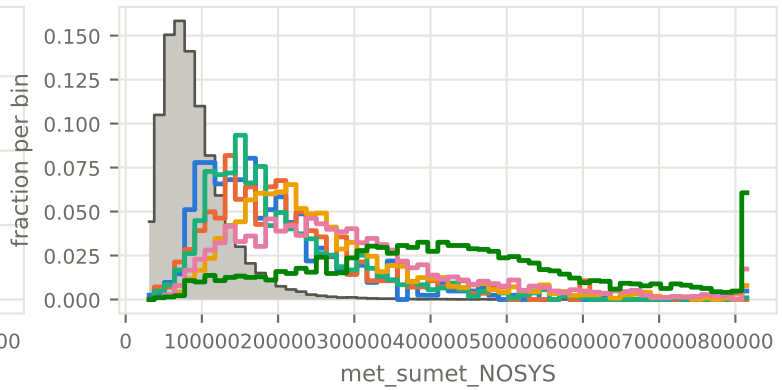
region: ENDCAP · rank 9 / 98

D_equal 0.687 (noise 0.022) D_pooled 0.741 min 0.563 (mH125_mS15) max 0.859 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

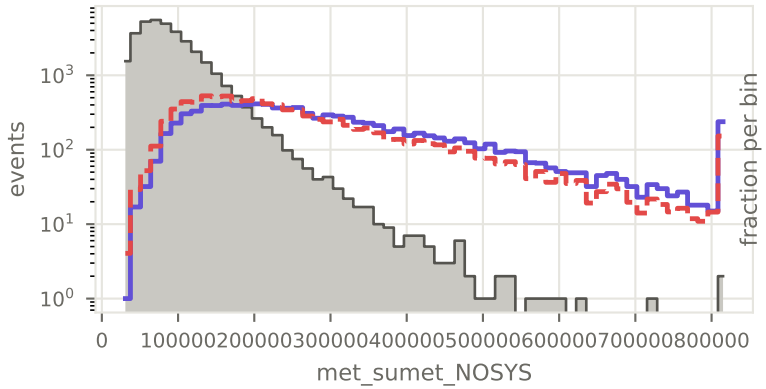


1b · each mass point vs background · normalized

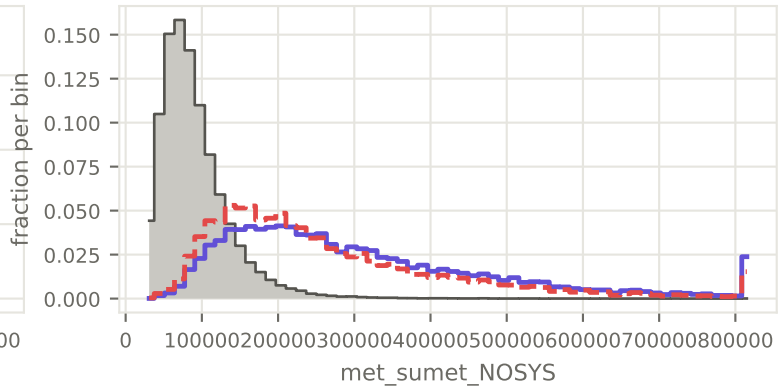


1 · each mass point
background n 34,970
mH125_mS15 D 0.563 n 411
mH125_mS55 D 0.663 n 281
mH200_mS50 D 0.626 n 1,071
mH400_mS100 D 0.762 n 2,648
mH600_mS150 D 0.743 n 2,882
mH1000_mS275 D 0.859 n 2,703

2a · combined signal vs background · raw

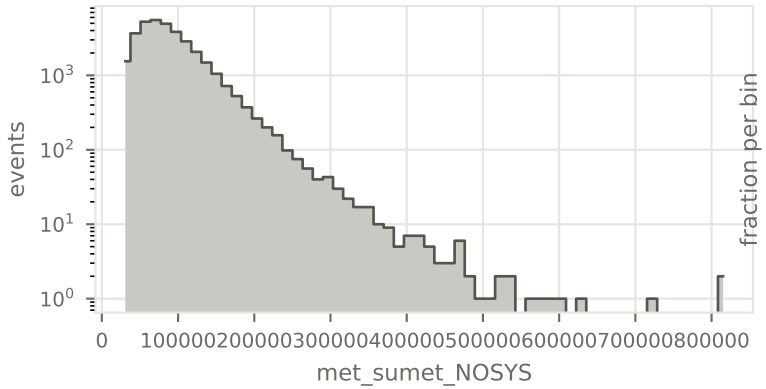


2b · combined signal vs background · normalized

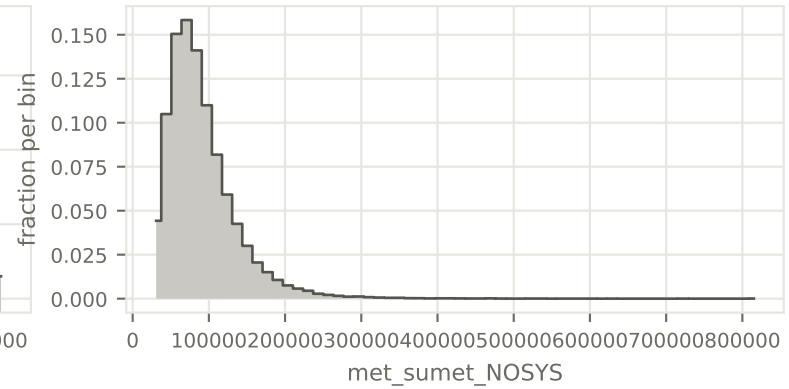


2 · combined signal
background n 34,970
combined, pooled D 0.741 n 9,996
equal weight per mass D 0.687

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

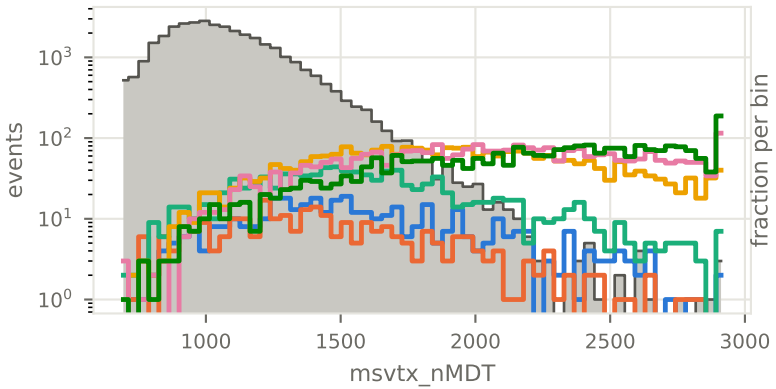
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nMDT

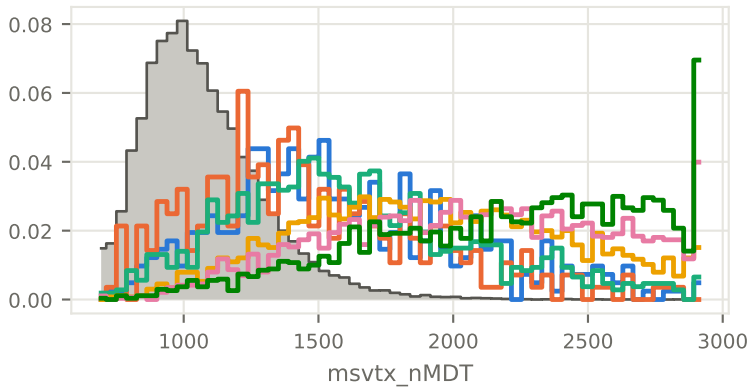
region: ENDCAP · rank 10 / 98

D_equal 0.672 (noise 0.022) D_pooled 0.759 min 0.491 (mH125_mS55) max 0.846 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

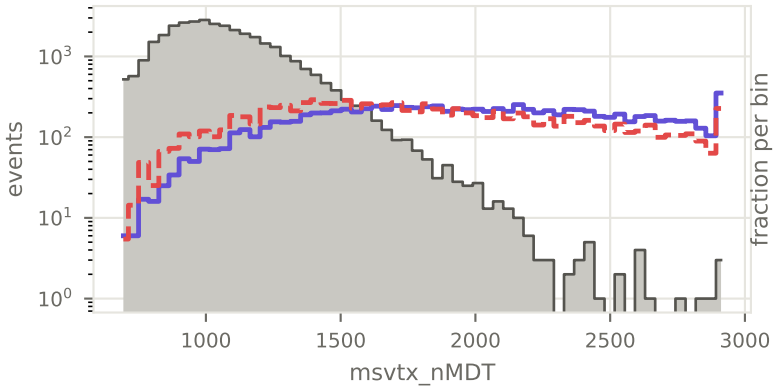


1b · each mass point vs background · normalized

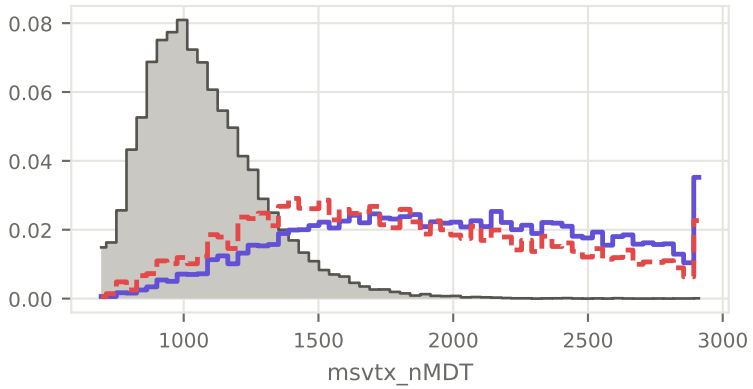


1 · each mass point
background n 34,970
mH125_mS15 D 0.603 n 411
mH125_mS55 D 0.491 n 281
mH200_mS50 D 0.613 n 1,071
mH400_mS100 D 0.767 n 2,648
mH600_mS150 D 0.808 n 2,882
mH1000_mS275 D 0.846 n 2,703

2a · combined signal vs background · raw

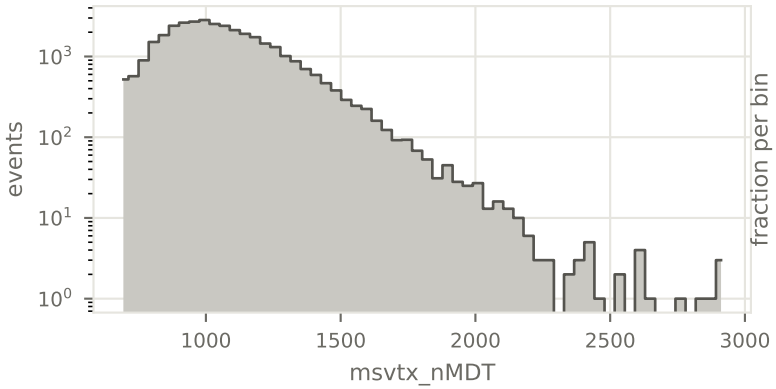


2b · combined signal vs background · normalized

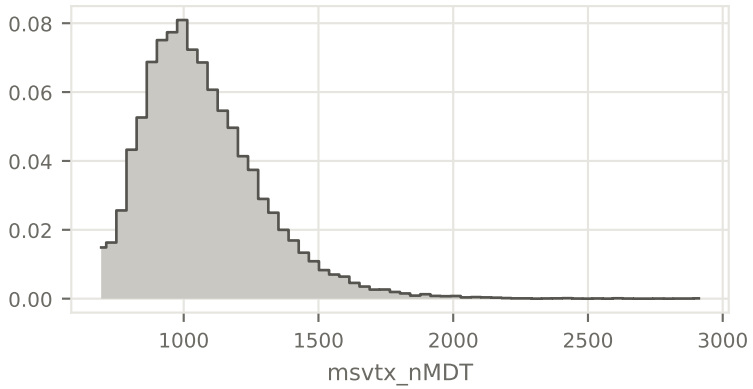


2 · combined signal
background n 34,970
combined, pooled D 0.759 n 9,996
equal weight per mass D 0.672

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

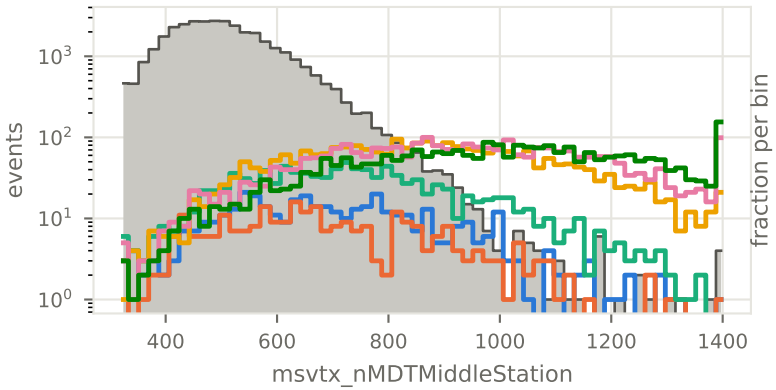
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nMDTMiddleStation

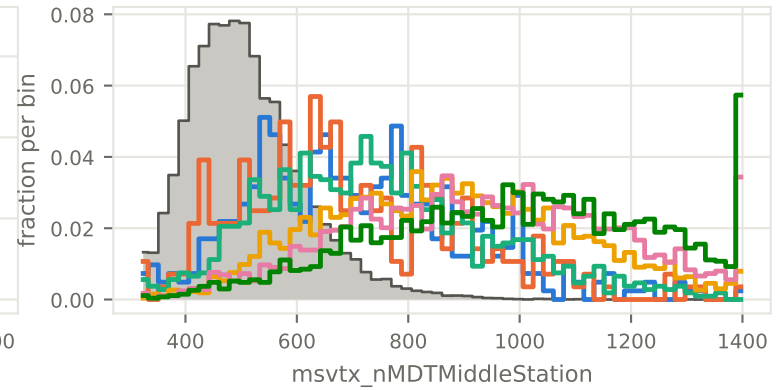
region: ENDCAP · rank 11 / 98

D_equal 0.648 (noise 0.022) D_pooled 0.736 min 0.504 (mH125_mS55) max 0.827 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

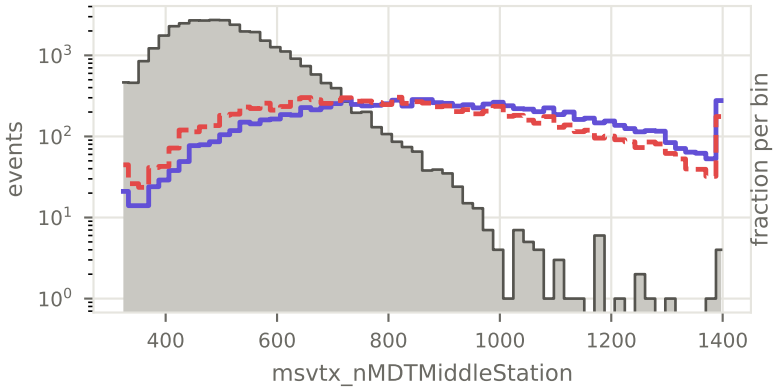


1b · each mass point vs background · normalized

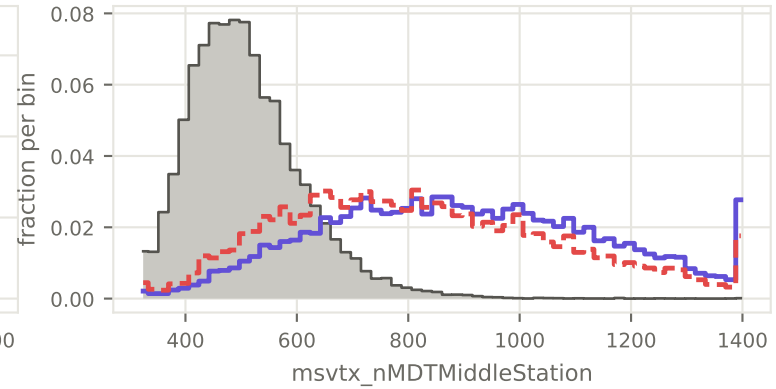


1 · each mass point
background n 34,970
mH125_mS15 D 0.519 n 411
mH125_mS55 D 0.504 n 281
mH200_mS50 D 0.579 n 1,071
mH400_mS100 D 0.738 n 2,648
mH600_mS150 D 0.789 n 2,882
mH1000_mS275 D 0.827 n 2,703

2a · combined signal vs background · raw

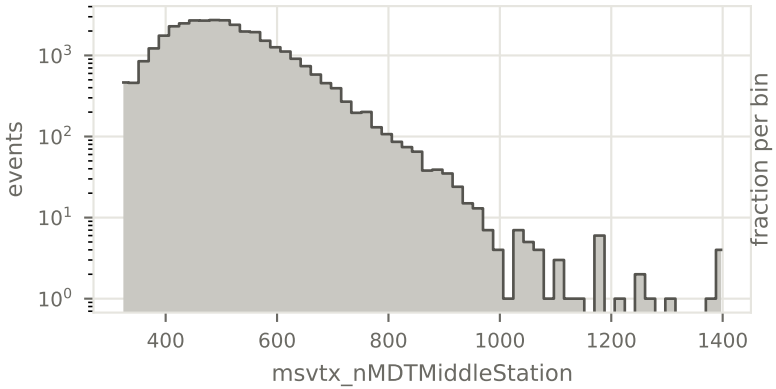


2b · combined signal vs background · normalized

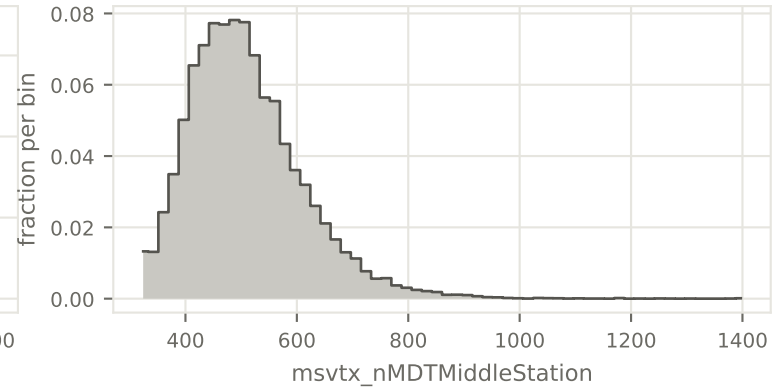


2 · combined signal
background n 34,970
combined, pooled D 0.736 n 9,996
equal weight per mass D 0.648

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

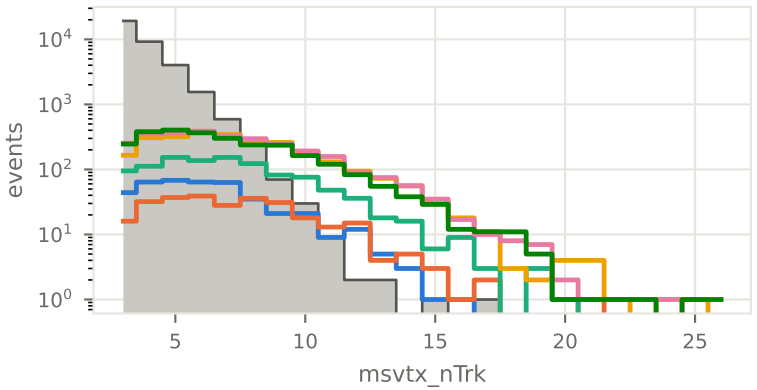
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTrk

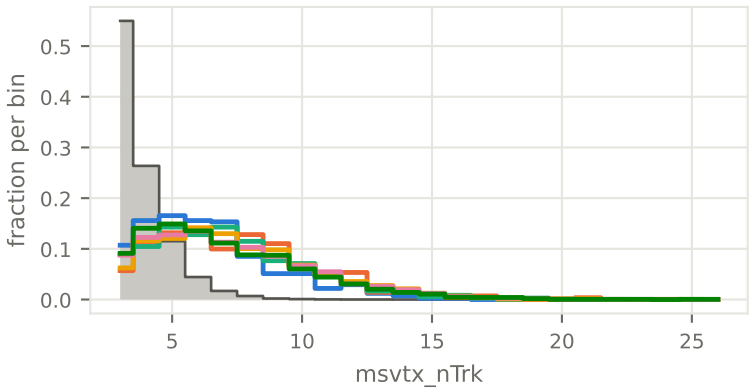
region: ENDCAP · rank 12 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.606 (noise 0.022) D_pooled 0.607 min 0.551 (mH125_mS15) max 0.643 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

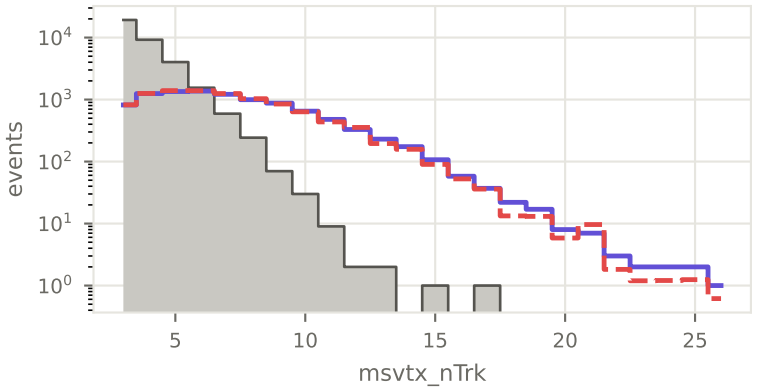


1b · each mass point vs background · normalized

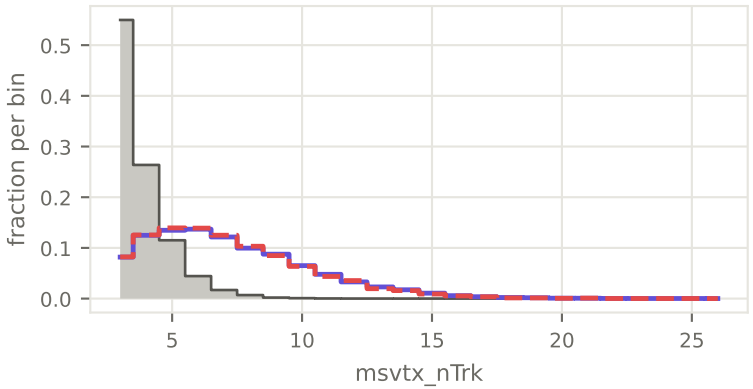


1 · each mass point
background n 34,970
mH125_mS15 D 0.551 n 411
mH125_mS55 D 0.643 n 281
mH200_mS50 D 0.620 n 1,071
mH400_mS100 D 0.635 n 2,648
mH600_mS150 D 0.603 n 2,882
mH1000_mS275 D 0.582 n 2,703

2a · combined signal vs background · raw

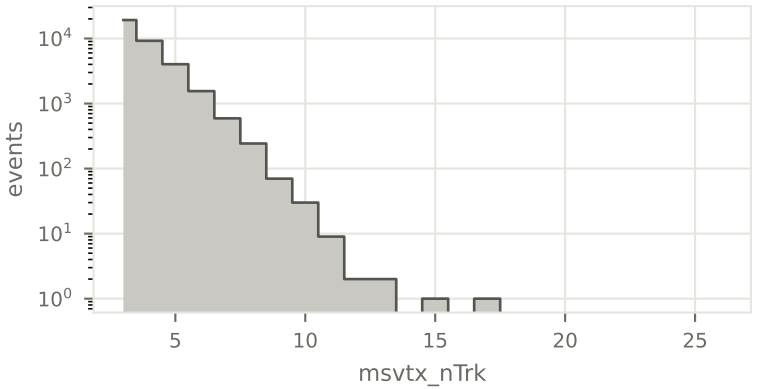


2b · combined signal vs background · normalized

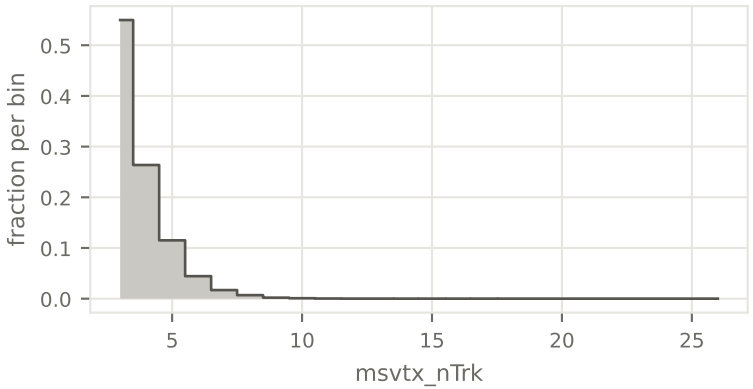


2 · combined signal
background n 34,970
combined, pooled D 0.607 n 9,996
equal weight per mass D 0.606

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

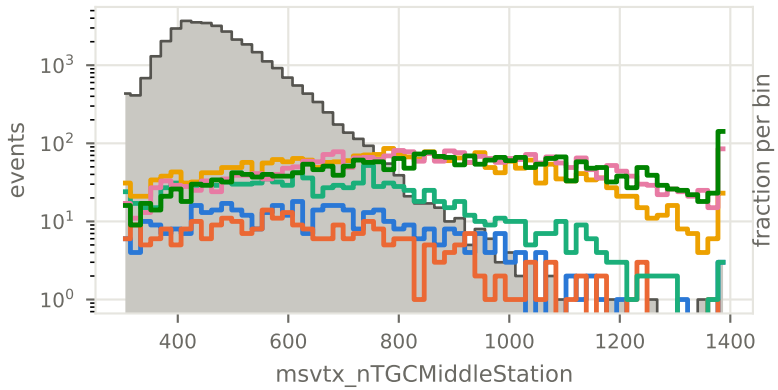
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCMiddleStation

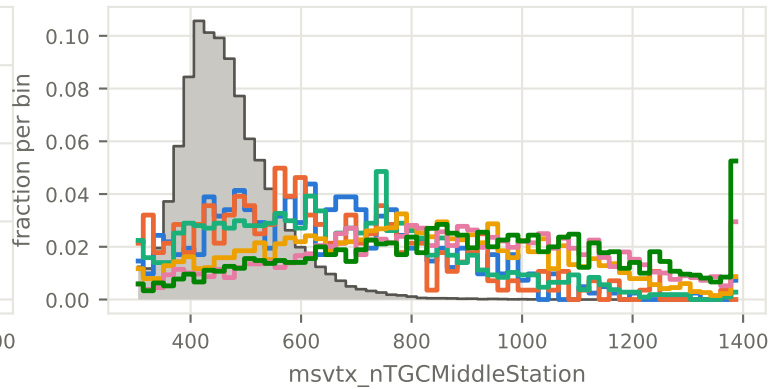
region: ENDCAP · rank 13 / 98

D_equal 0.601 (noise 0.022) D_pooled 0.683 min 0.471 (mH125_mS55) max 0.755 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

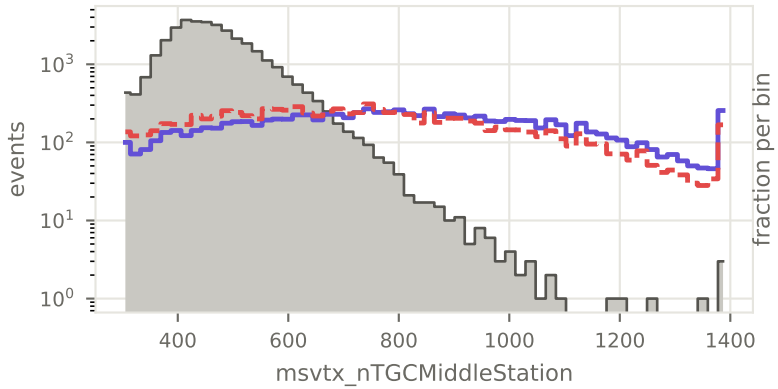


1b · each mass point vs background · normalized

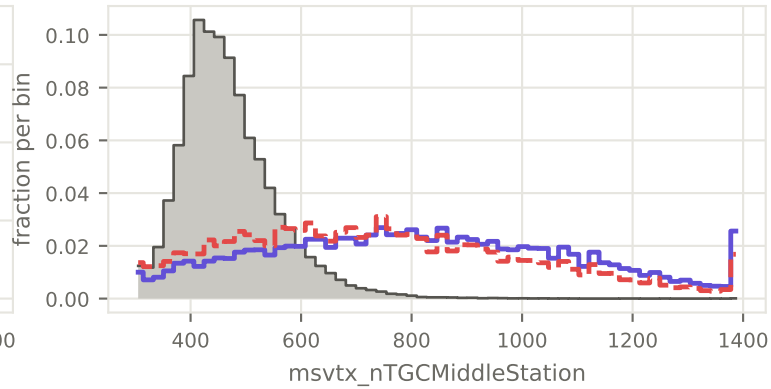


1 · each mass point
background n 34,970
mH125_mS15 D 0.504 n 411
mH125_mS55 D 0.471 n 281
mH200_mS50 D 0.511 n 1,071
mH400_mS100 D 0.672 n 2,648
mH600_mS150 D 0.754 n 2,882
mH1000_mS275 D 0.755 n 2,703

2a · combined signal vs background · raw

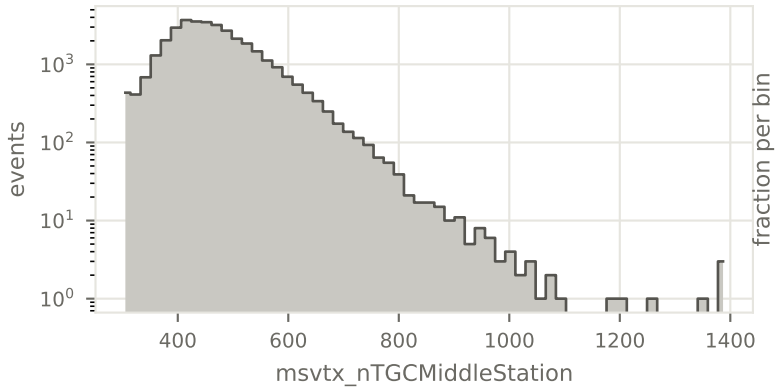


2b · combined signal vs background · normalized

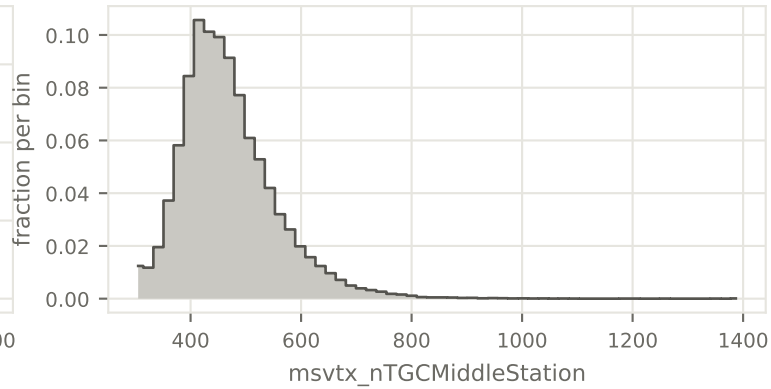


2 · combined signal
background n 34,970
combined, pooled D 0.683 n 9,996
equal weight per mass D 0.601

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

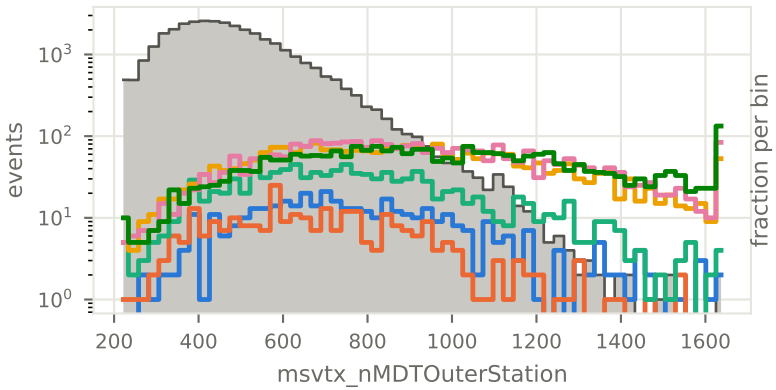
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nMDTOuterStation

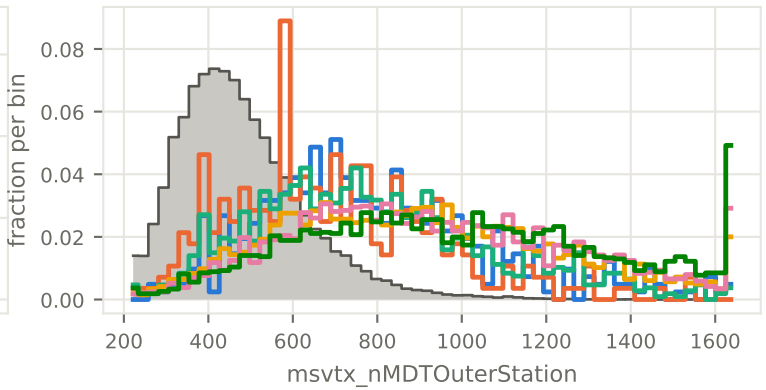
region: ENDCAP · rank 14 / 98

D_equal 0.583 (noise 0.022) D_pooled 0.636 min 0.465 (mH125_mS55) max 0.686 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

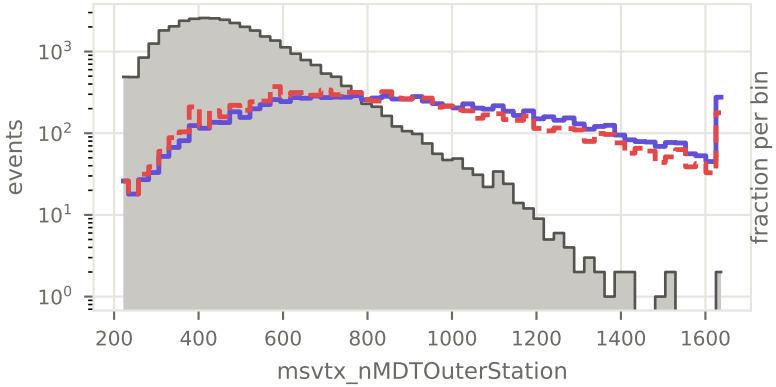


1b · each mass point vs background · normalized

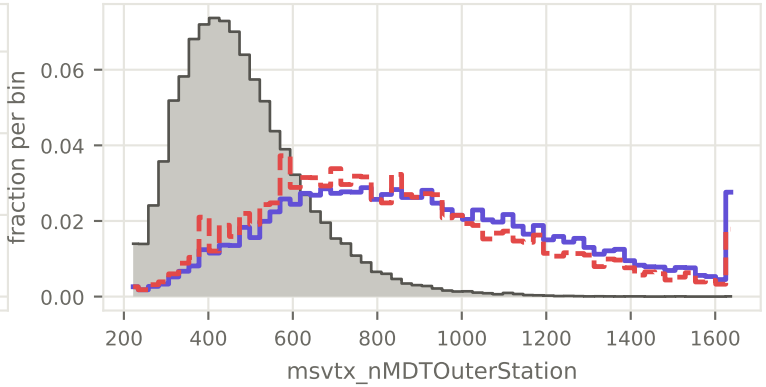


1 · each mass point
background n 34,970
mH125_mS15 D 0.576 n 411
mH125_mS55 D 0.465 n 281
mH200_mS50 D 0.545 n 1,071
mH400_mS100 D 0.631 n 2,648
mH600_mS150 D 0.669 n 2,882
mH1000_mS275 D 0.686 n 2,703

2a · combined signal vs background · raw

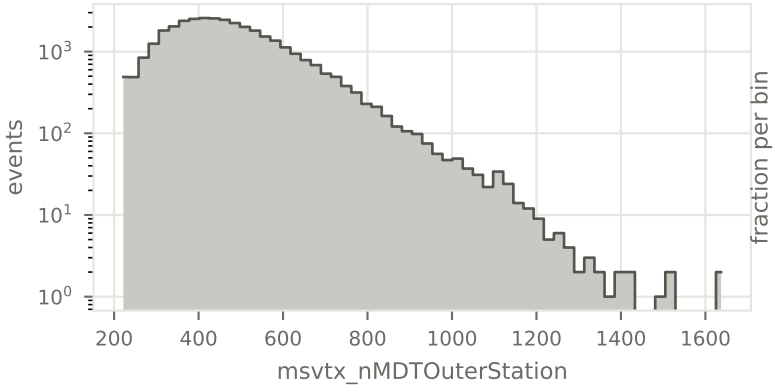


2b · combined signal vs background · normalized

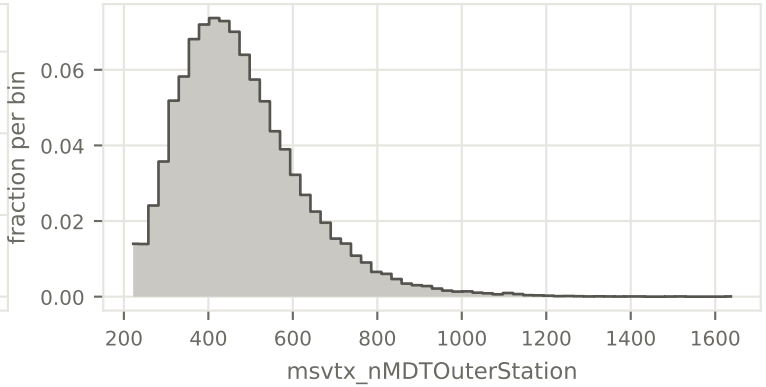


2 · combined signal
background n 34,970
combined, pooled D 0.636 n 9,996
equal weight per mass D 0.583

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

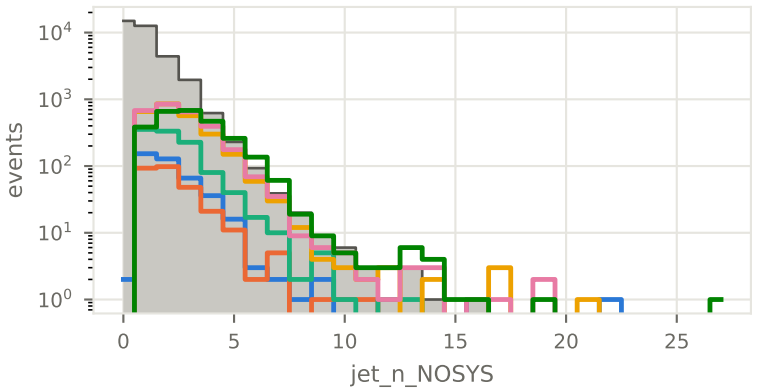
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

jet_n_NOSYS

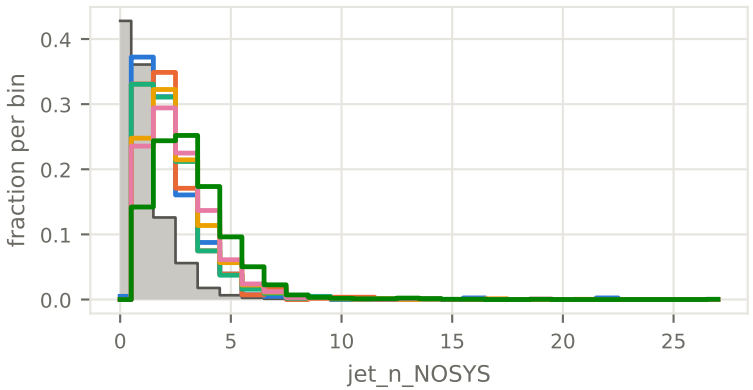
region: ENDCAP · rank 15 / 98

D_equal 0.511 (noise 0.022) D_pooled 0.556 min 0.423 (mH125_mS15) max 0.647 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

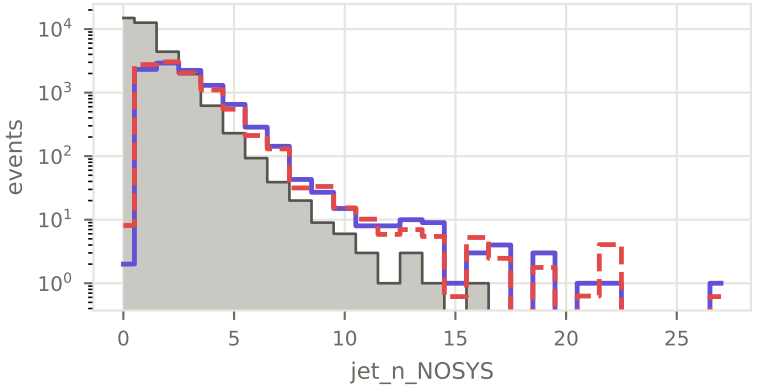


1b · each mass point vs background · normalized

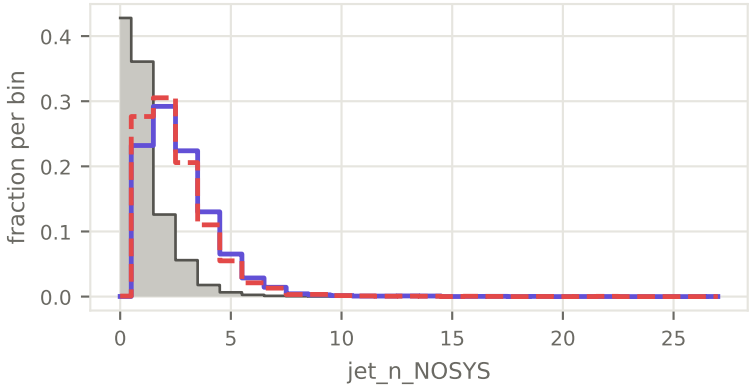


1 · each mass point
background n 34,970
mH125_mS15 D 0.423 n 411
mH125_mS55 D 0.458 n 281
mH200_mS50 D 0.458 n 1,071
mH400_mS100 D 0.541 n 2,648
mH600_mS150 D 0.553 n 2,882
mH1000_mS275 D 0.647 n 2,703

2a · combined signal vs background · raw

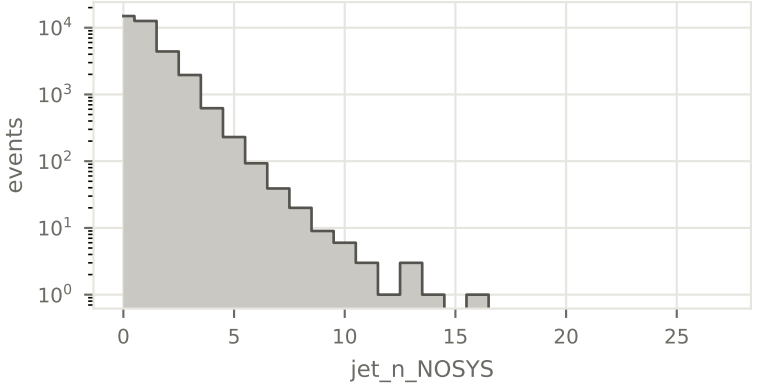


2b · combined signal vs background · normalized

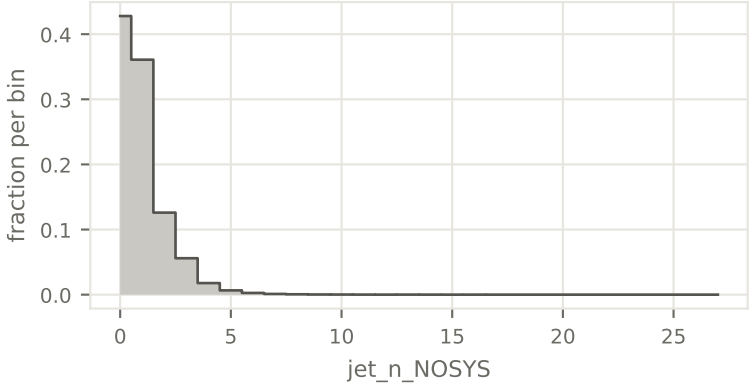


2 · combined signal
background n 34,970
combined, pooled D 0.556 n 9,996
equal weight per mass D 0.511

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

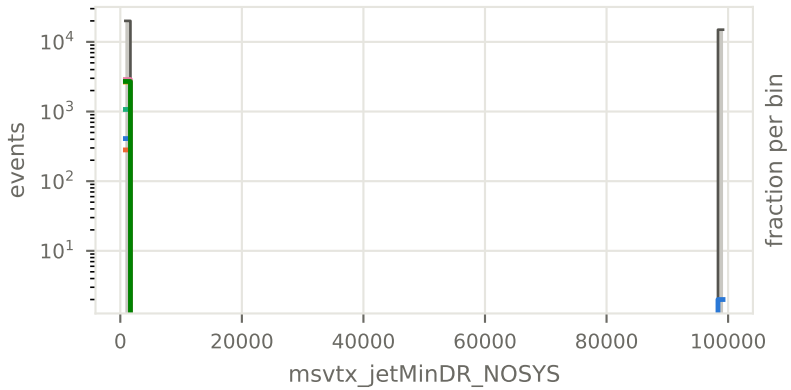
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_jetMinDR_NOSYS

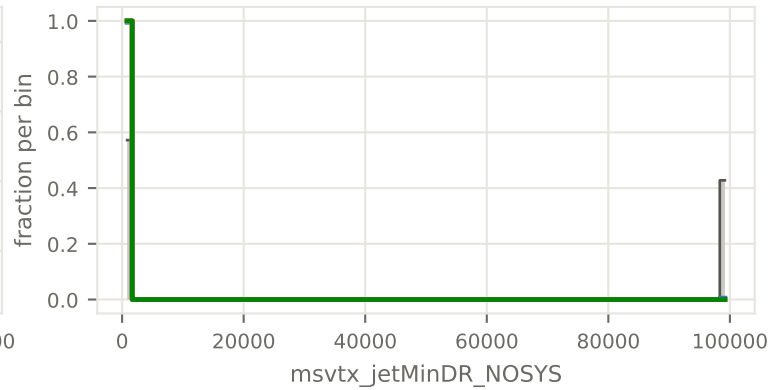
region: ENDCAP · rank 16 / 98

D_equal 0.493 (noise 0.022) D_pooled 0.512 min 0.468 (mH125_mS55) max 0.538 (mH400_mS100) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

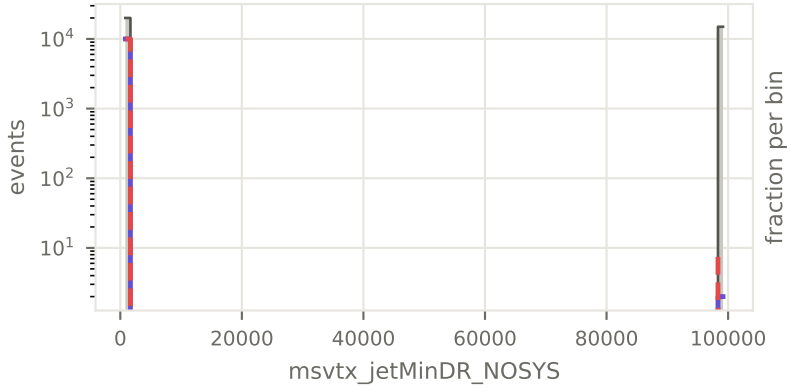


1b · each mass point vs background · normalized

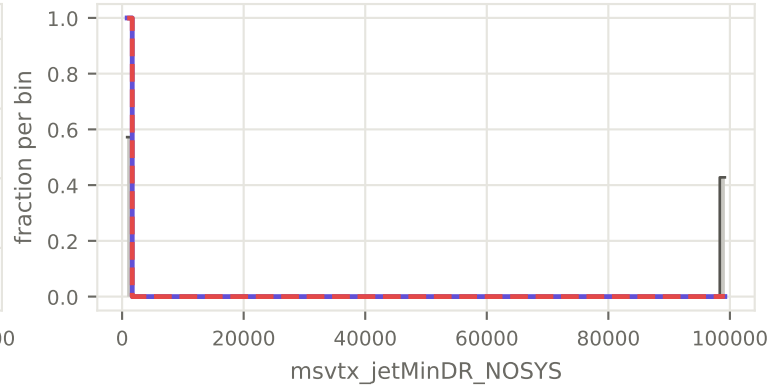


1 · each mass point
background n 34,970
mH125_mS15 D 0.469 n 411
mH125_mS55 D 0.468 n 281
mH200_mS50 D 0.514 n 1,071
mH400_mS100 D 0.538 n 2,648
mH600_mS150 D 0.502 n 2,882
mH1000_mS275 D 0.519 n 2,703

2a · combined signal vs background · raw

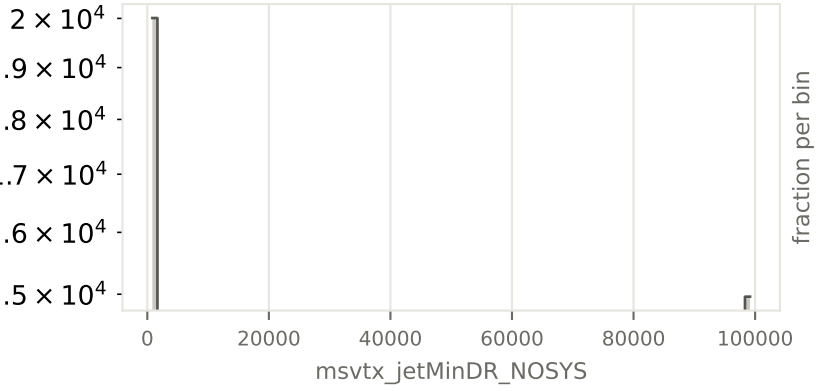


2b · combined signal vs background · normalized

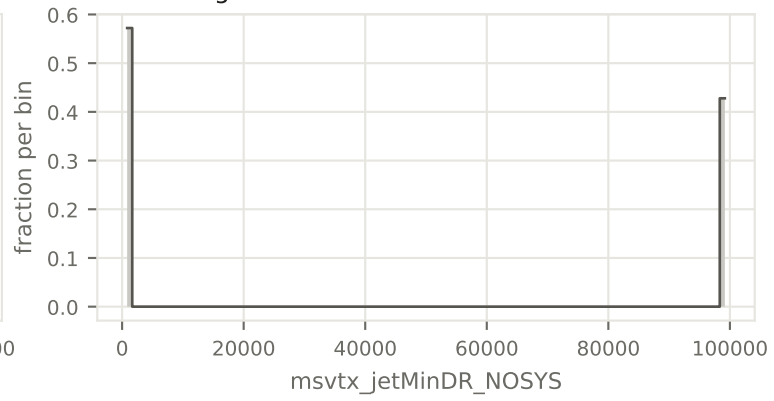


2 · combined signal
background n 34,970
combined, pooled D 0.512 n 9,996
equal weight per mass D 0.493

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

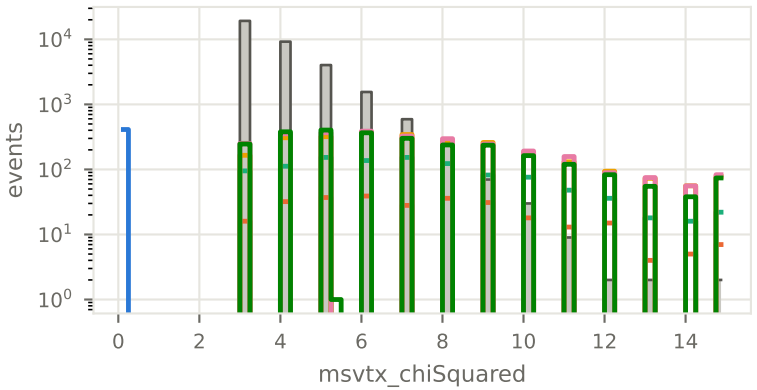
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_chiSquared

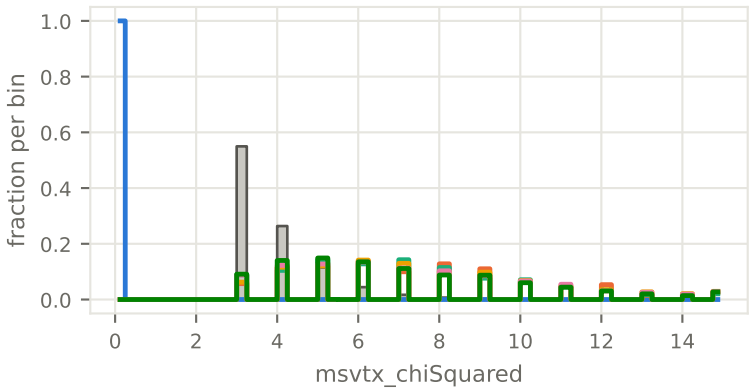
region: ENDCAP · rank 17 / 98

D_equal 0.486 (noise 0.022) D_pooled 0.576 min 0.582 (mH1000_mS275) max 1.000 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

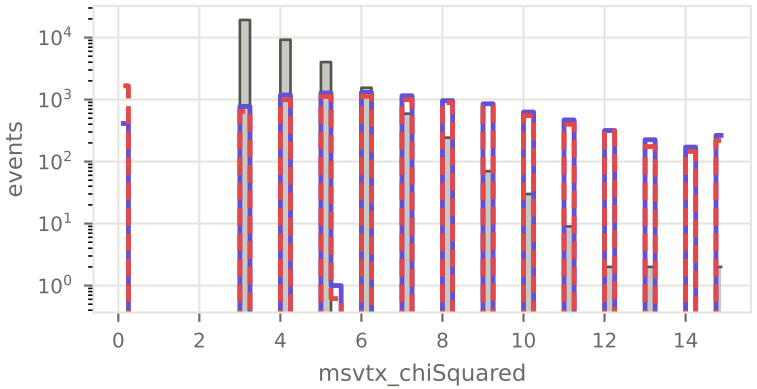


1b · each mass point vs background · normalized

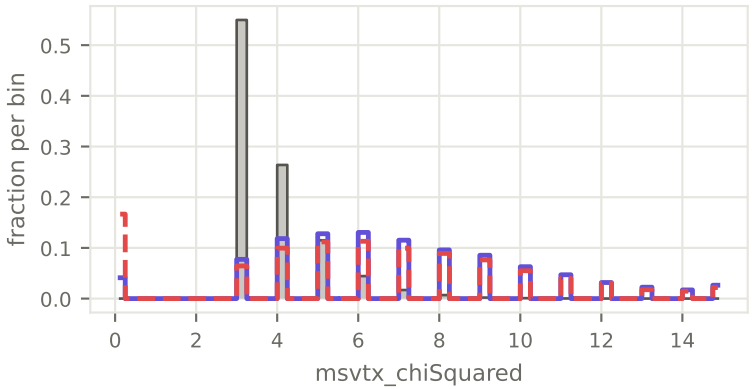


1 · each mass point
background n 34,970
mH125_mS15 D 1.000 n 411
mH125_mS55 D 0.643 n 281
mH200_mS50 D 0.620 n 1,071
mH400_mS100 D 0.635 n 2,648
mH600_mS150 D 0.603 n 2,882
mH1000_mS275 D 0.582 n 2,703

2a · combined signal vs background · raw

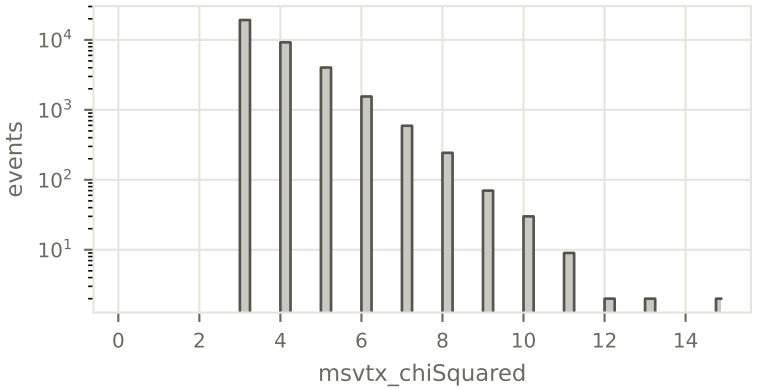


2b · combined signal vs background · normalized

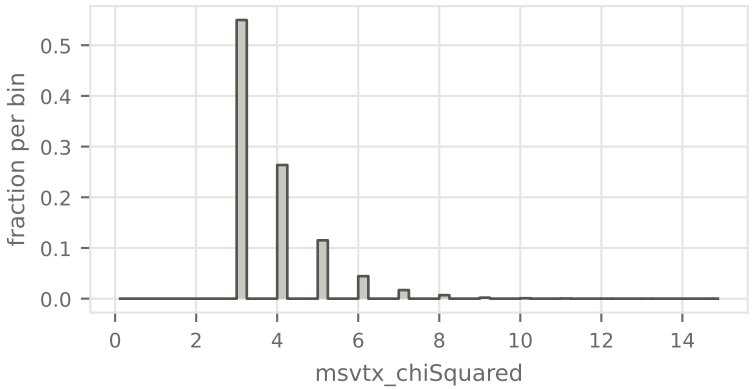


2 · combined signal
background n 34,970
combined, pooled D 0.576 n 9,996
equal weight per mass D 0.486

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

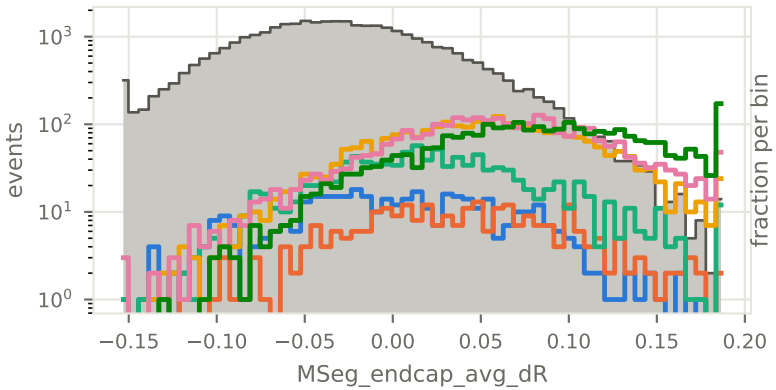
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MSeg_endcap_avg_dR

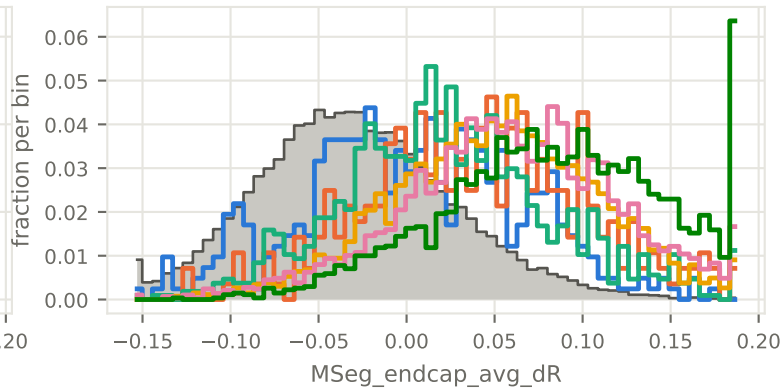
region: ENDCAP · rank 18 / 98 · in: NN2 endcap

D_equal 0.462 (noise 0.022) D_pooled 0.541 min 0.235 (mH125_mS15) max 0.667 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

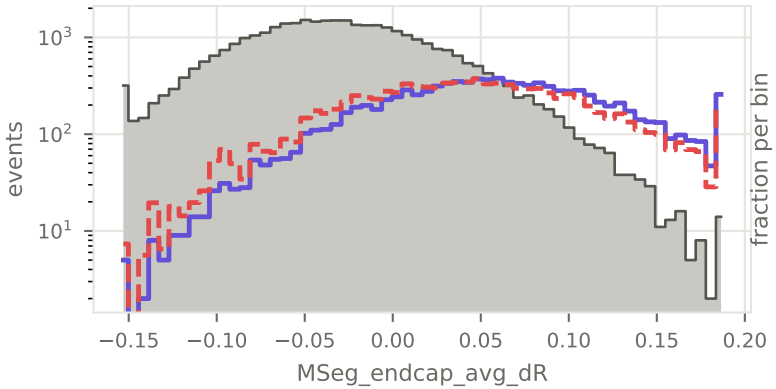


1b · each mass point vs background · normalized

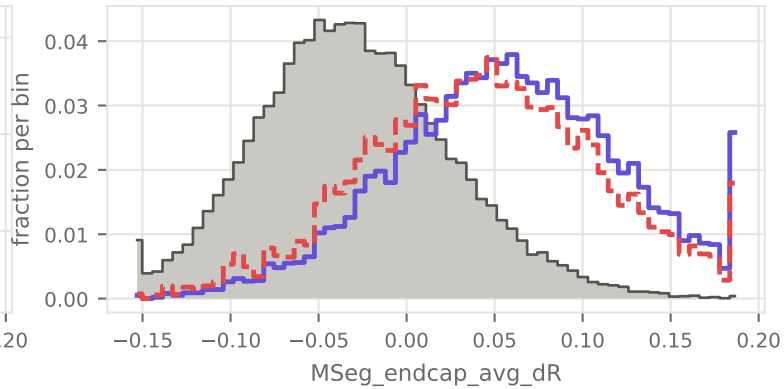


1 · each mass point
background n 34,970
mH125_mS15 D 0.235 n 411
mH125_mS55 D 0.457 n 281
mH200_mS50 D 0.369 n 1,071
mH400_mS100 D 0.532 n 2,648
mH600_mS150 D 0.577 n 2,882
mH1000_mS275 D 0.667 n 2,703

2a · combined signal vs background · raw

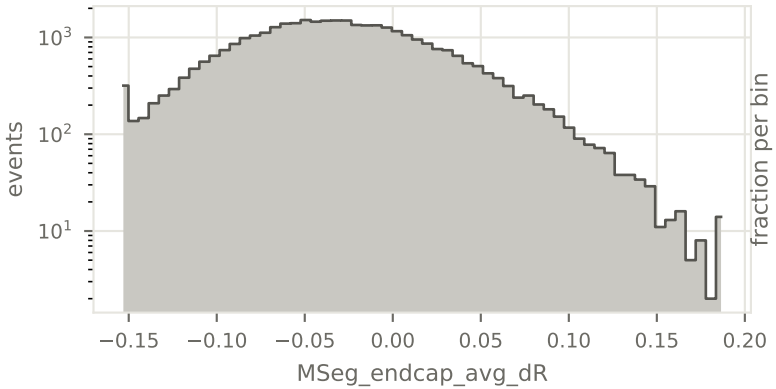


2b · combined signal vs background · normalized

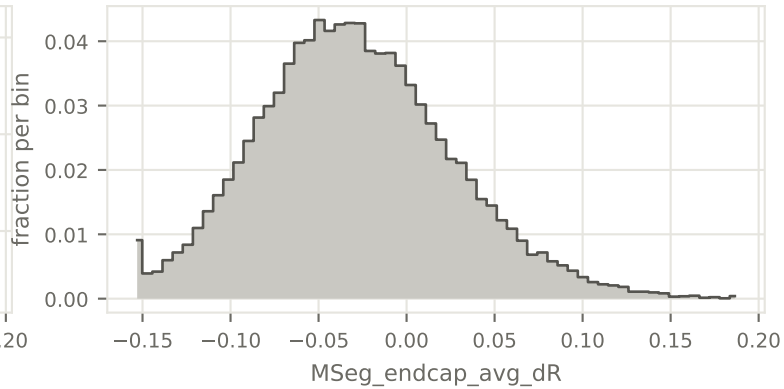


2 · combined signal
background n 34,970
combined, pooled D 0.541 n 9,996
equal weight per mass D 0.462

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

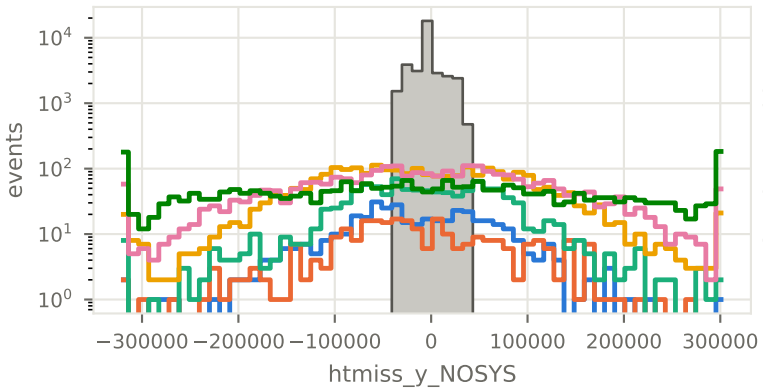
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

htmiss_y_NOSYS

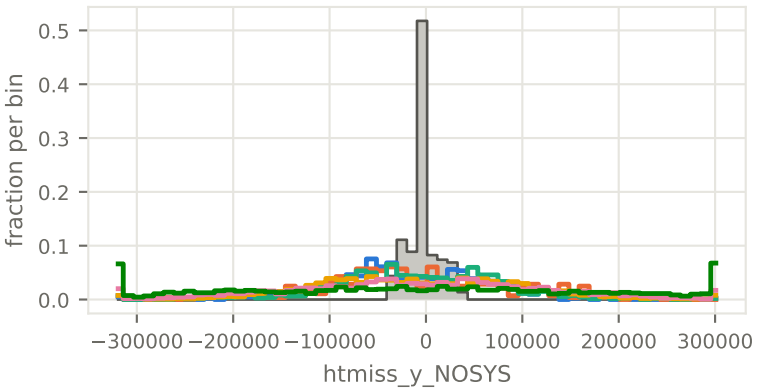
region: ENDCAP · rank 19 / 98

D_equal 0.413 (noise 0.022) D_pooled 0.417 min 0.368 (mH200_mS50) max 0.449 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

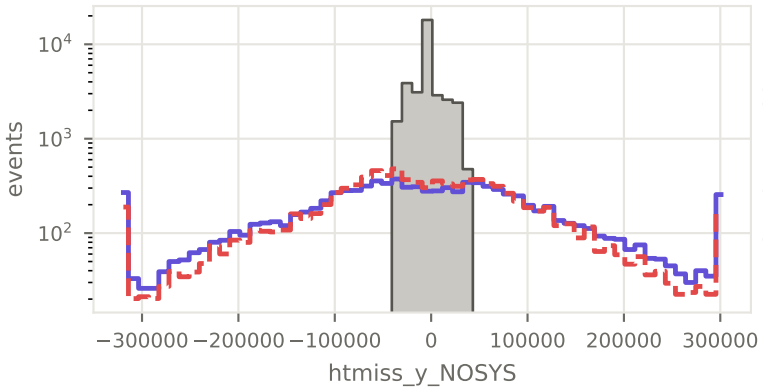


1b · each mass point vs background · normalized

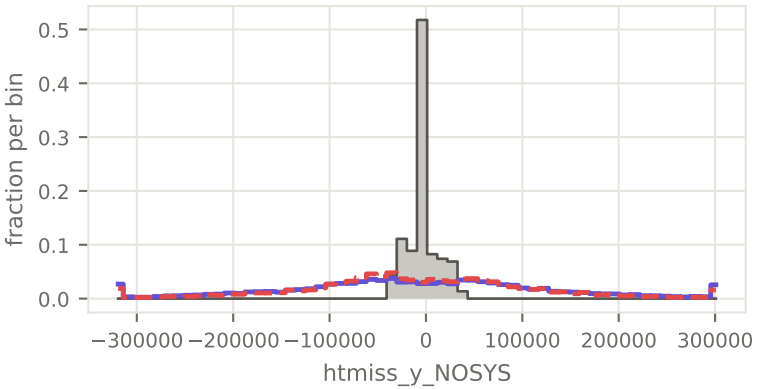


1 · each mass point
background n 34,970
mH125_mS15 D 0.440 n 411
mH125_mS55 D 0.425 n 281
mH200_mS50 D 0.368 n 1,071
mH400_mS100 D 0.420 n 2,648
mH600_mS150 D 0.403 n 2,882
mH1000_mS275 D 0.449 n 2,703

2a · combined signal vs background · raw

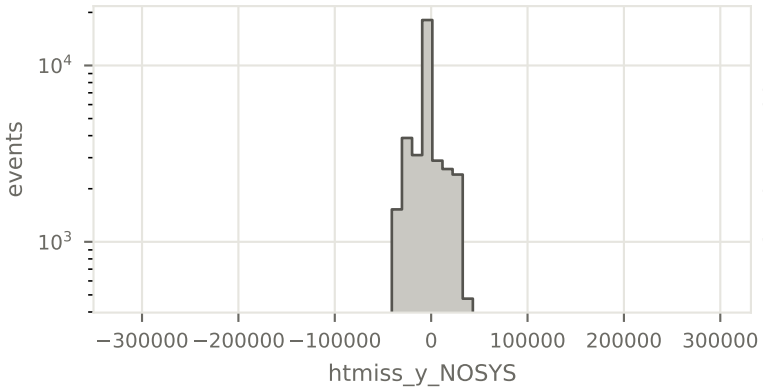


2b · combined signal vs background · normalized

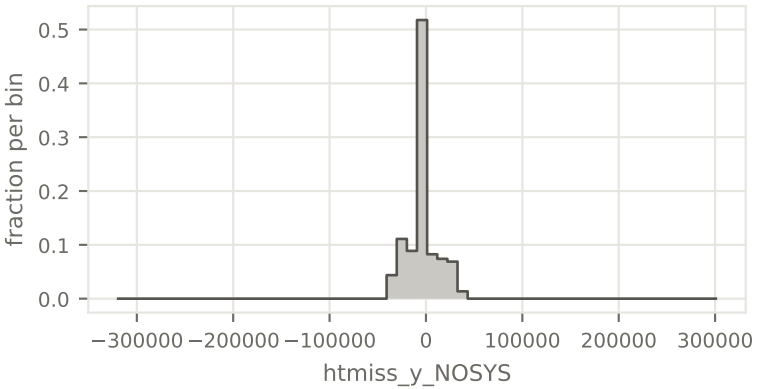


2 · combined signal
background n 34,970
combined, pooled D 0.417 n 9,996
equal weight per mass D 0.413

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

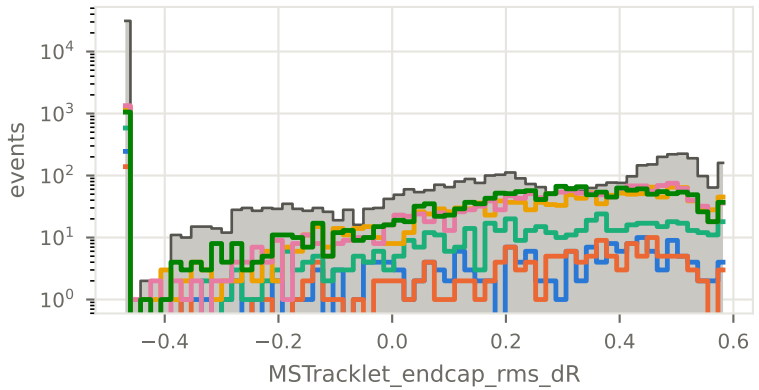
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MSTracklet_endcap_rms_dR

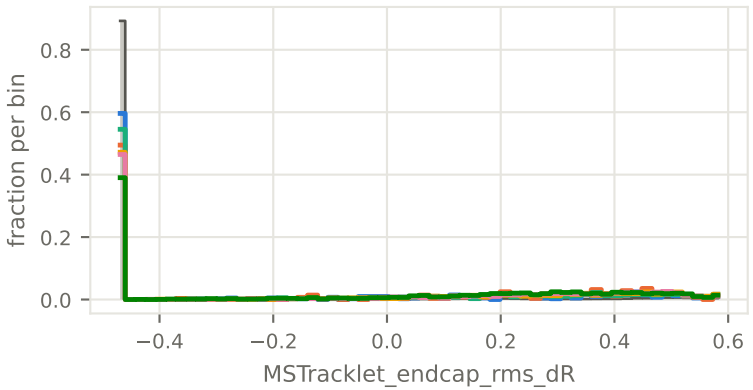
region: ENDCAP · rank 21 / 98 · in: NN2 endcap

D_equal 0.399 (noise 0.022) D_pooled 0.431 min 0.297 (mH125_mS15) max 0.502 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

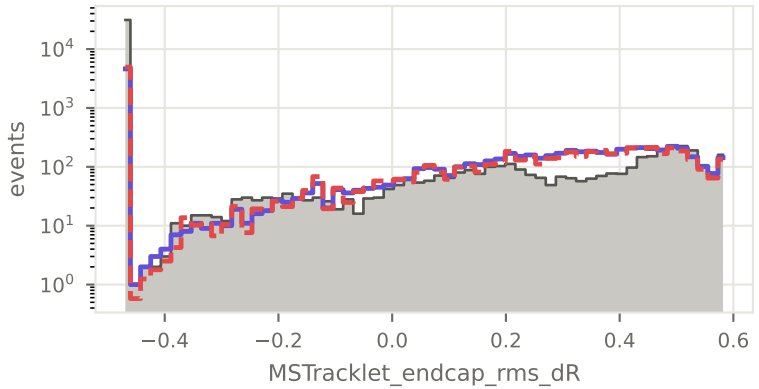


1b · each mass point vs background · normalized

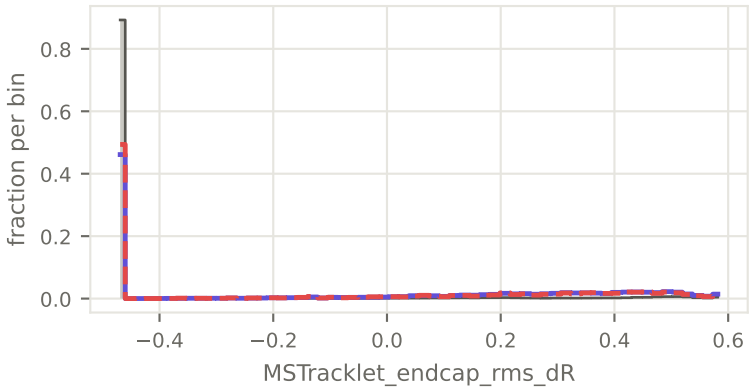


1 · each mass point
background n 34,970
mH125_mS15 D 0.297 n 411
mH125_mS55 D 0.399 n 281
mH200_mS50 D 0.349 n 1,071
mH400_mS100 D 0.421 n 2,648
mH600_mS150 D 0.428 n 2,882
mH1000_mS275 D 0.502 n 2,703

2a · combined signal vs background · raw

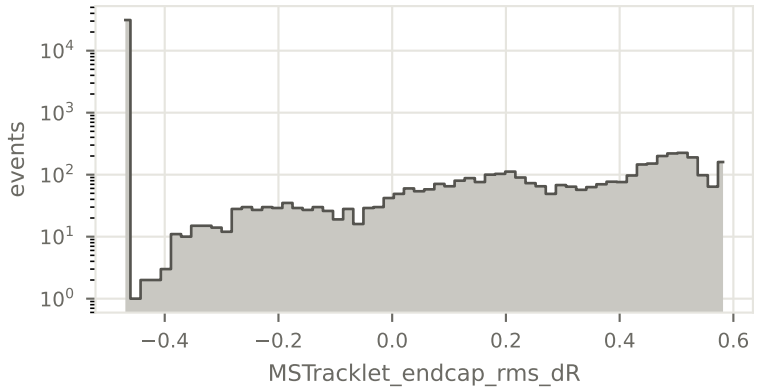


2b · combined signal vs background · normalized

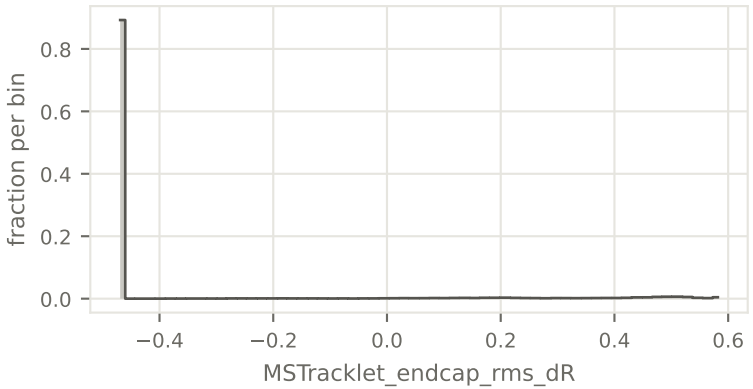


2 · combined signal
background n 34,970
combined, pooled D 0.431 n 9,996
equal weight per mass D 0.399

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

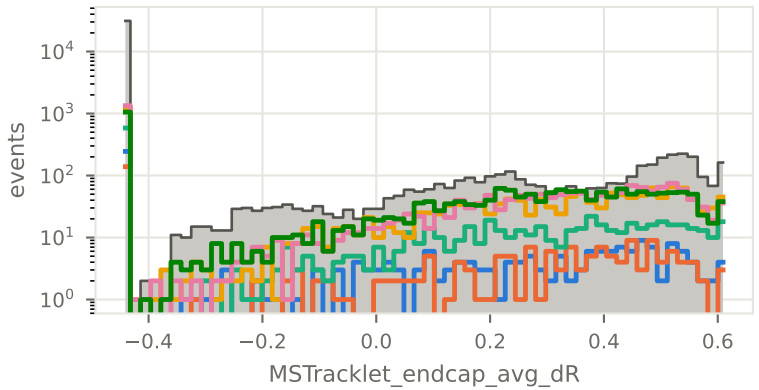
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MSTracklet_endcap_avg_dR

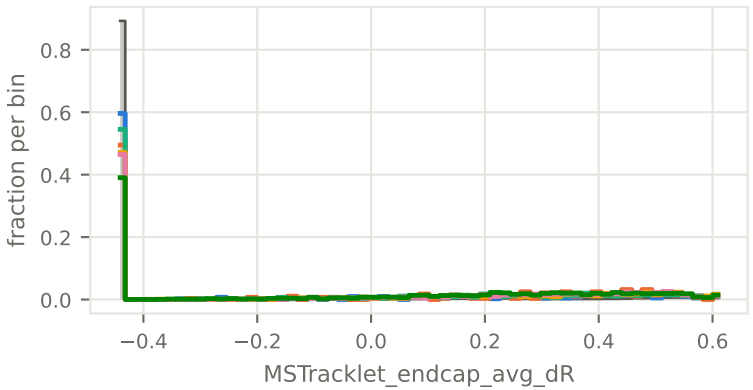
region: ENDCAP · rank 20 / 98 · in: NN2 endcap

D_equal 0.399 (noise 0.022) D_pooled 0.431 min 0.297 (mH125_mS15) max 0.502 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

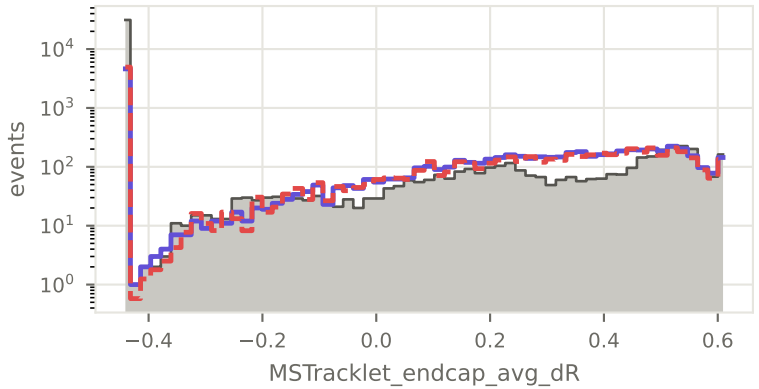


1b · each mass point vs background · normalized

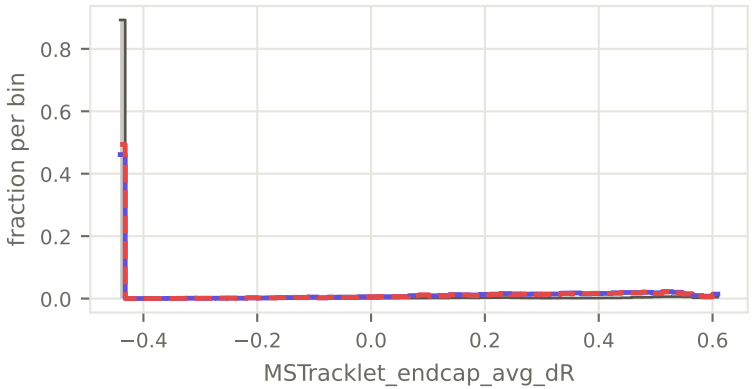


1 · each mass point
background n 34,970
mH125_mS15 D 0.297 n 411
mH125_mS55 D 0.399 n 281
mH200_mS50 D 0.349 n 1,071
mH400_mS100 D 0.421 n 2,648
mH600_mS150 D 0.428 n 2,882
mH1000_mS275 D 0.502 n 2,703

2a · combined signal vs background · raw

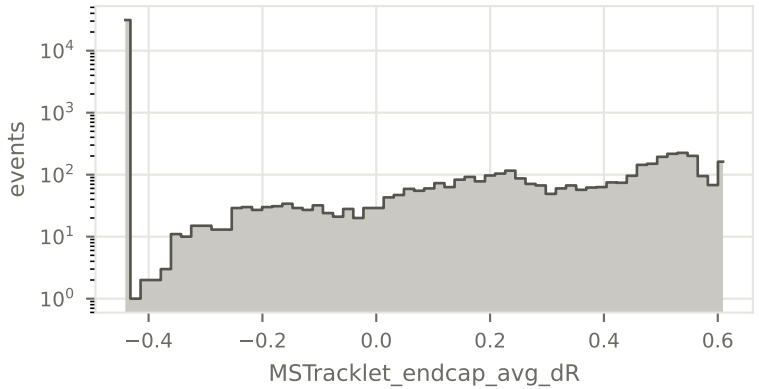


2b · combined signal vs background · normalized

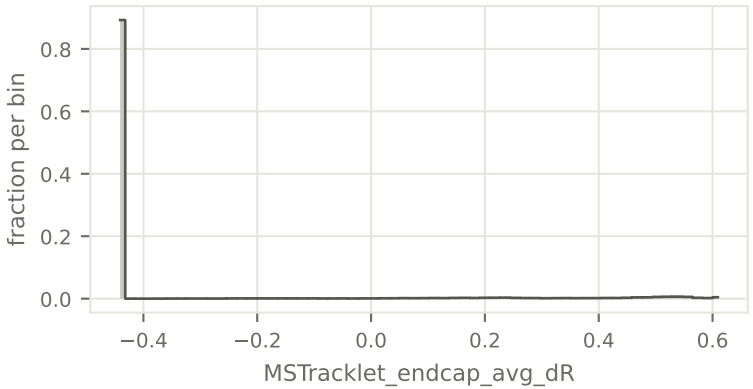


2 · combined signal
background n 34,970
combined, pooled D 0.431 n 9,996
equal weight per mass D 0.399

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

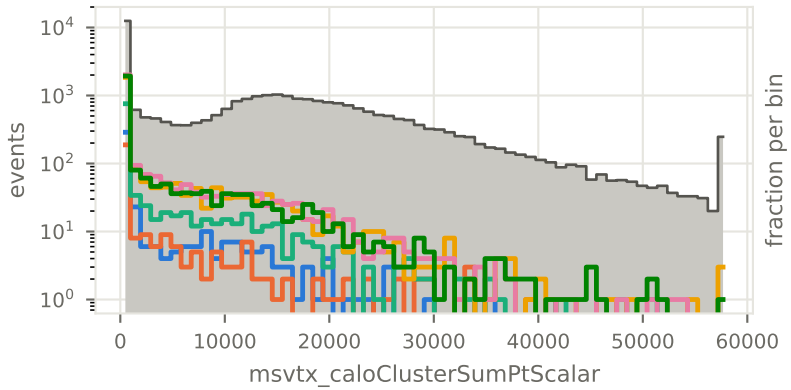
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_caloClusterSumPtScalar

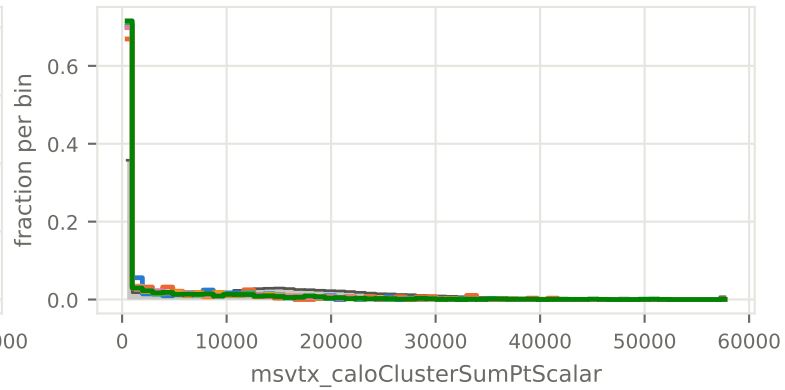
region: ENDCAP · rank 22 / 98 · in: NN1 barrel

D_equal 0.393 (noise 0.022) D_pooled 0.394 min 0.389 (mH125_mS55) max 0.404 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

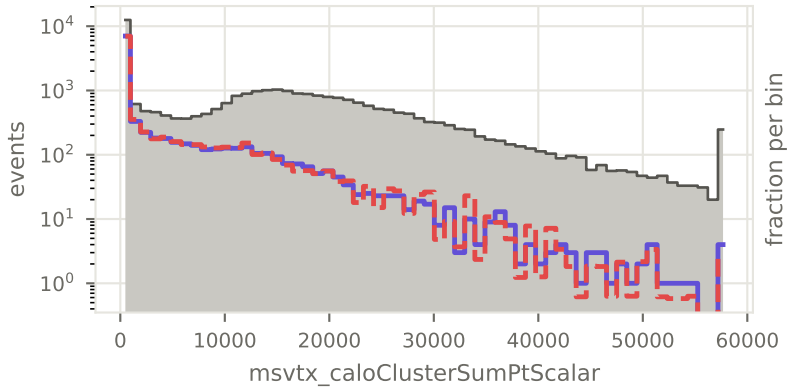


1b · each mass point vs background · normalized

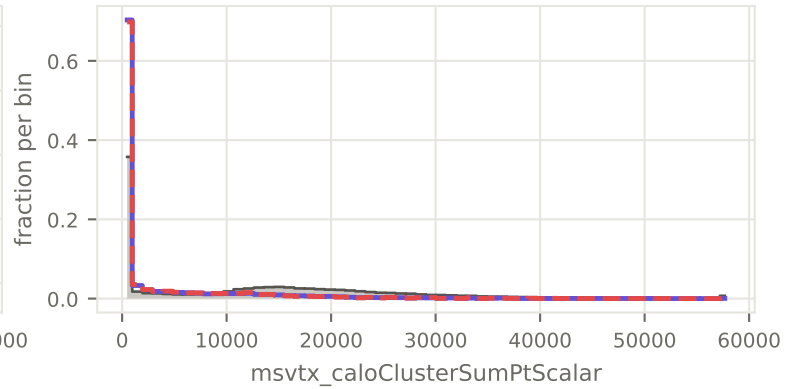


1 · each mass point
background n 34,970
mH125_mS15 D 0.404 n 411
mH125_mS55 D 0.389 n 281
mH200_mS50 D 0.396 n 1,071
mH400_mS100 D 0.390 n 2,648
mH600_mS150 D 0.395 n 2,882
mH1000_mS275 D 0.400 n 2,703

2a · combined signal vs background · raw

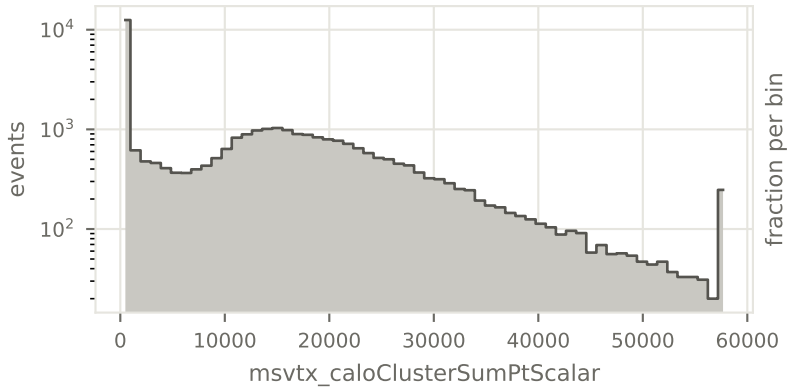


2b · combined signal vs background · normalized

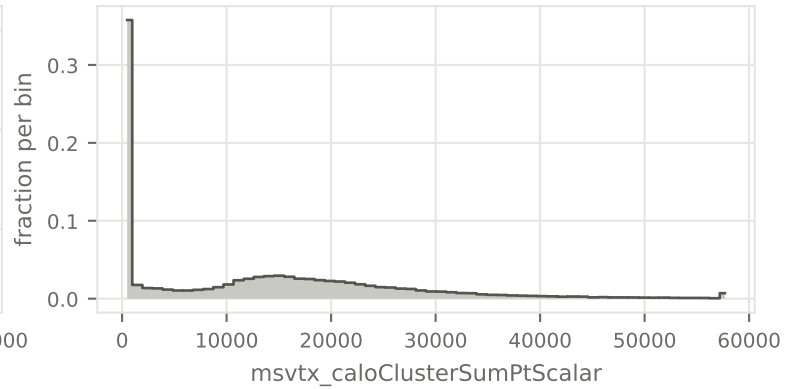


2 · combined signal
background n 34,970
combined, pooled D 0.394 n 9,996
equal weight per mass D 0.393

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

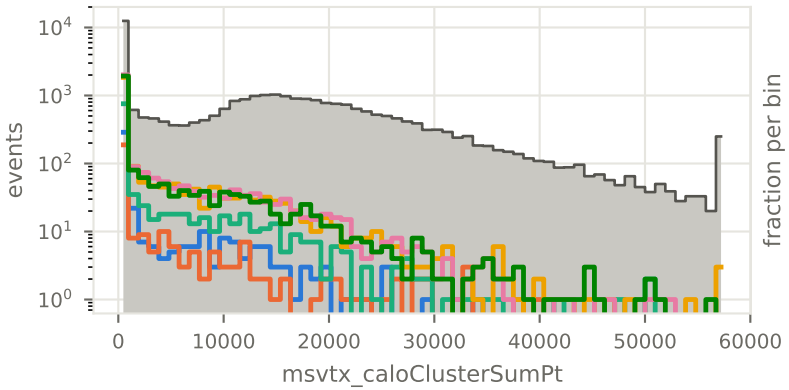
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_caloClusterSumPt

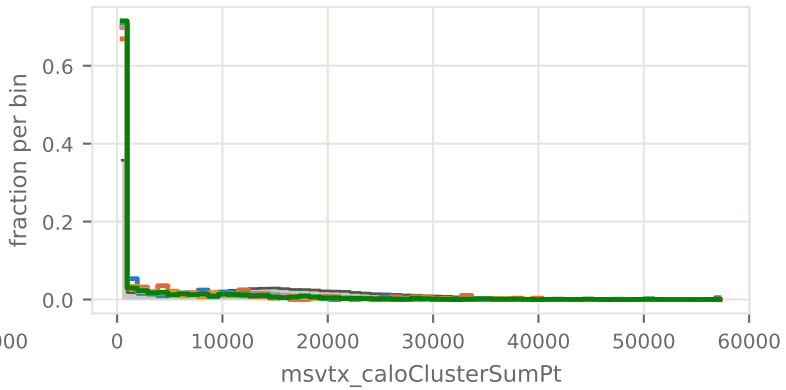
region: ENDCAP · rank 23 / 98

D_equal 0.393 (noise 0.022) D_pooled 0.393 min 0.388 (mH125_mS55) max 0.405 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

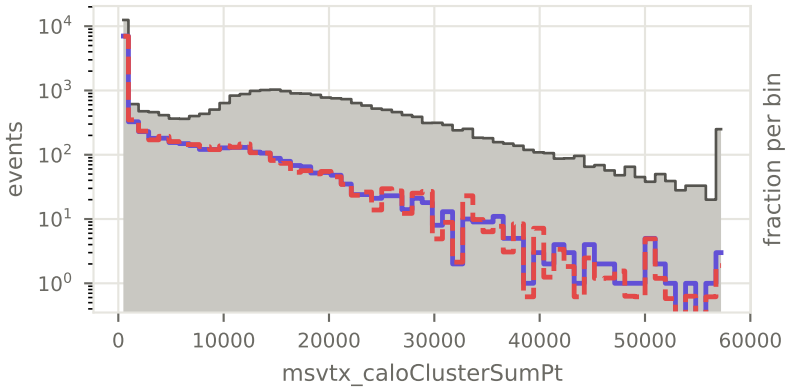


1b · each mass point vs background · normalized

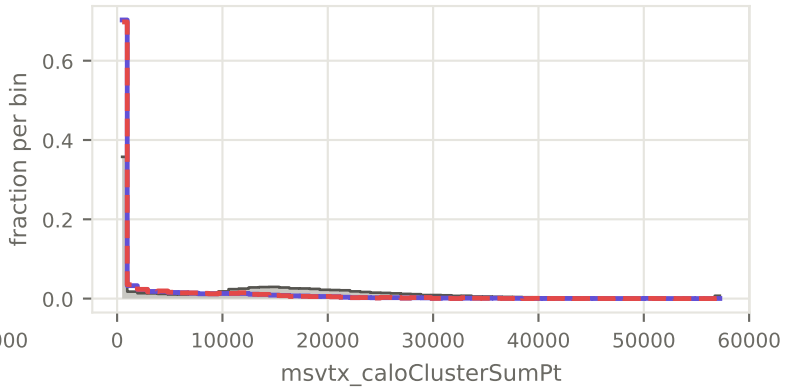


1 · each mass point
background n 34,970
mH125_mS15 D 0.405 n 411
mH125_mS55 D 0.388 n 281
mH200_mS50 D 0.395 n 1,071
mH400_mS100 D 0.391 n 2,648
mH600_mS150 D 0.395 n 2,882
mH1000_mS275 D 0.400 n 2,703

2a · combined signal vs background · raw

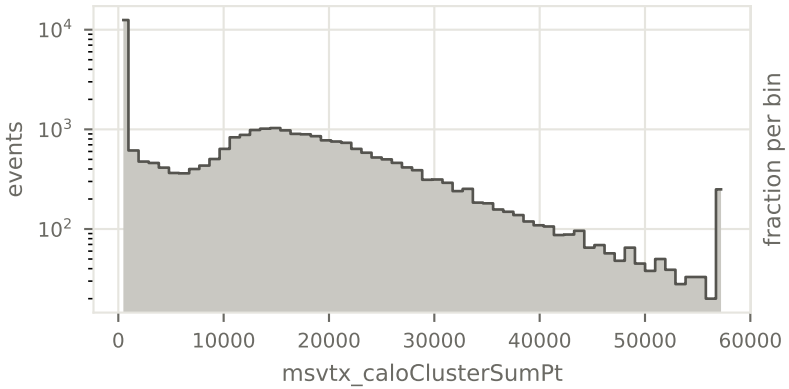


2b · combined signal vs background · normalized

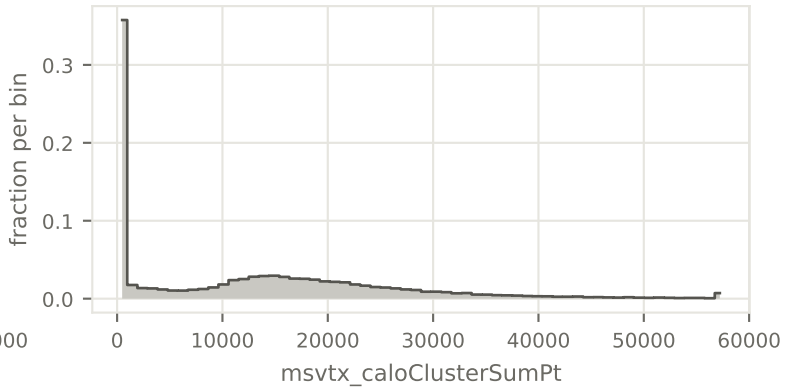


2 · combined signal
background n 34,970
combined, pooled D 0.393 n 9,996
equal weight per mass D 0.393

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

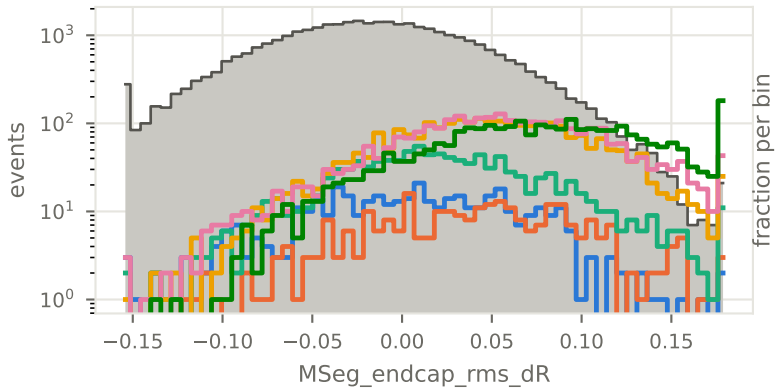
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MSeg_endcap_rms_dR

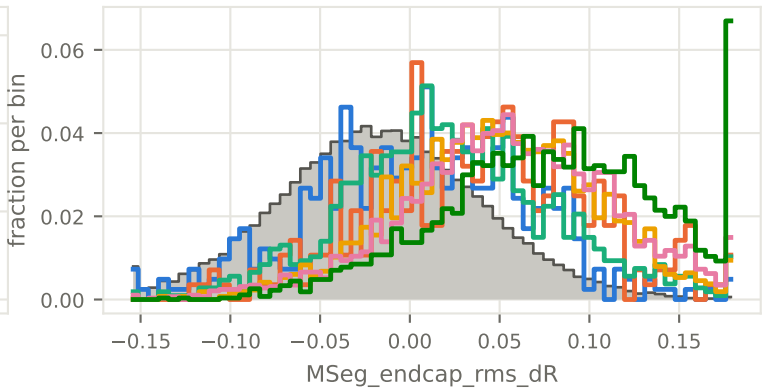
region: ENDCAP · rank 24 / 98 · in: NN2 endcap

D_equal 0.384 (noise 0.022) D_pooled 0.457 min 0.180 (mH125_mS15) max 0.594 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

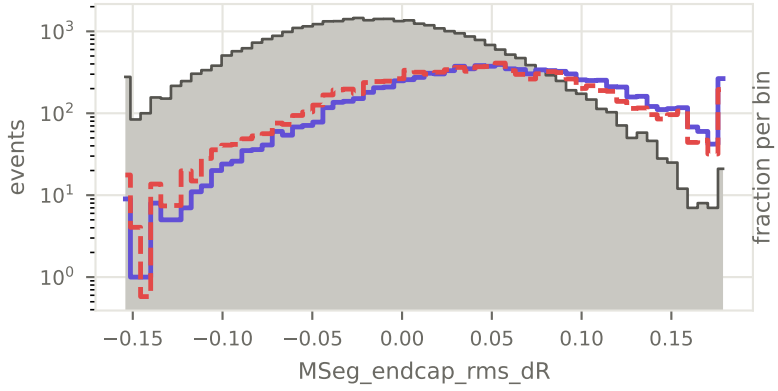


1b · each mass point vs background · normalized

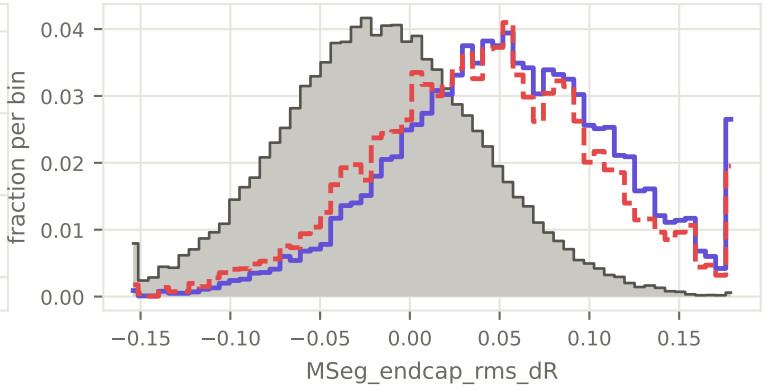


1 · each mass point
background n 34,970
mH125_mS15 D 0.180 n 411
mH125_mS55 D 0.391 n 281
mH200_mS50 D 0.276 n 1,071
mH400_mS100 D 0.447 n 2,648
mH600_mS150 D 0.487 n 2,882
mH1000_mS275 D 0.594 n 2,703

2a · combined signal vs background · raw

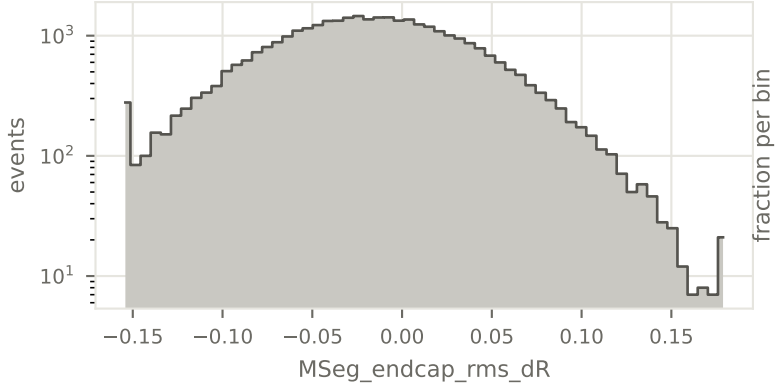


2b · combined signal vs background · normalized

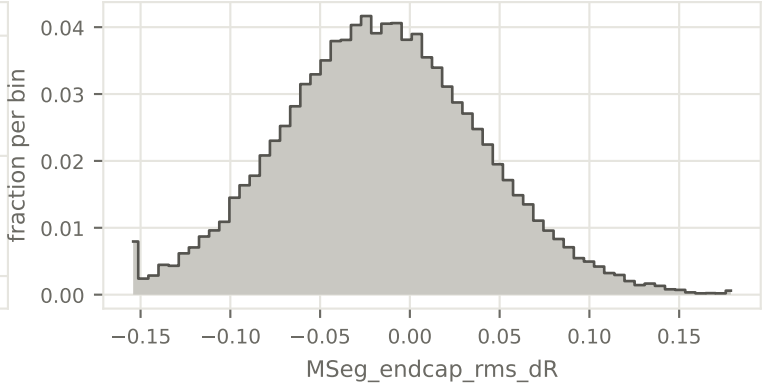


2 · combined signal
background n 34,970
combined, pooled D 0.457 n 9,996
equal weight per mass D 0.384

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

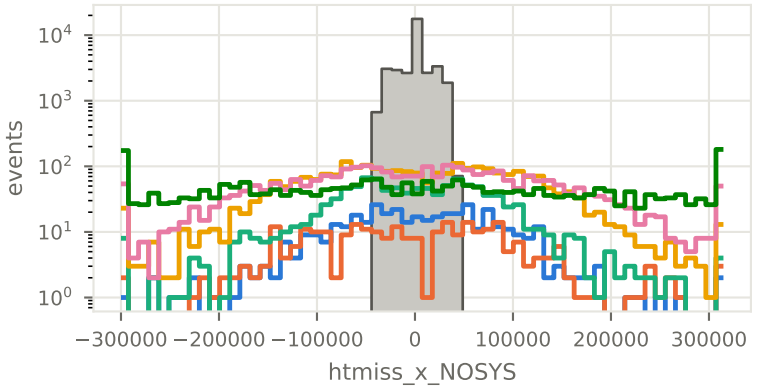
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

htmiss_x_NOSYS

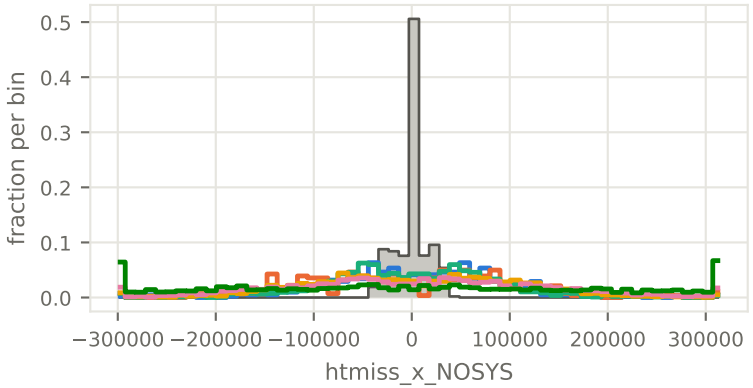
region: ENDCAP · rank 25 / 98

D_equal 0.383 (noise 0.022) D_pooled 0.398 min 0.354 (mH200_mS50) max 0.438 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

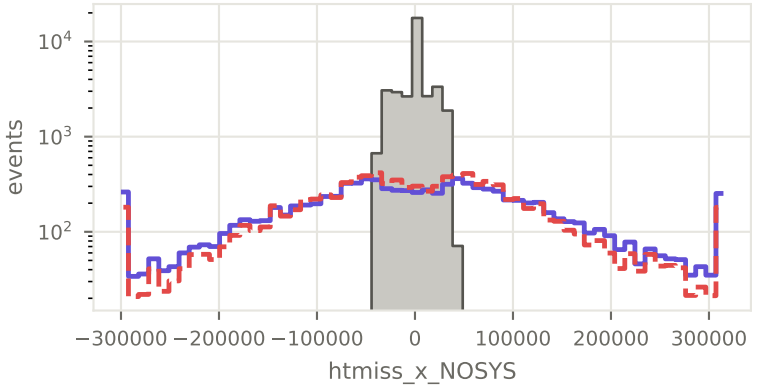


1b · each mass point vs background · normalized

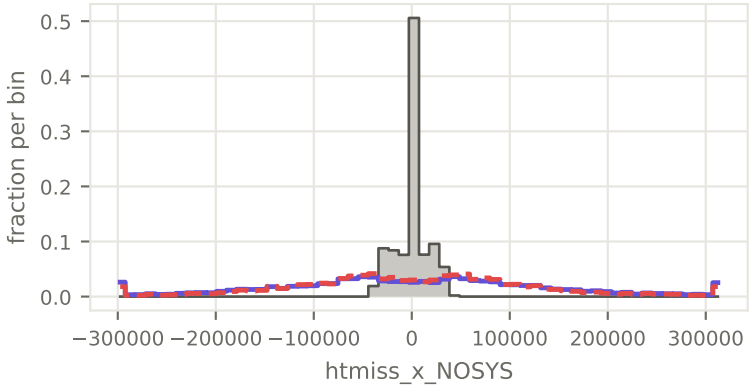


1 · each mass point
background n 34,970
mH125_mS15 D 0.370 n 411
mH125_mS55 D 0.401 n 281
mH200_mS50 D 0.354 n 1,071
mH400_mS100 D 0.397 n 2,648
mH600_mS150 D 0.400 n 2,882
mH1000_mS275 D 0.438 n 2,703

2a · combined signal vs background · raw

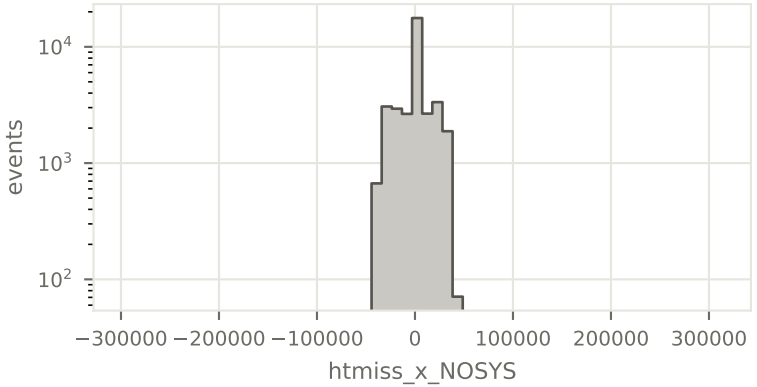


2b · combined signal vs background · normalized

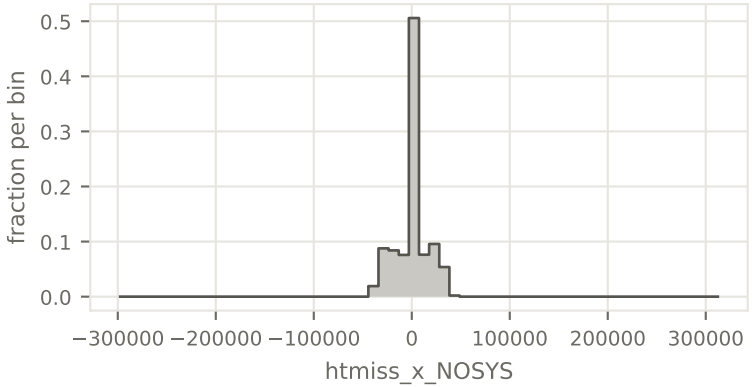


2 · combined signal
background n 34,970
combined, pooled D 0.398 n 9,996
equal weight per mass D 0.383

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

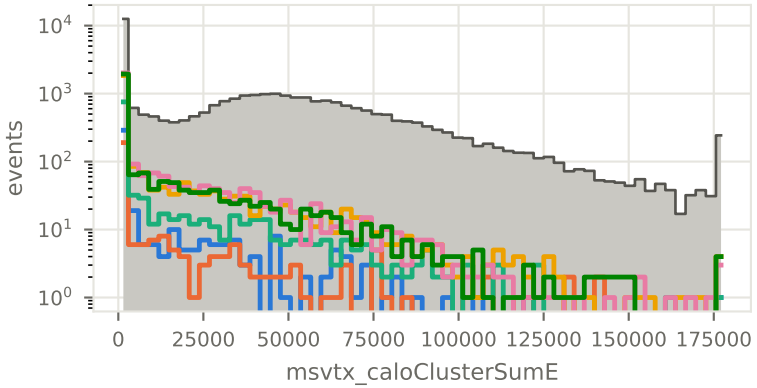
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_caloClusterSumE

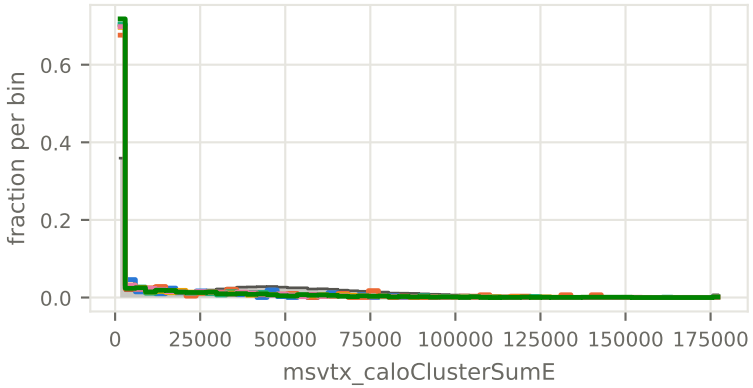
region: ENDCAP · rank 26 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.383 (noise 0.022) D_pooled 0.386 min 0.372 (mH125_mS55) max 0.396 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

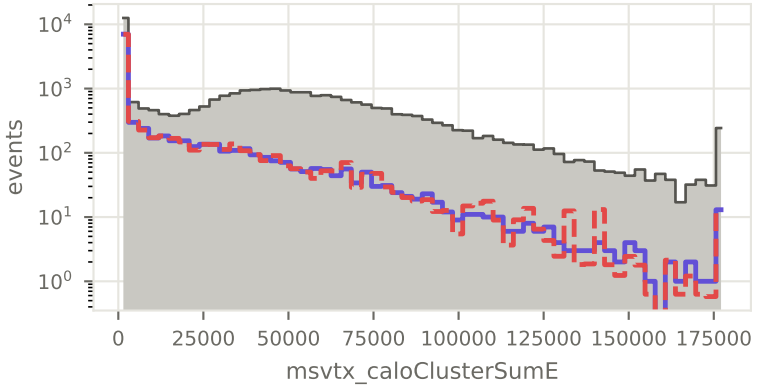


1b · each mass point vs background · normalized

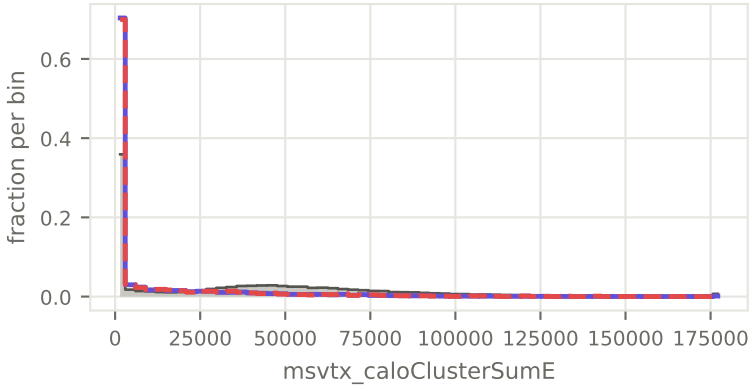


1 · each mass point
background n 34,970
mH125_mS15 D 0.392 n 411
mH125_mS55 D 0.372 n 281
mH200_mS50 D 0.381 n 1,071
mH400_mS100 D 0.379 n 2,648
mH600_mS150 D 0.389 n 2,882
mH1000_mS275 D 0.396 n 2,703

2a · combined signal vs background · raw

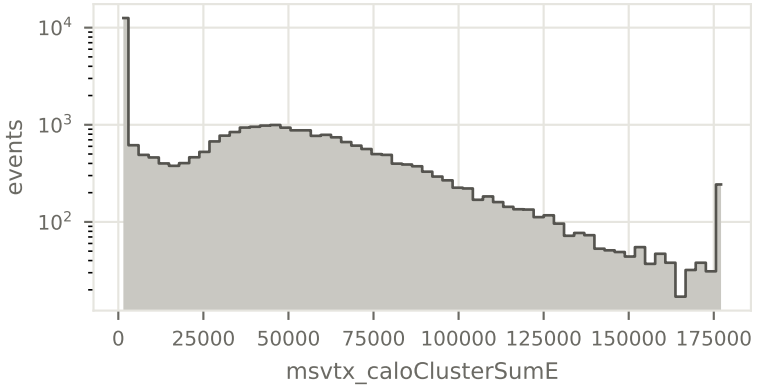


2b · combined signal vs background · normalized

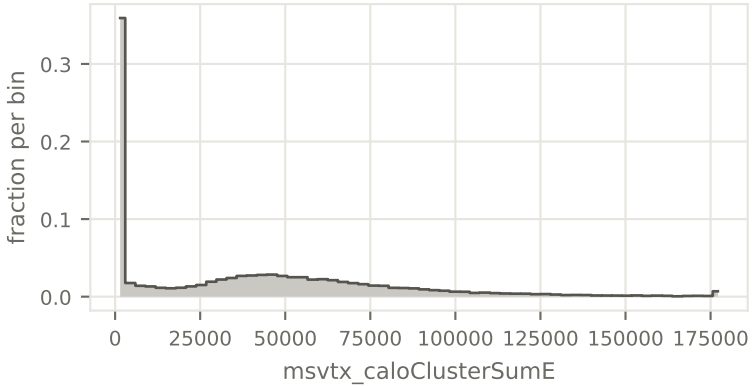


2 · combined signal
background n 34,970
combined, pooled D 0.386 n 9,996
equal weight per mass D 0.383

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

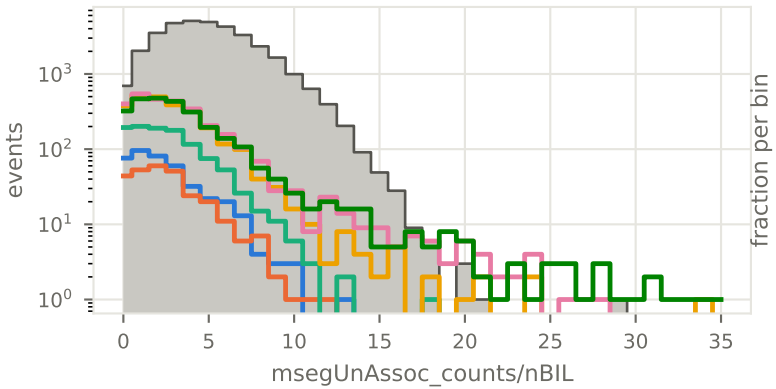
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msegUnAssoc_counts/nBIL

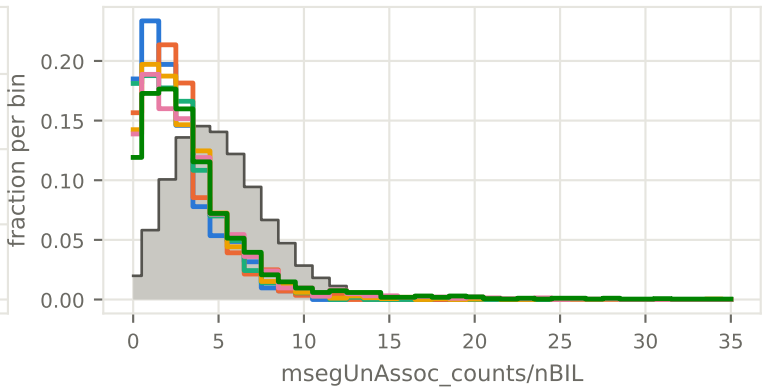
region: ENDCAP · rank 27 / 98 · in: NN1 barrel

D_equal 0.378 (noise 0.022) D_pooled 0.346 min 0.314 (mH1000_mS275) max 0.447 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

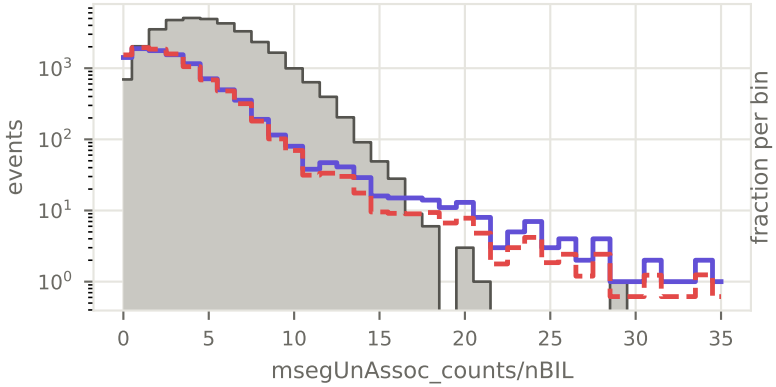


1b · each mass point vs background · normalized

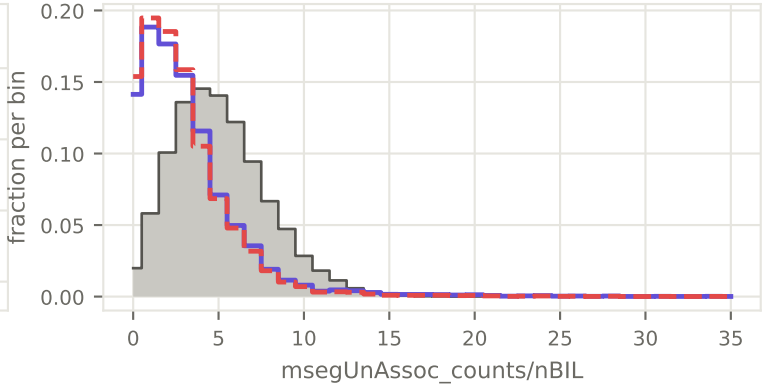


1 · each mass point
background n 34,970
mH125_mS15 D 0.447 n 411
mH125_mS55 D 0.426 n 281
mH200_mS50 D 0.398 n 1,071
mH400_mS100 D 0.359 n 2,648
mH600_mS150 D 0.325 n 2,882
mH1000_mS275 D 0.314 n 2,703

2a · combined signal vs background · raw

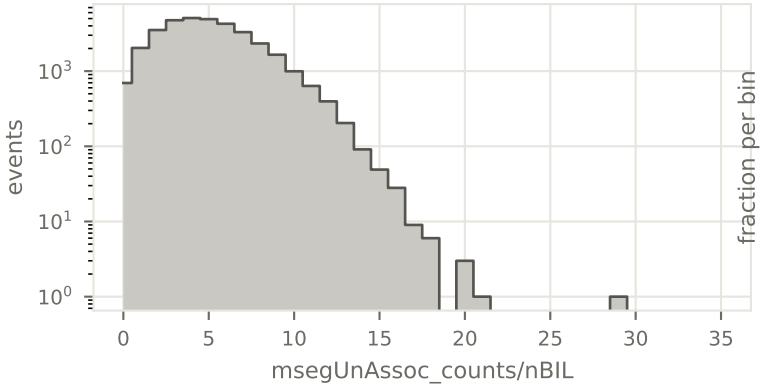


2b · combined signal vs background · normalized

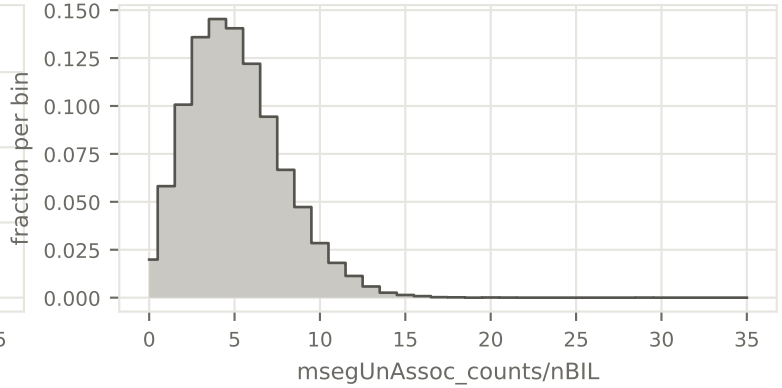


2 · combined signal
background n 34,970
combined, pooled D 0.346 n 9,996
equal weight per mass D 0.378

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

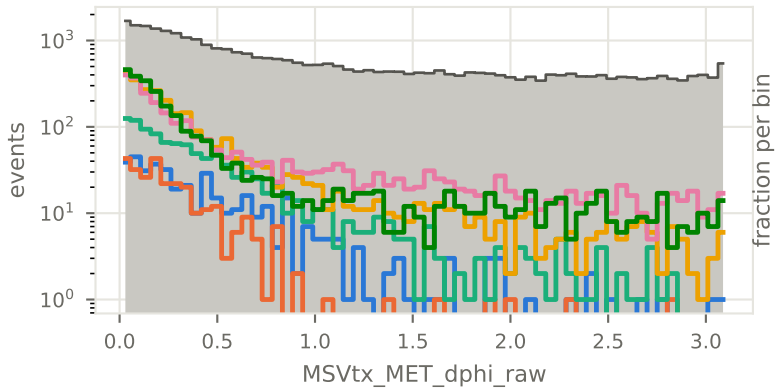
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MSVtx_MET_dphi_raw

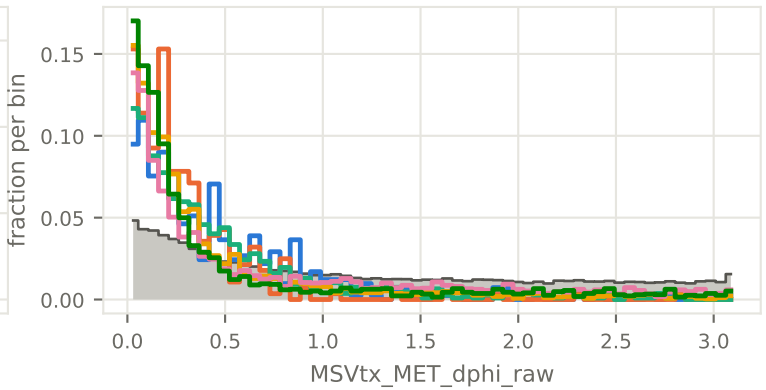
region: ENDCAP · rank 28 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.375 (noise 0.022) D_pooled 0.357 min 0.273 (mH600_mS150) max 0.508 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

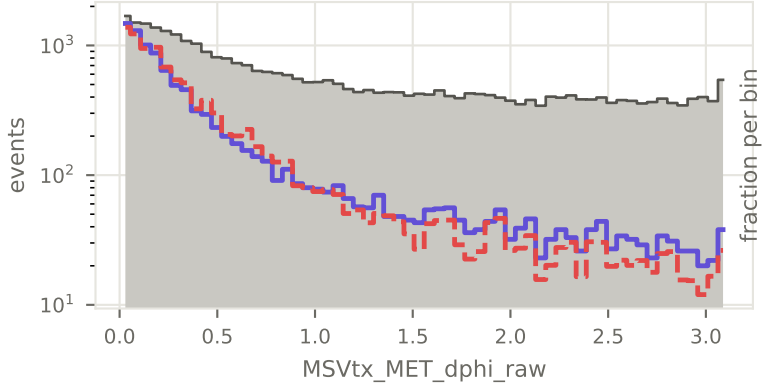


1b · each mass point vs background · normalized

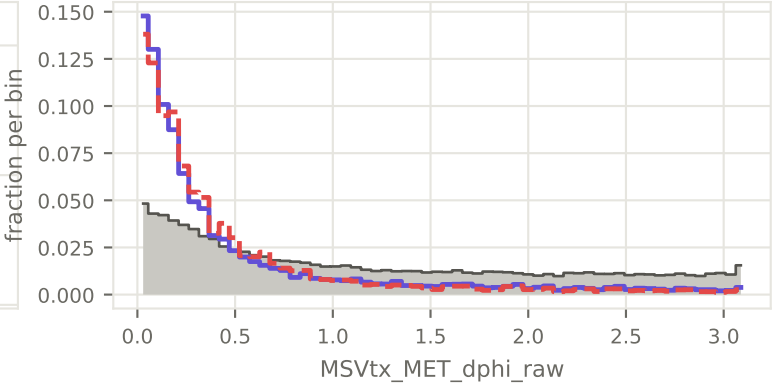


1 · each mass point
background n 34,970
mH125_mS15 D 0.377 n 411
mH125_mS55 D 0.508 n 281
mH200_mS50 D 0.377 n 1,071
mH400_mS100 D 0.411 n 2,648
mH600_mS150 D 0.273 n 2,882
mH1000_mS275 D 0.408 n 2,703

2a · combined signal vs background · raw

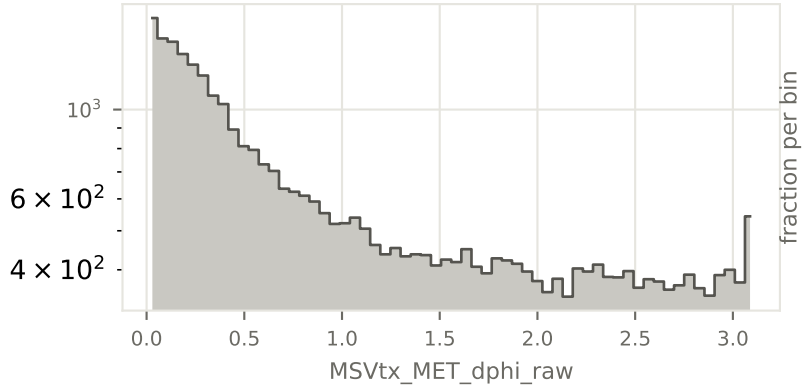


2b · combined signal vs background · normalized

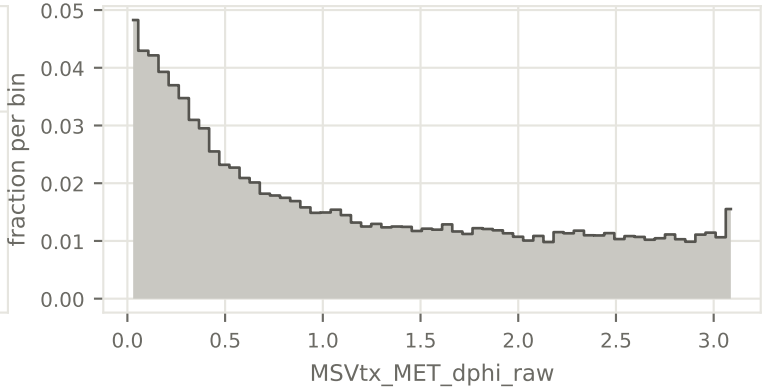


2 · combined signal
background n 34,970
combined, pooled D 0.357 n 9,996
equal weight per mass D 0.375

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

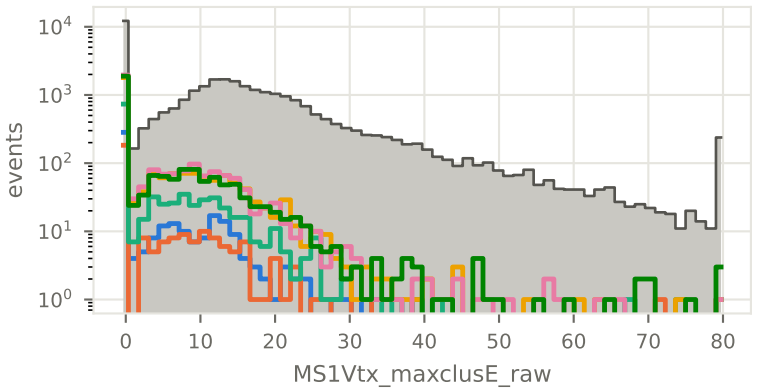
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_maxclusE_raw

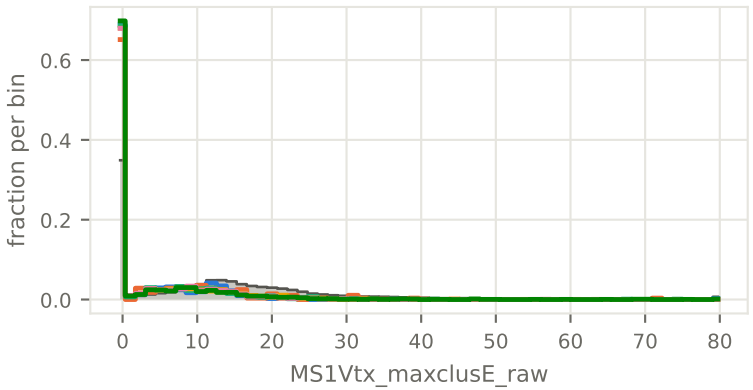
region: ENDCAP · rank 29 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.374 (noise 0.022) D_pooled 0.378 min 0.351 (mH125_mS55) max 0.386 (mH200_mS50) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

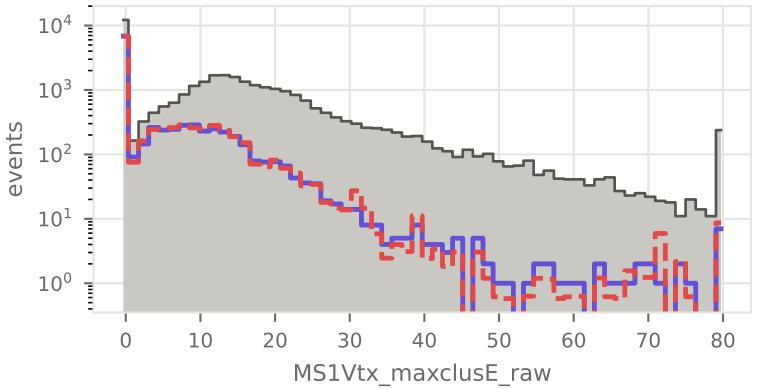


1b · each mass point vs background · normalized

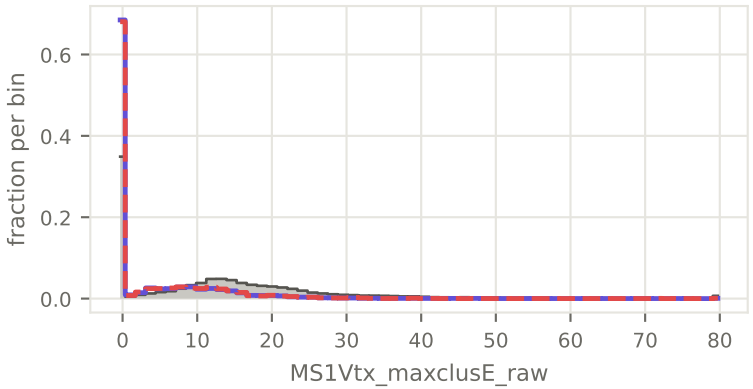


1 · each mass point
background n 34,970
mH125_mS15 D 0.386 n 411
mH125_mS55 D 0.351 n 281
mH200_mS50 D 0.386 n 1,071
mH400_mS100 D 0.375 n 2,648
mH600_mS150 D 0.376 n 2,882
mH1000_mS275 D 0.386 n 2,703

2a · combined signal vs background · raw

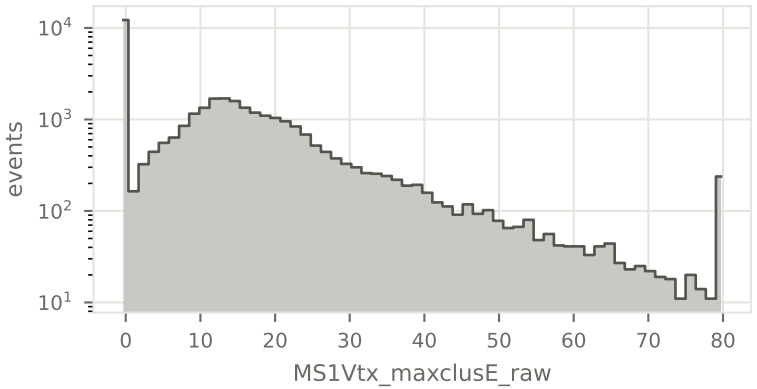


2b · combined signal vs background · normalized

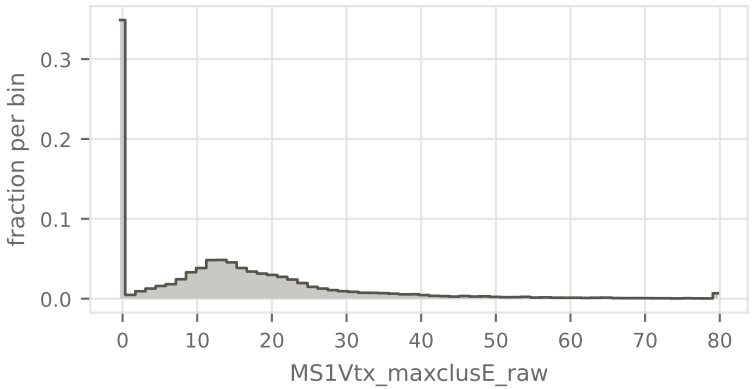


2 · combined signal
background n 34,970
combined, pooled D 0.378 n 9,996
equal weight per mass D 0.374

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

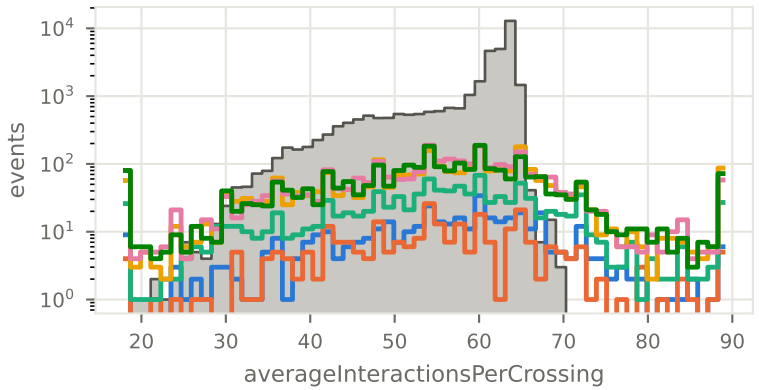
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

averageInteractionsPerCrossing

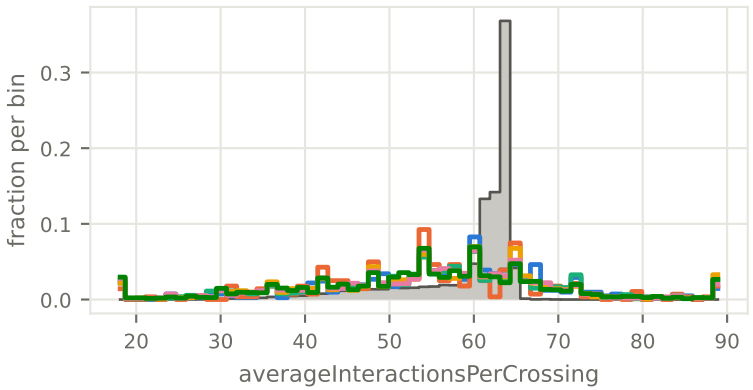
region: ENDCAP · rank 30 / 98

D_equal 0.368 (noise 0.022) D_pooled 0.373 min 0.338 (mH125_mS15) max 0.391 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

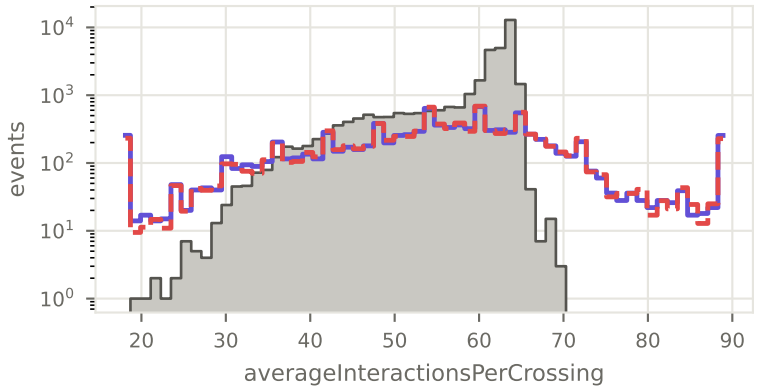


1b · each mass point vs background · normalized

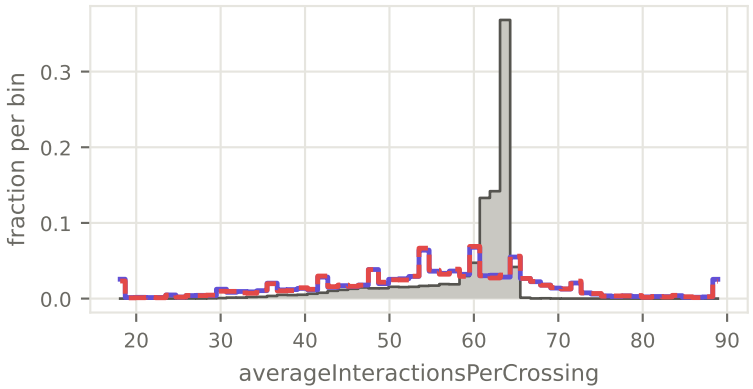


1 · each mass point
background n 34,970
mH125_mS15 D 0.338 n 411
mH125_mS55 D 0.388 n 281
mH200_mS50 D 0.367 n 1,071
mH400_mS100 D 0.355 n 2,648
mH600_mS150 D 0.380 n 2,882
mH1000_mS275 D 0.391 n 2,703

2a · combined signal vs background · raw

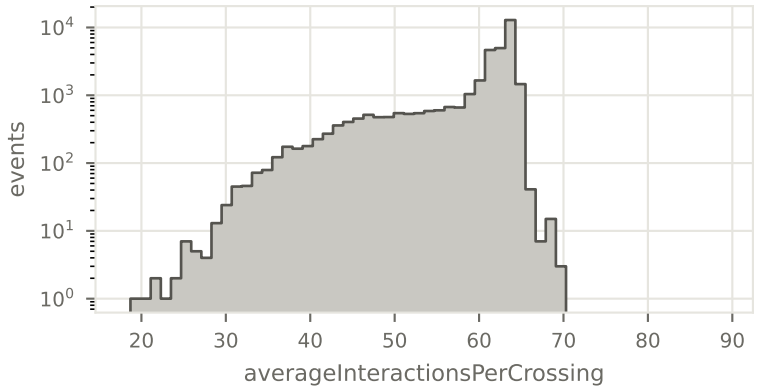


2b · combined signal vs background · normalized

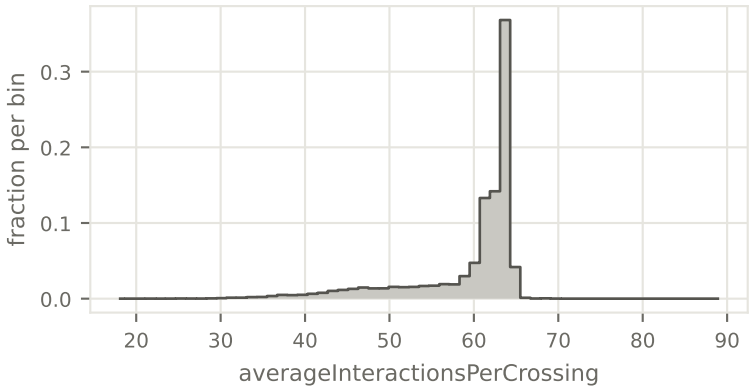


2 · combined signal
background n 34,970
combined, pooled D 0.373 n 9,996
equal weight per mass D 0.368

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

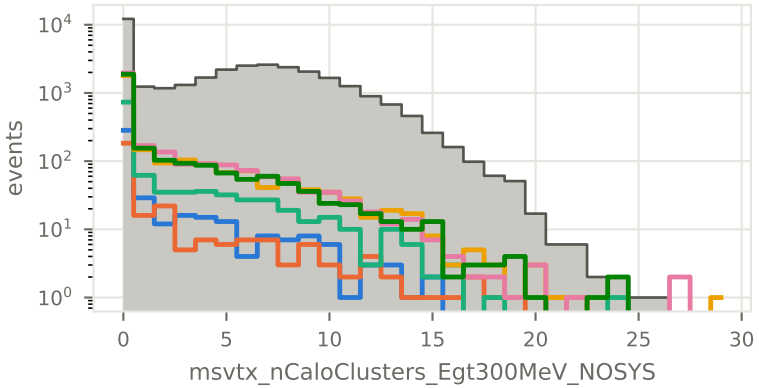
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nCaloClusters_Egt300MeV_NOSYS

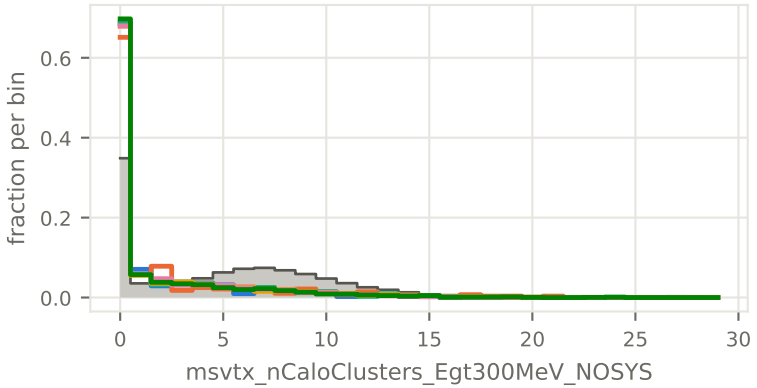
region: ENDCAP · rank 31 / 98

D_equal 0.366 (noise 0.022) D_pooled 0.365 min 0.355 (mH400_mS100) max 0.375 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

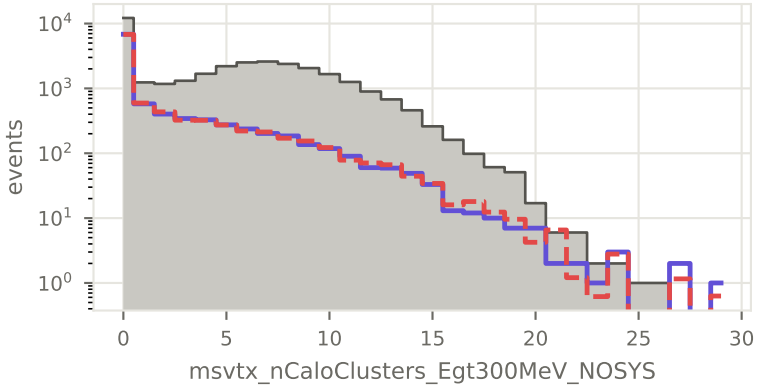


1b · each mass point vs background · normalized

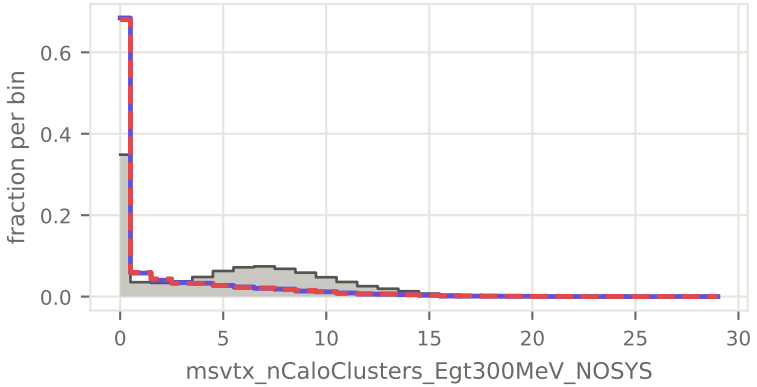


1 · each mass point
background n 34,970
mH125_mS15 D 0.375 n 411
mH125_mS55 D 0.369 n 281
mH200_mS50 D 0.359 n 1,071
mH400_mS100 D 0.355 n 2,648
mH600_mS150 D 0.368 n 2,882
mH1000_mS275 D 0.375 n 2,703

2a · combined signal vs background · raw

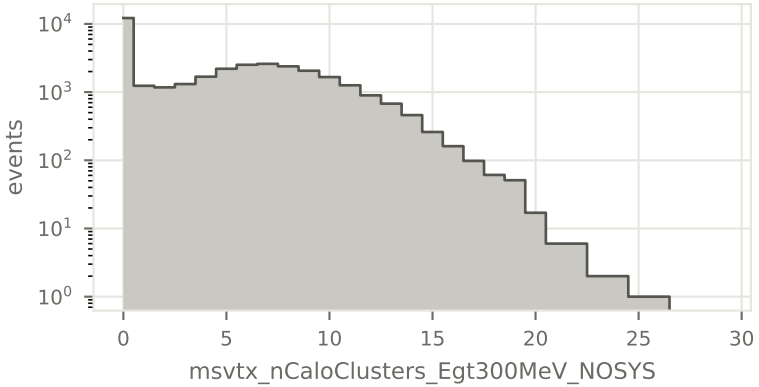


2b · combined signal vs background · normalized

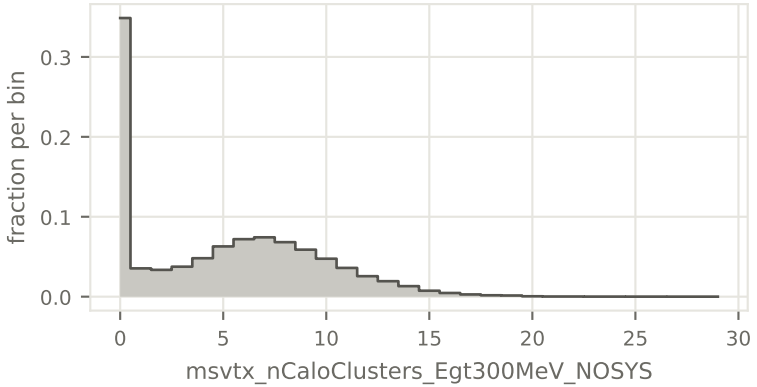


2 · combined signal
background n 34,970
combined, pooled D 0.365 n 9,996
equal weight per mass D 0.366

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

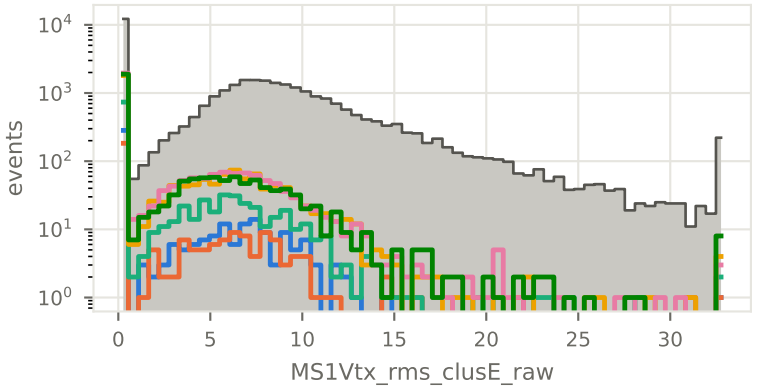
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_rms_clusE_raw

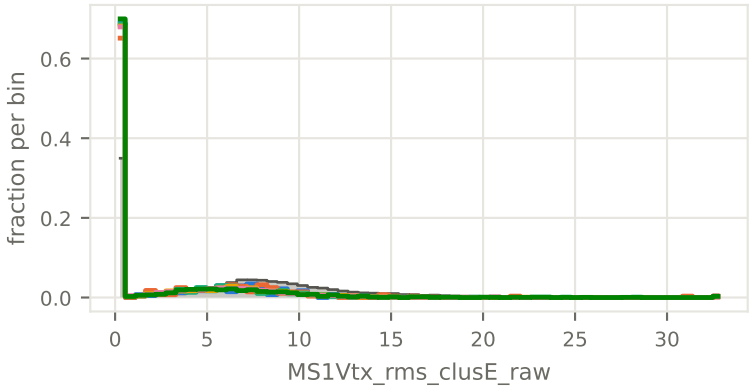
region: ENDCAP · rank 32 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.365 (noise 0.022) D_pooled 0.372 min 0.342 (mH125_mS55) max 0.384 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

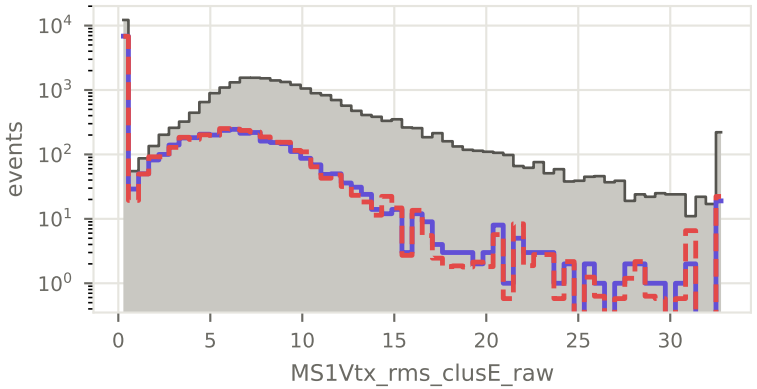


1b · each mass point vs background · normalized

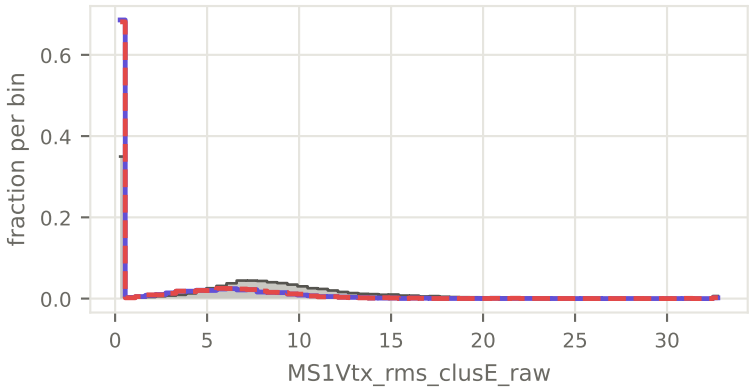


1 · each mass point
background n 34,970
mH125_mS15 D 0.362 n 411
mH125_mS55 D 0.342 n 281
mH200_mS50 D 0.372 n 1,071
mH400_mS100 D 0.368 n 2,648
mH600_mS150 D 0.373 n 2,882
mH1000_mS275 D 0.384 n 2,703

2a · combined signal vs background · raw

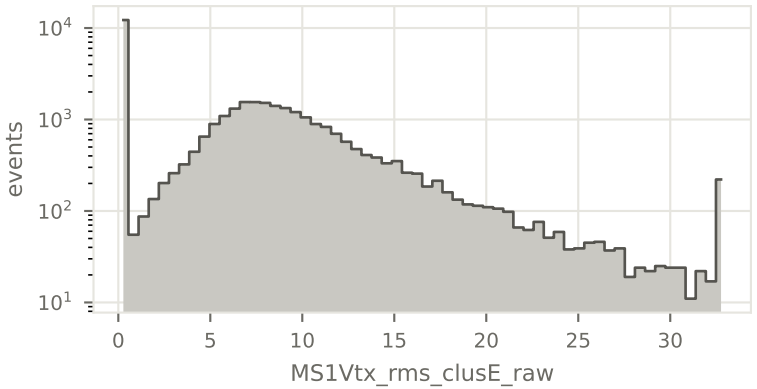


2b · combined signal vs background · normalized

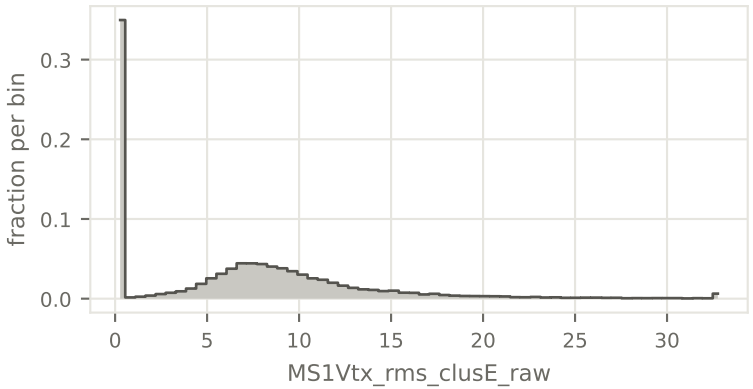


2 · combined signal
background n 34,970
combined, pooled D 0.372 n 9,996
equal weight per mass D 0.365

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

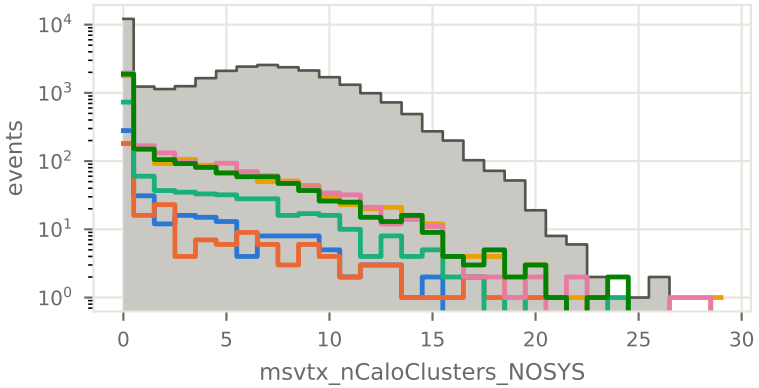
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nCaloClusters_NOSYS

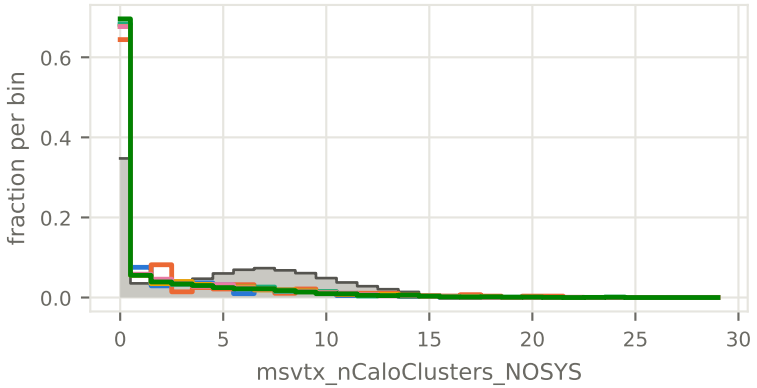
region: ENDCAP · rank 33 / 98

D_equal 0.365 (noise 0.022) D_pooled 0.364 min 0.356 (mH400_mS100) max 0.375 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

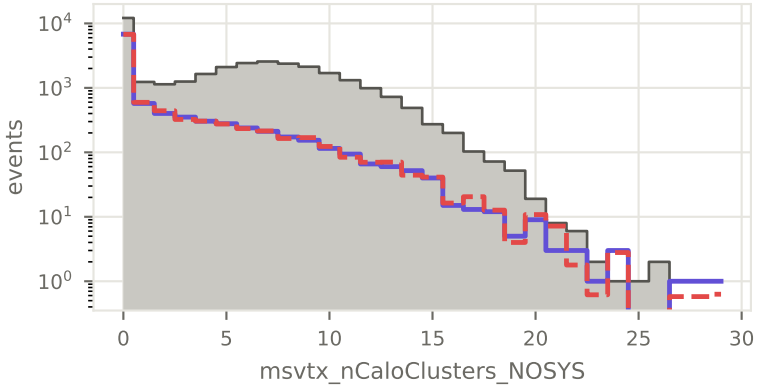


1b · each mass point vs background · normalized

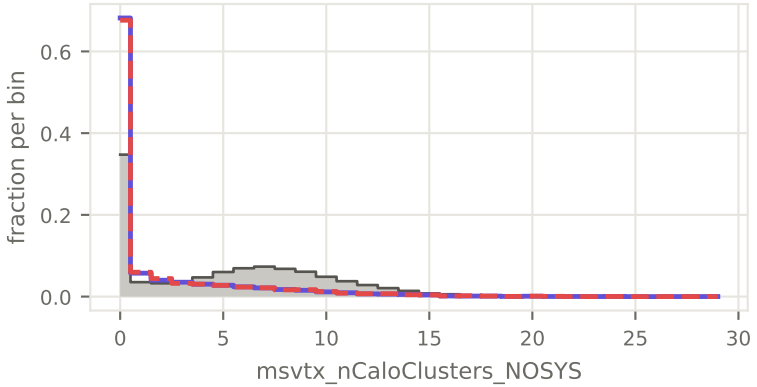


1 · each mass point
background n 34,970
mH125_mS15 D 0.374 n 411
mH125_mS55 D 0.368 n 281
mH200_mS50 D 0.359 n 1,071
mH400_mS100 D 0.356 n 2,648
mH600_mS150 D 0.365 n 2,882
mH1000_mS275 D 0.375 n 2,703

2a · combined signal vs background · raw

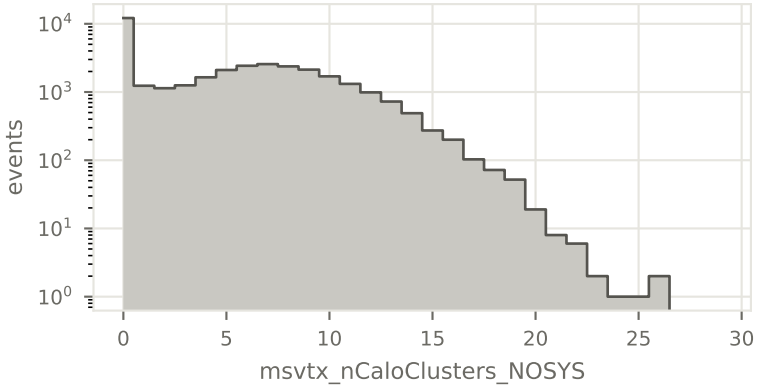


2b · combined signal vs background · normalized

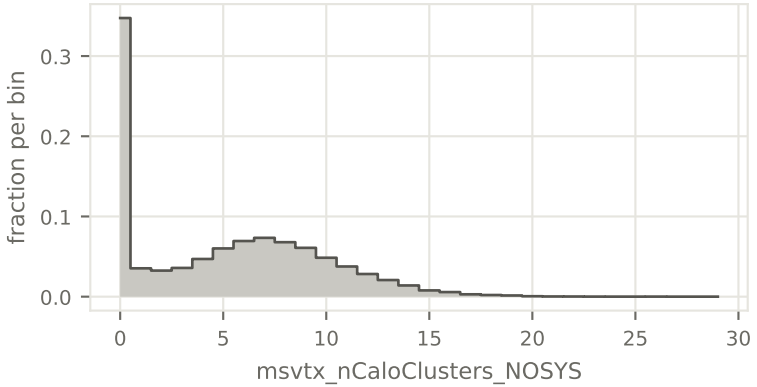


2 · combined signal
background n 34,970
combined, pooled D 0.364 n 9,996
equal weight per mass D 0.365

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

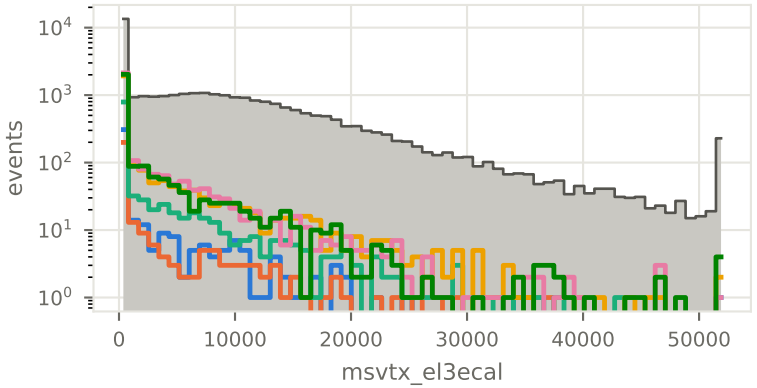
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el3ecal

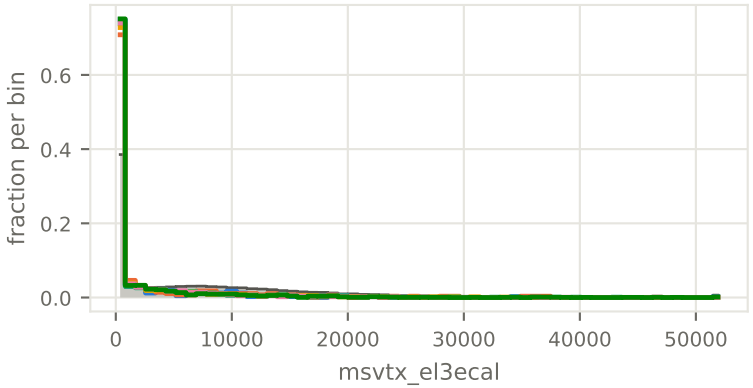
region: ENDCAP · rank 34 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.364 (noise 0.022) D_pooled 0.366 min 0.349 (mH125_mS55) max 0.382 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

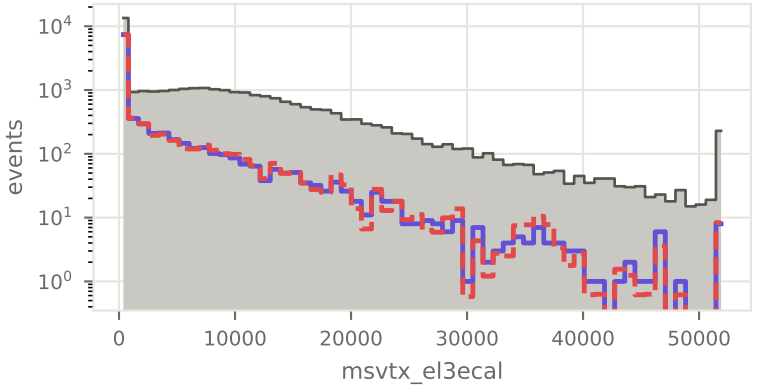


1b · each mass point vs background · normalized

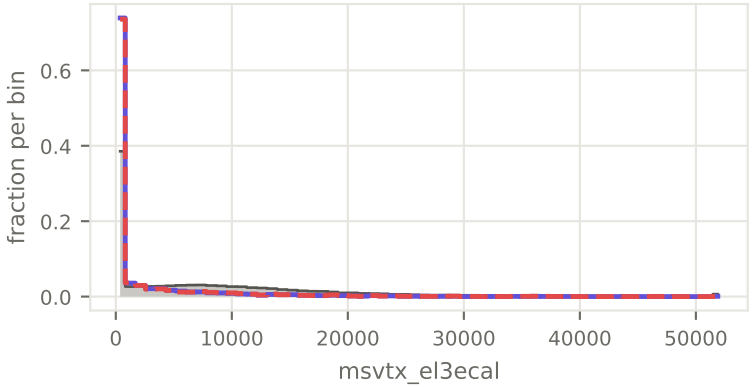


1 · each mass point
background n 34,970
mH125_mS15 D 0.382 n 411
mH125_mS55 D 0.349 n 281
mH200_mS50 D 0.357 n 1,071
mH400_mS100 D 0.360 n 2,648
mH600_mS150 D 0.367 n 2,882
mH1000_mS275 D 0.378 n 2,703

2a · combined signal vs background · raw

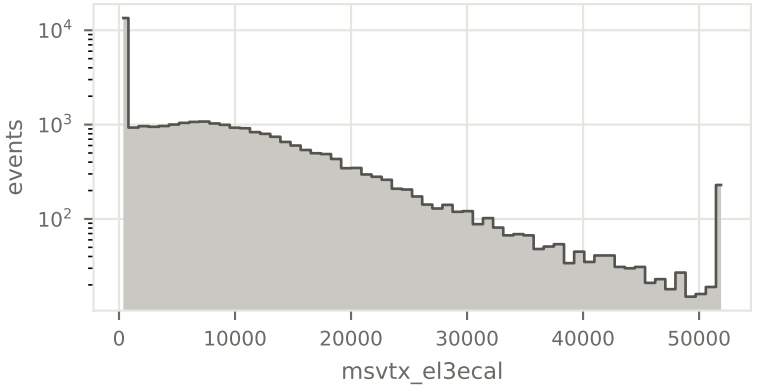


2b · combined signal vs background · normalized

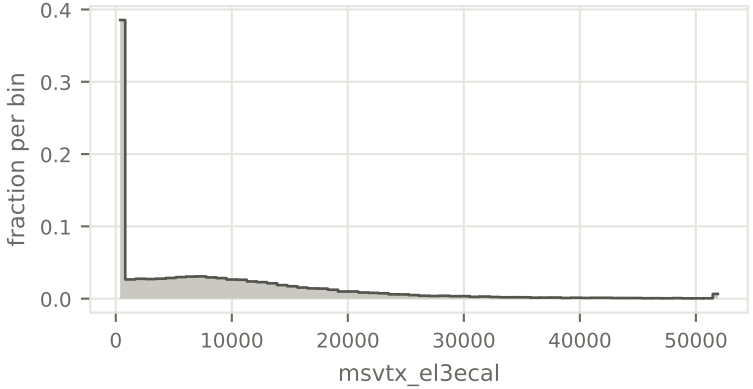


2 · combined signal
background n 34,970
combined, pooled D 0.366 n 9,996
equal weight per mass D 0.364

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

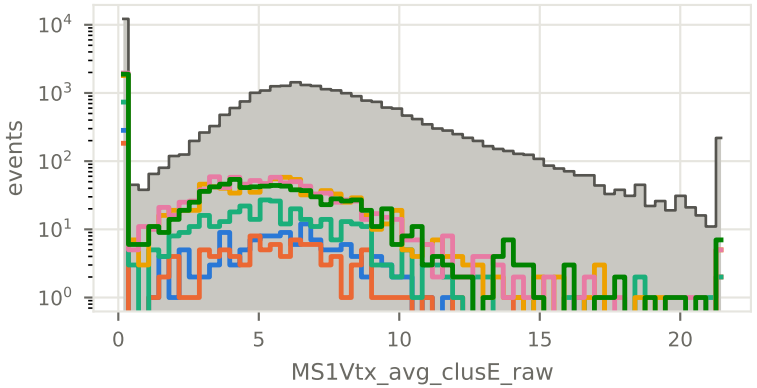
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_avg_clusE_raw

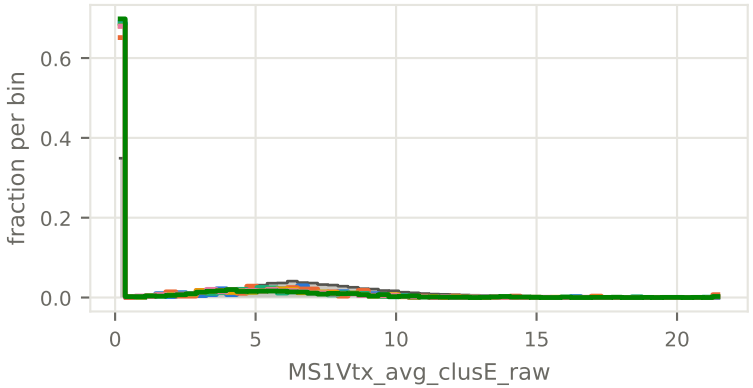
region: ENDCAP · rank 35 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.361 (noise 0.022) D_pooled 0.369 min 0.336 (mH125_mS55) max 0.382 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

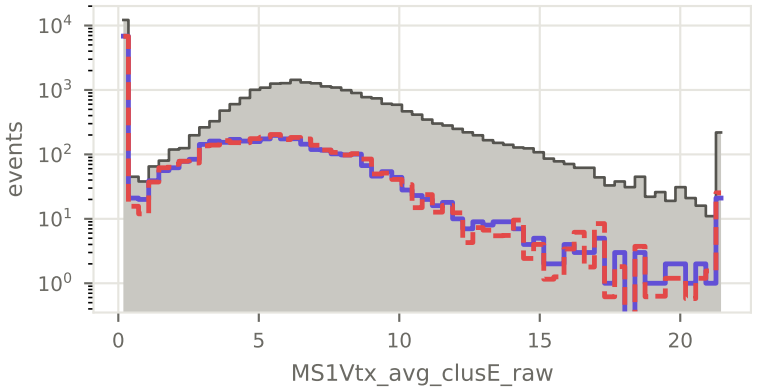


1b · each mass point vs background · normalized

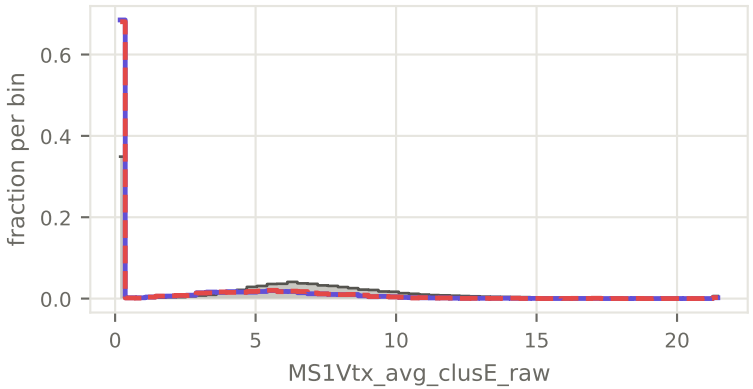


1 · each mass point
background n 34,970
mH125_mS15 D 0.357 n 411
mH125_mS55 D 0.336 n 281
mH200_mS50 D 0.366 n 1,071
mH400_mS100 D 0.366 n 2,648
mH600_mS150 D 0.372 n 2,882
mH1000_mS275 D 0.382 n 2,703

2a · combined signal vs background · raw

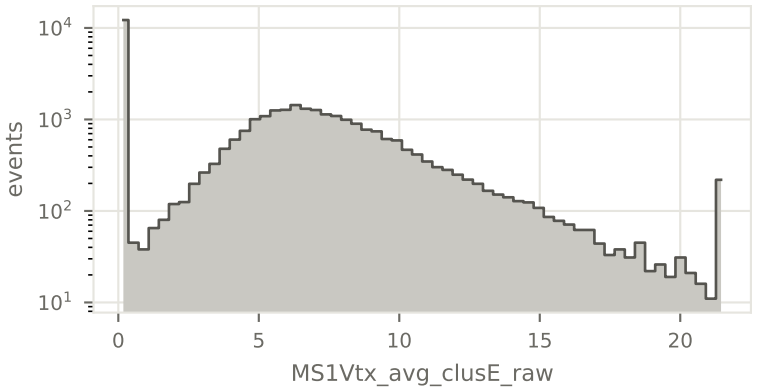


2b · combined signal vs background · normalized

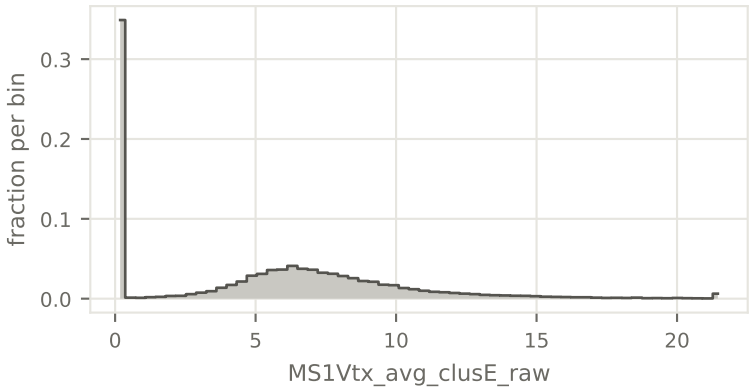


2 · combined signal
background n 34,970
combined, pooled D 0.369 n 9,996
equal weight per mass D 0.361

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

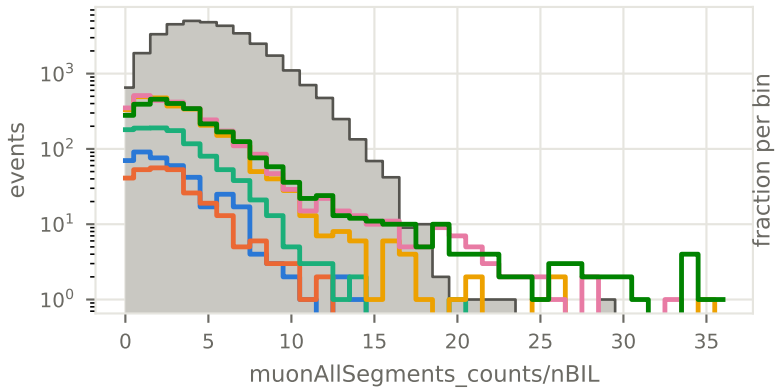
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

muonAllSegments_counts/nBIL

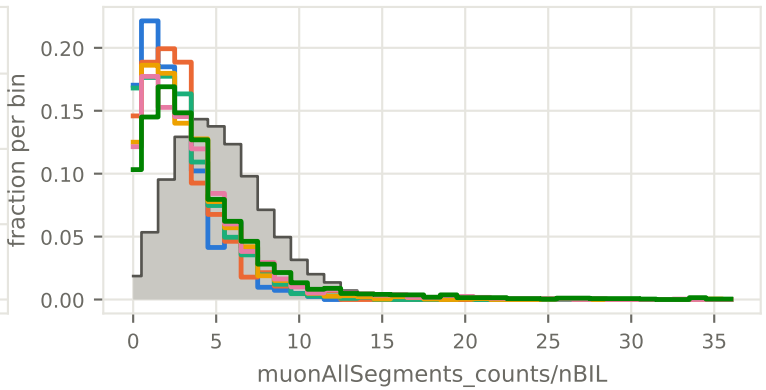
region: ENDCAP · rank 36 / 98

D_equal 0.357 (noise 0.022) D_pooled 0.319 min 0.269 (mH1000_mS275) max 0.426 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

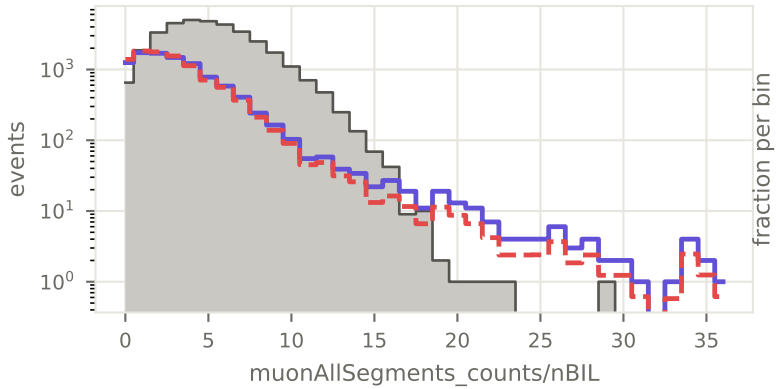


1b · each mass point vs background · normalized

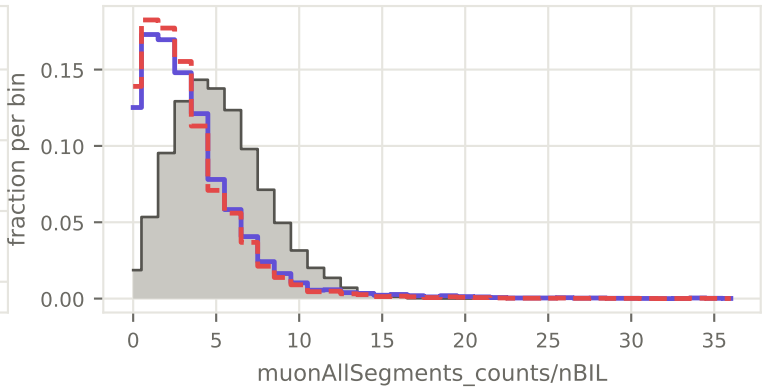


1 · each mass point
background n 34,970
mH125_mS15 D 0.426 n 411
mH125_mS55 D 0.426 n 281
mH200_mS50 D 0.389 n 1,071
mH400_mS100 D 0.334 n 2,648
mH600_mS150 D 0.300 n 2,882
mH1000_mS275 D 0.269 n 2,703

2a · combined signal vs background · raw

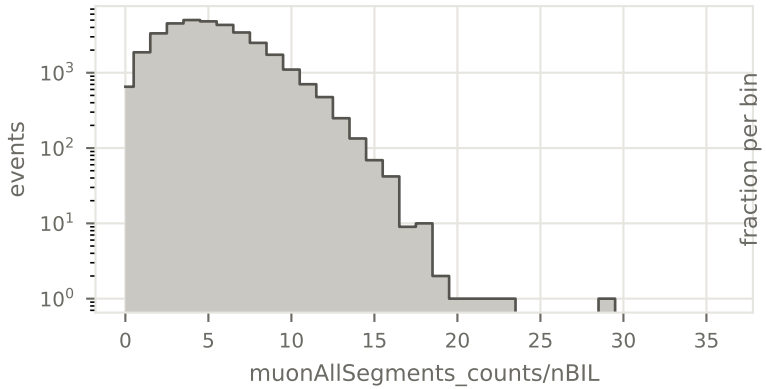


2b · combined signal vs background · normalized

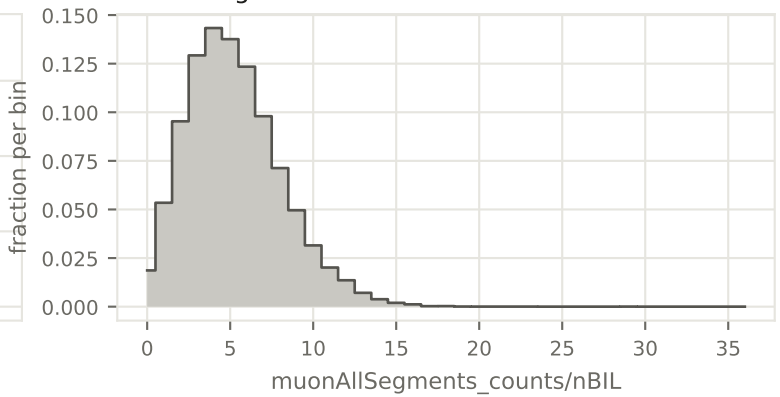


2 · combined signal
background n 34,970
combined, pooled D 0.319 n 9,996
equal weight per mass D 0.357

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

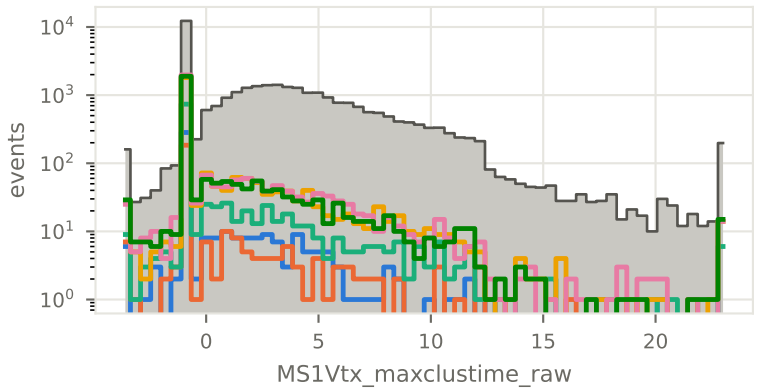
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_maxclustime_raw

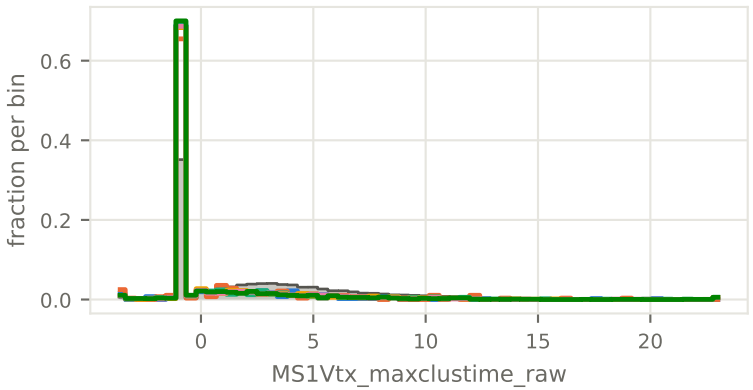
region: ENDCAP · rank 37 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.356 (noise 0.022) D_pooled 0.358 min 0.345 (mH125_mS55) max 0.371 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

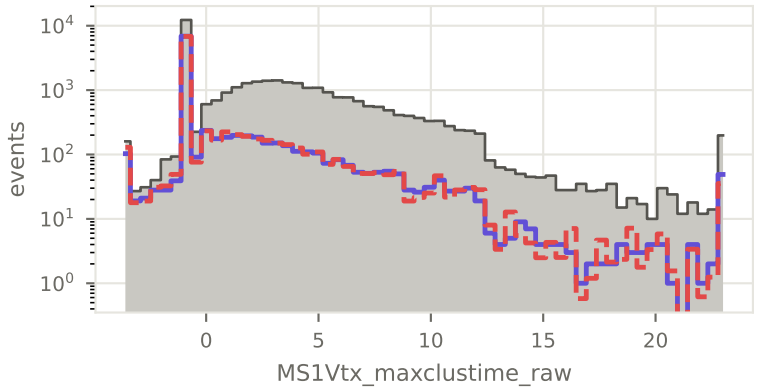


1b · each mass point vs background · normalized

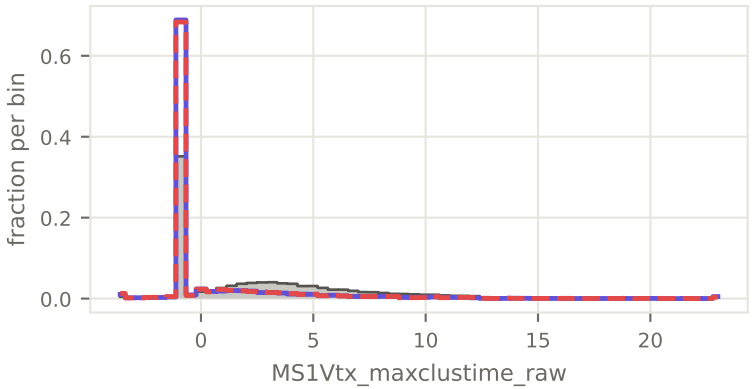


1 · each mass point
background n 34,970
mH125_mS15 D 0.359 n 411
mH125_mS55 D 0.345 n 281
mH200_mS50 D 0.358 n 1,071
mH400_mS100 D 0.352 n 2,648
mH600_mS150 D 0.356 n 2,882
mH1000_mS275 D 0.371 n 2,703

2a · combined signal vs background · raw

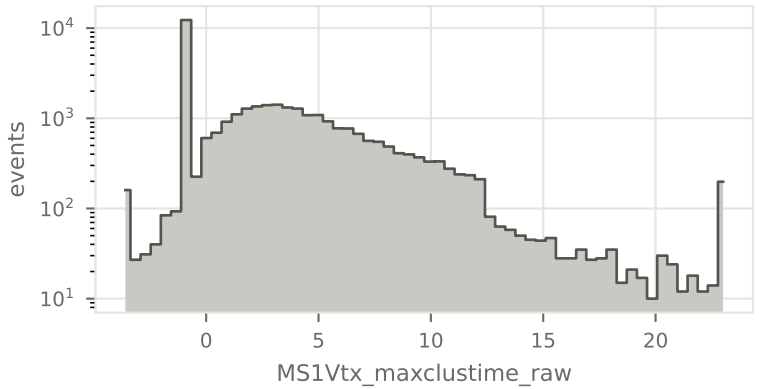


2b · combined signal vs background · normalized

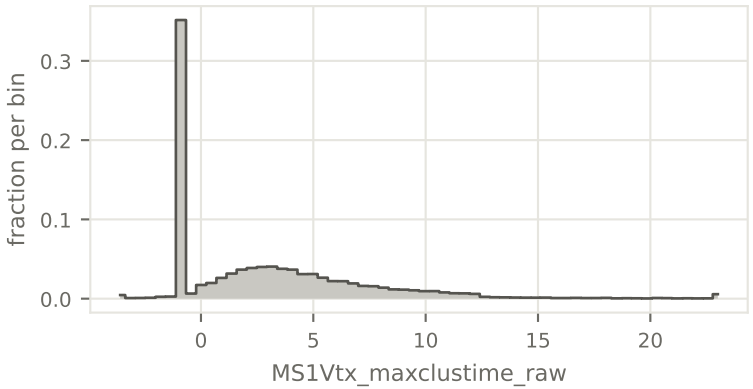


2 · combined signal
background n 34,970
combined, pooled D 0.358 n 9,996
equal weight per mass D 0.356

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

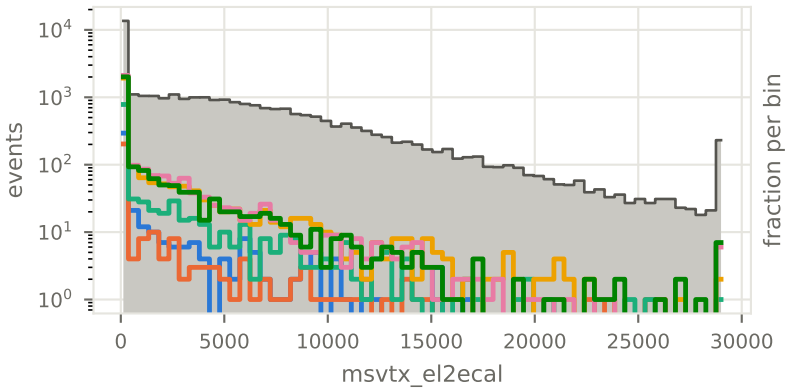
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el2ecal

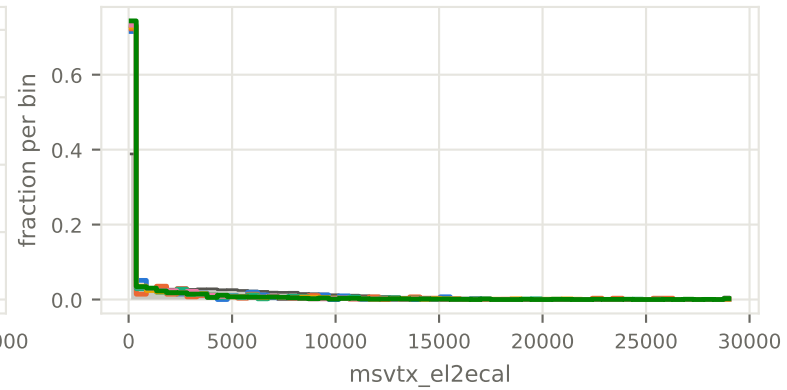
region: ENDCAP · rank 38 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.343 (noise 0.022) D_pooled 0.349 min 0.336 (mH125_mS55) max 0.363 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

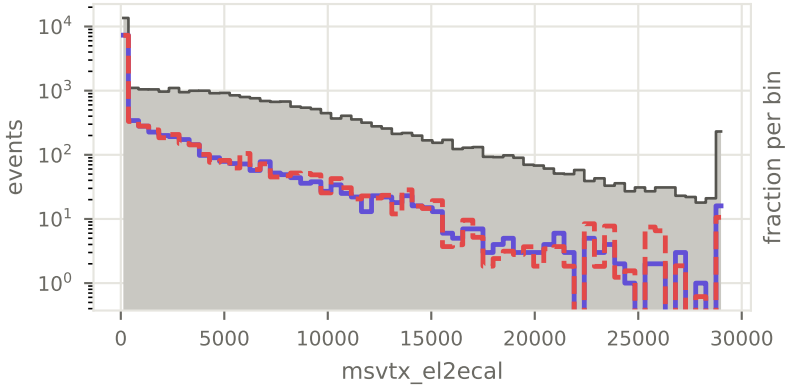


1b · each mass point vs background · normalized

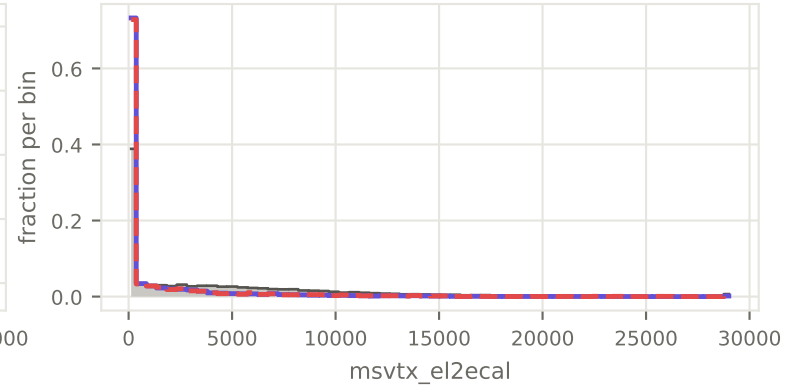


1 · each mass point
background n 34,970
mH125_mS15 D 0.352 n 411
mH125_mS55 D 0.336 n 281
mH200_mS50 D 0.344 n 1,071
mH400_mS100 D 0.344 n 2,648
mH600_mS150 D 0.349 n 2,882
mH1000_mS275 D 0.363 n 2,703

2a · combined signal vs background · raw

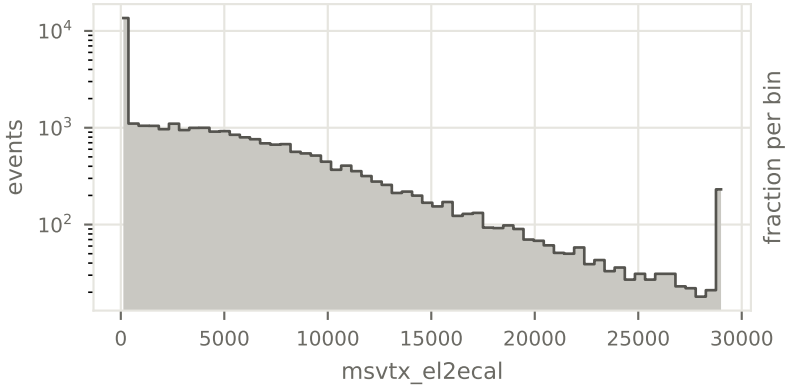


2b · combined signal vs background · normalized

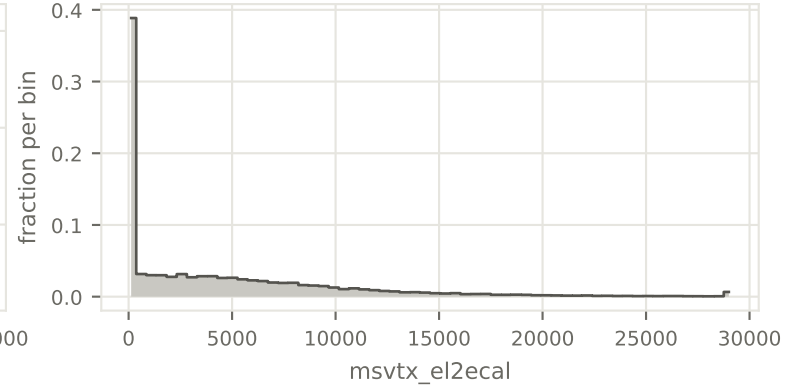


2 · combined signal
background n 34,970
combined, pooled D 0.349 n 9,996
equal weight per mass D 0.343

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

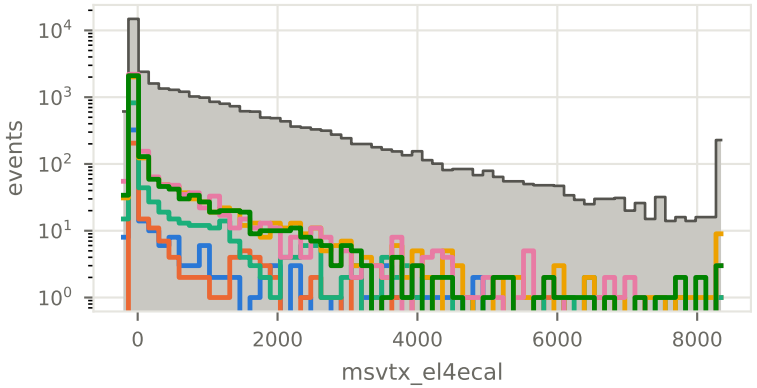
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el4ecal

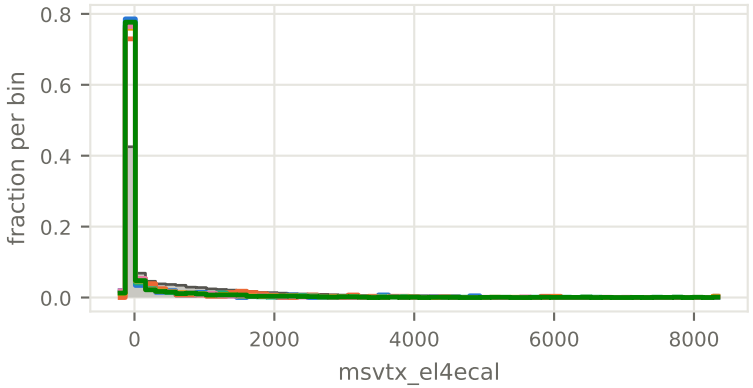
region: ENDCAP · rank 39 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.337 (noise 0.022) D_pooled 0.340 min 0.293 (mH125_mS55) max 0.368 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

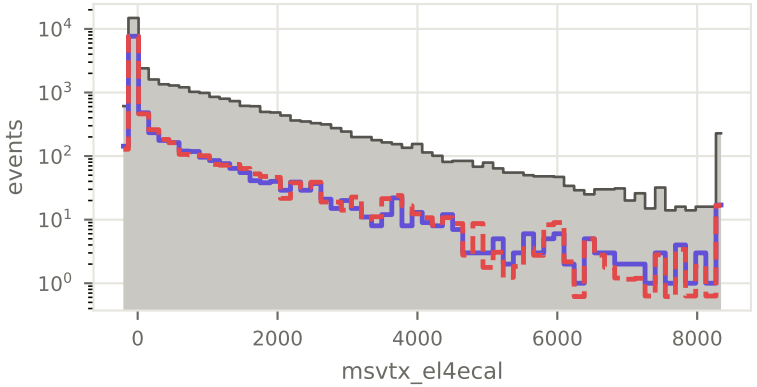


1b · each mass point vs background · normalized

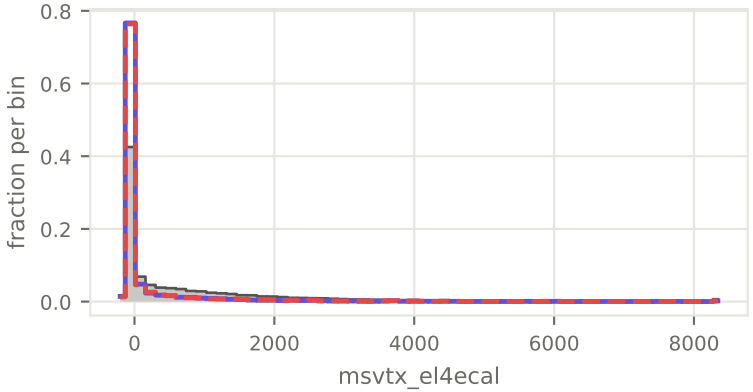


1 · each mass point
background n 34,970
mH125_mS15 D 0.368 n 411
mH125_mS55 D 0.293 n 281
mH200_mS50 D 0.342 n 1,071
mH400_mS100 D 0.331 n 2,648
mH600_mS150 D 0.345 n 2,882
mH1000_mS275 D 0.348 n 2,703

2a · combined signal vs background · raw

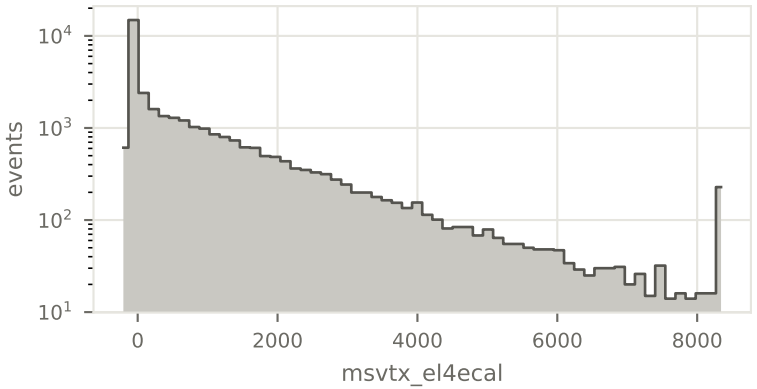


2b · combined signal vs background · normalized

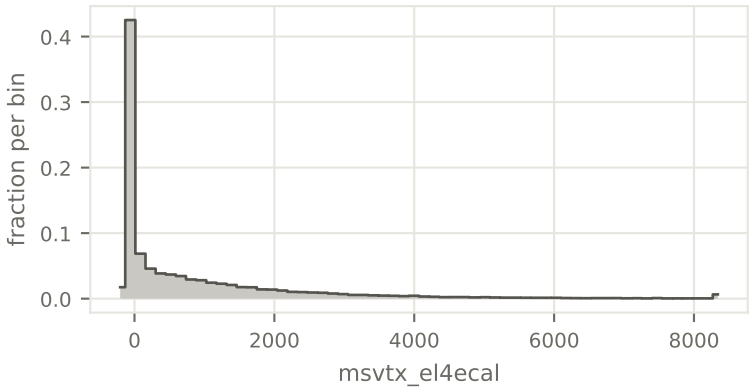


2 · combined signal
background n 34,970
combined, pooled D 0.340 n 9,996
equal weight per mass D 0.337

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

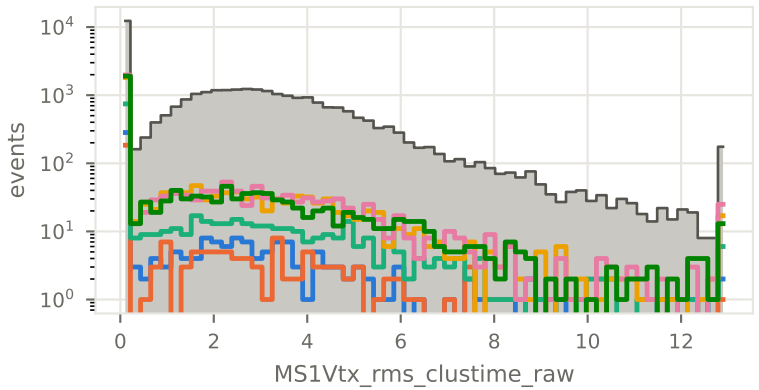
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_rms_clustime_raw

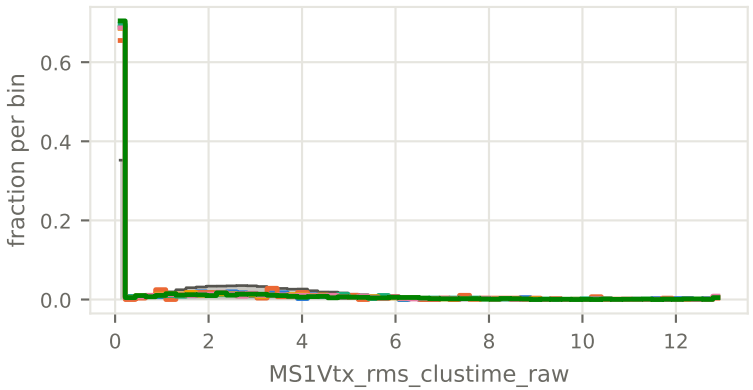
region: ENDCAP · rank 40 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.335 (noise 0.022) D_pooled 0.341 min 0.309 (mH125_mS55) max 0.356 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

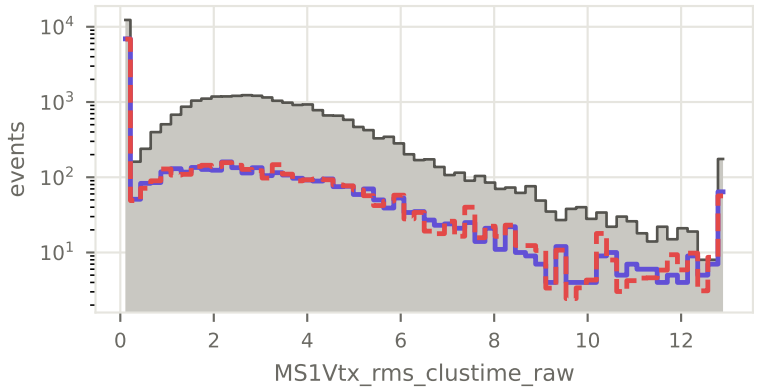


1b · each mass point vs background · normalized

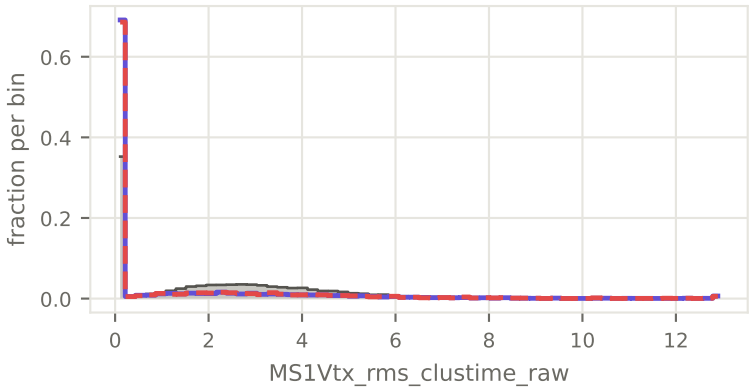


1 · each mass point
background n 34,970
mH125_mS15 D 0.342 n 411
mH125_mS55 D 0.309 n 281
mH200_mS50 D 0.348 n 1,071
mH400_mS100 D 0.338 n 2,648
mH600_mS150 D 0.335 n 2,882
mH1000_mS275 D 0.356 n 2,703

2a · combined signal vs background · raw

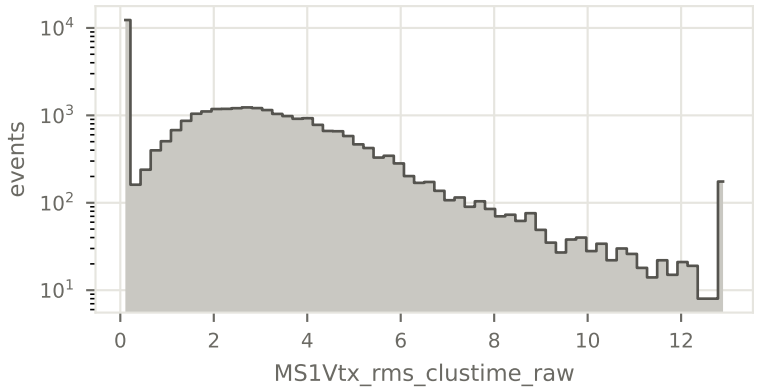


2b · combined signal vs background · normalized

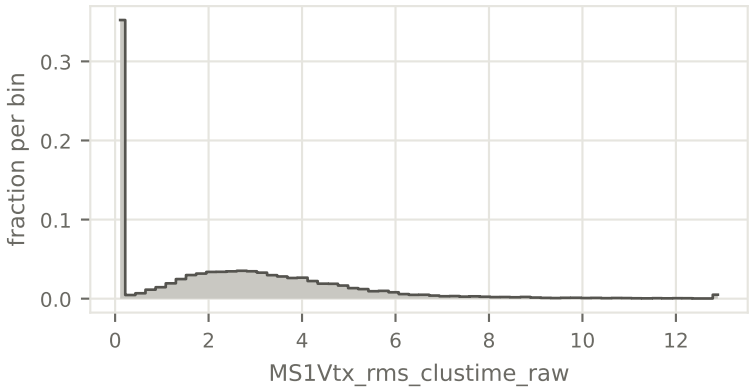


2 · combined signal
background n 34,970
combined, pooled D 0.341 n 9,996
equal weight per mass D 0.335

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

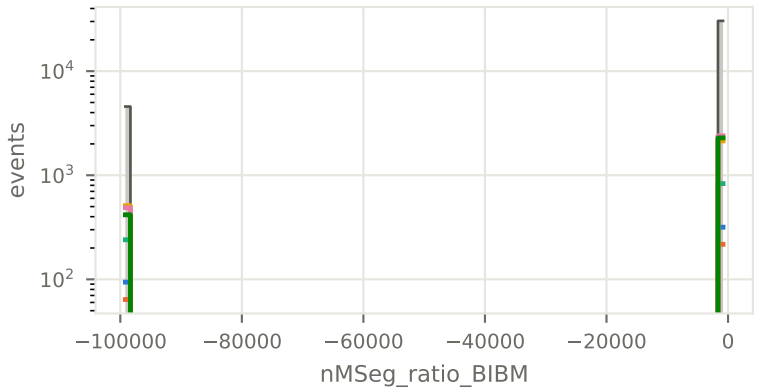
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

nMSeg_ratio_BIBM

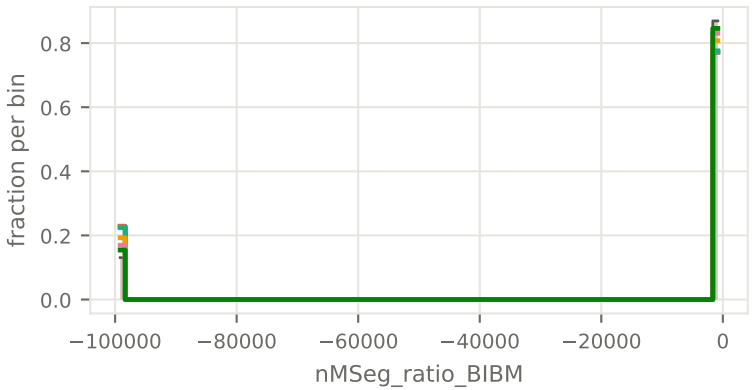
region: ENDCAP · rank 41 / 98 · in: NN1 barrel

D_equal 0.330 (noise 0.022) D_pooled 0.332 min 0.306 (mH200_mS50) max 0.349 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

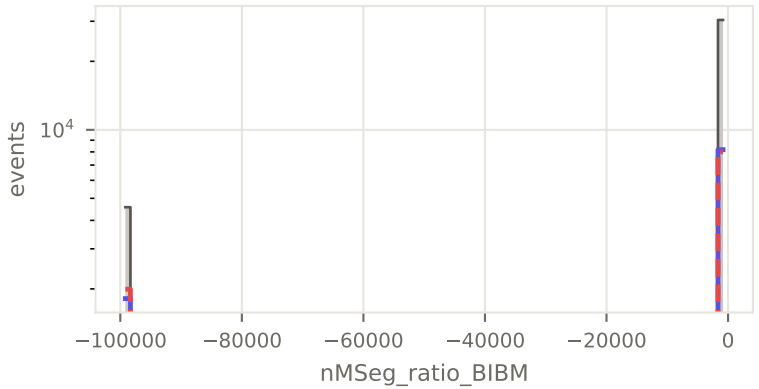


1b · each mass point vs background · normalized

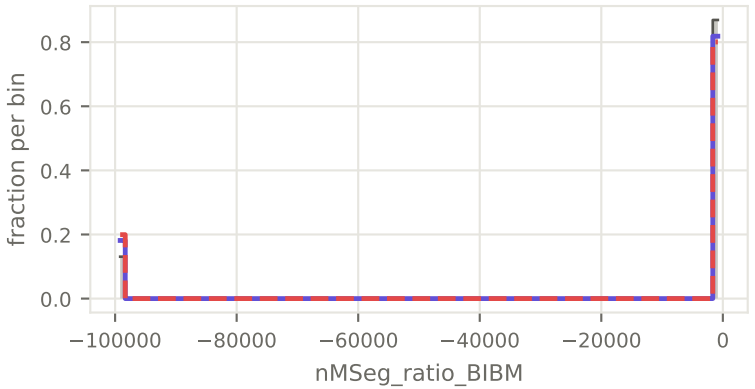


1 · each mass point
background n 34,970
mH125_mS15 D 0.349 n 411
mH125_mS55 D 0.324 n 281
mH200_mS50 D 0.306 n 1,071
mH400_mS100 D 0.320 n 2,648
mH600_mS150 D 0.345 n 2,882
mH1000_mS275 D 0.338 n 2,703

2a · combined signal vs background · raw

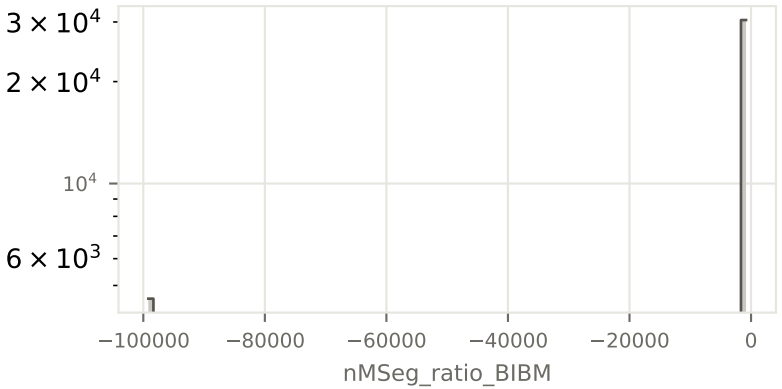


2b · combined signal vs background · normalized

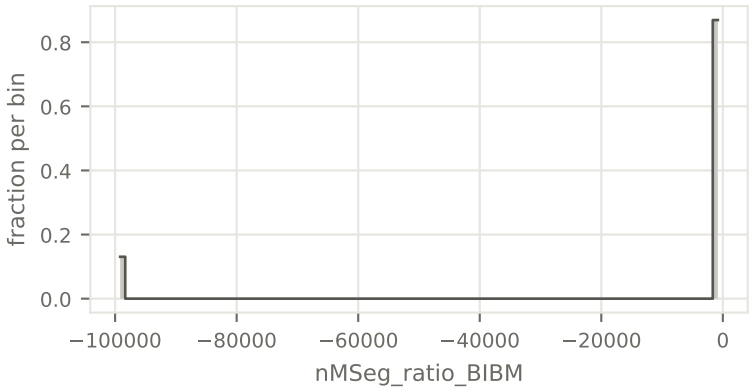


2 · combined signal
background n 34,970
combined, pooled D 0.332 n 9,996
equal weight per mass D 0.330

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

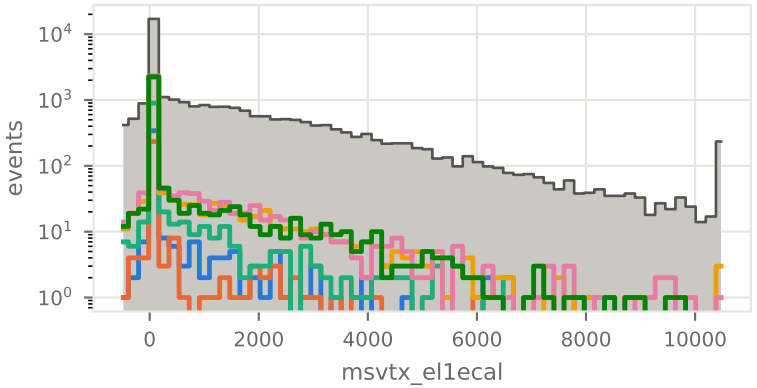
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el1ecal

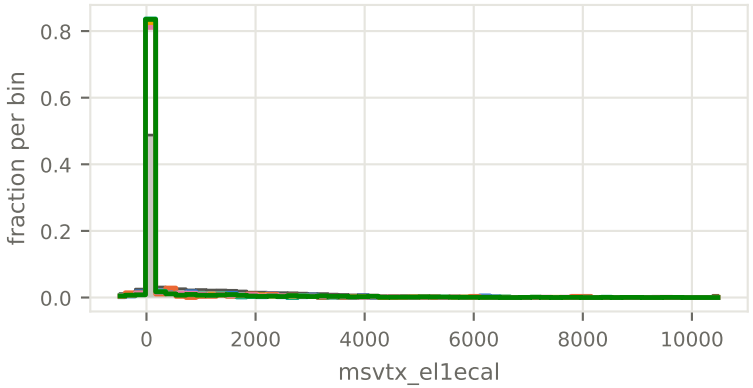
region: ENDCAP · rank 42 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.324 (noise 0.022) D_pooled 0.320 min 0.307 (mH600_mS150) max 0.337 (mH200_mS50) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

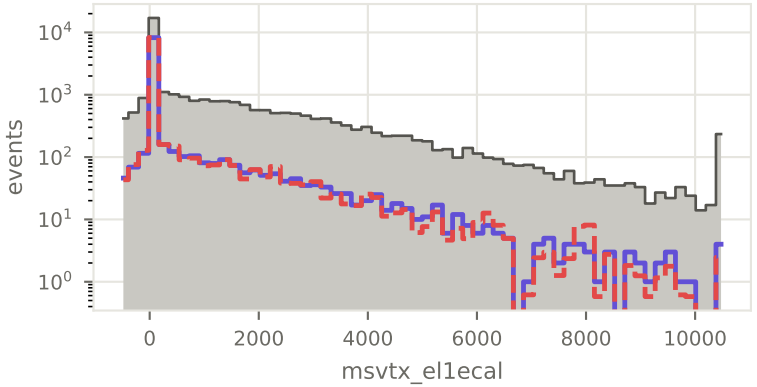


1b · each mass point vs background · normalized

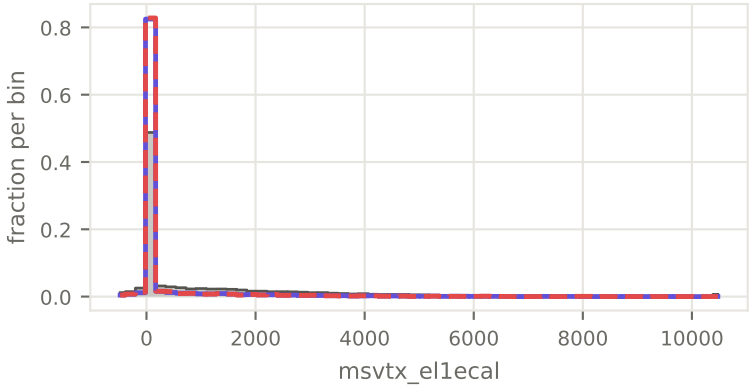


1 · each mass point
background n 34,970
mH125_mS15 D 0.331 n 411
mH125_mS55 D 0.328 n 281
mH200_mS50 D 0.337 n 1,071
mH400_mS100 D 0.321 n 2,648
mH600_mS150 D 0.307 n 2,882
mH1000_mS275 D 0.326 n 2,703

2a · combined signal vs background · raw

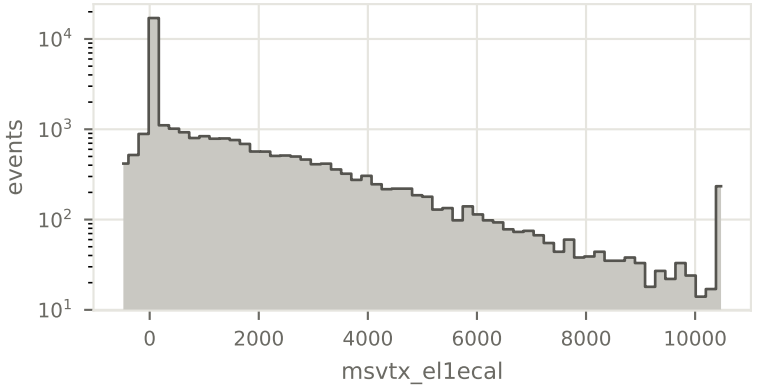


2b · combined signal vs background · normalized

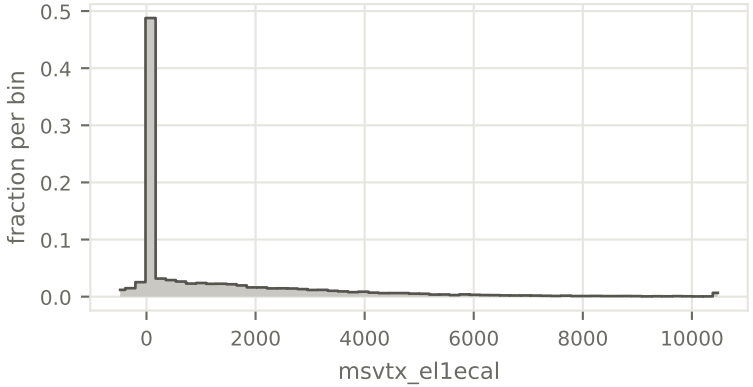


2 · combined signal
background n 34,970
combined, pooled D 0.320 n 9,996
equal weight per mass D 0.324

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

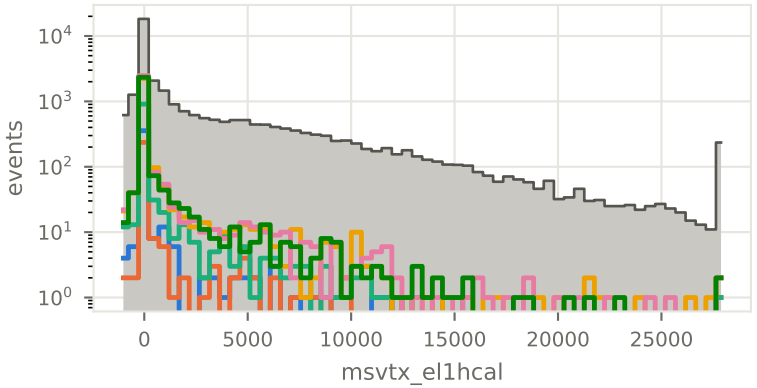
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el1hcal

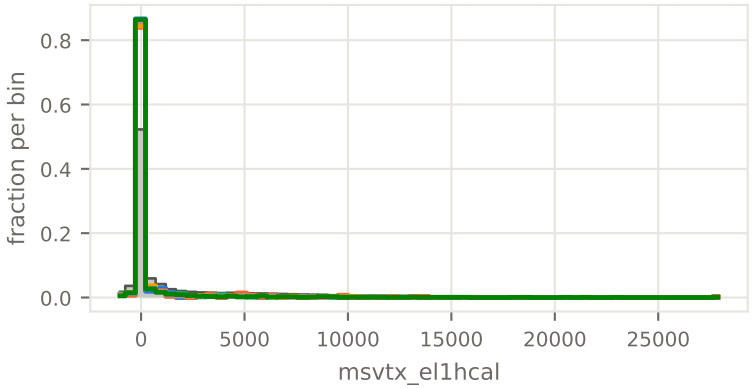
region: ENDCAP · rank 43 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.313 (noise 0.022) D_pooled 0.316 min 0.288 (mH125_mS55) max 0.331 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

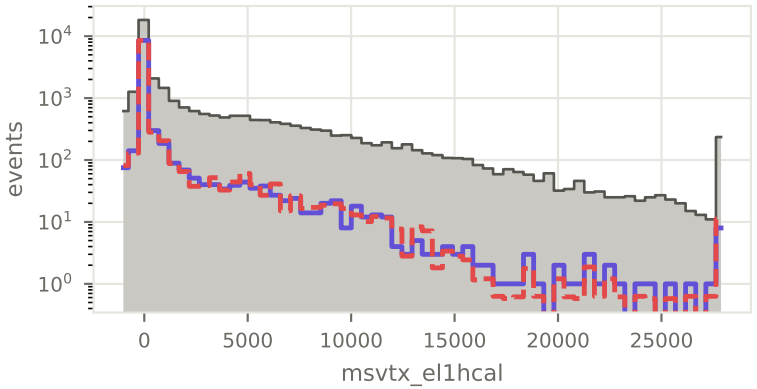


1b · each mass point vs background · normalized

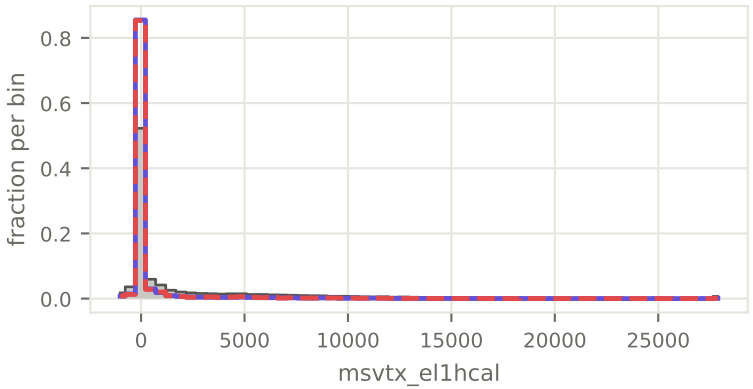


1 · each mass point
background n 34,970
mH125_mS15 D 0.331 n 411
mH125_mS55 D 0.288 n 281
mH200_mS50 D 0.310 n 1,071
mH400_mS100 D 0.305 n 2,648
mH600_mS150 D 0.321 n 2,882
mH1000_mS275 D 0.325 n 2,703

2a · combined signal vs background · raw

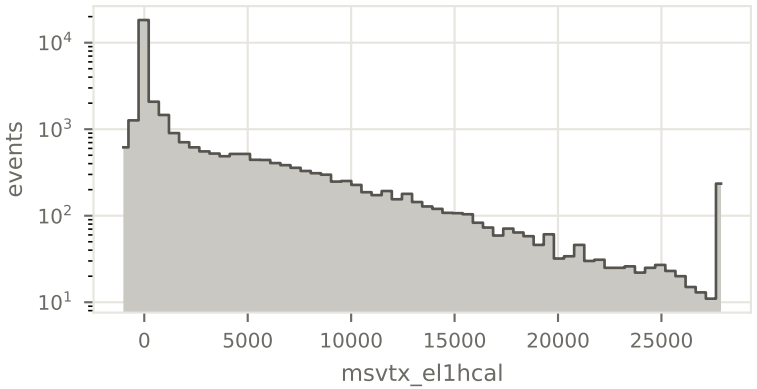


2b · combined signal vs background · normalized

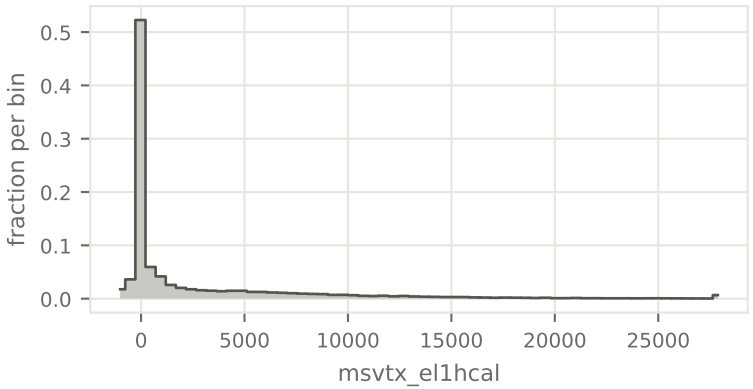


2 · combined signal
background n 34,970
combined, pooled D 0.316 n 9,996
equal weight per mass D 0.313

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

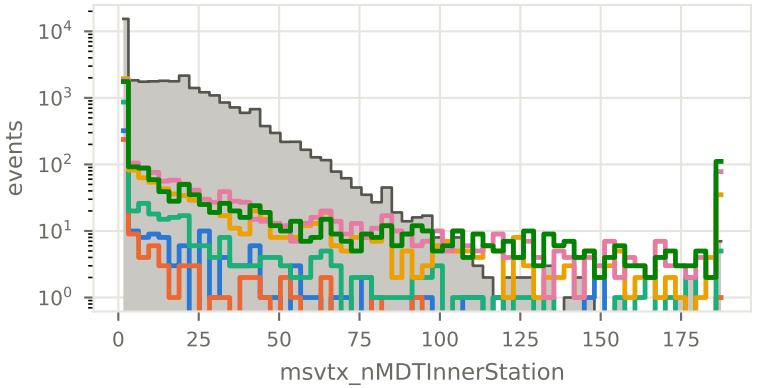
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nMDTInnerStation

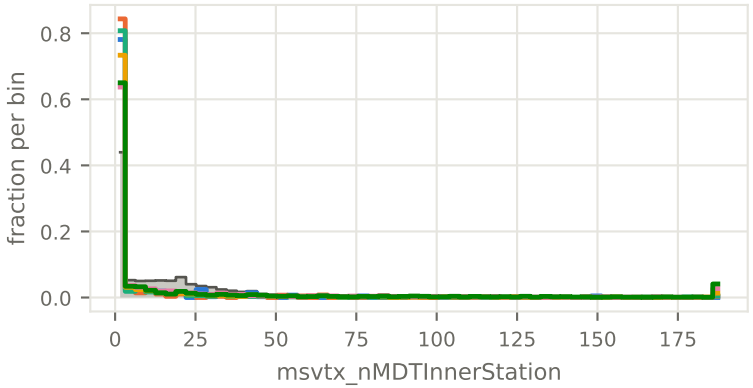
region: ENDCAP · rank 44 / 98

D_equal 0.312 (noise 0.022) D_pooled 0.270 min 0.205 (mH600_mS150) max 0.405 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

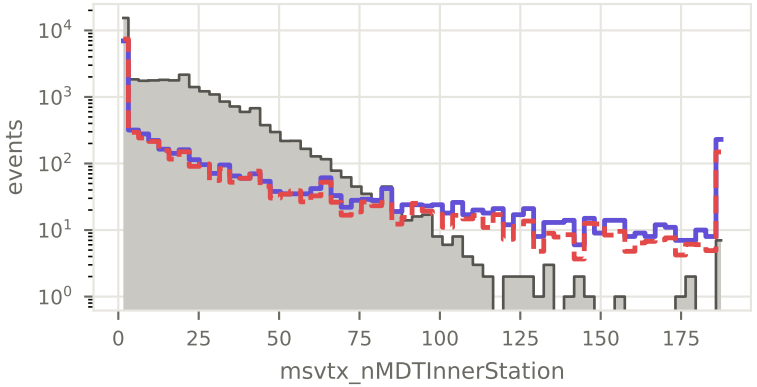


1b · each mass point vs background · normalized

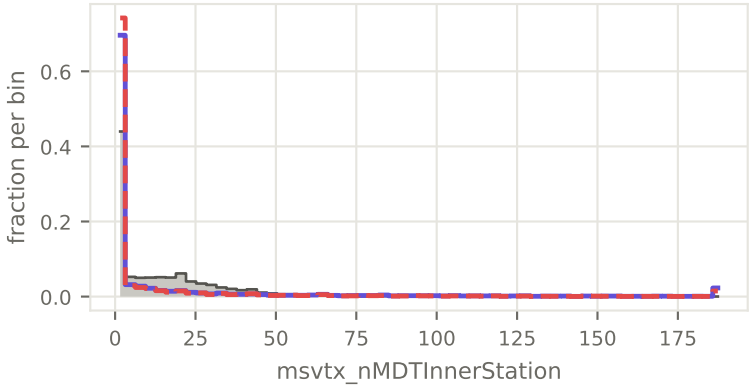


1 · each mass point
background n 34,970
mH125_mS15 D 0.363 n 411
mH125_mS55 D 0.405 n 281
mH200_mS50 D 0.382 n 1,071
mH400_mS100 D 0.310 n 2,648
mH600_mS150 D 0.205 n 2,882
mH1000_mS275 D 0.230 n 2,703

2a · combined signal vs background · raw

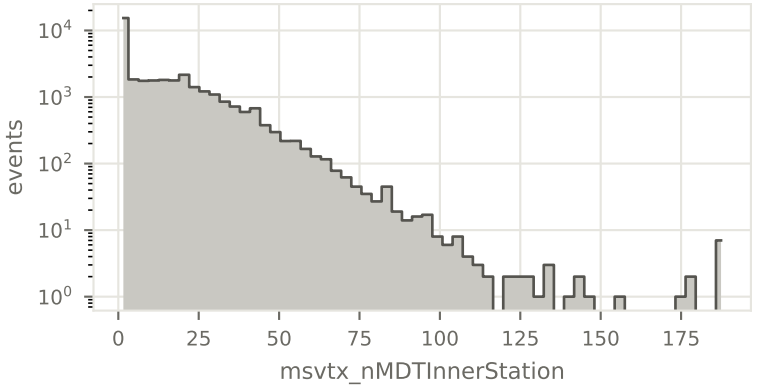


2b · combined signal vs background · normalized

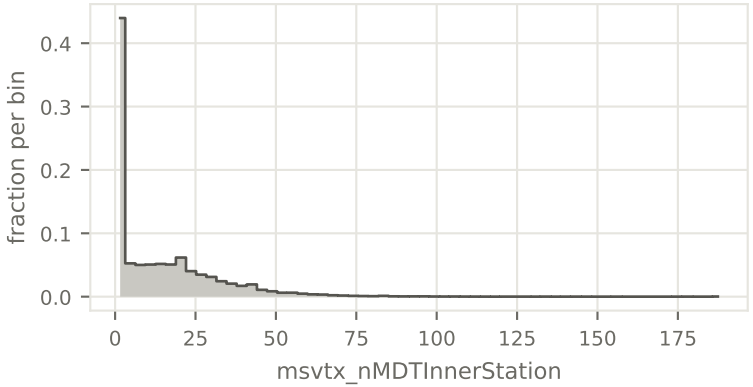


2 · combined signal
background n 34,970
combined, pooled D 0.270 n 9,996
equal weight per mass D 0.312

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

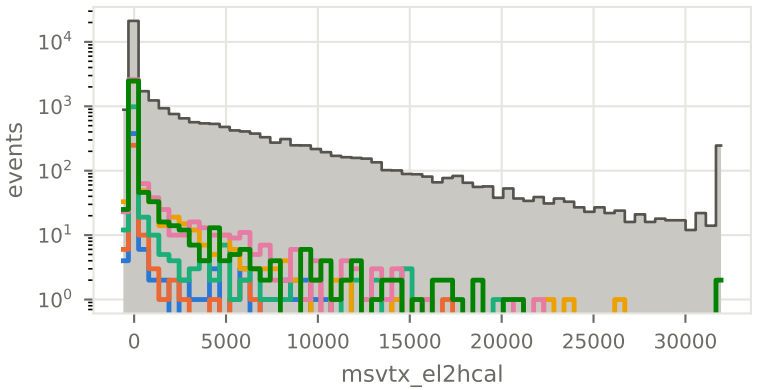
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el2hcal

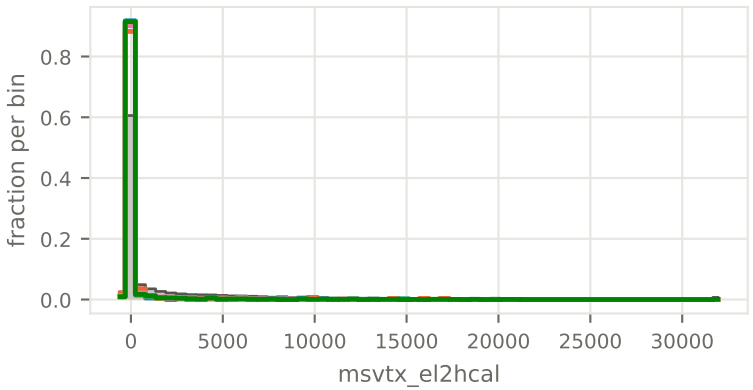
region: ENDCAP · rank 45 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.311 (noise 0.022) D_pooled 0.309 min 0.299 (mH600_mS150) max 0.323 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

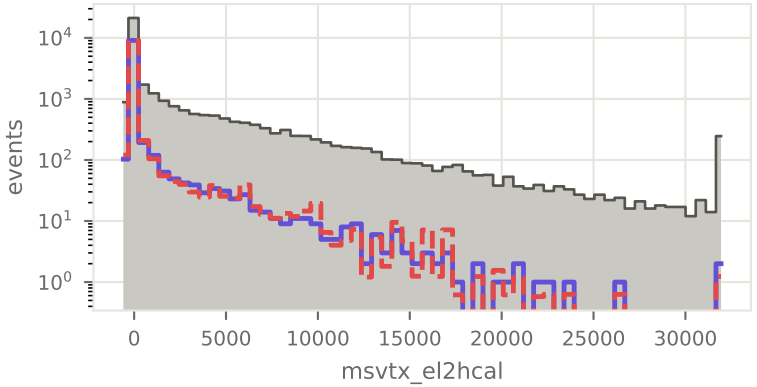


1b · each mass point vs background · normalized

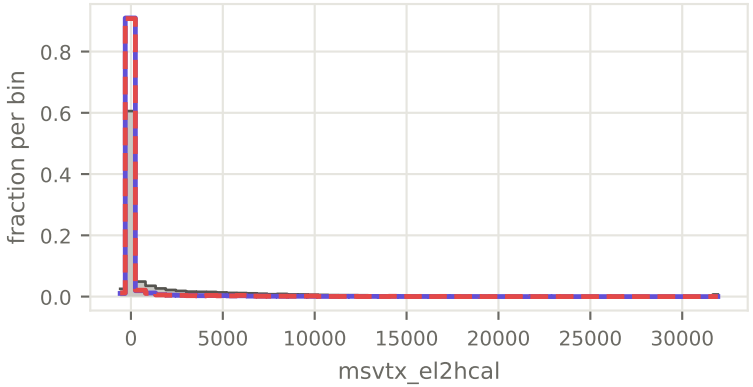


1 · each mass point
background n 34,970
mH125_mS15 D 0.323 n 411
mH125_mS55 D 0.310 n 281
mH200_mS50 D 0.316 n 1,071
mH400_mS100 D 0.311 n 2,648
mH600_mS150 D 0.299 n 2,882
mH1000_mS275 D 0.312 n 2,703

2a · combined signal vs background · raw

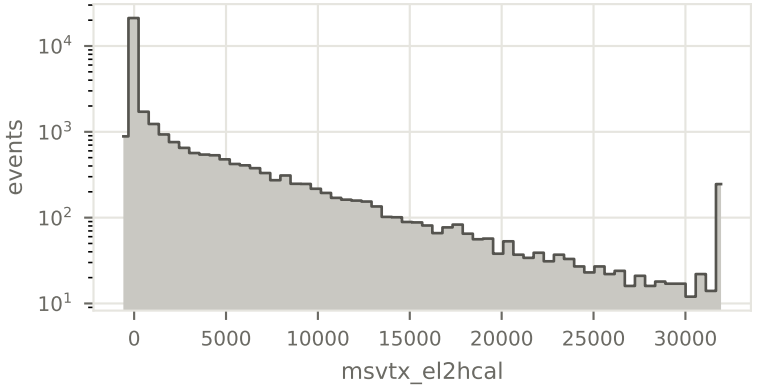


2b · combined signal vs background · normalized

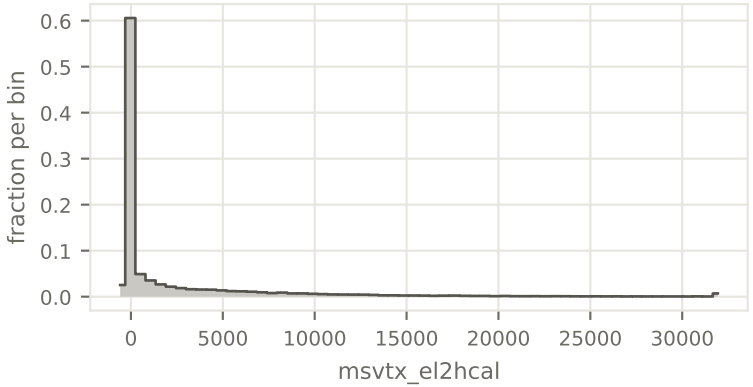


2 · combined signal
background n 34,970
combined, pooled D 0.309 n 9,996
equal weight per mass D 0.311

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

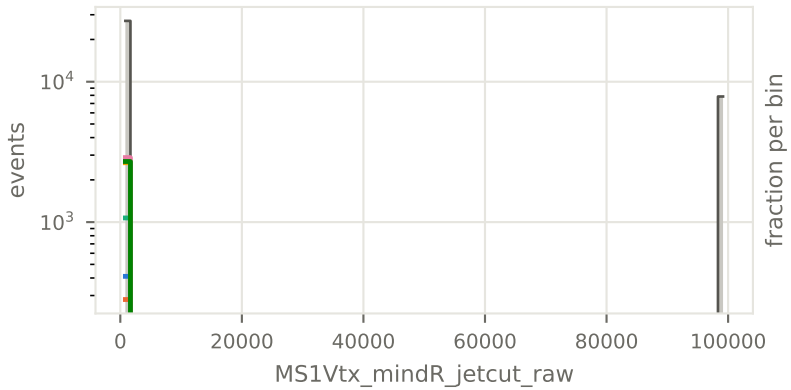
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_mindR_jetcut_raw

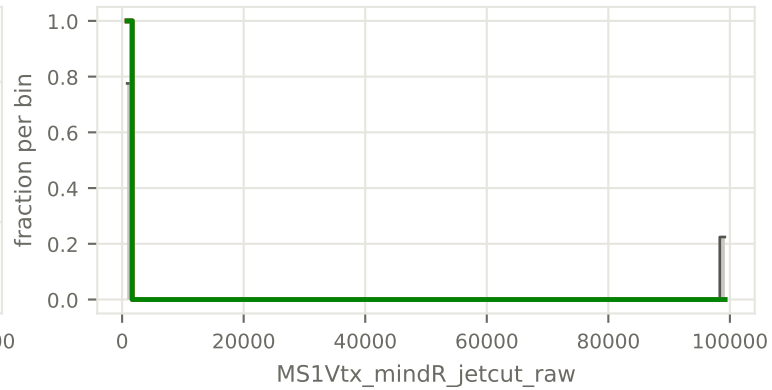
region: ENDCAP · rank 46 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.308 (noise 0.022) D_pooled 0.330 min 0.259 (mH125_mS55) max 0.356 (mH400_mS100) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

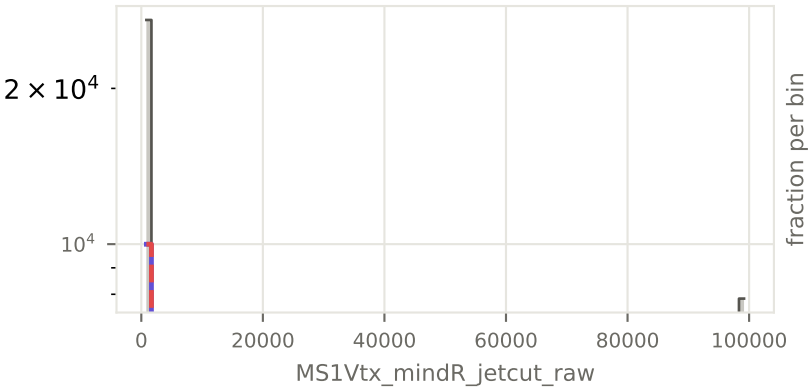


1b · each mass point vs background · normalized

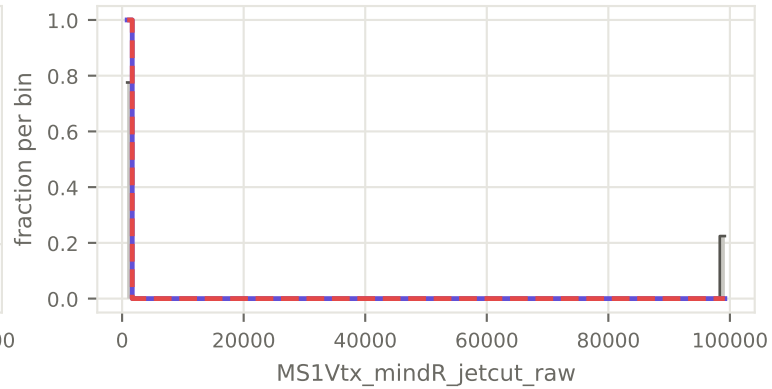


1 · each mass point
background n 34,970
mH125_mS15 D 0.282 n 411
mH125_mS55 D 0.259 n 281
mH200_mS50 D 0.330 n 1,071
mH400_mS100 D 0.356 n 2,648
mH600_mS150 D 0.324 n 2,882
mH1000_mS275 D 0.340 n 2,703

2a · combined signal vs background · raw

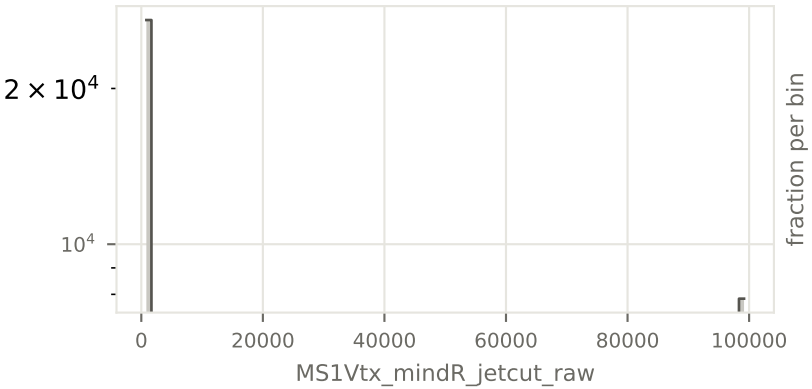


2b · combined signal vs background · normalized

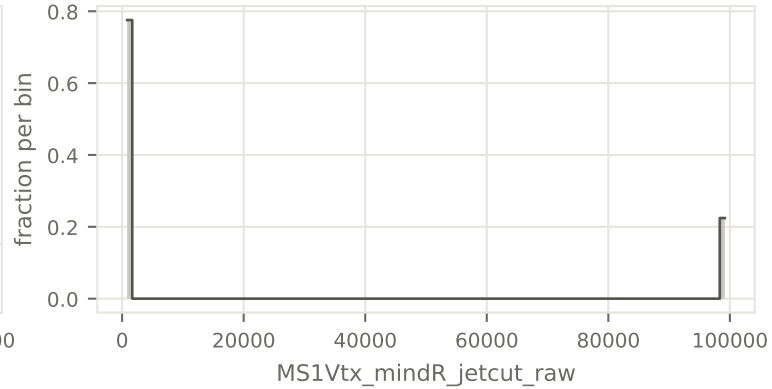


2 · combined signal
background n 34,970
combined, pooled D 0.330 n 9,996
equal weight per mass D 0.308

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

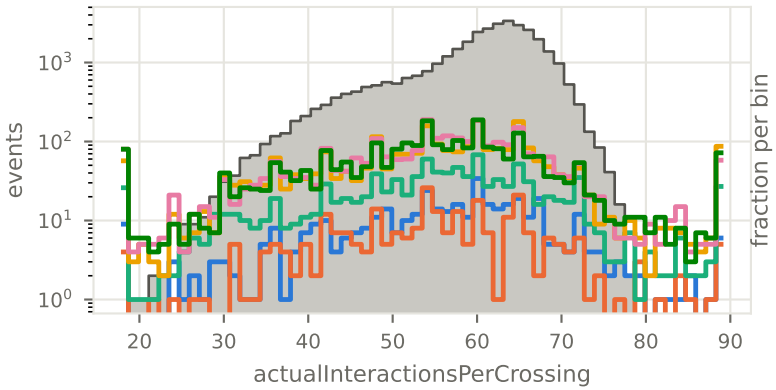
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

actualInteractionsPerCrossing

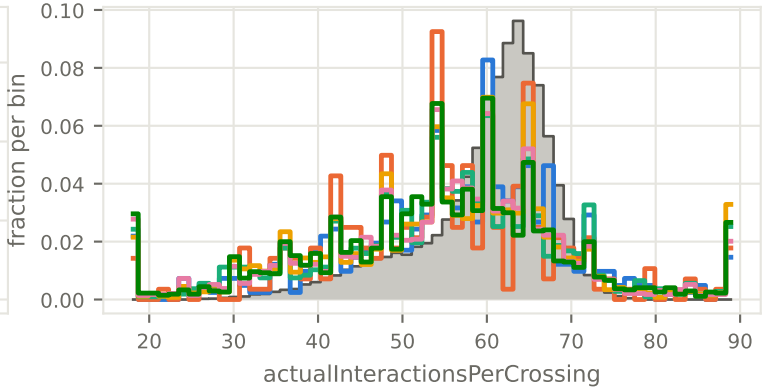
region: ENDCAP · rank 47 / 98

D_equal 0.298 (noise 0.022) D_pooled 0.300 min 0.253 (mH125_mS15) max 0.336 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

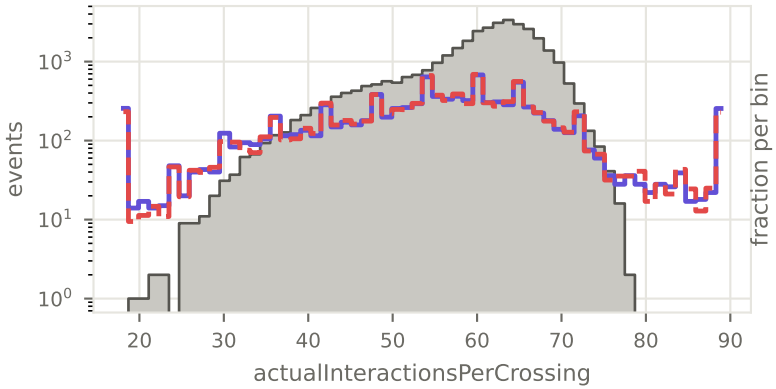


1b · each mass point vs background · normalized

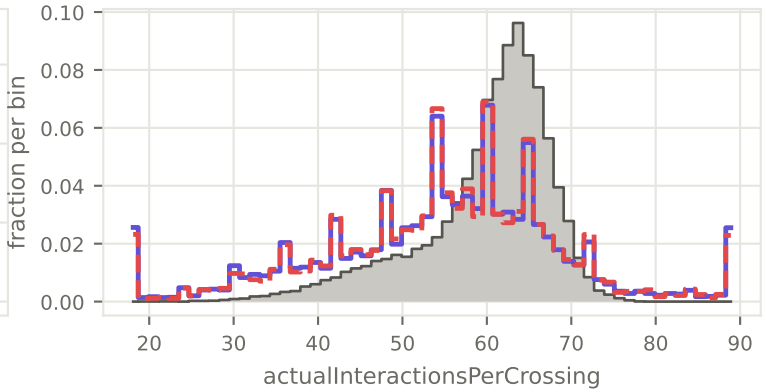


1 · each mass point
background n 34,970
mH125_mS15 D 0.253 n 411
mH125_mS55 D 0.336 n 281
mH200_mS50 D 0.300 n 1,071
mH400_mS100 D 0.282 n 2,648
mH600_mS150 D 0.306 n 2,882
mH1000_mS275 D 0.317 n 2,703

2a · combined signal vs background · raw

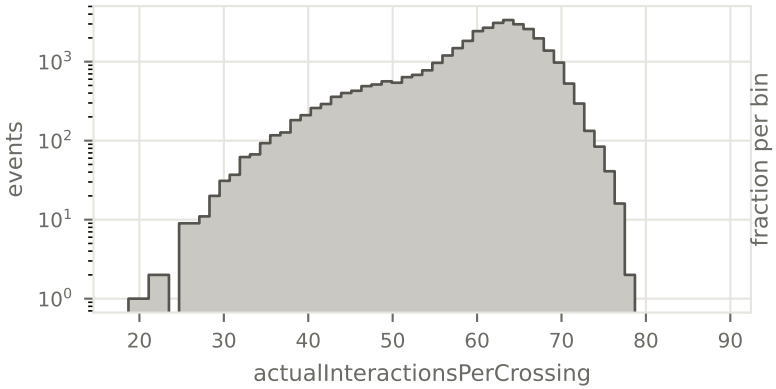


2b · combined signal vs background · normalized

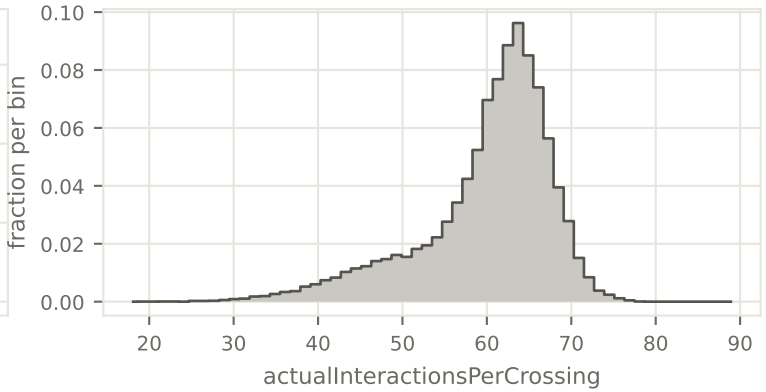


2 · combined signal
background n 34,970
combined, pooled D 0.300 n 9,996
equal weight per mass D 0.298

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

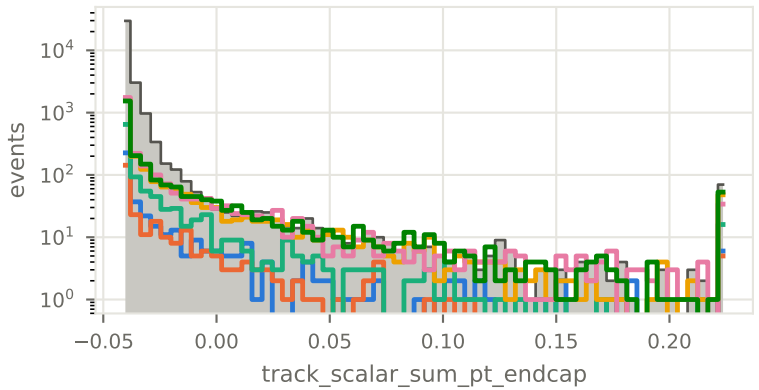
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

track_scalar_sum_pt_endcap

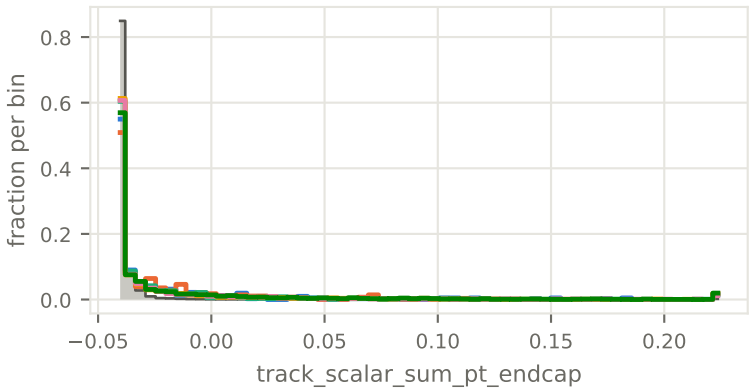
region: ENDCAP · rank 48 / 98 · in: NN1 endcap

D_equal 0.286 (noise 0.022) D_pooled 0.272 min 0.251 (mH400_mS100) max 0.347 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

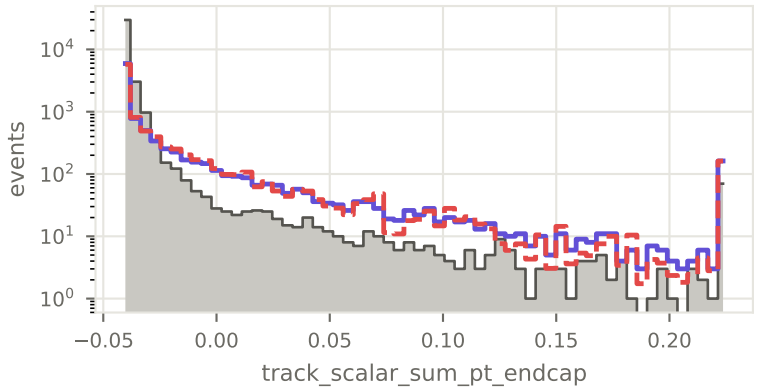


1b · each mass point vs background · normalized

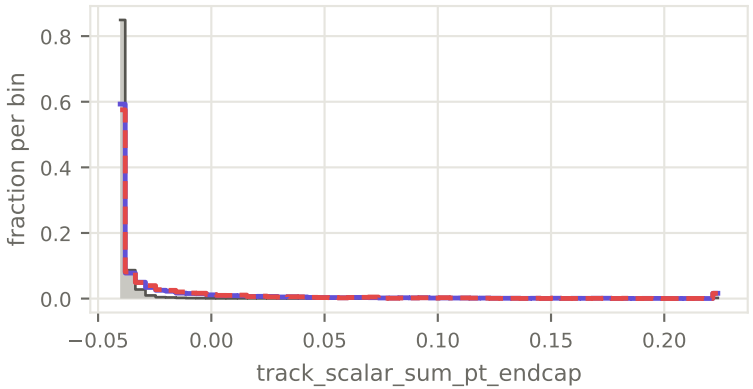


1 · each mass point
background n 34,970
mH125_mS15 D 0.316 n 411
mH125_mS55 D 0.347 n 281
mH200_mS50 D 0.256 n 1,071
mH400_mS100 D 0.251 n 2,648
mH600_mS150 D 0.262 n 2,882
mH1000_mS275 D 0.302 n 2,703

2a · combined signal vs background · raw

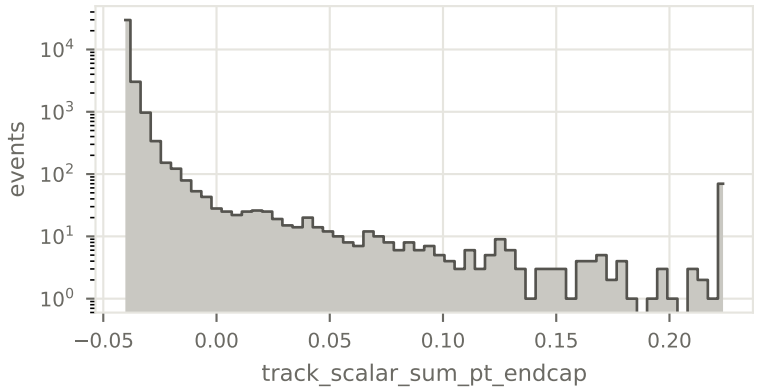


2b · combined signal vs background · normalized

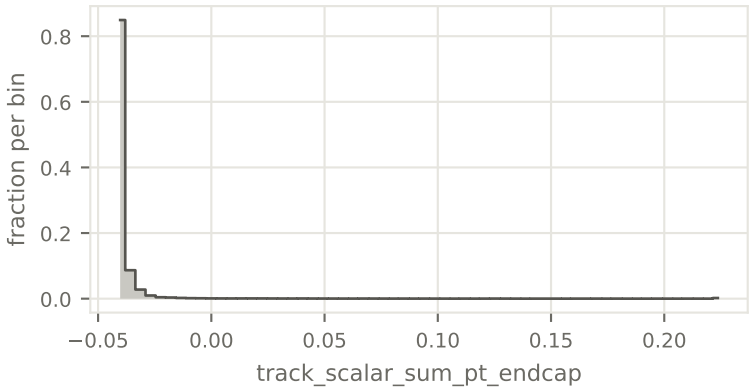


2 · combined signal
background n 34,970
combined, pooled D 0.272 n 9,996
equal weight per mass D 0.286

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

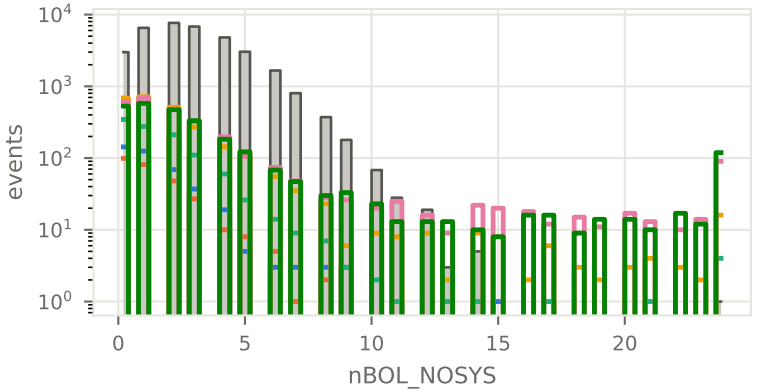
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

nBOL_NOSYS

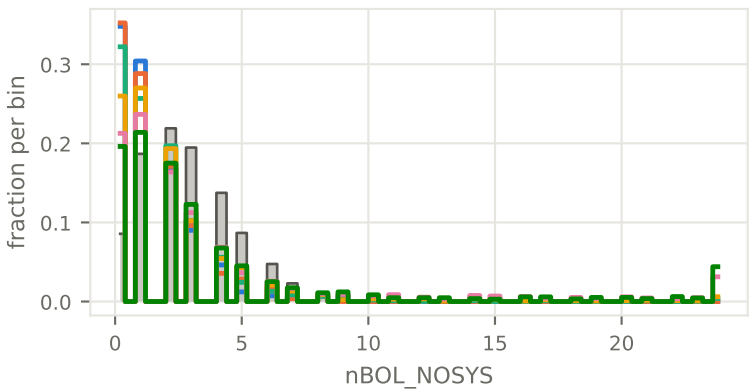
region: ENDCAP · rank 49 / 98 · in: NN1 barrel

D_equal 0.271 (noise 0.022) D_pooled 0.215 min 0.138 (mH1000_mS275) max 0.380 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

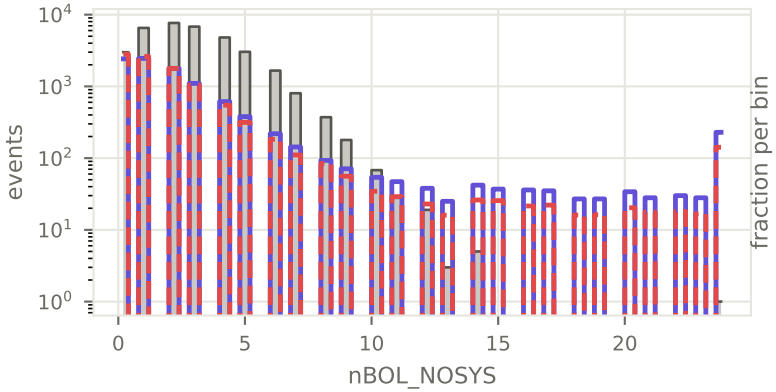


1b · each mass point vs background · normalized

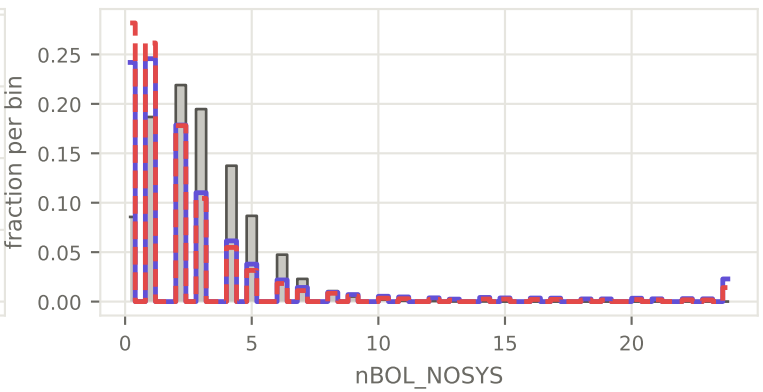


1 · each mass point
background n 34,970
mH125_mS15 D 0.380 n 411
mH125_mS55 D 0.368 n 281
mH200_mS50 D 0.307 n 1,071
mH400_mS100 D 0.257 n 2,648
mH600_mS150 D 0.177 n 2,882
mH1000_mS275 D 0.138 n 2,703

2a · combined signal vs background · raw

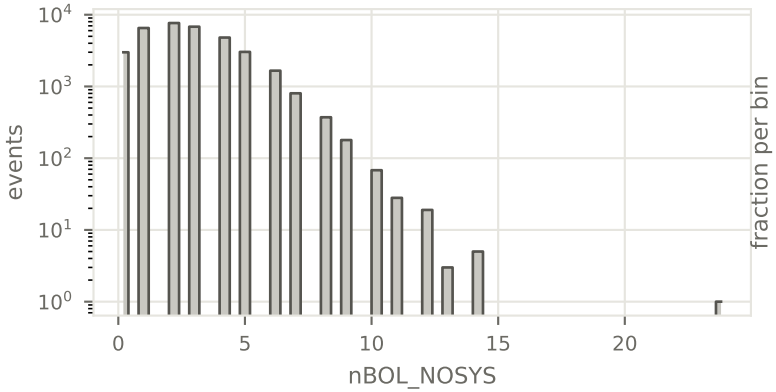


2b · combined signal vs background · normalized

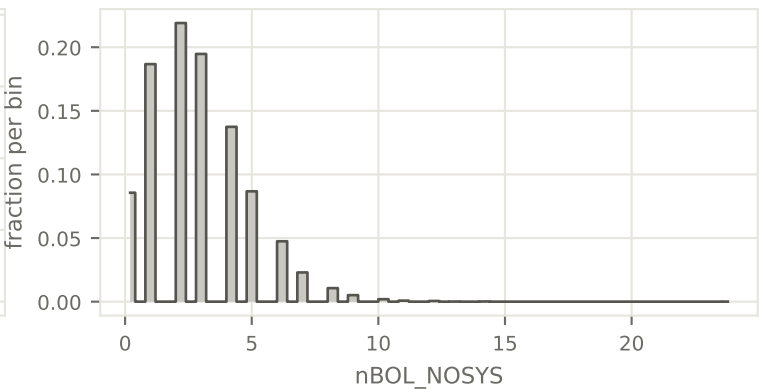


2 · combined signal
background n 34,970
combined, pooled D 0.215 n 9,996
equal weight per mass D 0.271

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

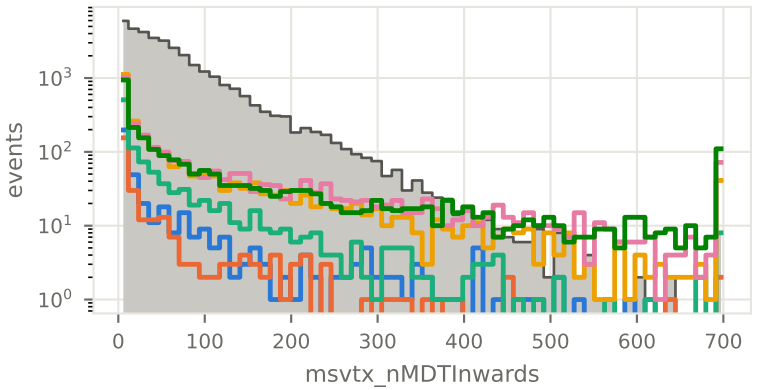
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nMDTInwards

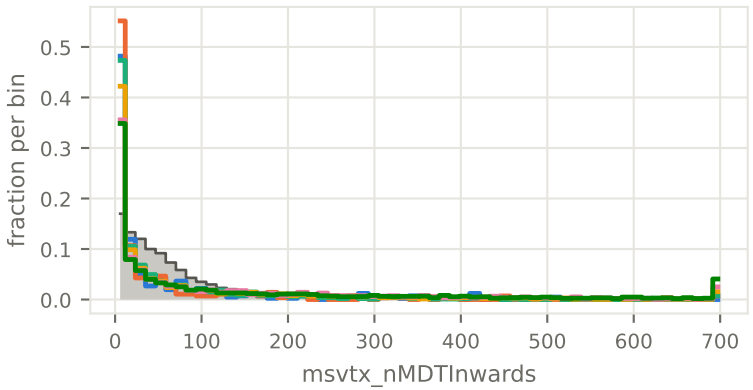
region: ENDCAP · rank 50 / 98

D_equal 0.269 (noise 0.022) D_pooled 0.227 min 0.196 (mH600_mS150) max 0.382 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

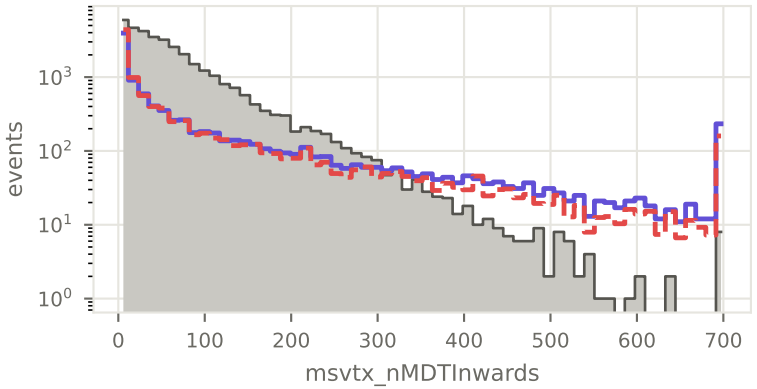


1b · each mass point vs background · normalized

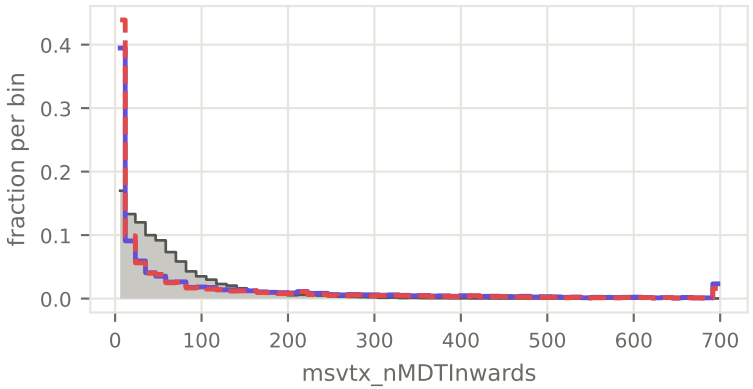


1 · each mass point
background n 34,970
mH125_mS15 D 0.320 n 411
mH125_mS55 D 0.382 n 281
mH200_mS50 D 0.311 n 1,071
mH400_mS100 D 0.253 n 2,648
mH600_mS150 D 0.196 n 2,882
mH1000_mS275 D 0.207 n 2,703

2a · combined signal vs background · raw

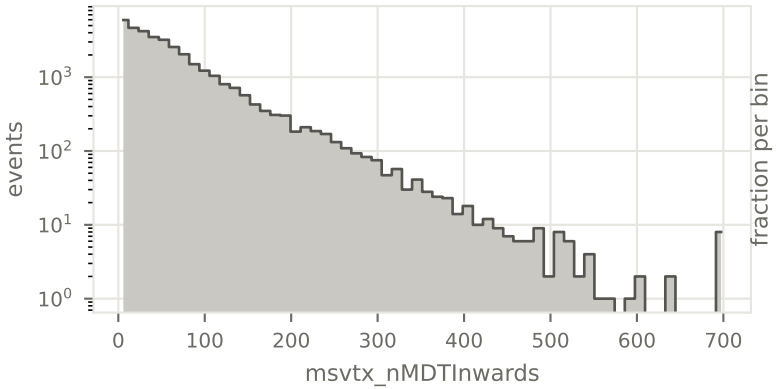


2b · combined signal vs background · normalized

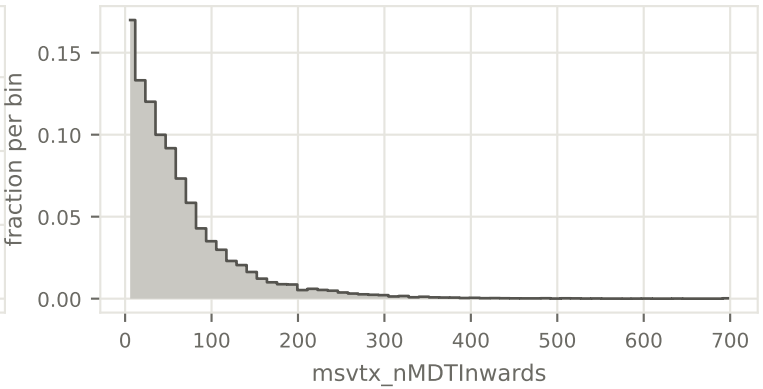


2 · combined signal
background n 34,970
combined, pooled D 0.227 n 9,996
equal weight per mass D 0.269

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

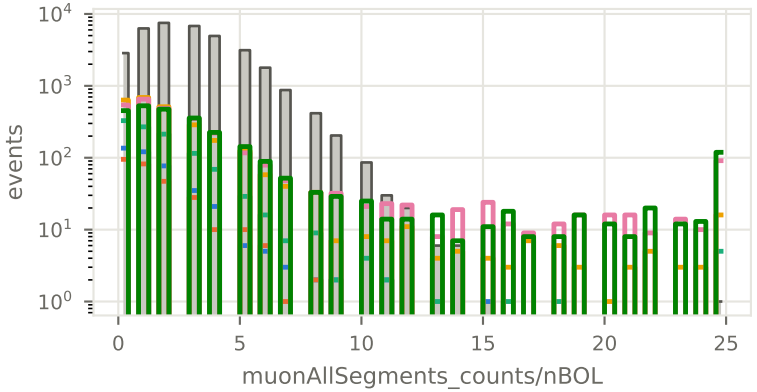
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

muonAllSegments_counts/nBOL

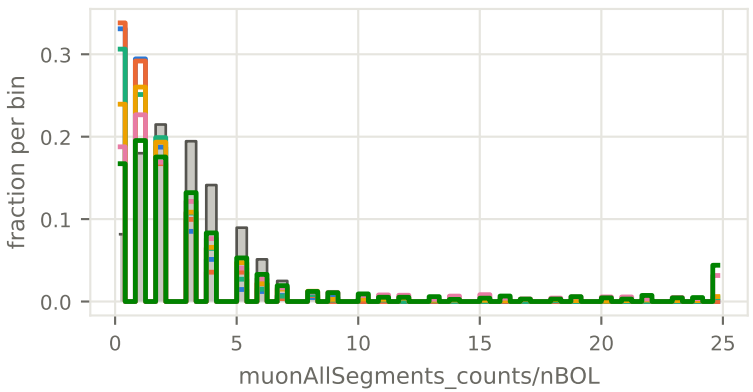
region: ENDCAP · rank 51 / 98

D_equal 0.253 (noise 0.022) D_pooled 0.191 min 0.120 (mH1000_mS275) max 0.368 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

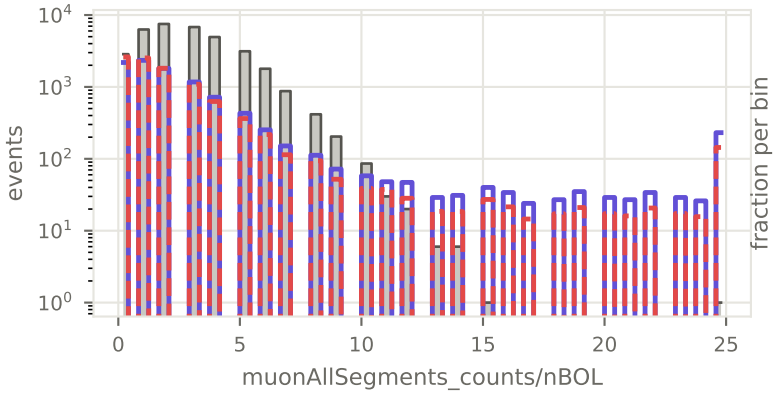


1b · each mass point vs background · normalized

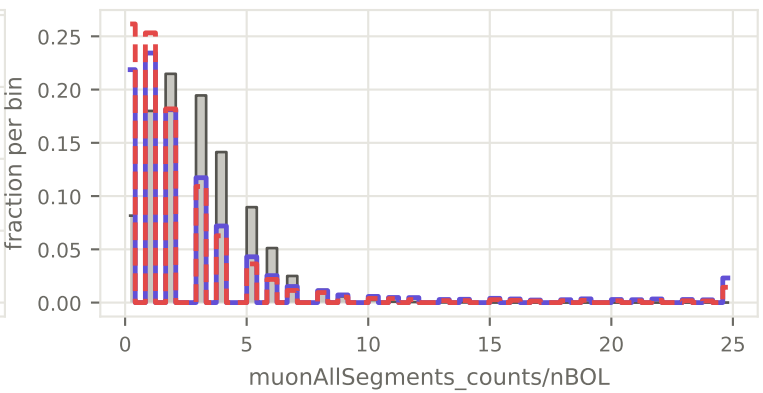


1 · each mass point
background n 34,970
mH125_mS15 D 0.364 n 411
mH125_mS55 D 0.368 n 281
mH200_mS50 D 0.296 n 1,071
mH400_mS100 D 0.238 n 2,648
mH600_mS150 D 0.153 n 2,882
mH1000_mS275 D 0.120 n 2,703

2a · combined signal vs background · raw

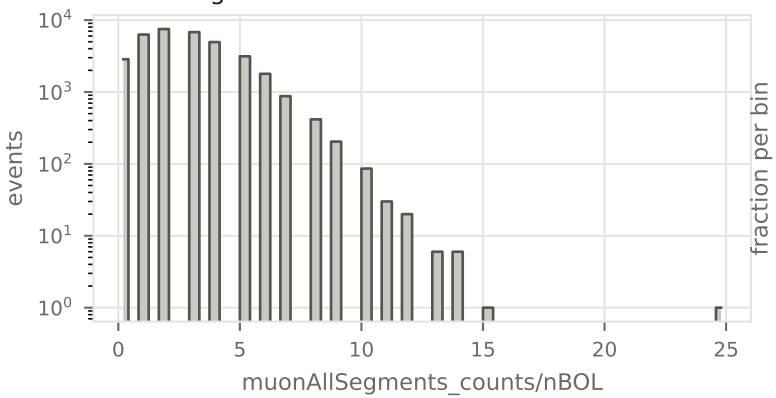


2b · combined signal vs background · normalized

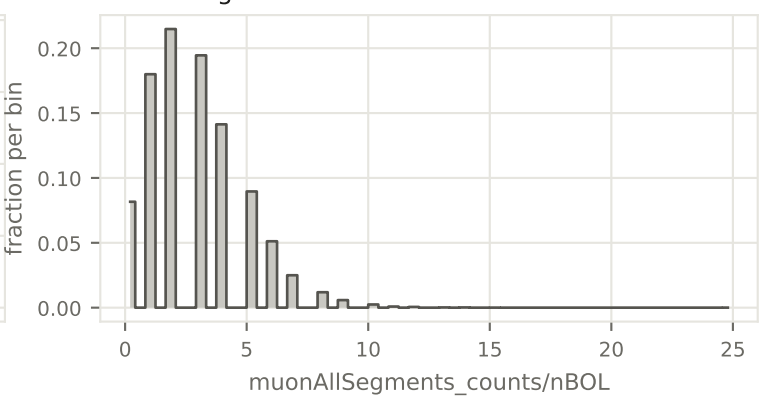


2 · combined signal
background n 34,970
combined, pooled D 0.191 n 9,996
equal weight per mass D 0.253

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

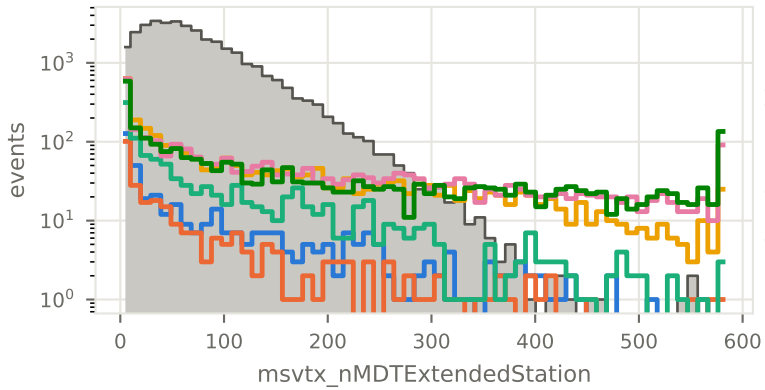
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nMDTExtendedStation

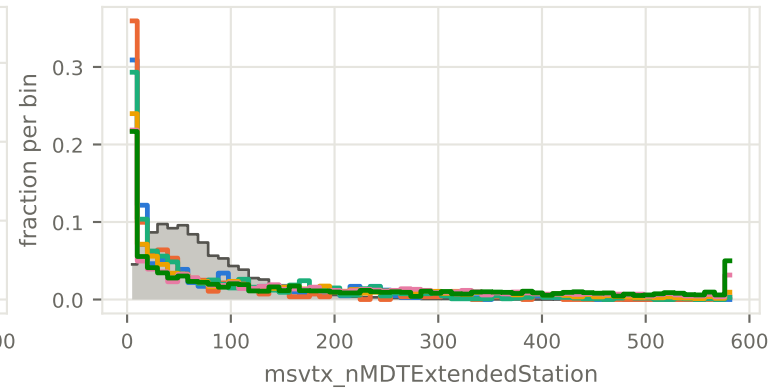
region: ENDCAP · rank 52 / 98

D_equal 0.243 (noise 0.022) D_pooled 0.286 min 0.261 (mH400_mS100) max 0.351 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

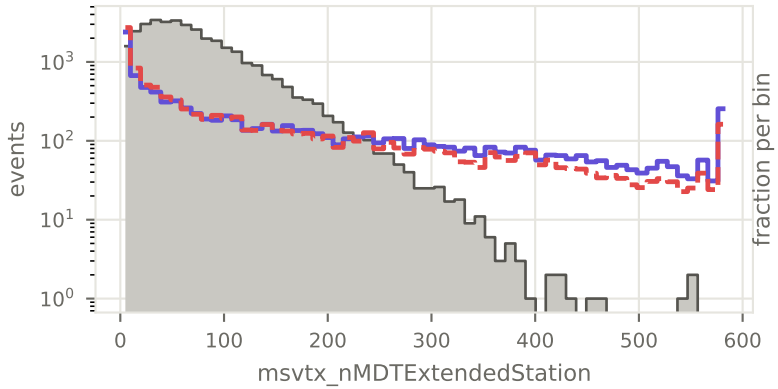


1b · each mass point vs background · normalized

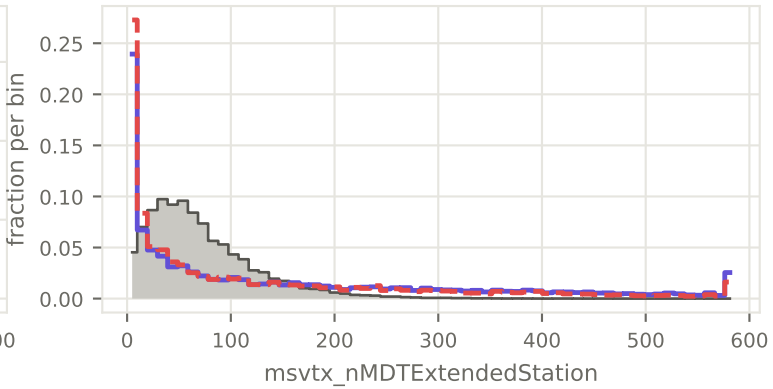


1 · each mass point
background n 34,970
mH125_mS15 D 0.319 n 411
mH125_mS55 D 0.347 n 281
mH200_mS50 D 0.281 n 1,071
mH400_mS100 D 0.261 n 2,648
mH600_mS150 D 0.343 n 2,882
mH1000_mS275 D 0.351 n 2,703

2a · combined signal vs background · raw

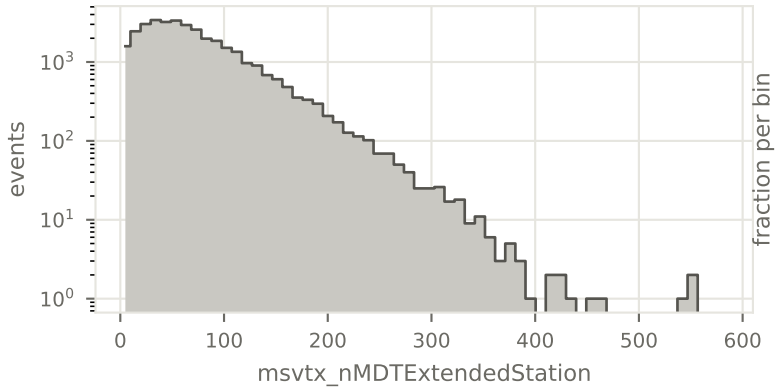


2b · combined signal vs background · normalized

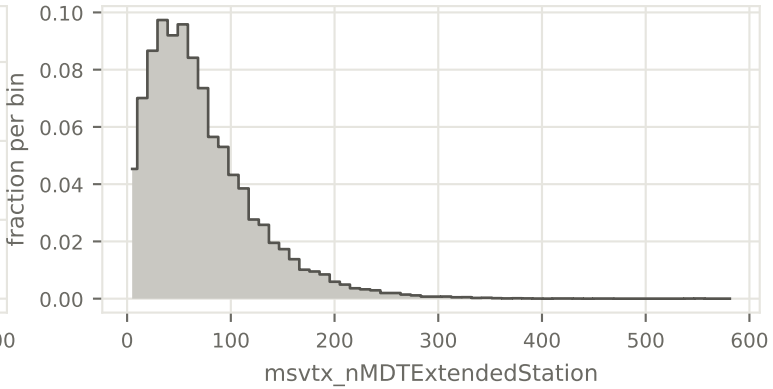


2 · combined signal
background n 34,970
combined, pooled D 0.286 n 9,996
equal weight per mass D 0.243

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

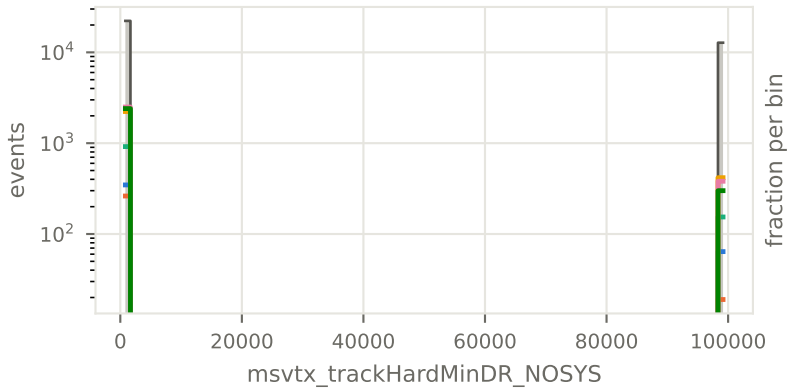
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_trackHardMinDR_NOSYS

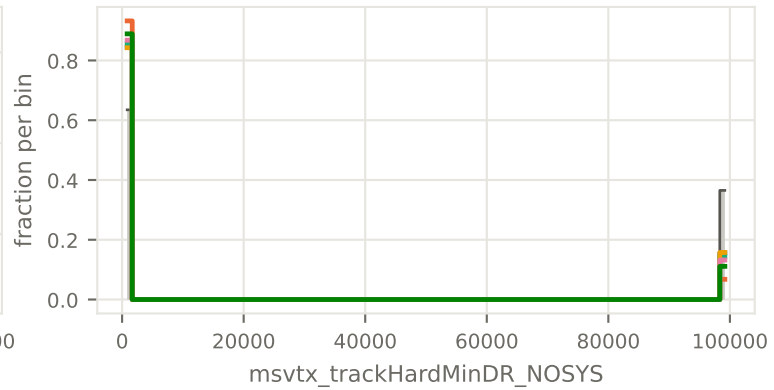
region: ENDCAP · rank 53 / 98

D_equal 0.243 (noise 0.022) D_pooled 0.254 min 0.210 (mH125_mS15) max 0.300 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

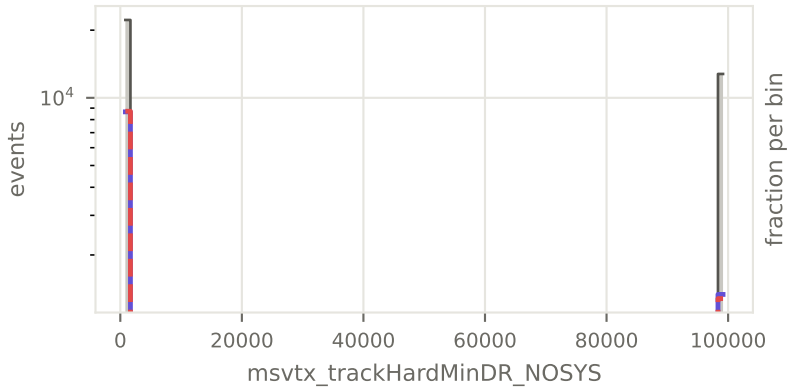


1b · each mass point vs background · normalized

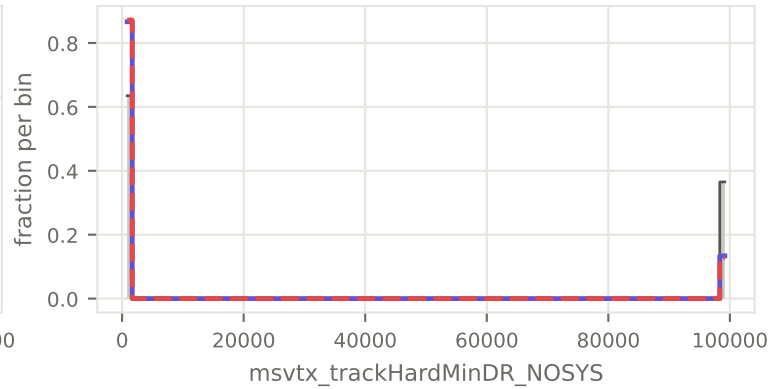


1 · each mass point
background n 34,970
mH125_mS15 D 0.210 n 411
mH125_mS55 D 0.300 n 281
mH200_mS50 D 0.224 n 1,071
mH400_mS100 D 0.242 n 2,648
mH600_mS150 D 0.265 n 2,882
mH1000_mS275 D 0.291 n 2,703

2a · combined signal vs background · raw

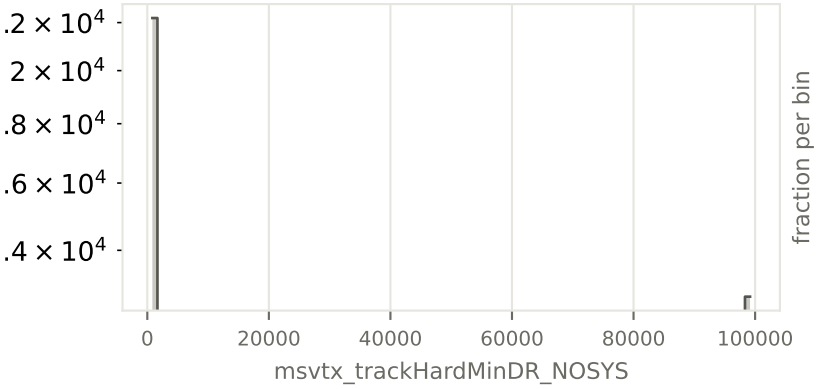


2b · combined signal vs background · normalized

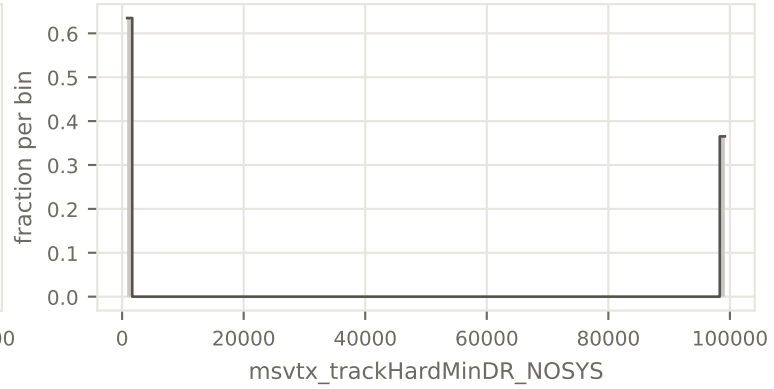


2 · combined signal
background n 34,970
combined, pooled D 0.254 n 9,996
equal weight per mass D 0.243

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

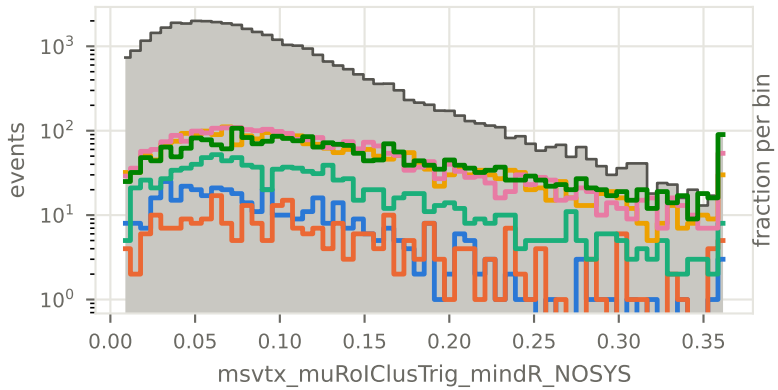
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_muRoIClusTrig_mindR_NOSYS

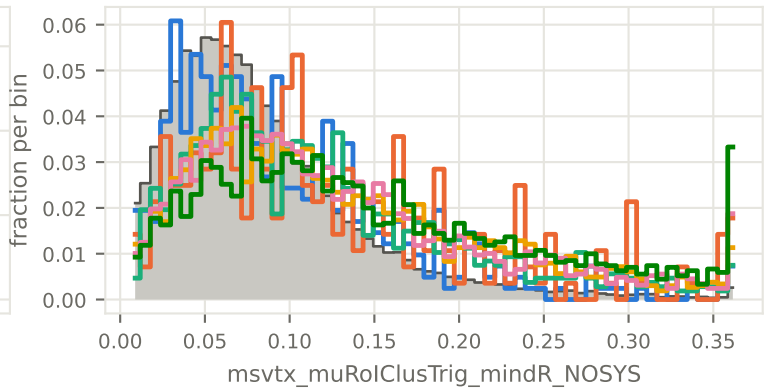
region: ENDCAP · rank 54 / 98

D_equal 0.232 (noise 0.022) D_pooled 0.262 min 0.119 (mH125_mS15) max 0.332 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

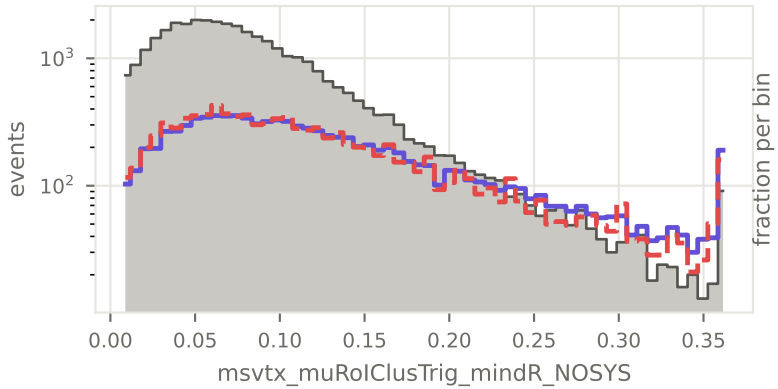


1b · each mass point vs background · normalized

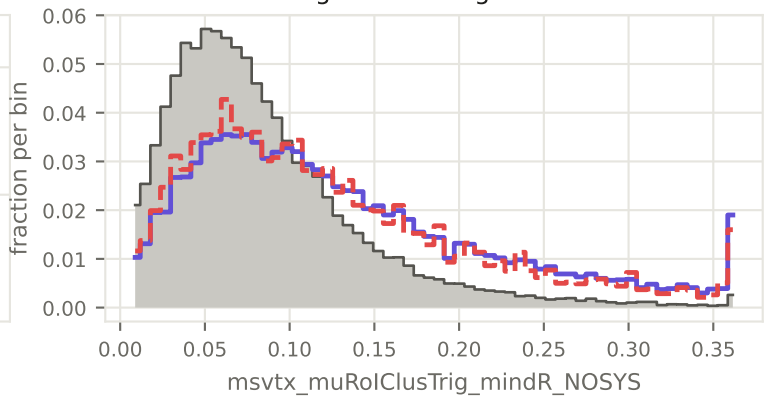


1 · each mass point
background n 34,970
mH125_mS15 D 0.119 n 411
mH125_mS55 D 0.262 n 281
mH200_mS50 D 0.218 n 1,071
mH400_mS100 D 0.259 n 2,648
mH600_mS150 D 0.249 n 2,882
mH1000_mS275 D 0.332 n 2,703

2a · combined signal vs background · raw

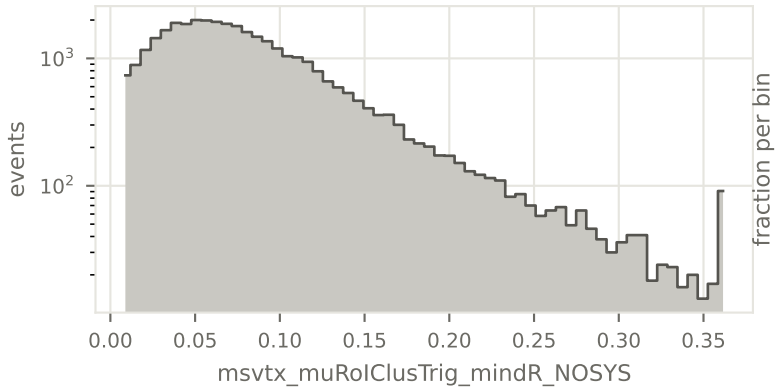


2b · combined signal vs background · normalized

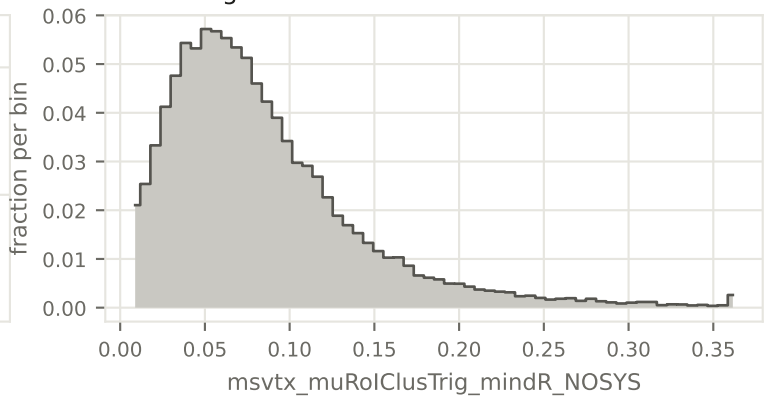


2 · combined signal
background n 34,970
combined, pooled D 0.262 n 9,996
equal weight per mass D 0.232

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

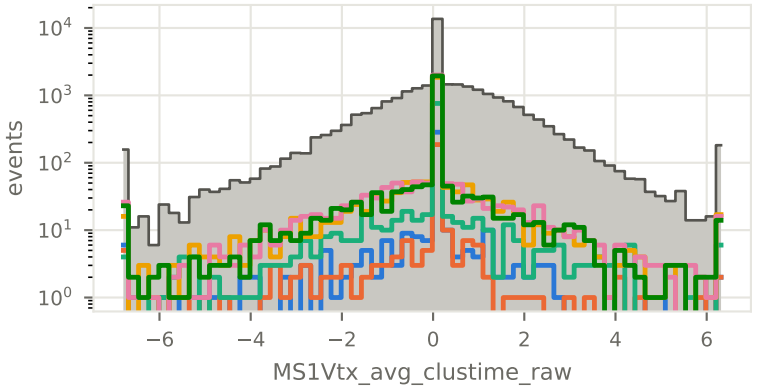
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_avg_clustime_raw

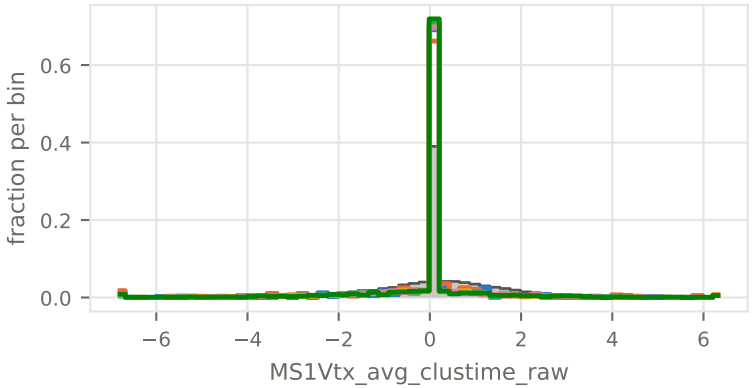
region: ENDCAP · rank 55 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.229 (noise 0.022) D_pooled 0.231 min 0.218 (mH125_mS55) max 0.238 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

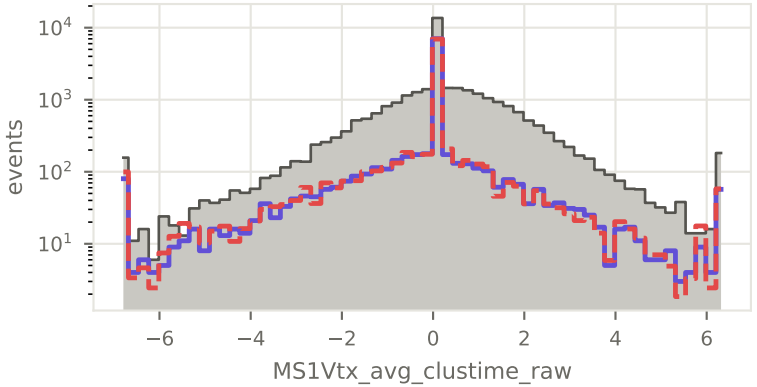


1b · each mass point vs background · normalized

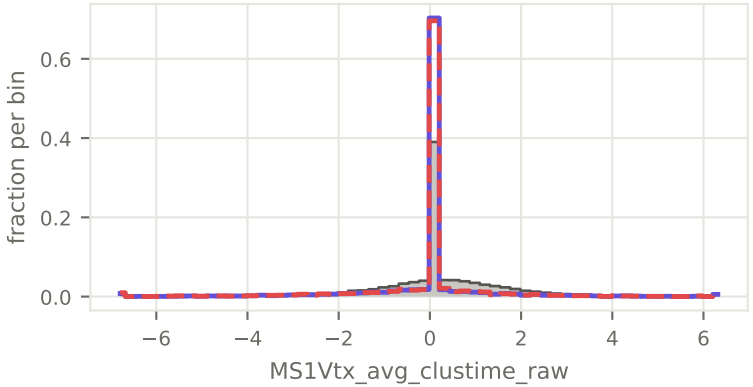


1 · each mass point
background n 34,970
mH125_mS15 D 0.234 n 411
mH125_mS55 D 0.218 n 281
mH200_mS50 D 0.234 n 1,071
mH400_mS100 D 0.225 n 2,648
mH600_mS150 D 0.231 n 2,882
mH1000_mS275 D 0.238 n 2,703

2a · combined signal vs background · raw

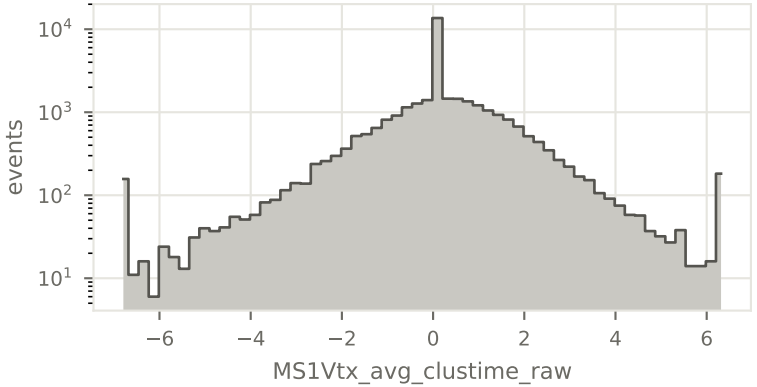


2b · combined signal vs background · normalized

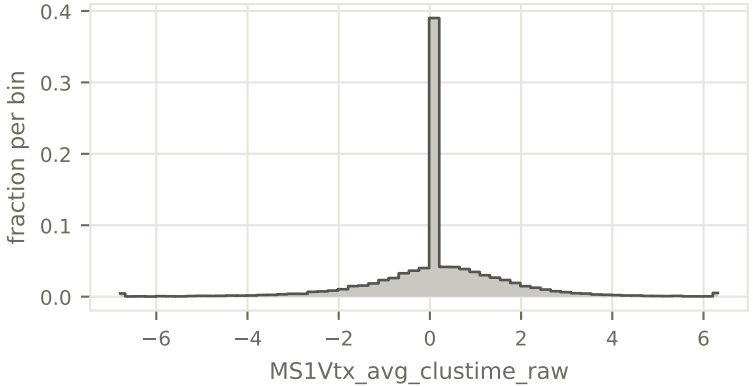


2 · combined signal
background n 34,970
combined, pooled D 0.231 n 9,996
equal weight per mass D 0.229

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

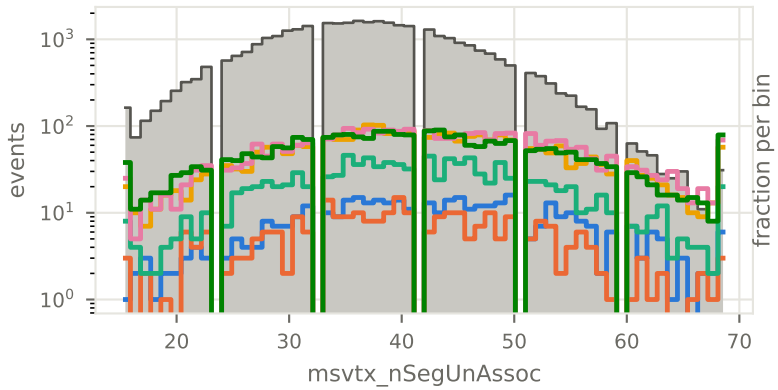
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nSegUnAssoc

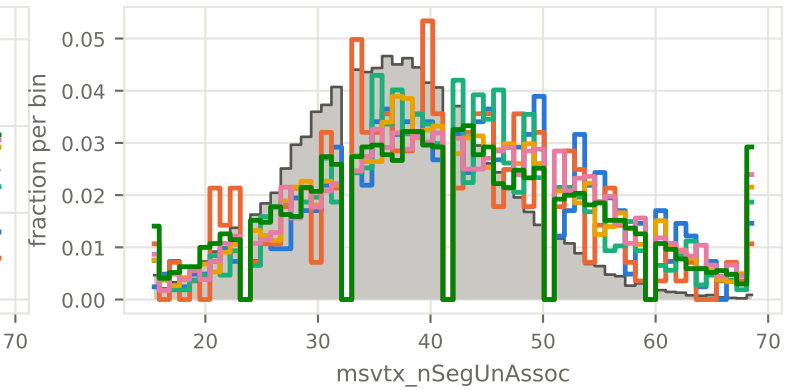
region: ENDCAP · rank 56 / 98

D_equal 0.214 (noise 0.022) D_pooled 0.217 min 0.169 (mH125_mS55) max 0.254 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

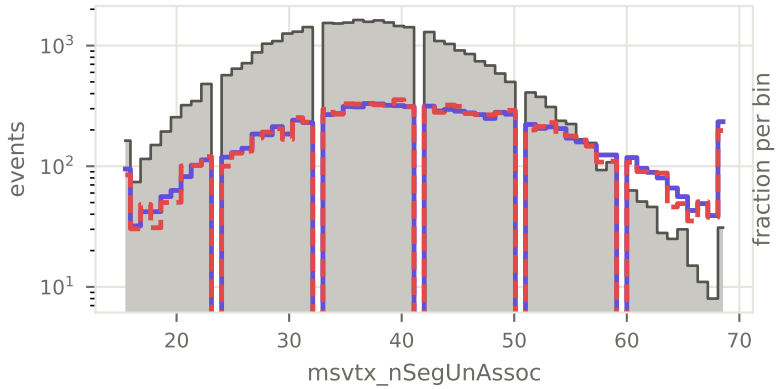


1b · each mass point vs background · normalized

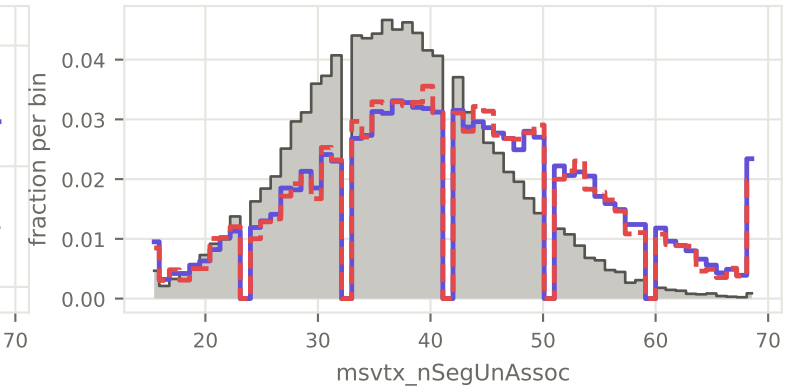


1 · each mass point
background n 34,970
mH125_mS15 D 0.254 n 411
mH125_mS55 D 0.169 n 281
mH200_mS50 D 0.208 n 1,071
mH400_mS100 D 0.214 n 2,648
mH600_mS150 D 0.248 n 2,882
mH1000_mS275 D 0.196 n 2,703

2a · combined signal vs background · raw

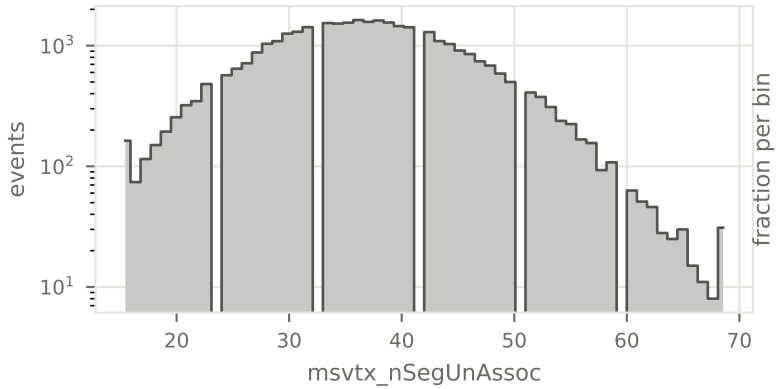


2b · combined signal vs background · normalized

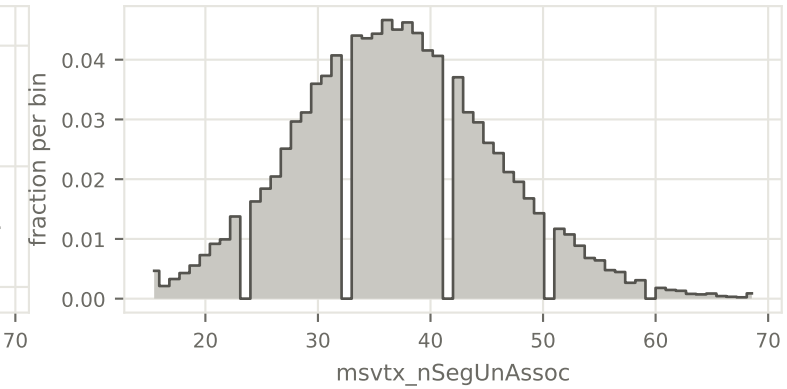


2 · combined signal
background n 34,970
combined, pooled D 0.217 n 9,996
equal weight per mass D 0.214

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

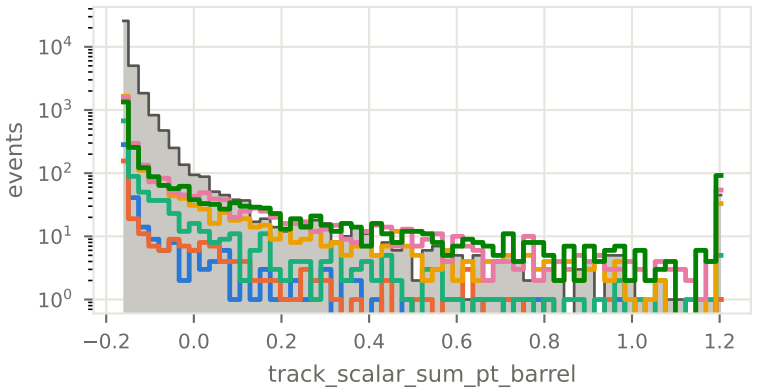
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

track_scalar_sum_pt_barrel

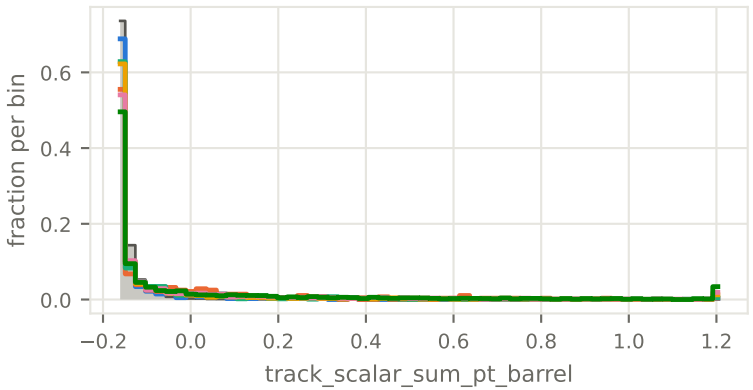
region: ENDCAP · rank 57 / 98 · in: NN1 barrel

D_equal 0.213 (noise 0.022) D_pooled 0.226 min 0.116 (mH125_mS15) max 0.299 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

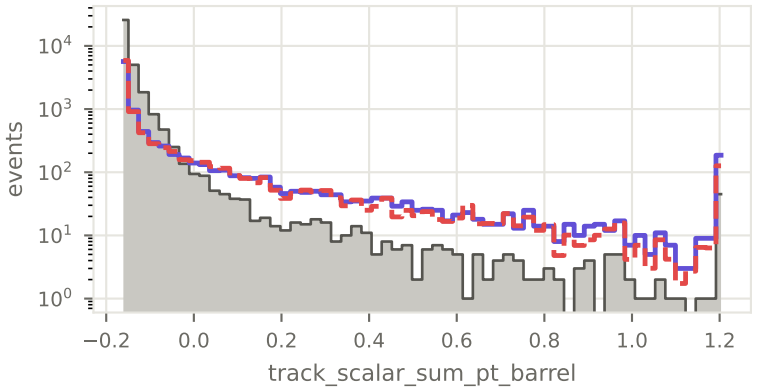


1b · each mass point vs background · normalized

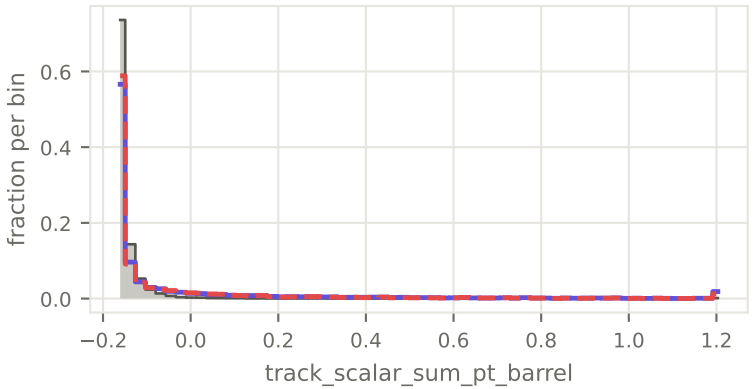


1 · each mass point
background n 34,970
mH125_mS15 D 0.116 n 411
mH125_mS55 D 0.280 n 281
mH200_mS50 D 0.176 n 1,071
mH400_mS100 D 0.170 n 2,648
mH600_mS150 D 0.245 n 2,882
mH1000_mS275 D 0.299 n 2,703

2a · combined signal vs background · raw

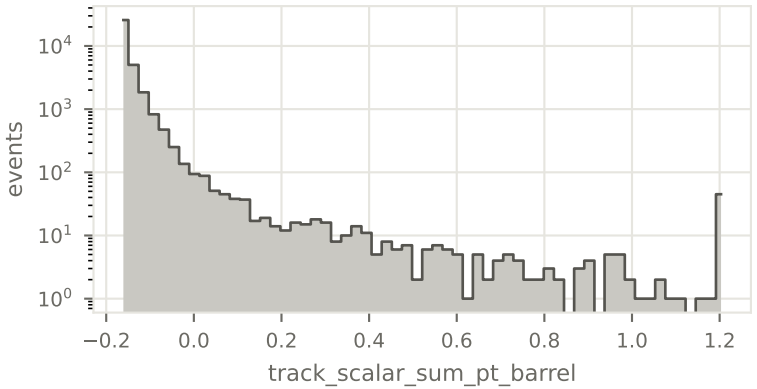


2b · combined signal vs background · normalized

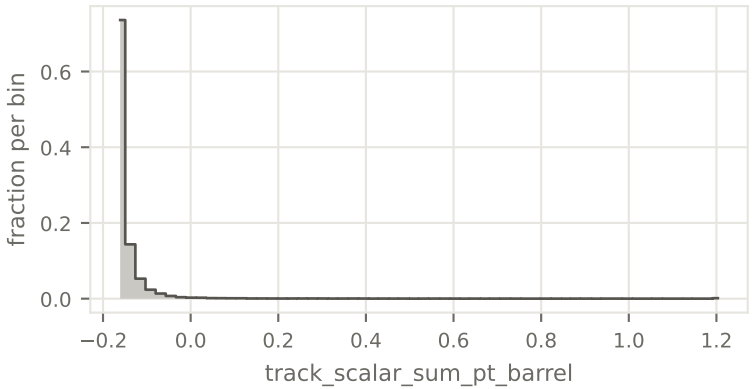


2 · combined signal
background n 34,970
combined, pooled D 0.226 n 9,996
equal weight per mass D 0.213

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

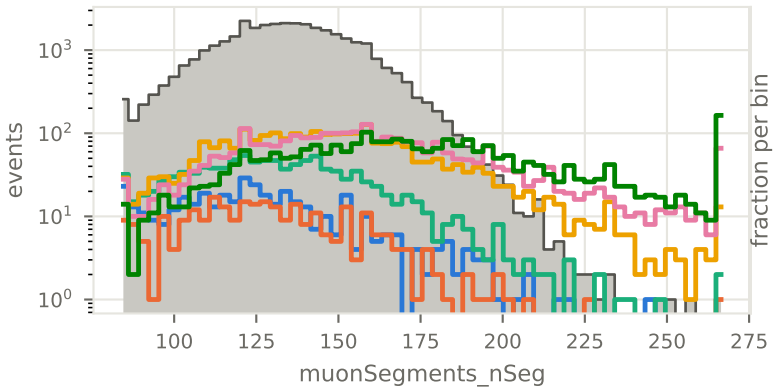
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

muonSegments_nSeg

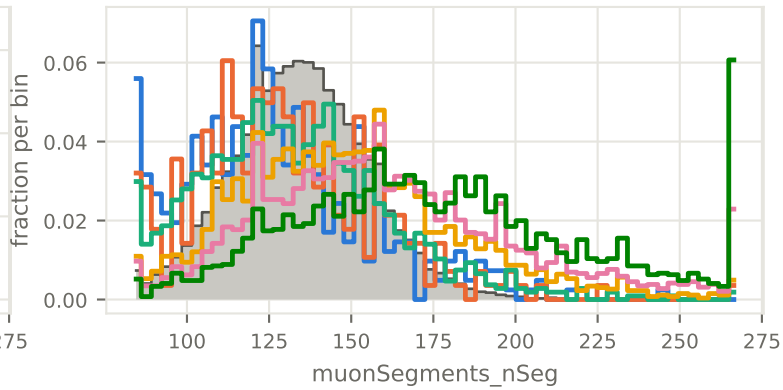
region: ENDCAP · rank 58 / 98

D_equal 0.201 (noise 0.022) D_pooled 0.310 min 0.122 (mH200_mS50) max 0.518 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

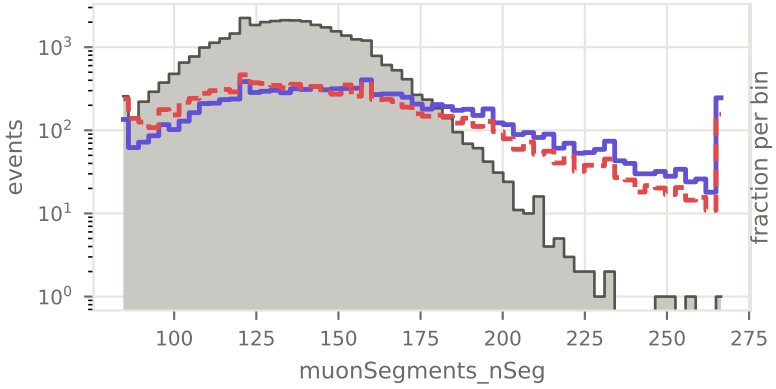


1b · each mass point vs background · normalized

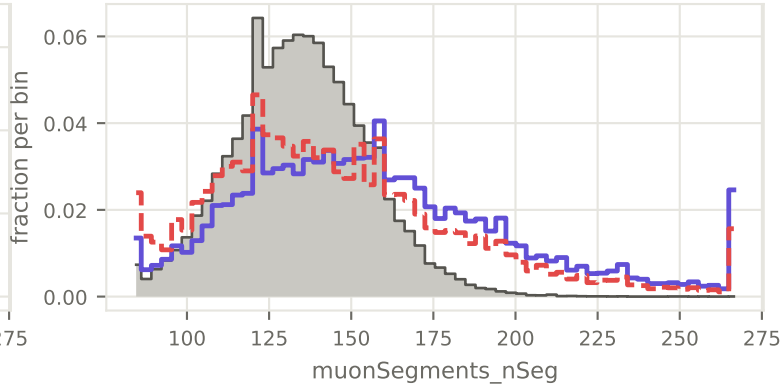


1 · each mass point
background n 34,970
mH125_mS15 D 0.200 n 411
mH125_mS55 D 0.159 n 281
mH200_mS50 D 0.122 n 1,071
mH400_mS100 D 0.232 n 2,648
mH600_mS150 D 0.363 n 2,882
mH1000_mS275 D 0.518 n 2,703

2a · combined signal vs background · raw

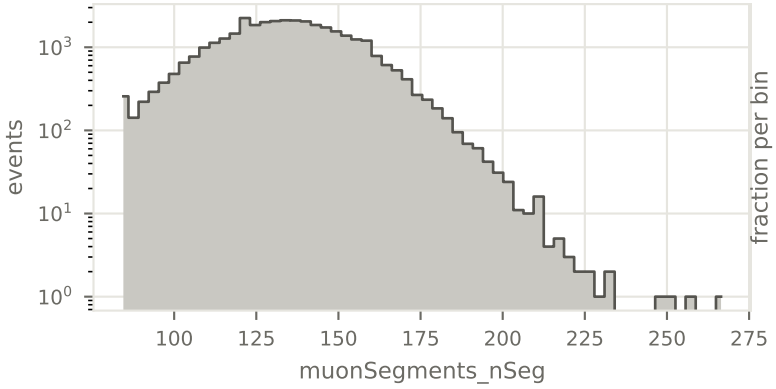


2b · combined signal vs background · normalized

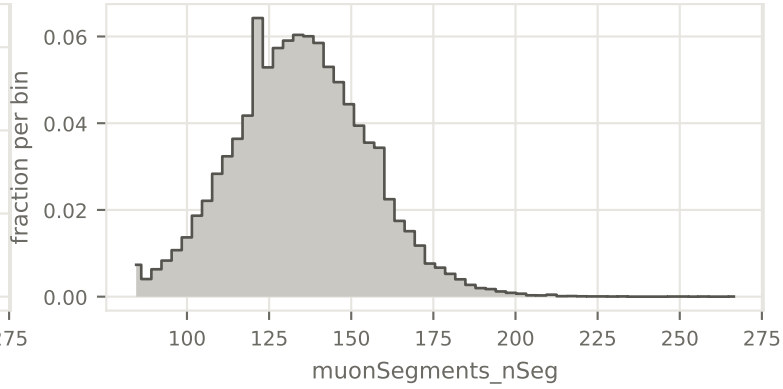


2 · combined signal
background n 34,970
combined, pooled D 0.310 n 9,996
equal weight per mass D 0.201

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

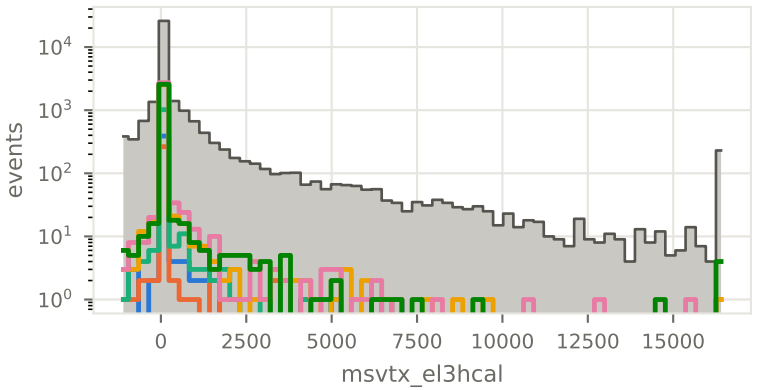
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el3hcal

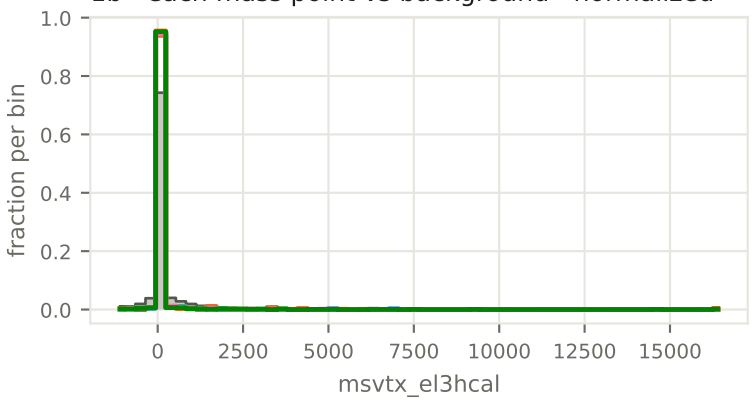
region: ENDCAP · rank 59 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.193 (noise 0.022) D_pooled 0.192 min 0.184 (mH600_mS150) max 0.197 (mH400_mS100) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

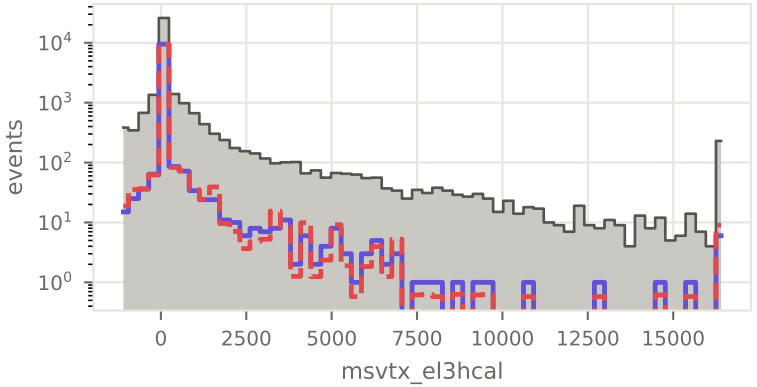


1b · each mass point vs background · normalized

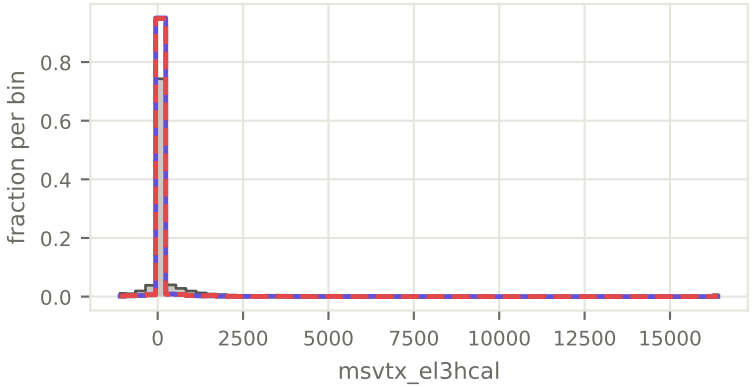


1 · each mass point
background n 34,970
mH125_mS15 D 0.196 n 411
mH125_mS55 D 0.190 n 281
mH200_mS50 D 0.196 n 1,071
mH400_mS100 D 0.197 n 2,648
mH600_mS150 D 0.184 n 2,882
mH1000_mS275 D 0.196 n 2,703

2a · combined signal vs background · raw

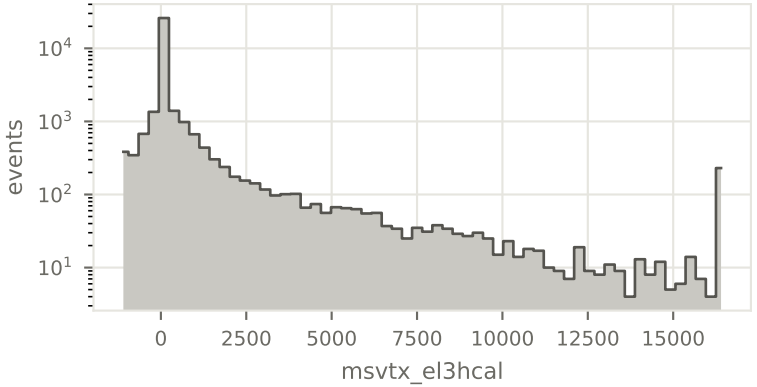


2b · combined signal vs background · normalized

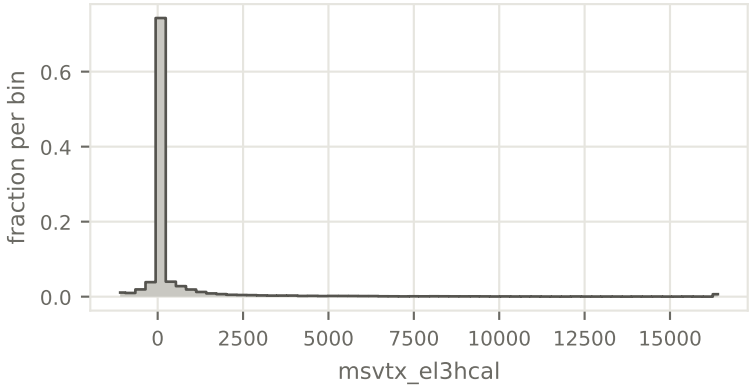


2 · combined signal
background n 34,970
combined, pooled D 0.192 n 9,996
equal weight per mass D 0.193

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

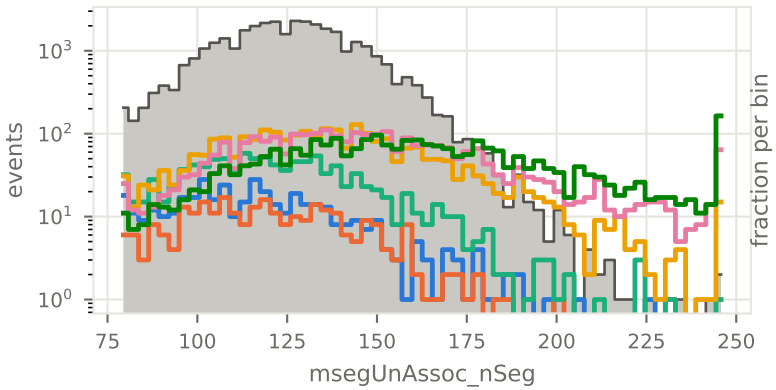
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msegUnAssoc_nSeg

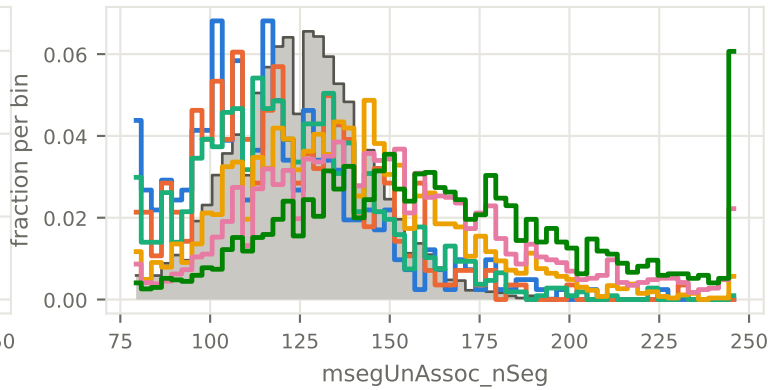
region: ENDCAP · rank 60 / 98

D_equal 0.187 (noise 0.022) D_pooled 0.297 min 0.134 (mH200_mS50) max 0.504 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

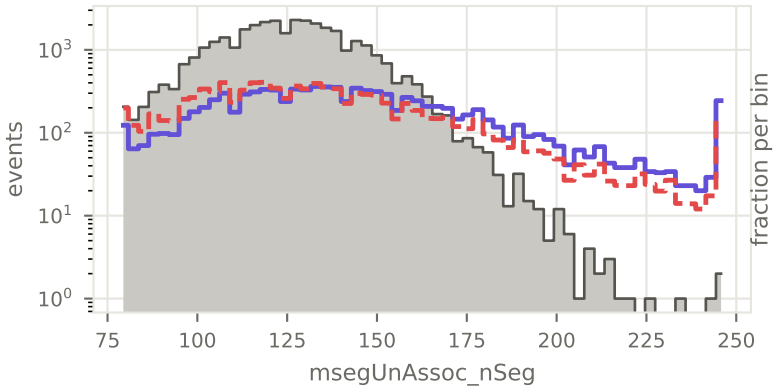


1b · each mass point vs background · normalized

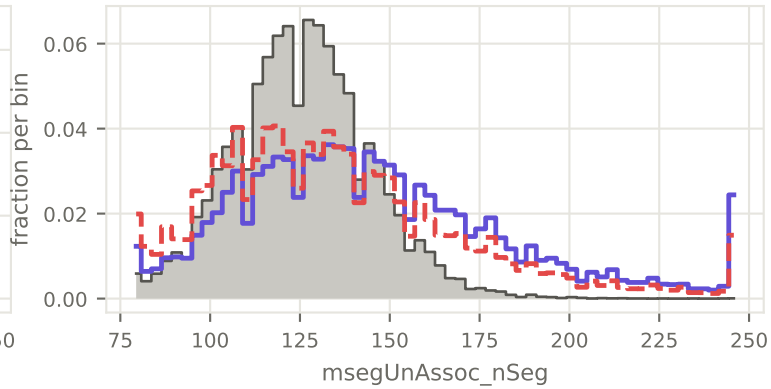


1 · each mass point
background n 34,970
mH125_mS15 D 0.229 n 411
mH125_mS55 D 0.171 n 281
mH200_mS50 D 0.134 n 1,071
mH400_mS100 D 0.222 n 2,648
mH600_mS150 D 0.355 n 2,882
mH1000_mS275 D 0.504 n 2,703

2a · combined signal vs background · raw

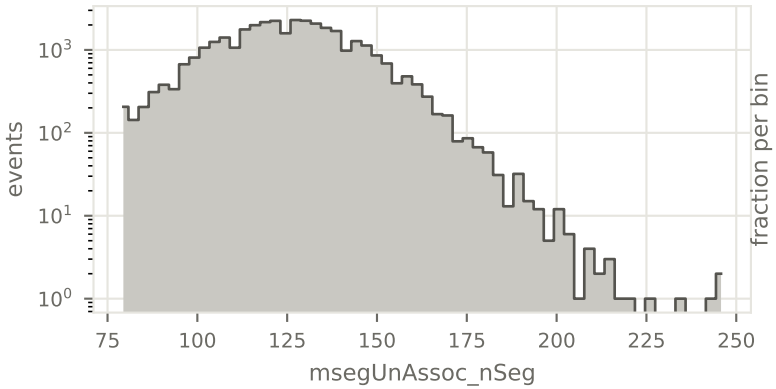


2b · combined signal vs background · normalized

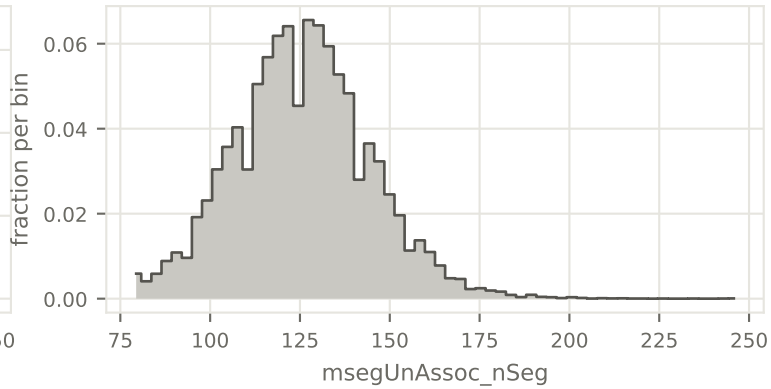


2 · combined signal
background n 34,970
combined, pooled D 0.297 n 9,996
equal weight per mass D 0.187

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

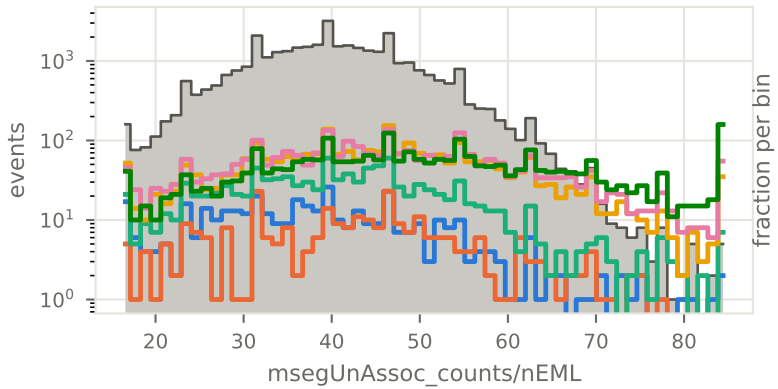
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msegUnAssoc_counts/nEML

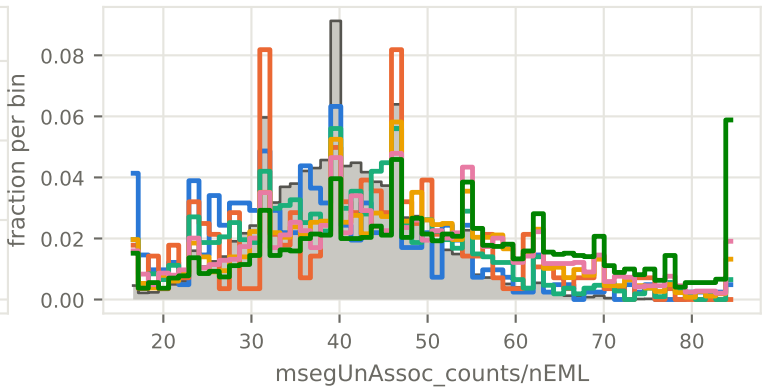
region: ENDCAP · rank 61 / 98

D_equal 0.187 (noise 0.022) D_pooled 0.247 min 0.100 (mH200_mS50) max 0.360 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

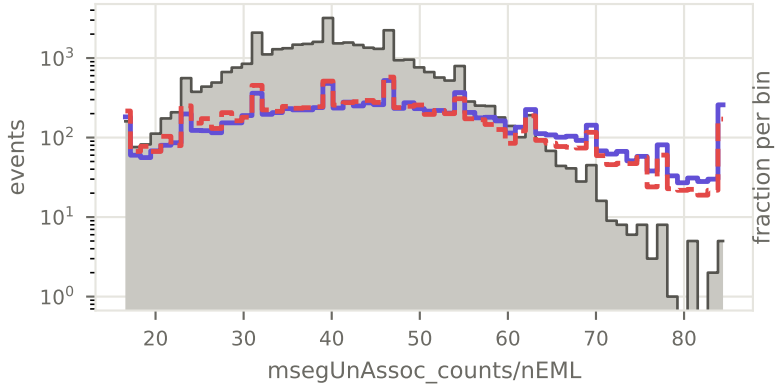


1b · each mass point vs background · normalized

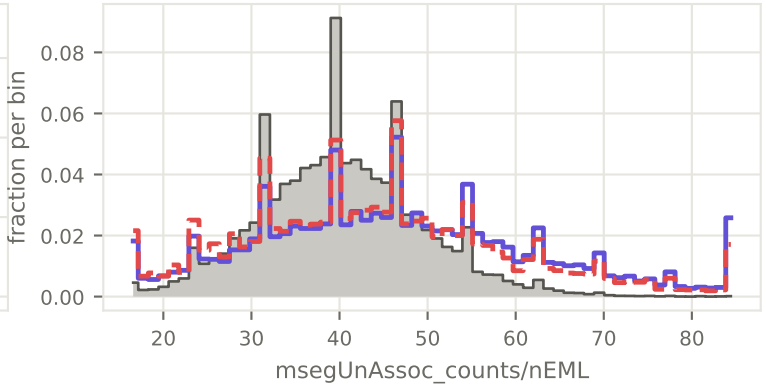


1 · each mass point
background n 34,970
mH125_mS15 D 0.157 n 411
mH125_mS55 D 0.149 n 281
mH200_mS50 D 0.100 n 1,071
mH400_mS100 D 0.234 n 2,648
mH600_mS150 D 0.258 n 2,882
mH1000_mS275 D 0.360 n 2,703

2a · combined signal vs background · raw

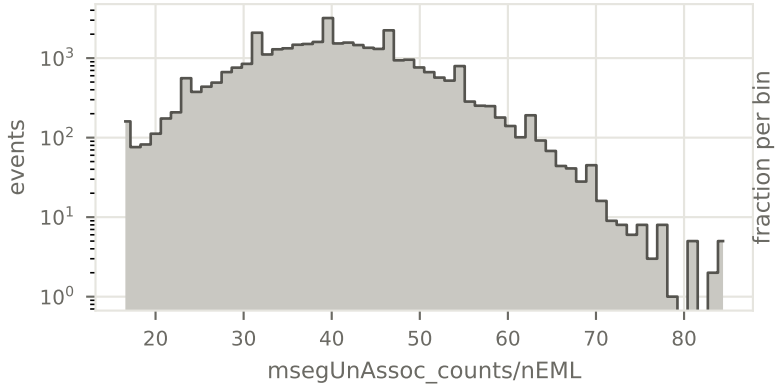


2b · combined signal vs background · normalized

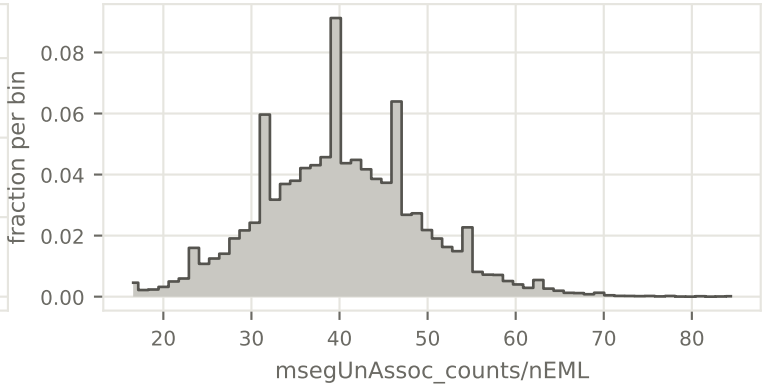


2 · combined signal
background n 34,970
combined, pooled D 0.247 n 9,996
equal weight per mass D 0.187

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

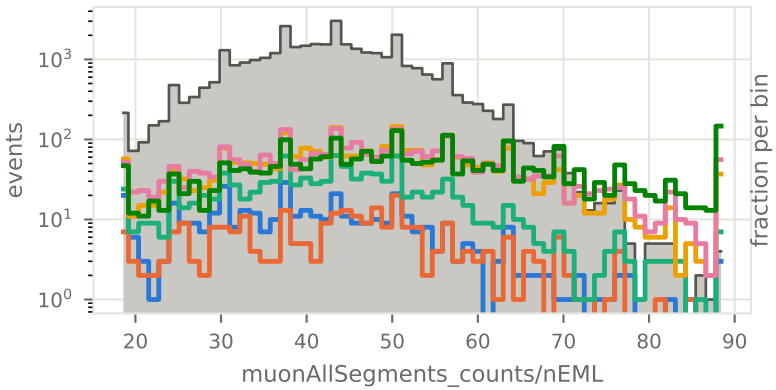
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

muonAllSegments_counts/nEML

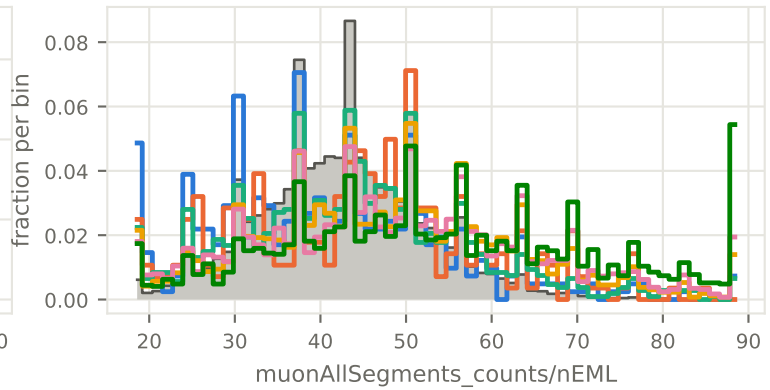
region: ENDCAP · rank 62 / 98

D_equal 0.179 (noise 0.022) D_pooled 0.239 min 0.089 (mH200_mS50) max 0.352 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

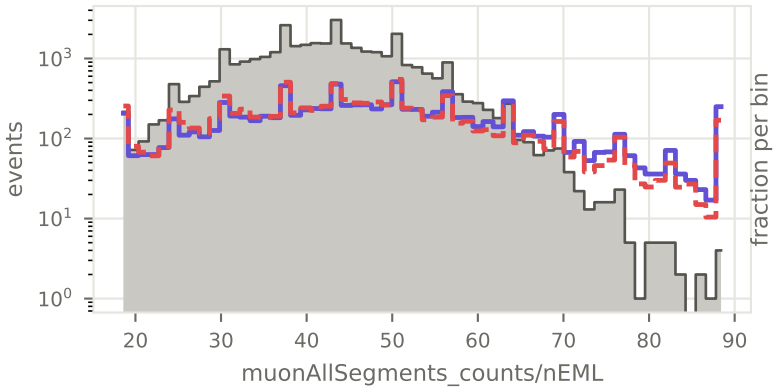


1b · each mass point vs background · normalized

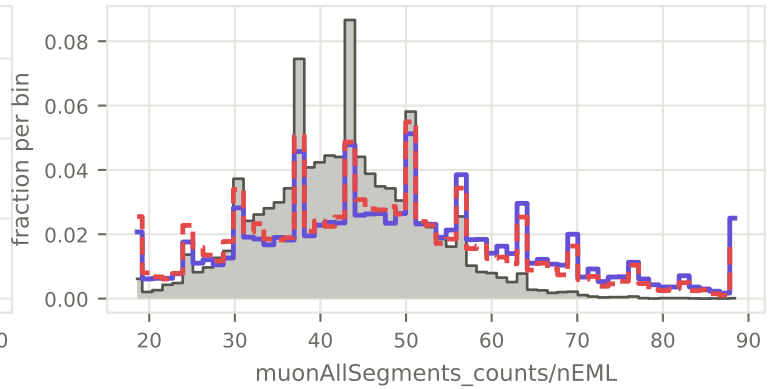


1 · each mass point
background n 34,970
mH125_mS15 D 0.158 n 411
mH125_mS55 D 0.133 n 281
mH200_mS50 D 0.089 n 1,071
mH400_mS100 D 0.225 n 2,648
mH600_mS150 D 0.253 n 2,882
mH1000_mS275 D 0.352 n 2,703

2a · combined signal vs background · raw

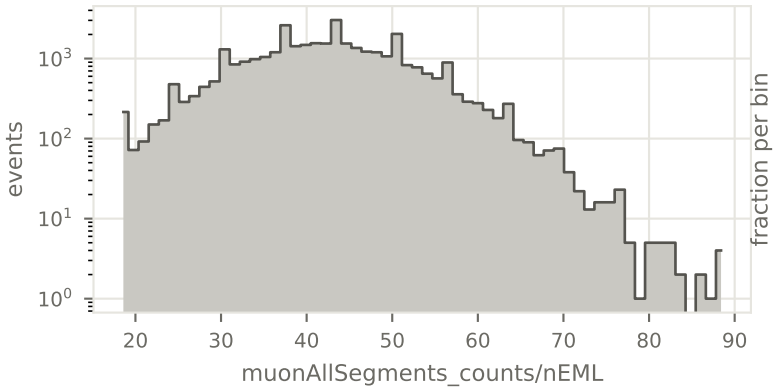


2b · combined signal vs background · normalized

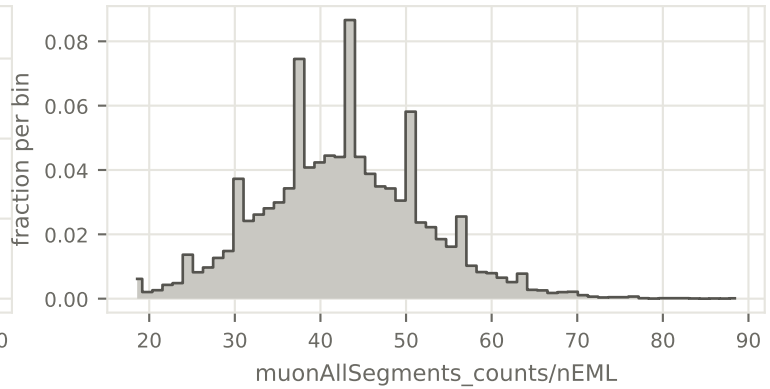


2 · combined signal
background n 34,970
combined, pooled D 0.239 n 9,996
equal weight per mass D 0.179

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

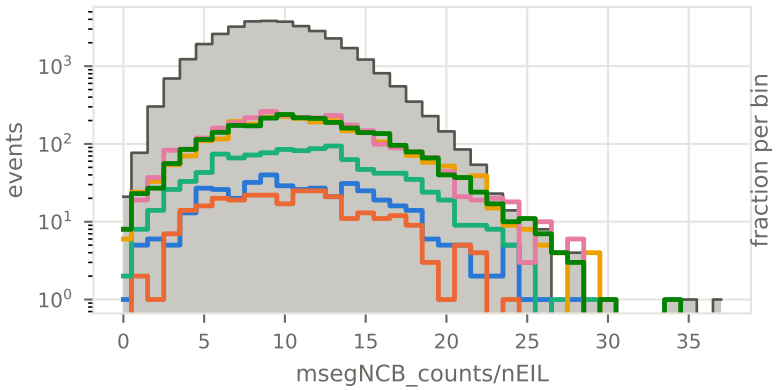
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msegNCB_counts/nEIL

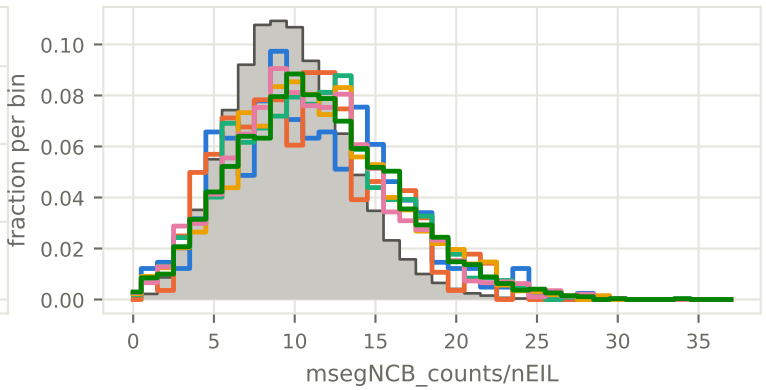
region: ENDCAP · rank 63 / 98

D_equal 0.151 (noise 0.022) D_pooled 0.157 min 0.118 (mH125_mS55) max 0.175 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

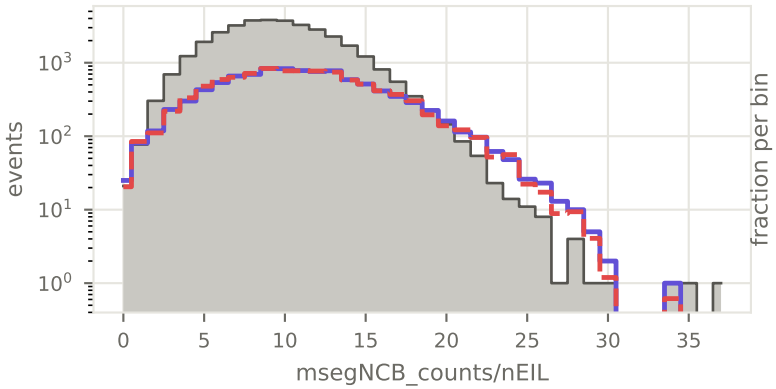


1b · each mass point vs background · normalized

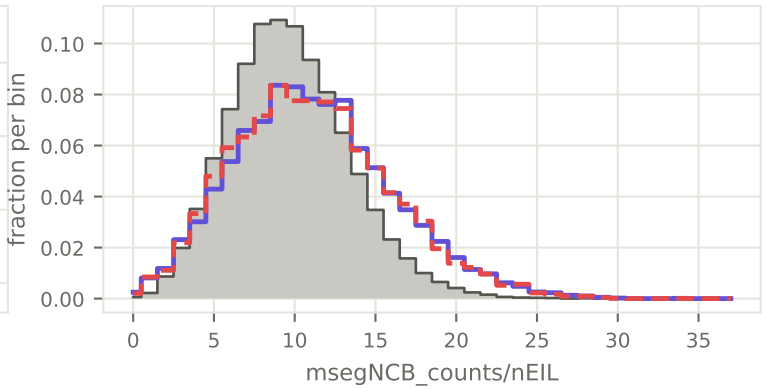


1 · each mass point
background n 34,970
mH125_mS15 D 0.175 n 411
mH125_mS55 D 0.118 n 281
mH200_mS50 D 0.162 n 1,071
mH400_mS100 D 0.167 n 2,648
mH600_mS150 D 0.145 n 2,882
mH1000_mS275 D 0.164 n 2,703

2a · combined signal vs background · raw

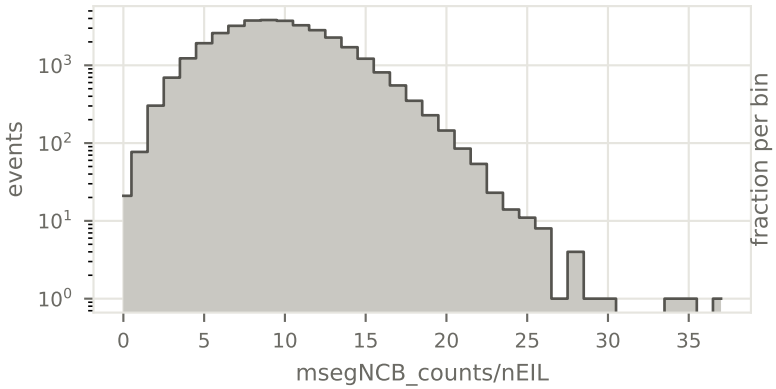


2b · combined signal vs background · normalized

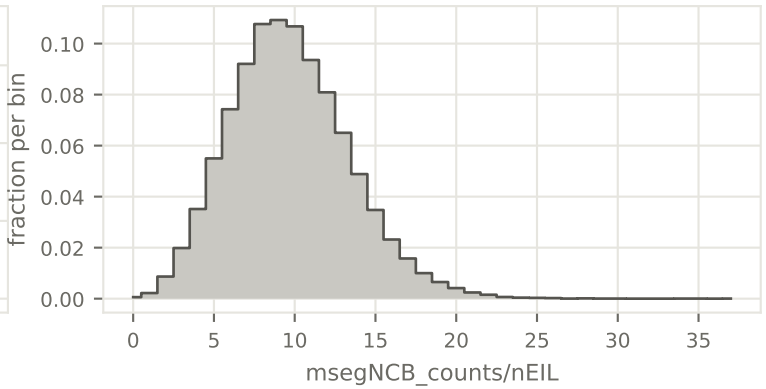


2 · combined signal
background n 34,970
combined, pooled D 0.157 n 9,996
equal weight per mass D 0.151

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

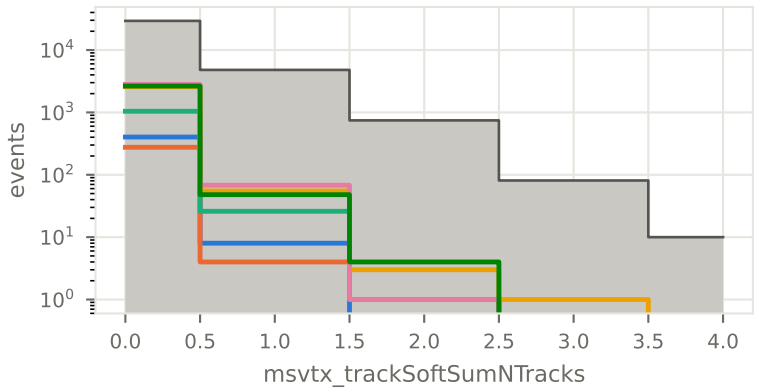
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_trackSoftSumNTracks

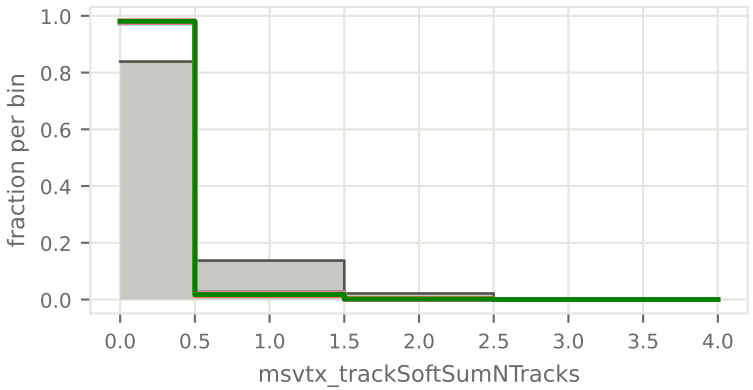
region: ENDCAP · rank 64 / 98

D_equal 0.140 (noise 0.022) D_pooled 0.139 min 0.136 (mH200_mS50) max 0.143 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

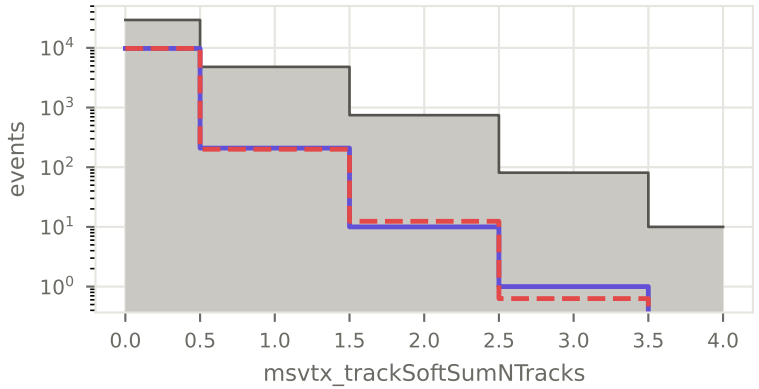


1b · each mass point vs background · normalized

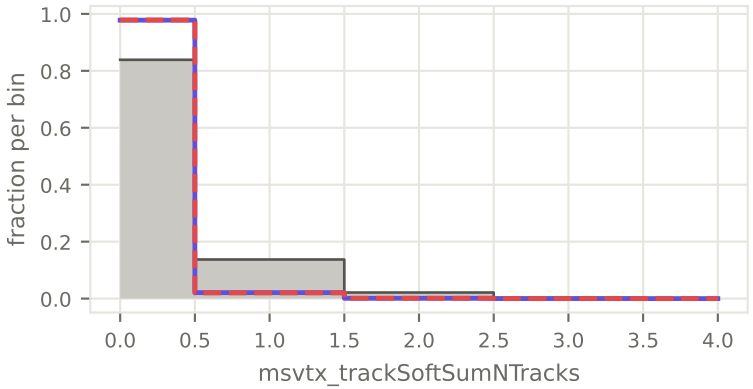


1 · each mass point
background n 34,970
mH125_mS15 D 0.142 n 411
mH125_mS55 D 0.143 n 281
mH200_mS50 D 0.136 n 1,071
mH400_mS100 D 0.139 n 2,648
mH600_mS150 D 0.137 n 2,882
mH1000_mS275 D 0.142 n 2,703

2a · combined signal vs background · raw

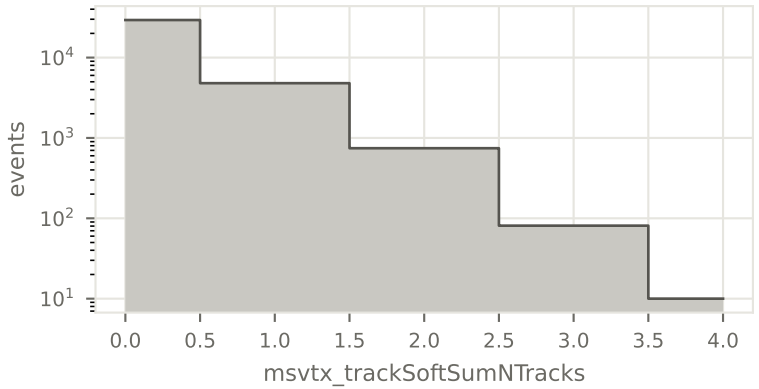


2b · combined signal vs background · normalized

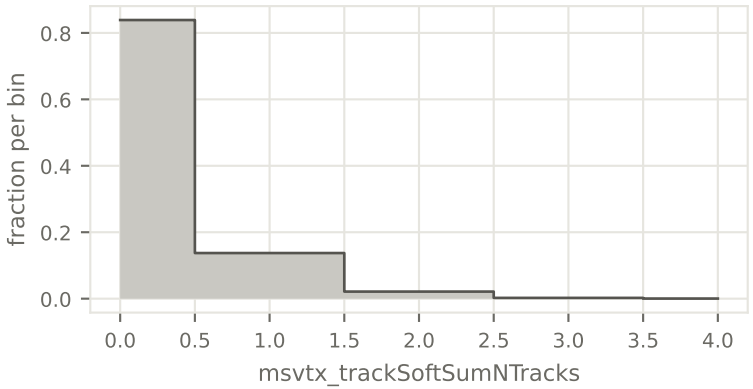


2 · combined signal
background n 34,970
combined, pooled D 0.139 n 9,996
equal weight per mass D 0.140

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

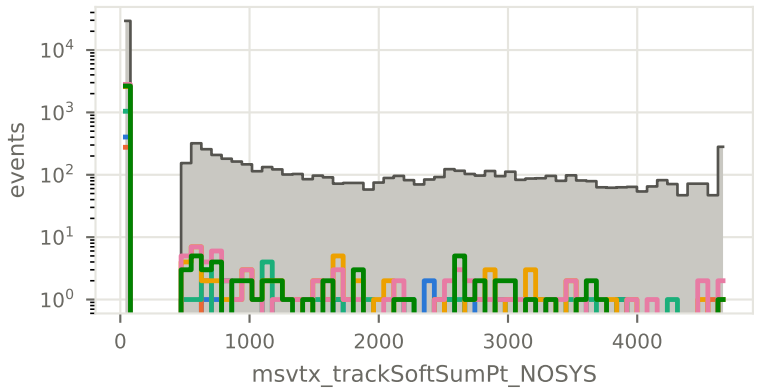
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_trackSoftSumPt_NOSYS

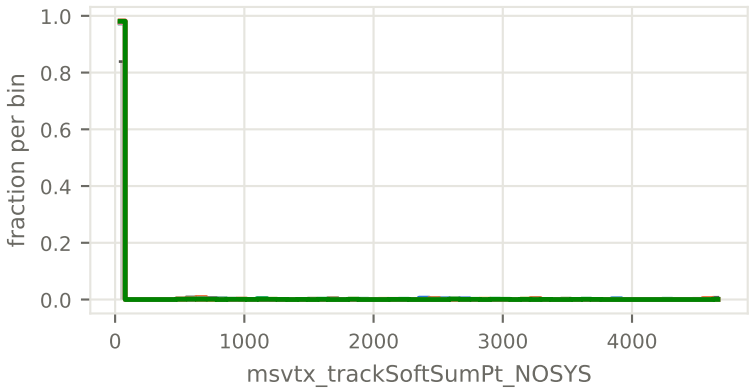
region: ENDCAP · rank 65 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.140 (noise 0.022) D_pooled 0.139 min 0.136 (mH200_mS50) max 0.143 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

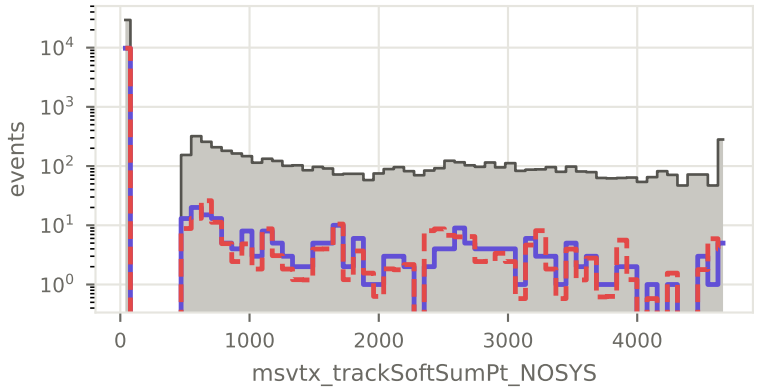


1b · each mass point vs background · normalized

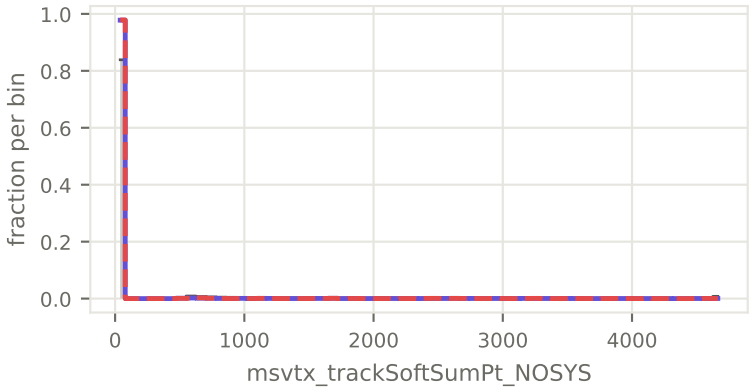


1 · each mass point
background n 34,970
mH125_mS15 D 0.142 n 411
mH125_mS55 D 0.143 n 281
mH200_mS50 D 0.136 n 1,071
mH400_mS100 D 0.139 n 2,648
mH600_mS150 D 0.138 n 2,882
mH1000_mS275 D 0.142 n 2,703

2a · combined signal vs background · raw

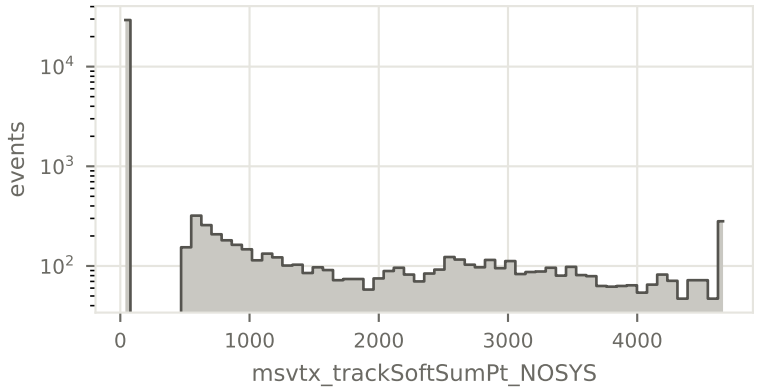


2b · combined signal vs background · normalized

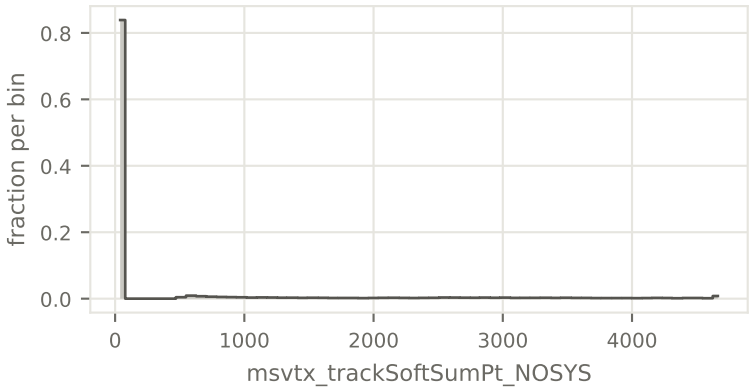


2 · combined signal
background n 34,970
combined, pooled D 0.139 n 9,996
equal weight per mass D 0.140

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

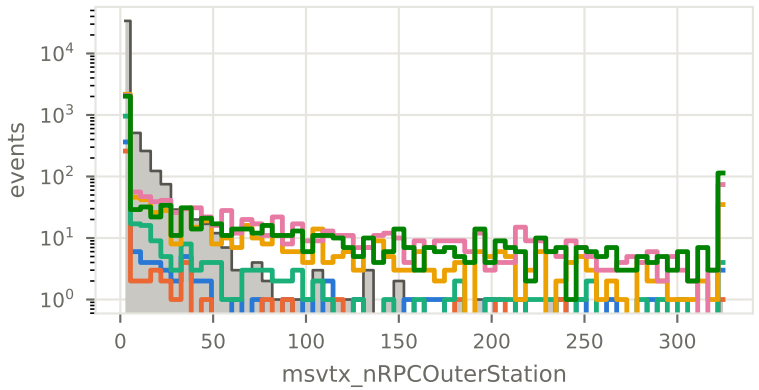
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nRPCOuterStation

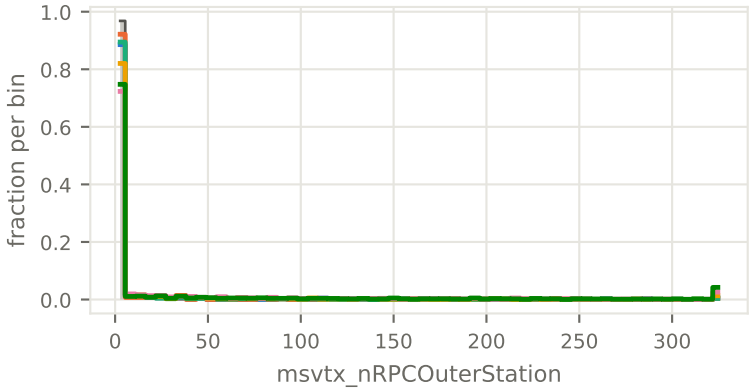
region: ENDCAP · rank 66 / 98

D_equal 0.138 (noise 0.022) D_pooled 0.183 min 0.055 (mH125_mS55) max 0.248 (mH600_mS150) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

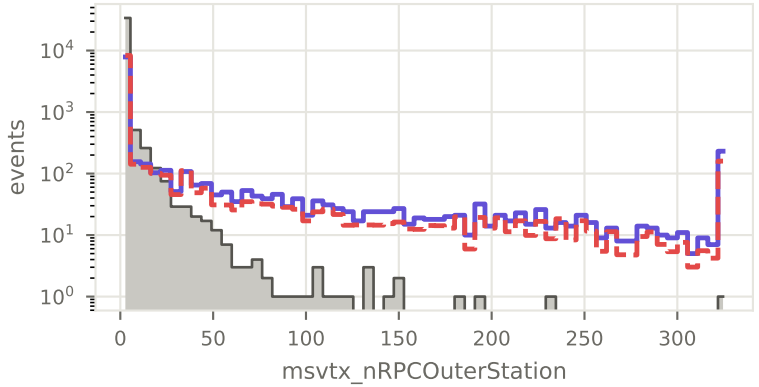


1b · each mass point vs background · normalized

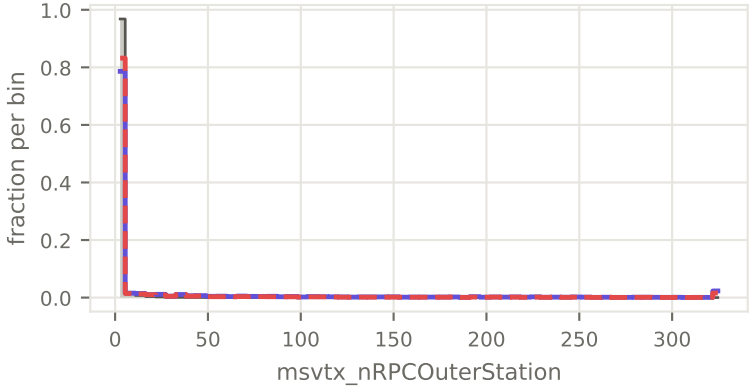


1 · each mass point
background n 34,970
mH125_mS15 D 0.086 n 411
mH125_mS55 D 0.055 n 281
mH200_mS50 D 0.077 n 1,071
mH400_mS100 D 0.148 n 2,648
mH600_mS150 D 0.248 n 2,882
mH1000_mS275 D 0.226 n 2,703

2a · combined signal vs background · raw

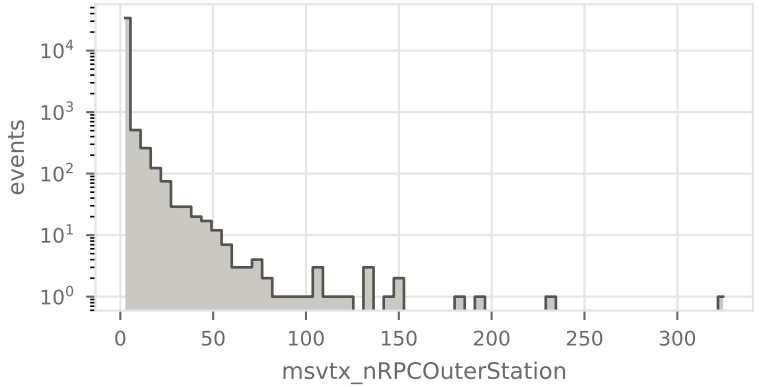


2b · combined signal vs background · normalized

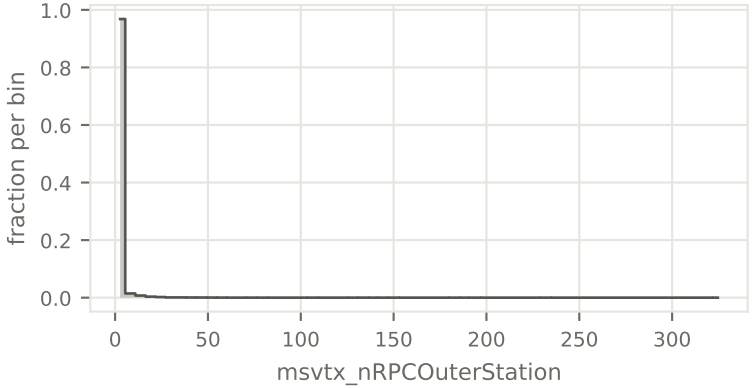


2 · combined signal
background n 34,970
combined, pooled D 0.183 n 9,996
equal weight per mass D 0.138

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

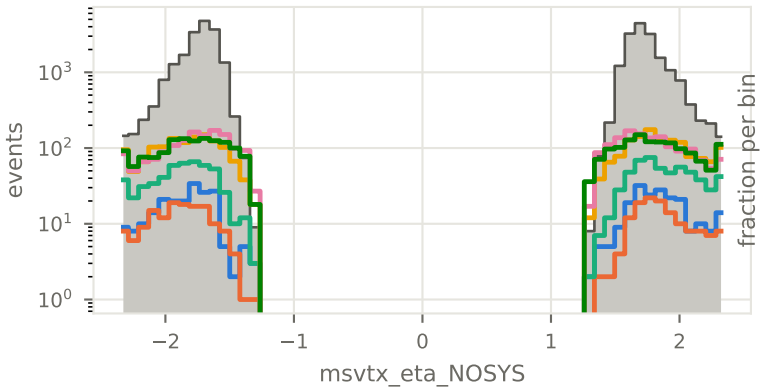
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_eta_NOSYS

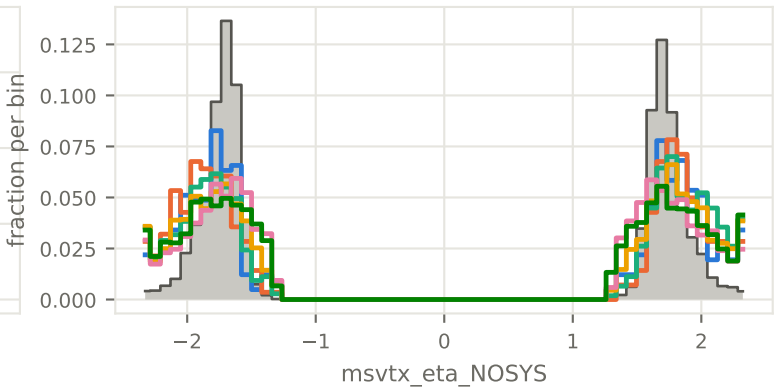
region: ENDCAP · rank 67 / 98

D_equal 0.136 (noise 0.022) D_pooled 0.125 min 0.102 (mH600_mS150) max 0.181 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

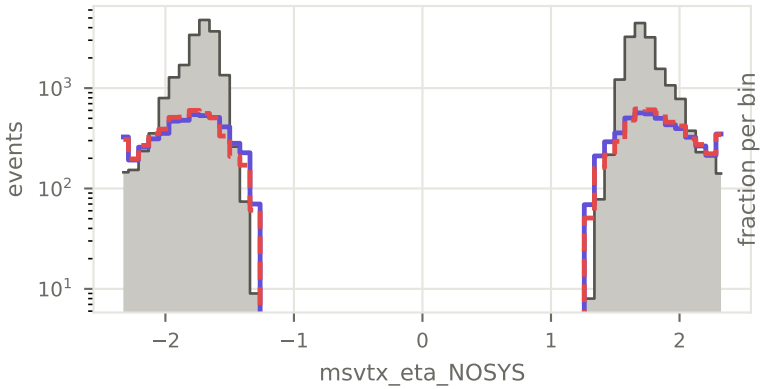


1b · each mass point vs background · normalized

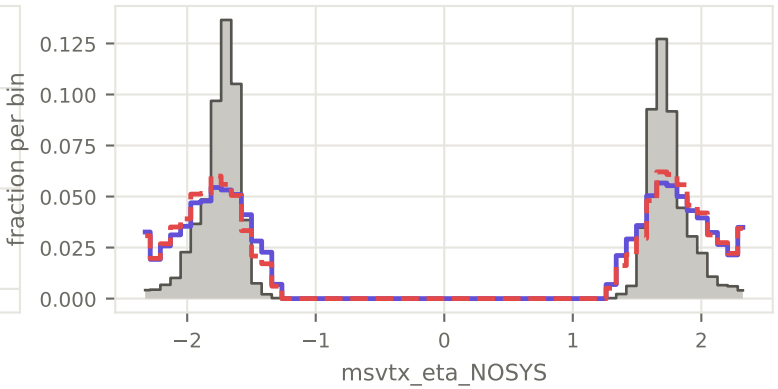


1 · each mass point
background n 34,970
mH125_mS15 D 0.153 n 411
mH125_mS55 D 0.181 n 281
mH200_mS50 D 0.171 n 1,071
mH400_mS100 D 0.141 n 2,648
mH600_mS150 D 0.102 n 2,882
mH1000_mS275 D 0.121 n 2,703

2a · combined signal vs background · raw

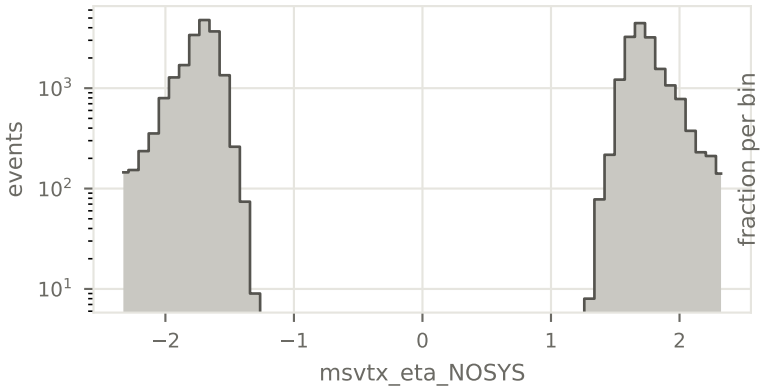


2b · combined signal vs background · normalized

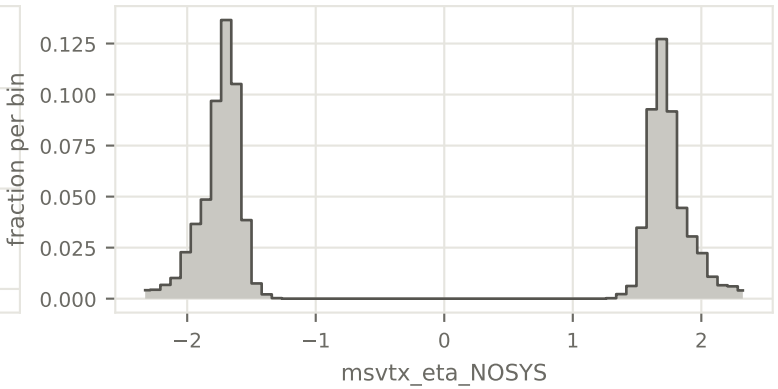


2 · combined signal
background n 34,970
combined, pooled D 0.125 n 9,996
equal weight per mass D 0.136

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

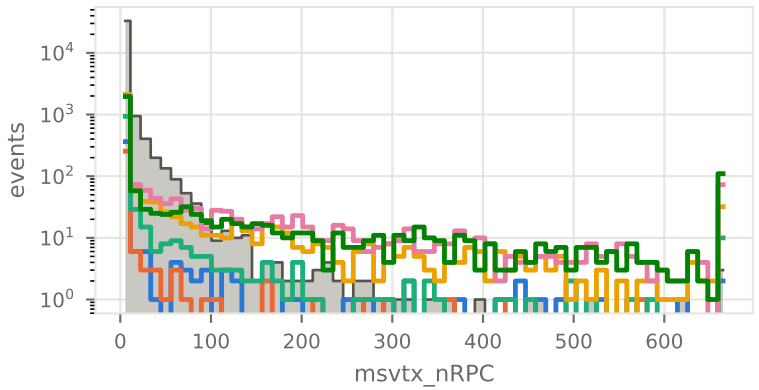
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nRPC

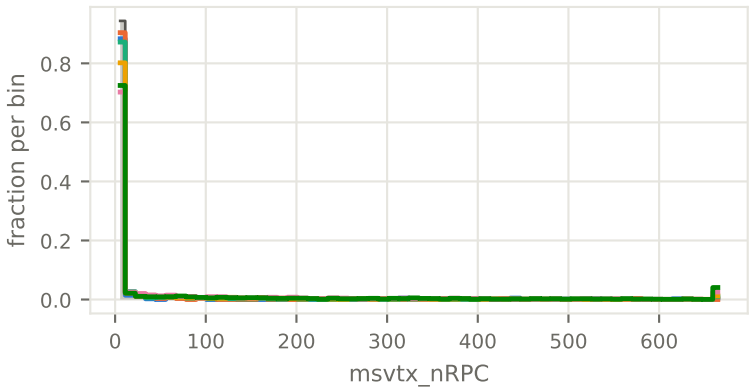
region: ENDCAP · rank 68 / 98

D_equal 0.134 (noise 0.022) D_pooled 0.181 min 0.051 (mH125_mS55) max 0.244 (mH600_mS150) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

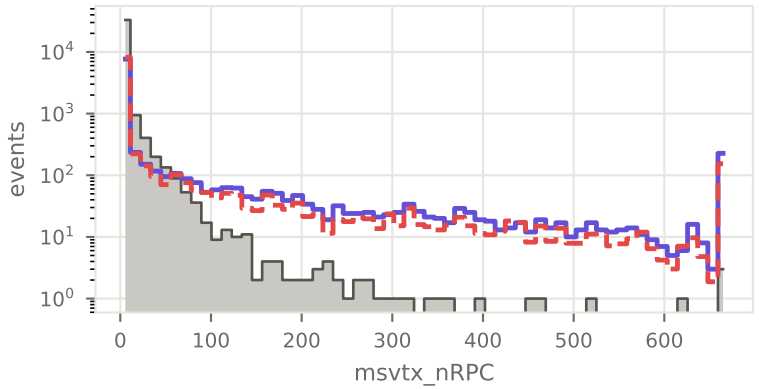


1b · each mass point vs background · normalized

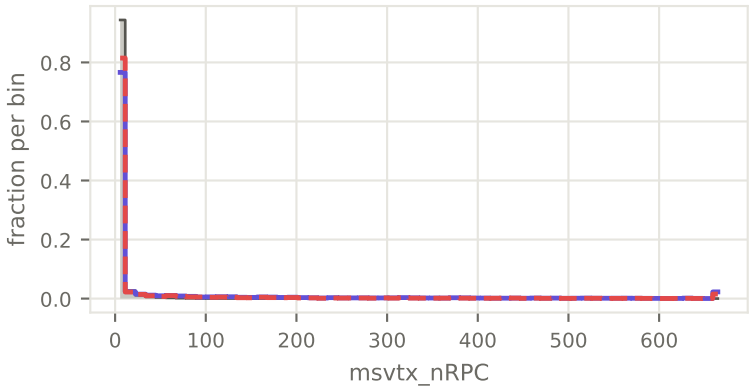


1 · each mass point
background n 34,970
mH125_mS15 D 0.077 n 411
mH125_mS55 D 0.051 n 281
mH200_mS50 D 0.076 n 1,071
mH400_mS100 D 0.147 n 2,648
mH600_mS150 D 0.244 n 2,882
mH1000_mS275 D 0.225 n 2,703

2a · combined signal vs background · raw

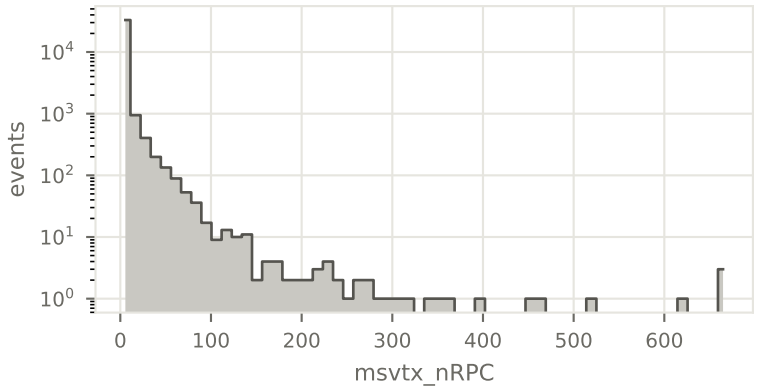


2b · combined signal vs background · normalized

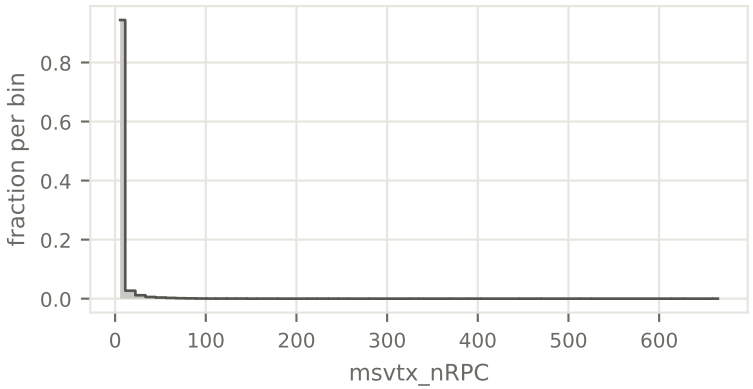


2 · combined signal
background n 34,970
combined, pooled D 0.181 n 9,996
equal weight per mass D 0.134

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

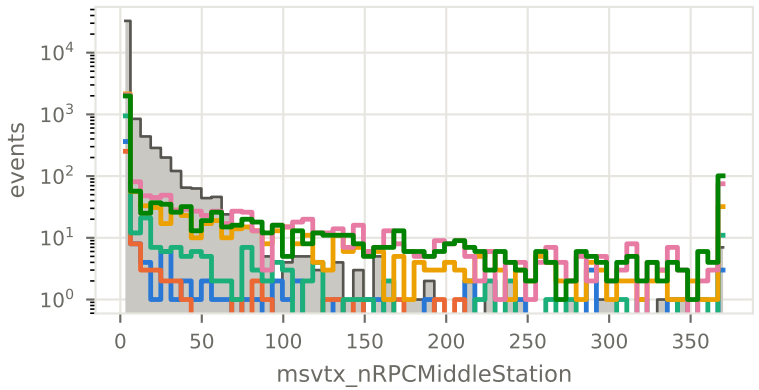
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nRPCMiddleStation

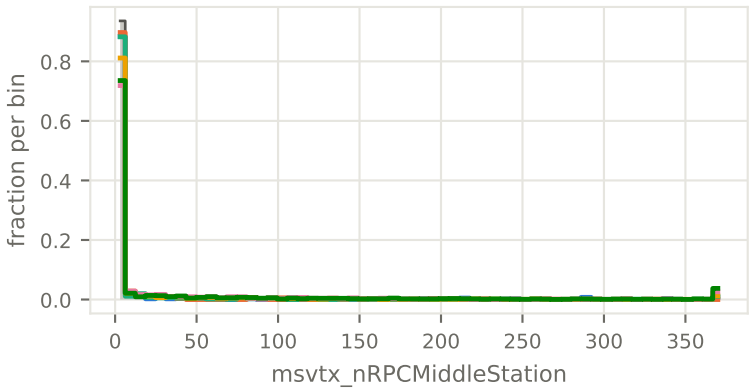
region: ENDCAP · rank 69 / 98

D_equal 0.119 (noise 0.022) D_pooled 0.162 min 0.047 (mH125_mS55) max 0.217 (mH600_mS150) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

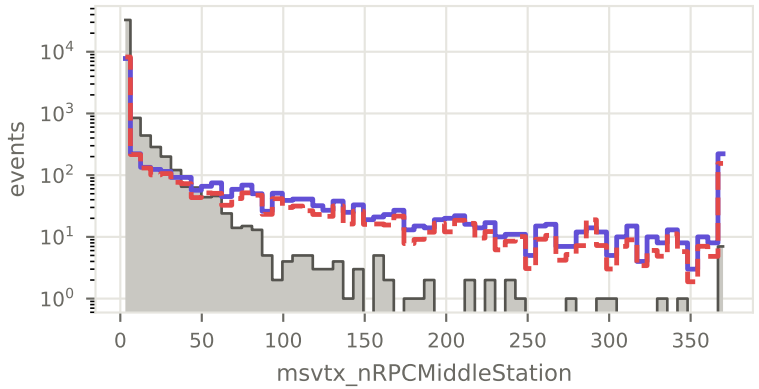


1b · each mass point vs background · normalized

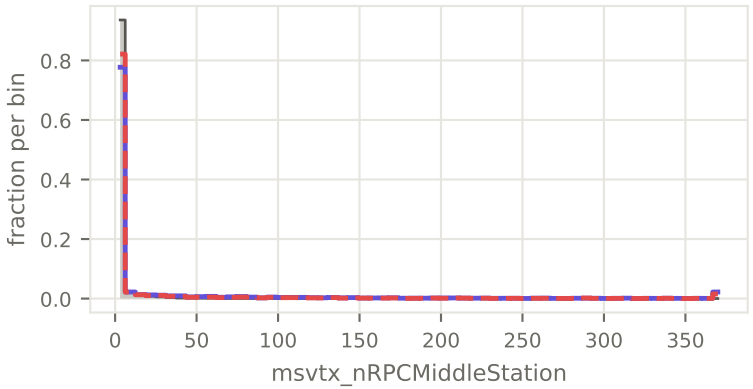


1 · each mass point
background n 34,970
mH125_mS15 D 0.067 n 411
mH125_mS55 D 0.047 n 281
mH200_mS50 D 0.067 n 1,071
mH400_mS100 D 0.129 n 2,648
mH600_mS150 D 0.217 n 2,882
mH1000_mS275 D 0.207 n 2,703

2a · combined signal vs background · raw

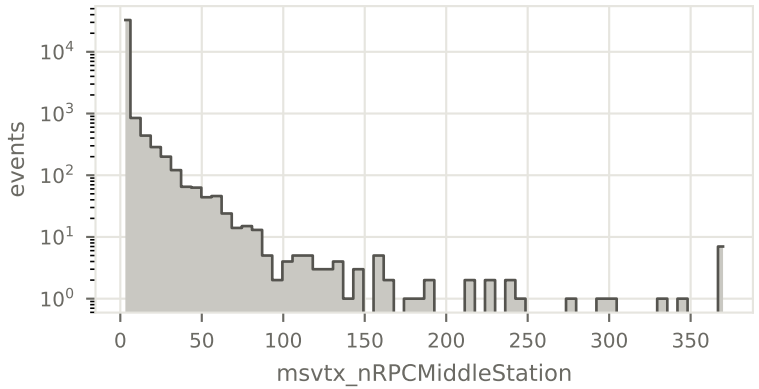


2b · combined signal vs background · normalized

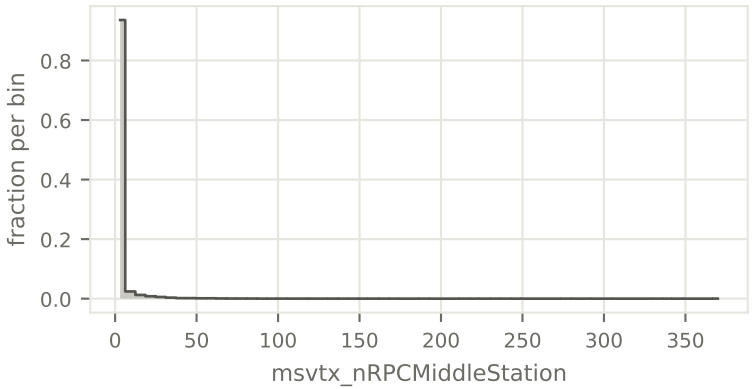


2 · combined signal
background n 34,970
combined, pooled D 0.162 n 9,996
equal weight per mass D 0.119

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

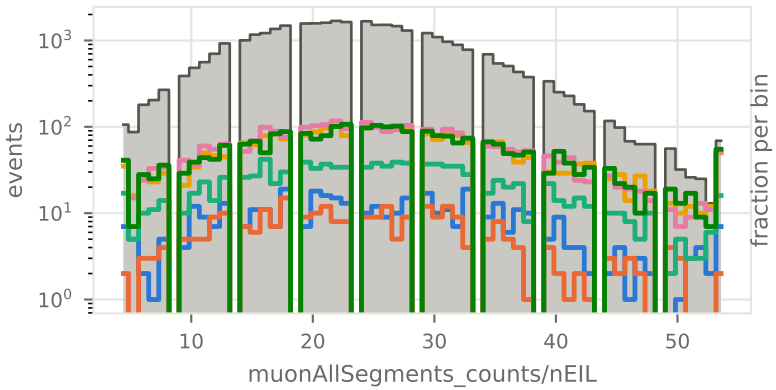
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

muonAllSegments_counts/nEIL

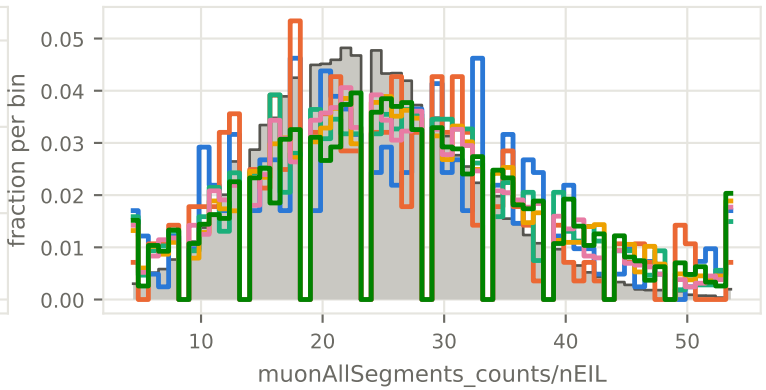
region: ENDCAP · rank 70 / 98

D_equal 0.119 (noise 0.022) D_pooled 0.125 min 0.083 (mH125_mS55) max 0.148 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

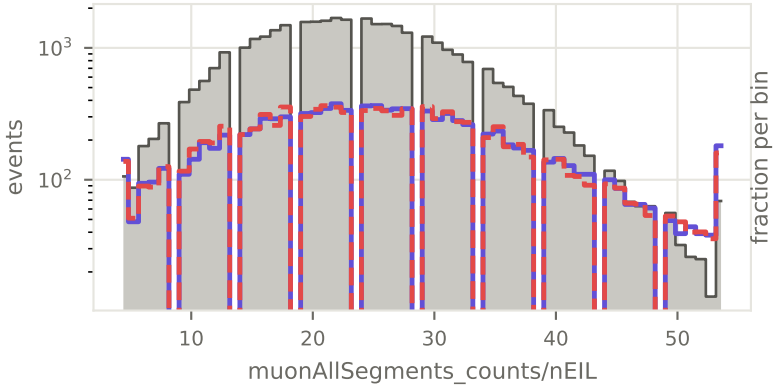


1b · each mass point vs background · normalized

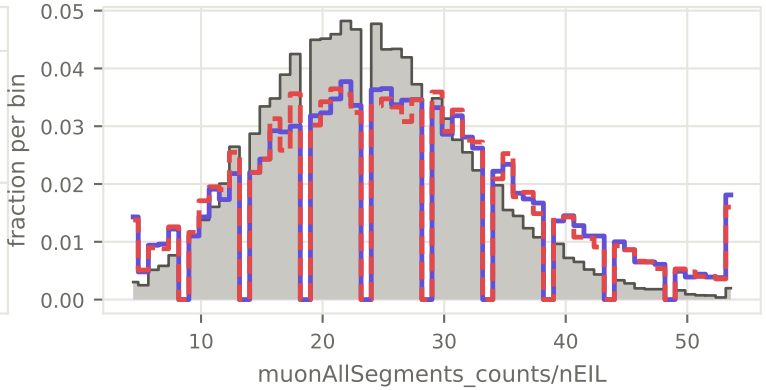


1 · each mass point
background n 34,970
mH125_mS15 D 0.148 n 411
mH125_mS55 D 0.083 n 281
mH200_mS50 D 0.123 n 1,071
mH400_mS100 D 0.133 n 2,648
mH600_mS150 D 0.114 n 2,882
mH1000_mS275 D 0.133 n 2,703

2a · combined signal vs background · raw

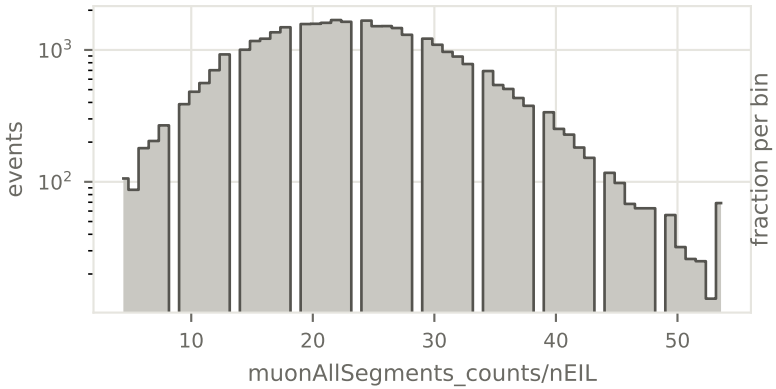


2b · combined signal vs background · normalized

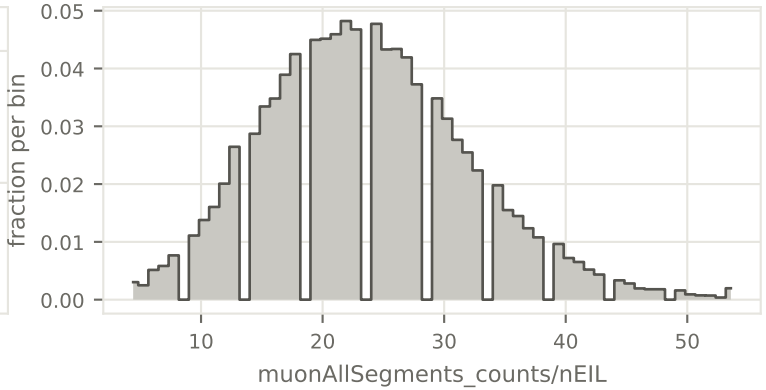


2 · combined signal
background n 34,970
combined, pooled D 0.125 n 9,996
equal weight per mass D 0.119

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

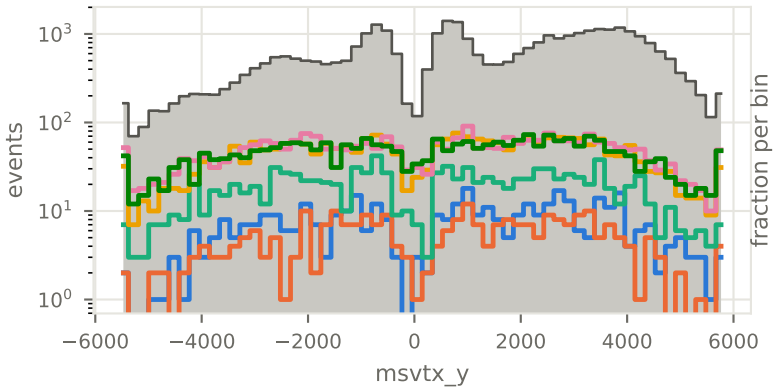
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_y

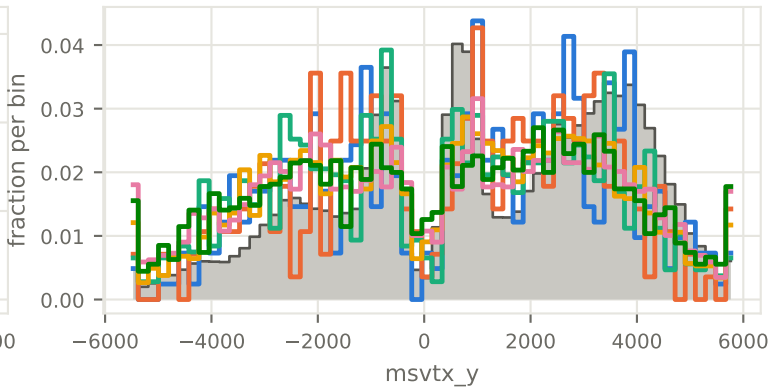
region: ENDCAP · rank 71 / 98

D_equal 0.118 (noise 0.022) D_pooled 0.127 min 0.100 (mH125_mS15) max 0.143 (mH600_mS150) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

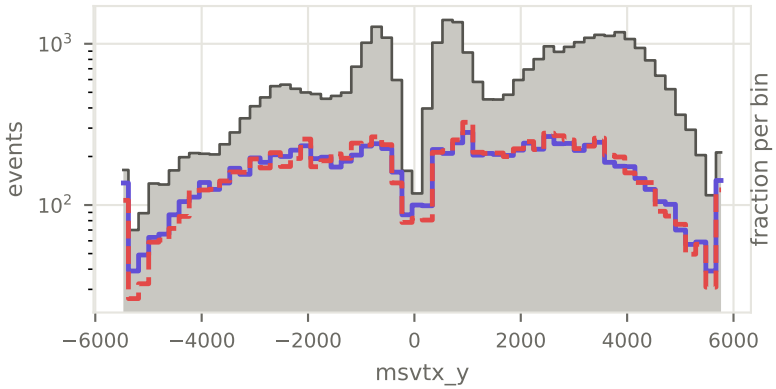


1b · each mass point vs background · normalized

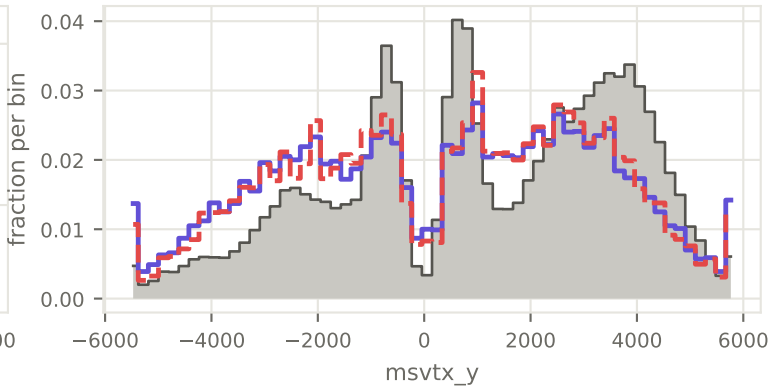


1 · each mass point
background n 34,970
mH125_mS15 D 0.100 n 411
mH125_mS55 D 0.124 n 281
mH200_mS50 D 0.130 n 1,071
mH400_mS100 D 0.121 n 2,648
mH600_mS150 D 0.143 n 2,882
mH1000_mS275 D 0.131 n 2,703

2a · combined signal vs background · raw

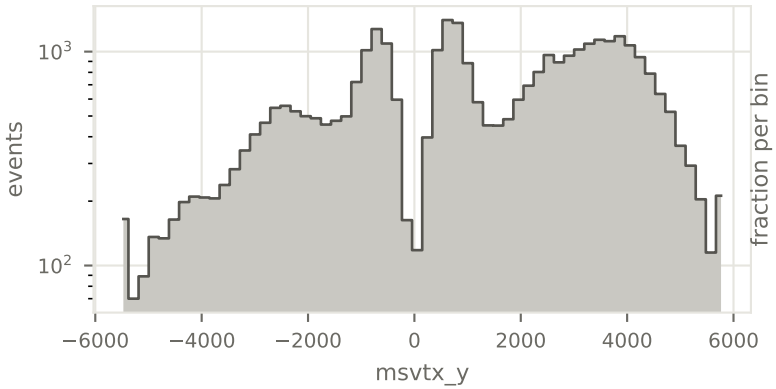


2b · combined signal vs background · normalized

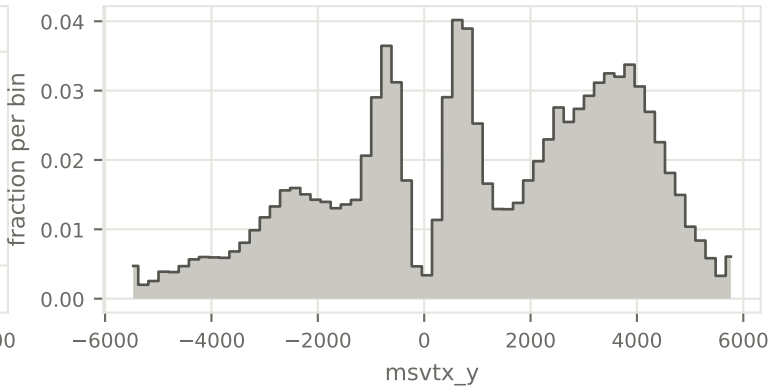


2 · combined signal
background n 34,970
combined, pooled D 0.127 n 9,996
equal weight per mass D 0.118

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

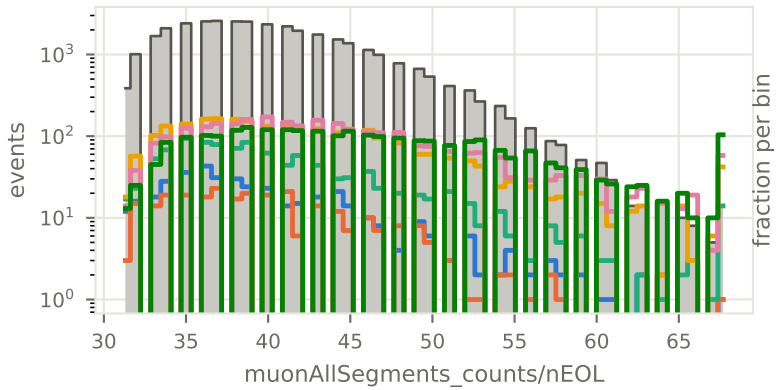
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

muonAllSegments_counts/nEOL

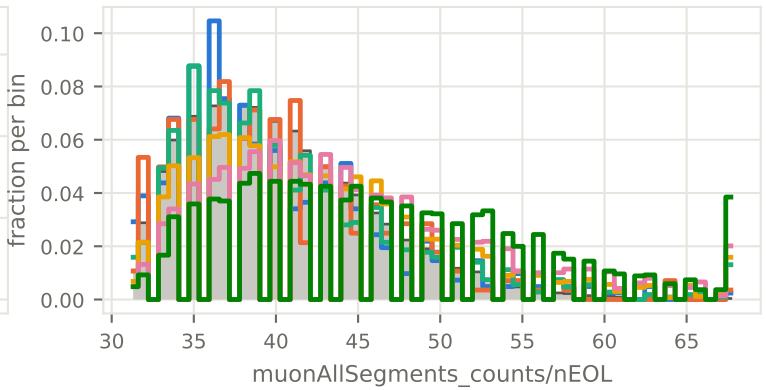
region: ENDCAP · rank 72 / 98

D_equal 0.116 (noise 0.022) D_pooled 0.183 min 0.034 (mH125_mS55) max 0.312 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

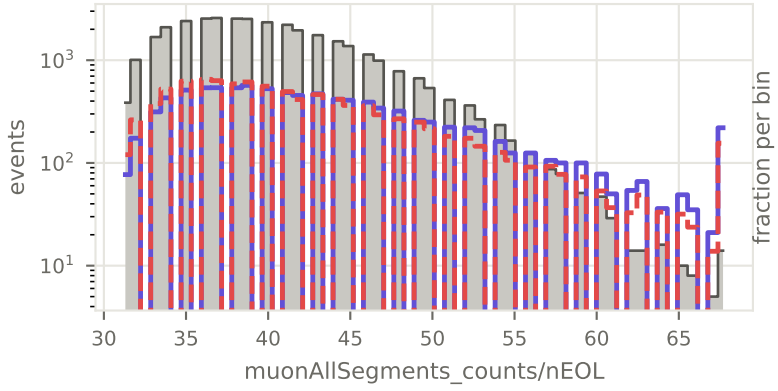


1b · each mass point vs background · normalized

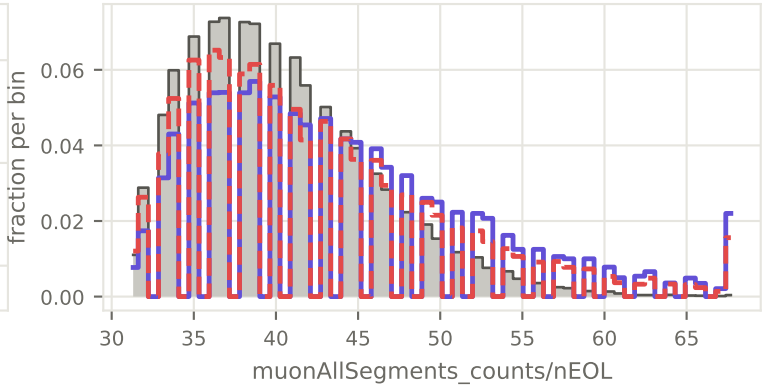


1 · each mass point
background n 34,970
mH125_mS15 D 0.085 n 411
mH125_mS55 D 0.034 n 281
mH200_mS50 D 0.051 n 1,071
mH400_mS100 D 0.139 n 2,648
mH600_mS150 D 0.213 n 2,882
mH1000_mS275 D 0.312 n 2,703

2a · combined signal vs background · raw

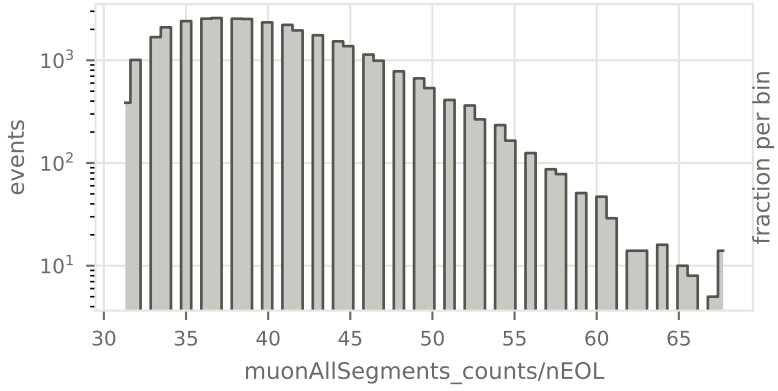


2b · combined signal vs background · normalized

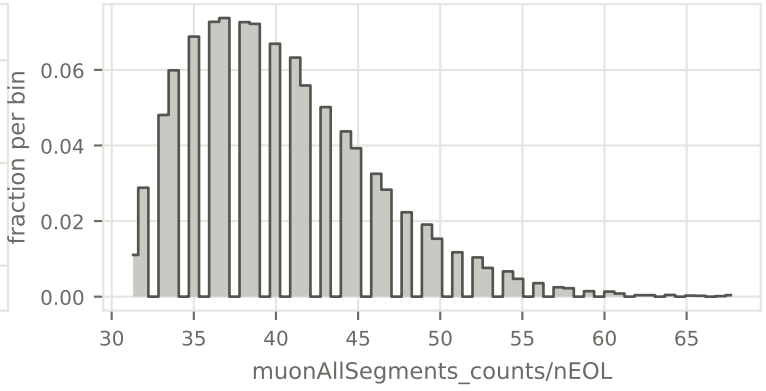


2 · combined signal
background n 34,970
combined, pooled D 0.183 n 9,996
equal weight per mass D 0.116

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

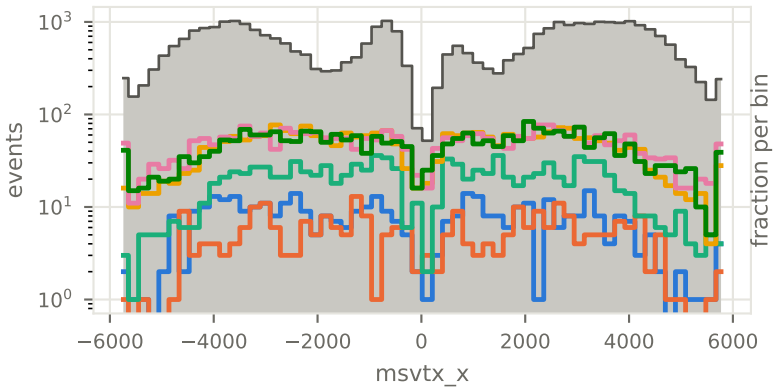
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_x

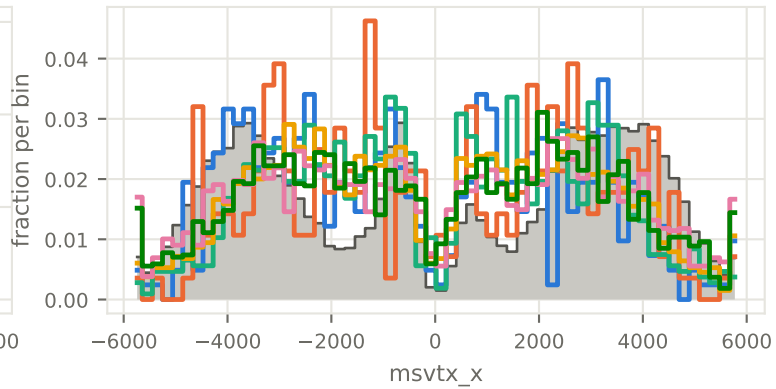
region: ENDCAP · rank 73 / 98

D_equal 0.115 (noise 0.022) D_pooled 0.111 min 0.100 (mH600_mS150) max 0.161 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

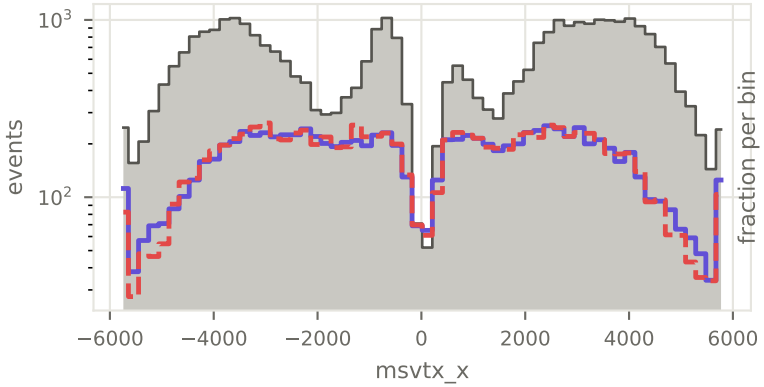


1b · each mass point vs background · normalized

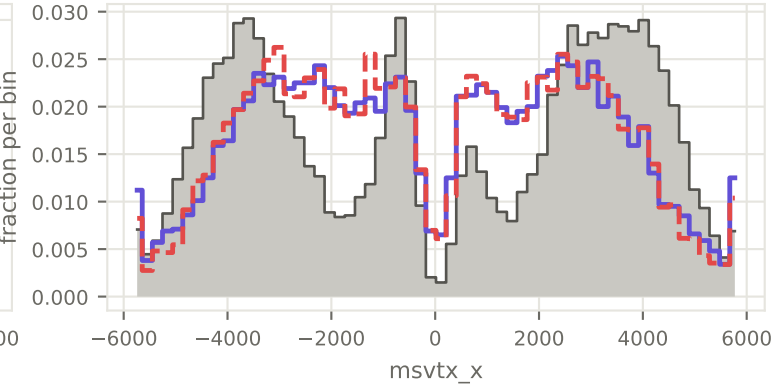


1 · each mass point
background n 34,970
mH125_mS15 D 0.161 n 411
mH125_mS55 D 0.113 n 281
mH200_mS50 D 0.129 n 1,071
mH400_mS100 D 0.118 n 2,648
mH600_mS150 D 0.100 n 2,882
mH1000_mS275 D 0.109 n 2,703

2a · combined signal vs background · raw

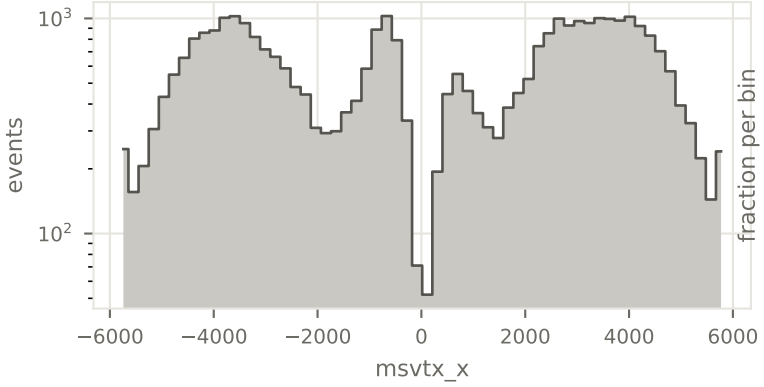


2b · combined signal vs background · normalized

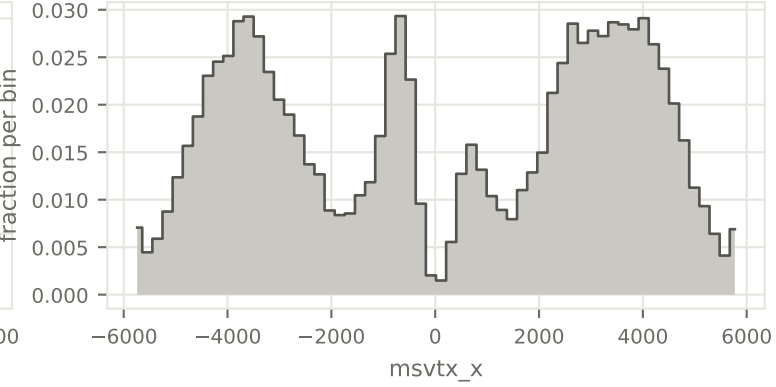


2 · combined signal
background n 34,970
combined, pooled D 0.111 n 9,996
equal weight per mass D 0.115

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

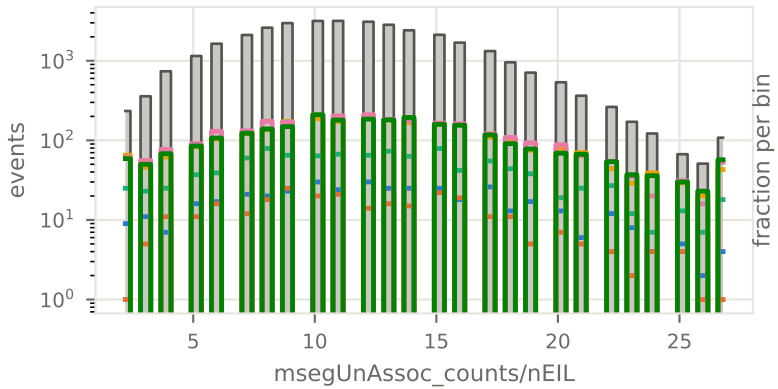
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msegUnAssoc_counts/nEIL

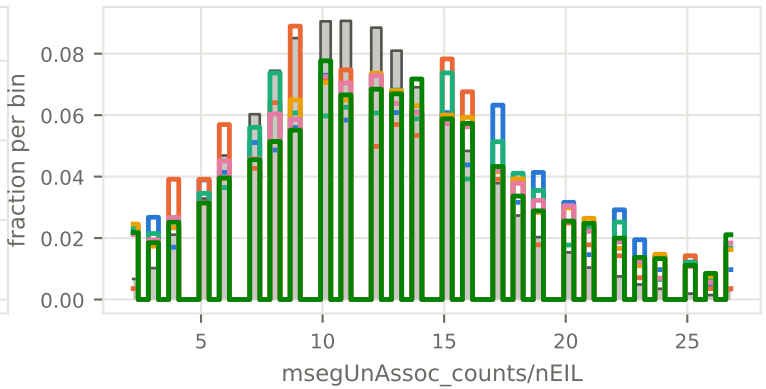
region: ENDCAP · rank 74 / 98

D_equal 0.115 (noise 0.022) D_pooled 0.115 min 0.099 (mH125_mS55) max 0.134 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

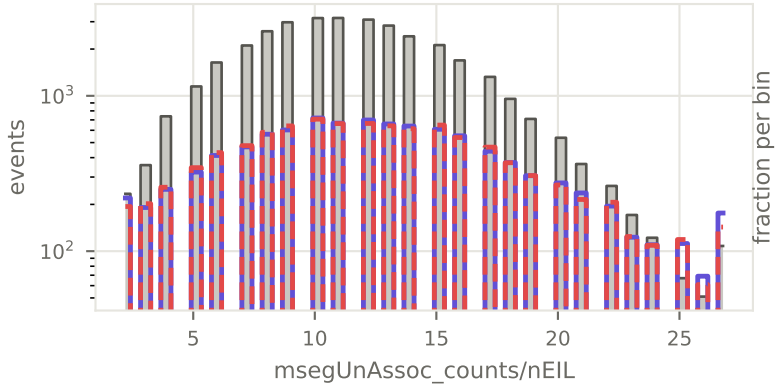


1b · each mass point vs background · normalized

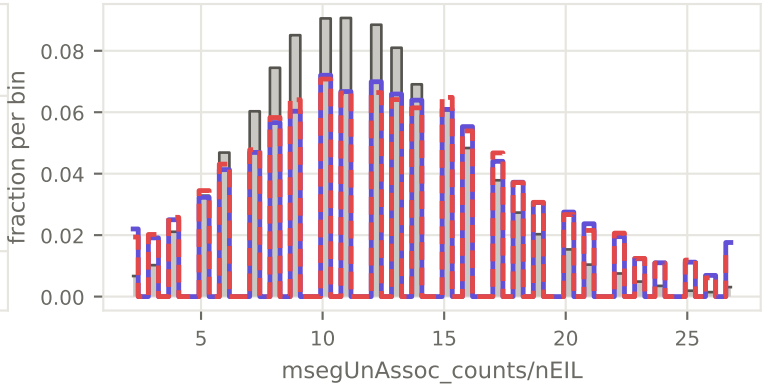


1 · each mass point
background n 34,970
mH125_mS15 D 0.134 n 411
mH125_mS55 D 0.099 n 281
mH200_mS50 D 0.118 n 1,071
mH400_mS100 D 0.120 n 2,648
mH600_mS150 D 0.112 n 2,882
mH1000_mS275 D 0.120 n 2,703

2a · combined signal vs background · raw

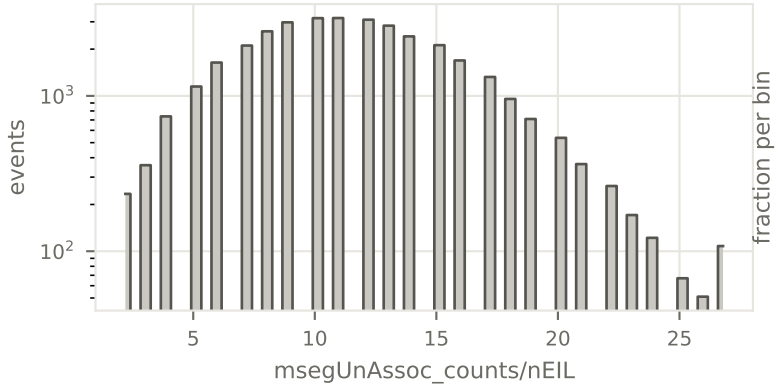


2b · combined signal vs background · normalized

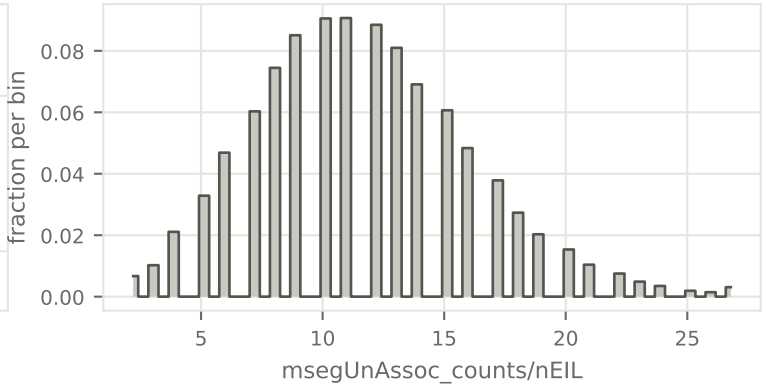


2 · combined signal
background n 34,970
combined, pooled D 0.115 n 9,996
equal weight per mass D 0.115

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

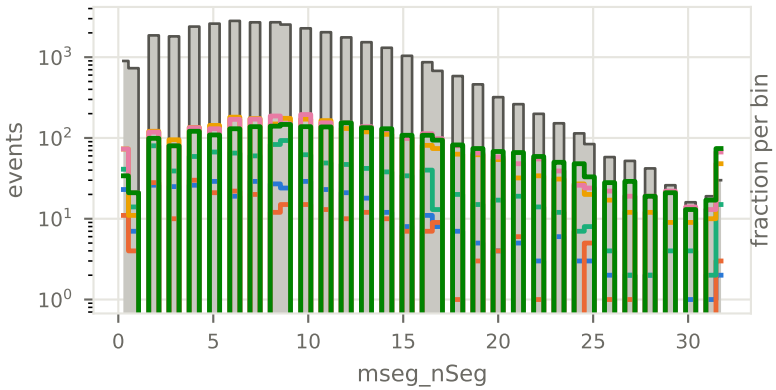
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_nSeg

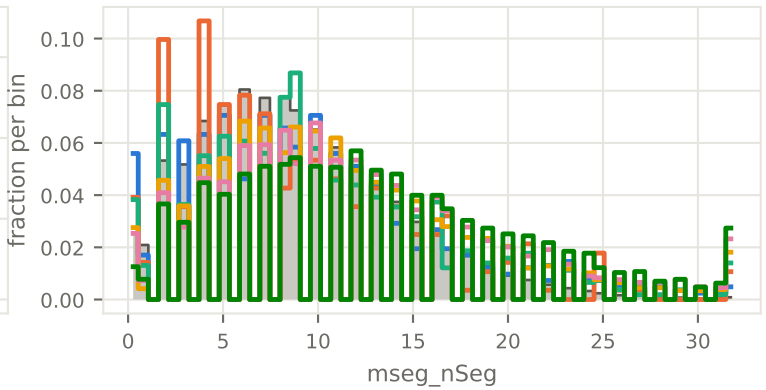
region: ENDCAP · rank 75 / 98

D_equal 0.109 (noise 0.022) D_pooled 0.158 min 0.045 (mH125_mS15) max 0.247 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

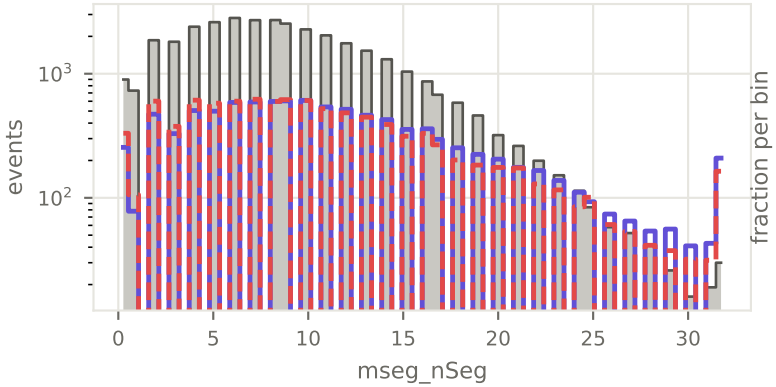


1b · each mass point vs background · normalized

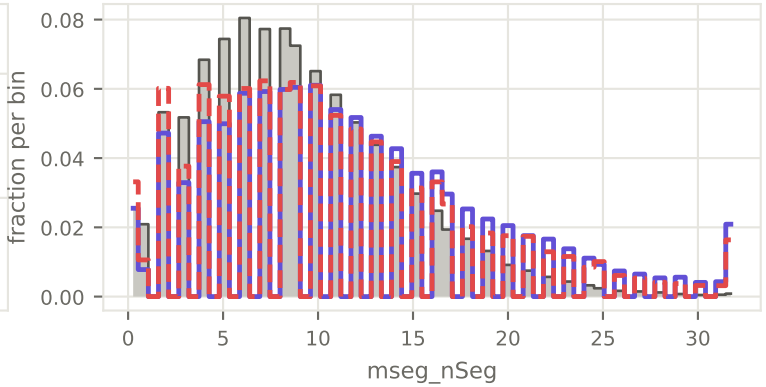


1 · each mass point
background n 34,970
mH125_mS15 D 0.045 n 411
mH125_mS55 D 0.076 n 281
mH200_mS50 D 0.074 n 1,071
mH400_mS100 D 0.128 n 2,648
mH600_mS150 D 0.176 n 2,882
mH1000_mS275 D 0.247 n 2,703

2a · combined signal vs background · raw

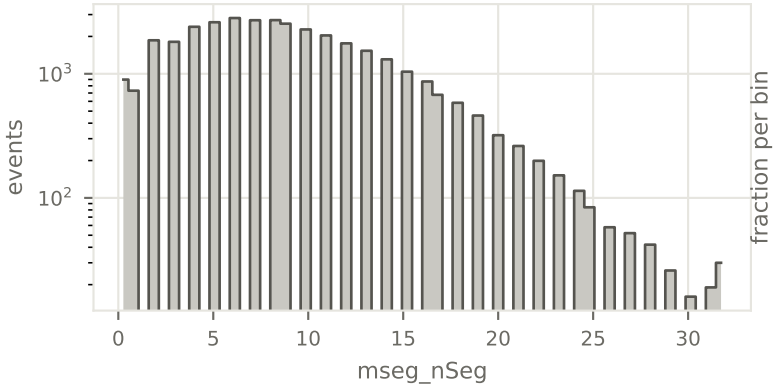


2b · combined signal vs background · normalized

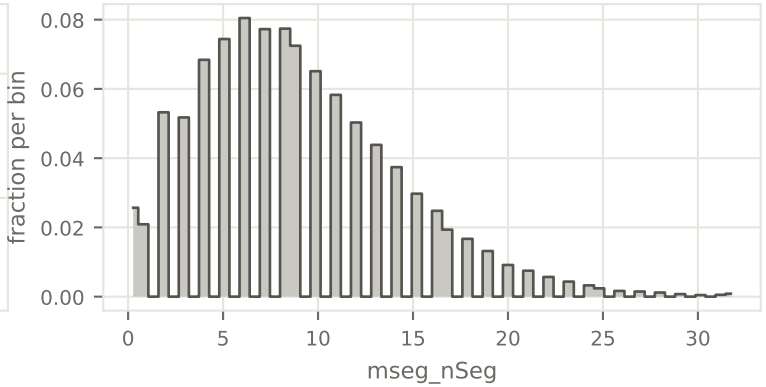


2 · combined signal
background n 34,970
combined, pooled D 0.158 n 9,996
equal weight per mass D 0.109

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

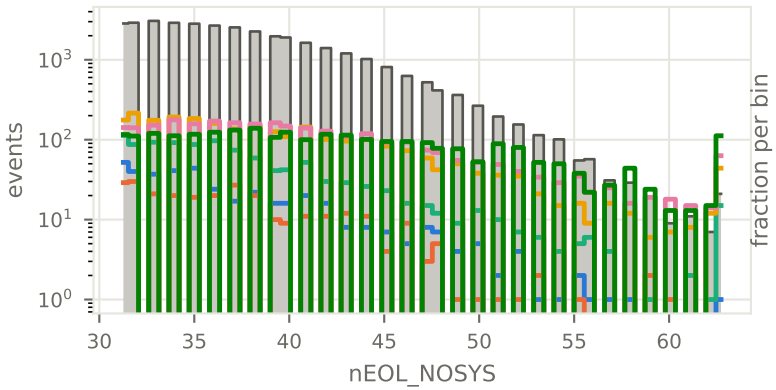
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

nEOL_NOSYS

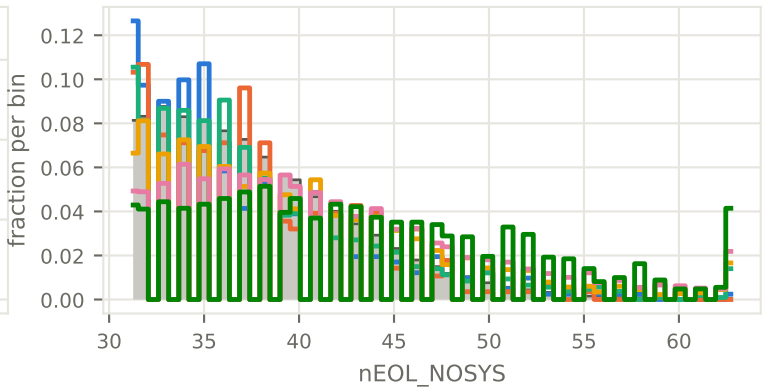
region: ENDCAP · rank 76 / 98

D_equal 0.107 (noise 0.022) D_pooled 0.171 min 0.039 (mH200_mS50) max 0.306 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

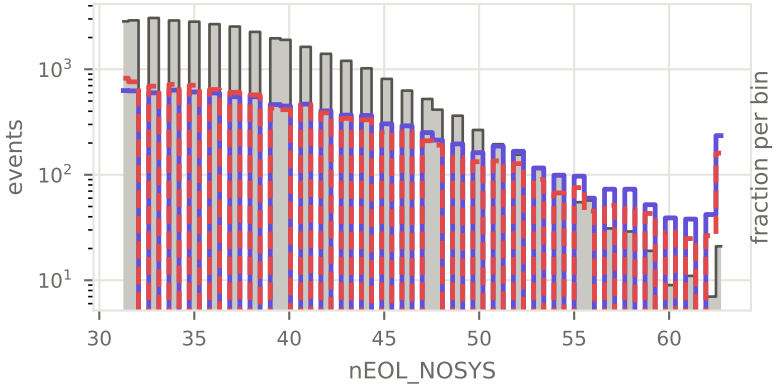


1b · each mass point vs background · normalized

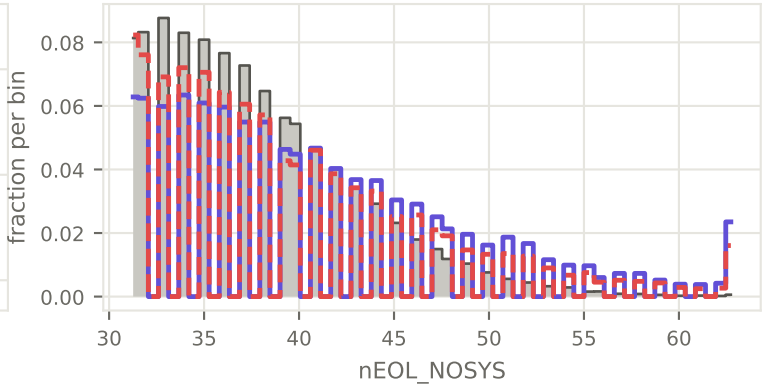


1 · each mass point
background n 34,970
mH125_mS15 D 0.105 n 411
mH125_mS55 D 0.045 n 281
mH200_mS50 D 0.039 n 1,071
mH400_mS100 D 0.127 n 2,648
mH600_mS150 D 0.195 n 2,882
mH1000_mS275 D 0.306 n 2,703

2a · combined signal vs background · raw

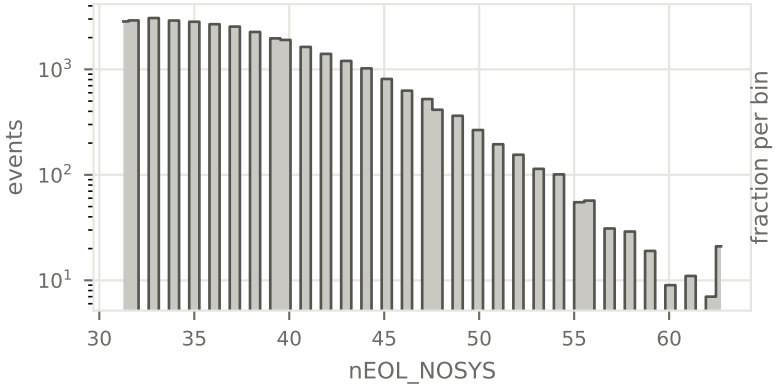


2b · combined signal vs background · normalized

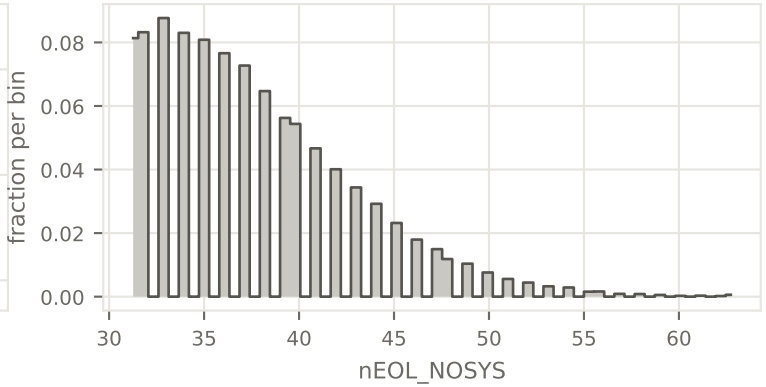


2 · combined signal
background n 34,970
combined, pooled D 0.171 n 9,996
equal weight per mass D 0.107

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

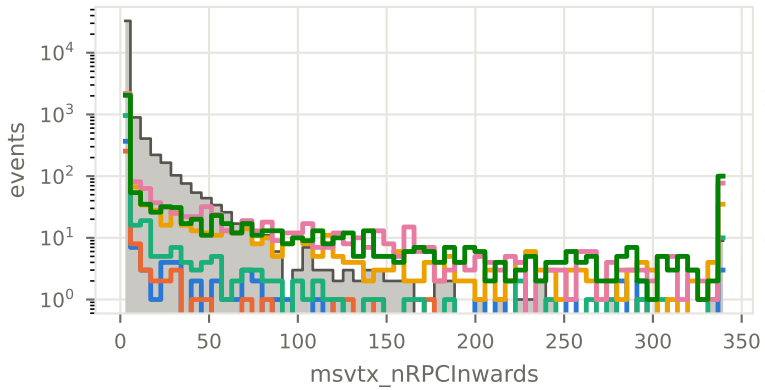
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nRPCInwards

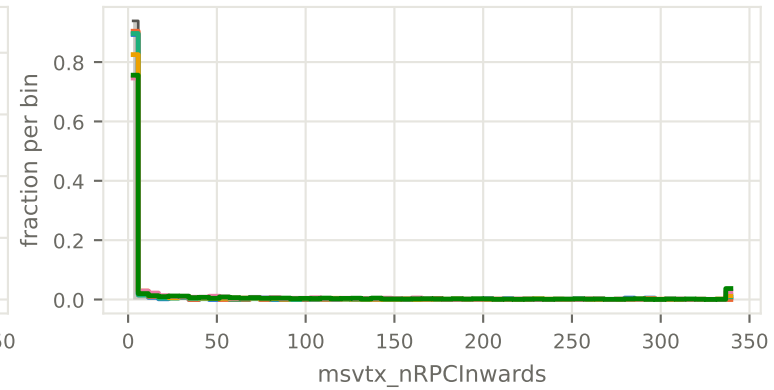
region: ENDCAP · rank 77 / 98

D_equal 0.105 (noise 0.022) D_pooled 0.145 min 0.037 (mH125_mS55) max 0.193 (mH600_mS150) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

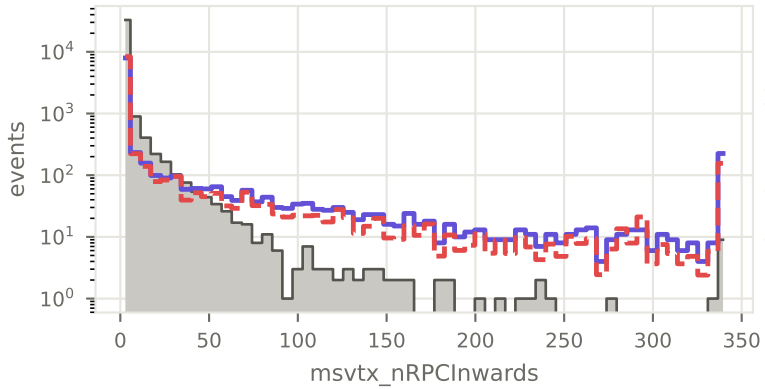


1b · each mass point vs background · normalized

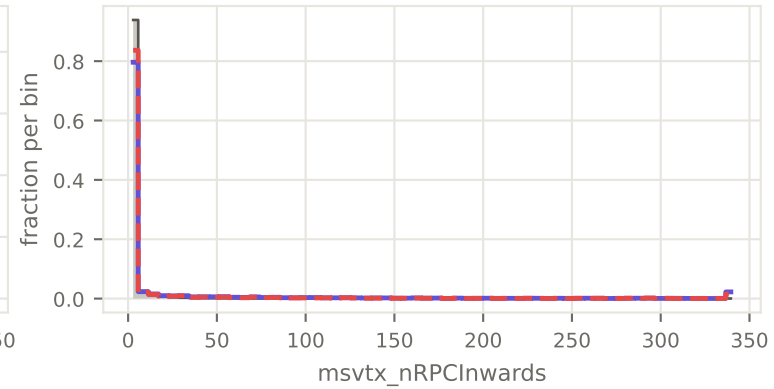


1 · each mass point
background n 34,970
mH125_mS15 D 0.065 n 411
mH125_mS55 D 0.037 n 281
mH200_mS50 D 0.054 n 1,071
mH400_mS100 D 0.115 n 2,648
mH600_mS150 D 0.193 n 2,882
mH1000_mS275 D 0.189 n 2,703

2a · combined signal vs background · raw

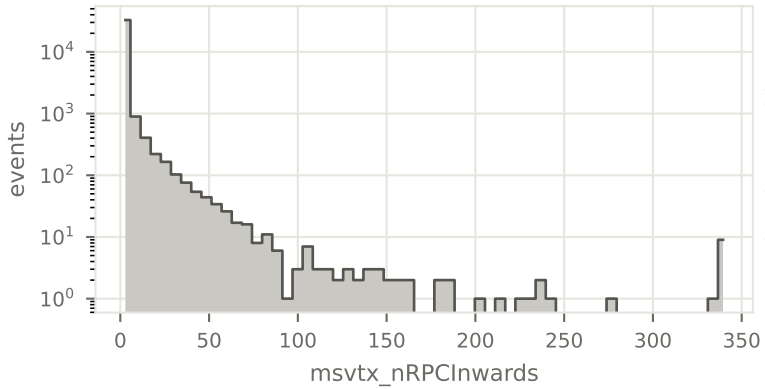


2b · combined signal vs background · normalized

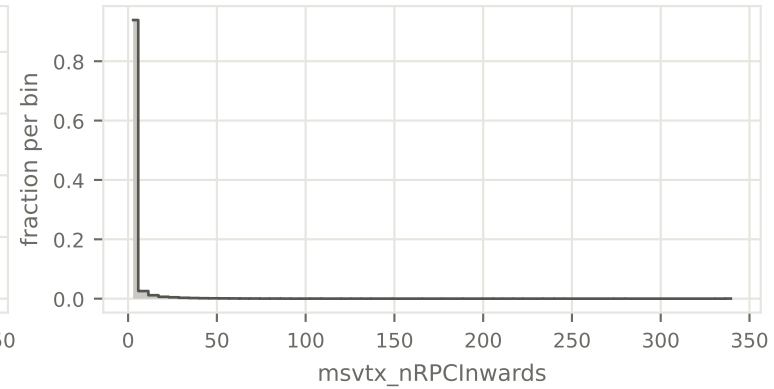


2 · combined signal
background n 34,970
combined, pooled D 0.145 n 9,996
equal weight per mass D 0.105

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

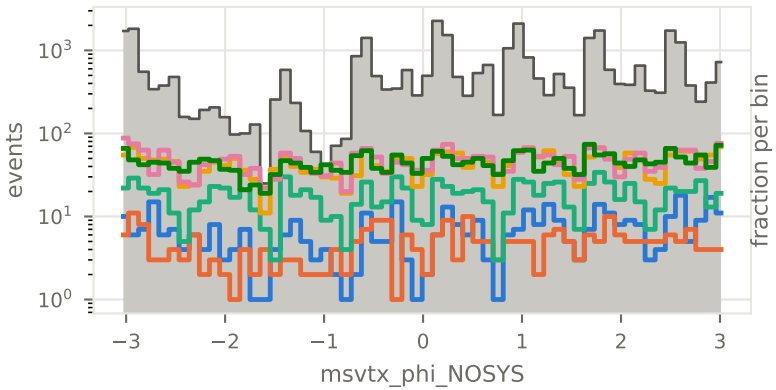
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_phi_NOSYS

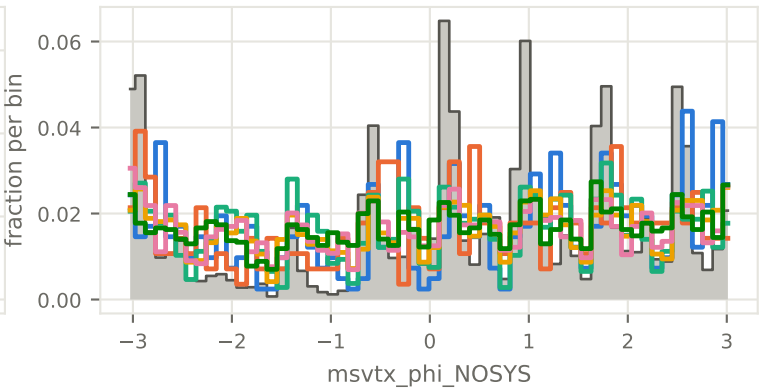
region: ENDCAP · rank 78 / 98

D_equal 0.099 (noise 0.022) D_pooled 0.111 min 0.083 (mH125_mS15) max 0.123 (mH600_mS150) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

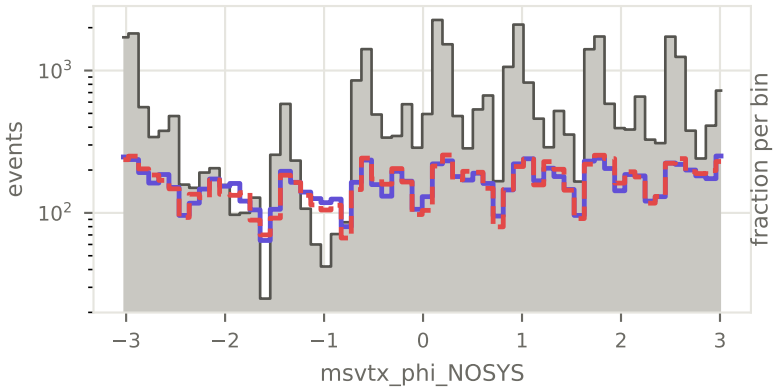


1b · each mass point vs background · normalized

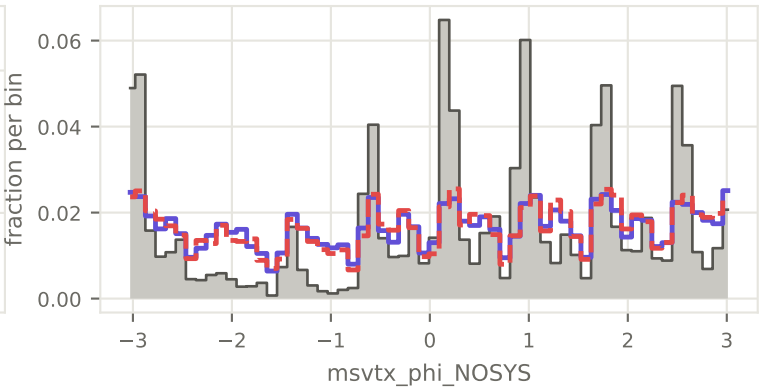


1 · each mass point
background n 34,970
mH125_mS15 D 0.083 n 411
mH125_mS55 D 0.088 n 281
mH200_mS50 D 0.120 n 1,071
mH400_mS100 D 0.109 n 2,648
mH600_mS150 D 0.123 n 2,882
mH1000_mS275 D 0.110 n 2,703

2a · combined signal vs background · raw

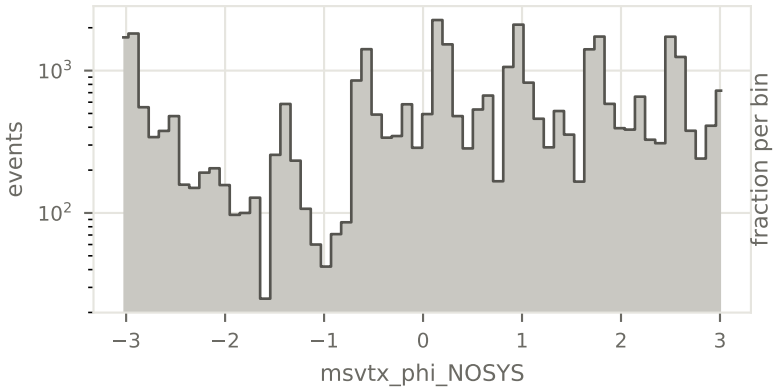


2b · combined signal vs background · normalized

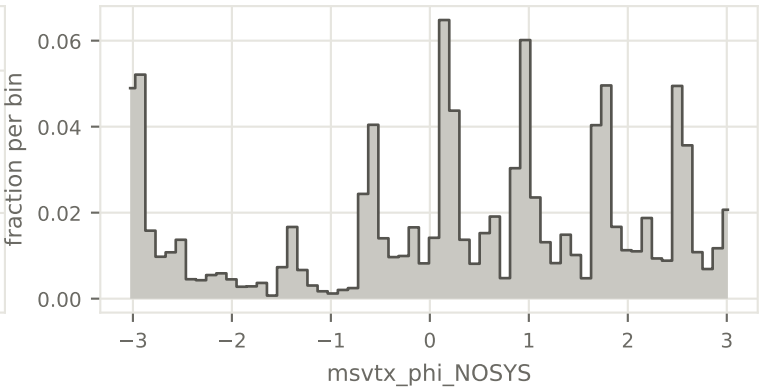


2 · combined signal
background n 34,970
combined, pooled D 0.111 n 9,996
equal weight per mass D 0.099

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

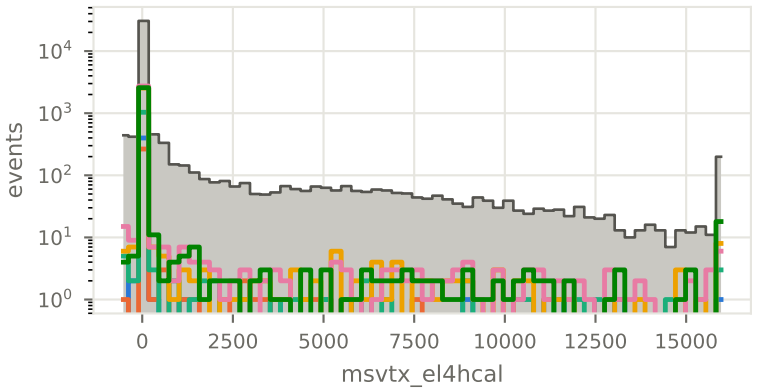
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el4hcal

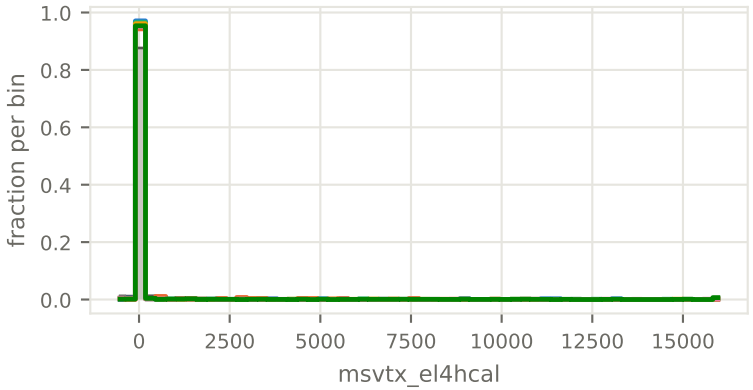
region: ENDCAP · rank 79 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.094 (noise 0.022) D_pooled 0.093 min 0.080 (mH125_mS55) max 0.110 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

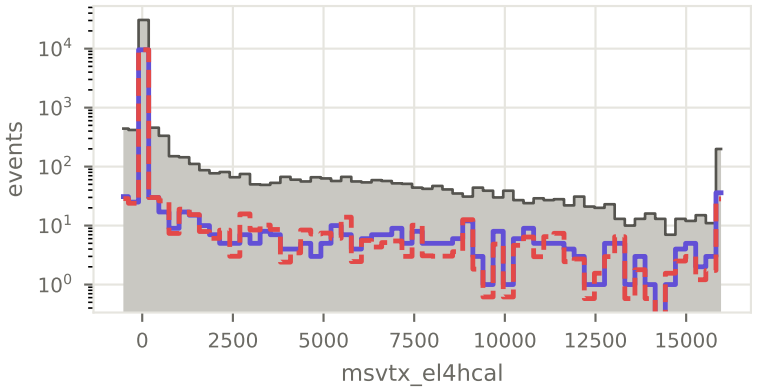


1b · each mass point vs background · normalized

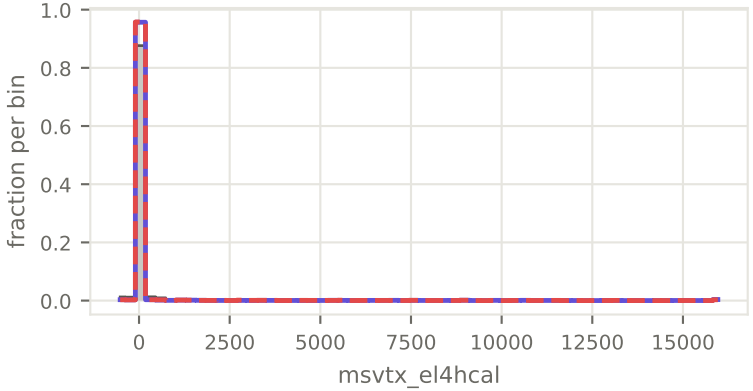


1 · each mass point
background n 34,970
mH125_mS15 D 0.110 n 411
mH125_mS55 D 0.080 n 281
mH200_mS50 D 0.104 n 1,071
mH400_mS100 D 0.096 n 2,648
mH600_mS150 D 0.089 n 2,882
mH1000_mS275 D 0.089 n 2,703

2a · combined signal vs background · raw

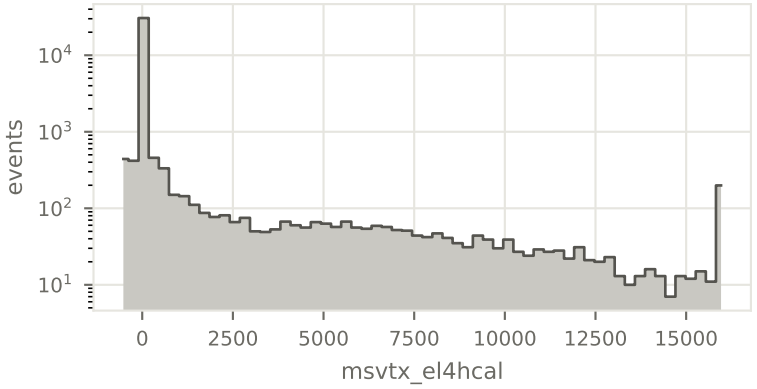


2b · combined signal vs background · normalized

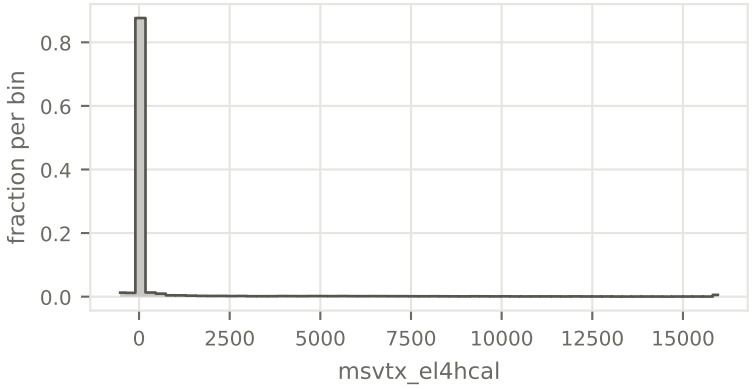


2 · combined signal
background n 34,970
combined, pooled D 0.093 n 9,996
equal weight per mass D 0.094

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

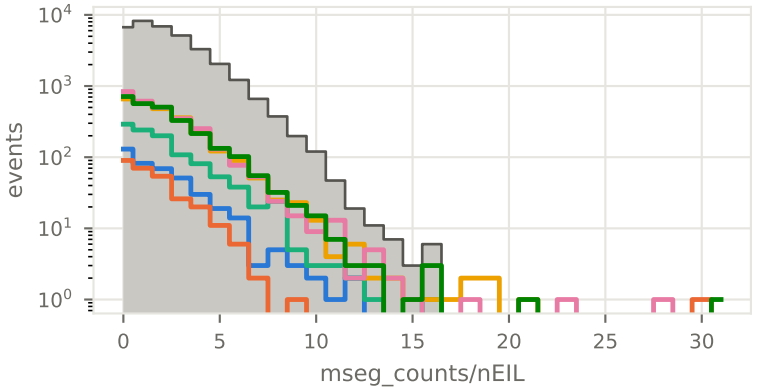
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_counts/nEIL

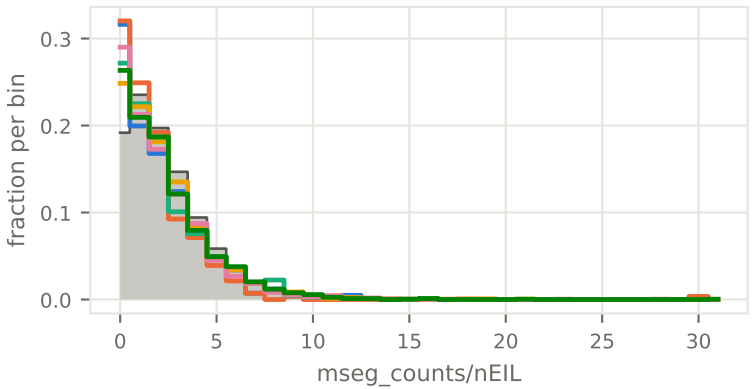
region: ENDCAP · rank 80 / 98

D_equal 0.093 (noise 0.022) D_pooled 0.080 min 0.057 (mH400_mS100) max 0.142 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

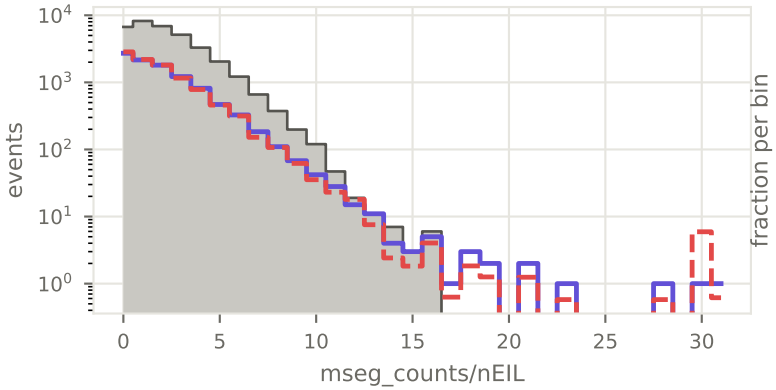


1b · each mass point vs background · normalized

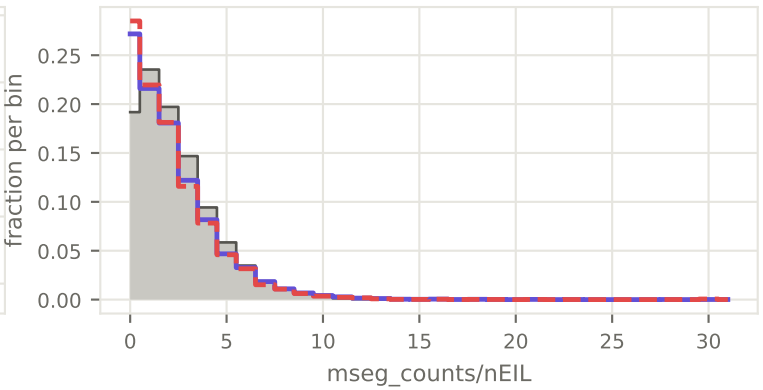


1 · each mass point
background n 34,970
mH125_mS15 D 0.125 n 411
mH125_mS55 D 0.142 n 281
mH200_mS50 D 0.080 n 1,071
mH400_mS100 D 0.057 n 2,648
mH600_mS150 D 0.098 n 2,882
mH1000_mS275 D 0.072 n 2,703

2a · combined signal vs background · raw

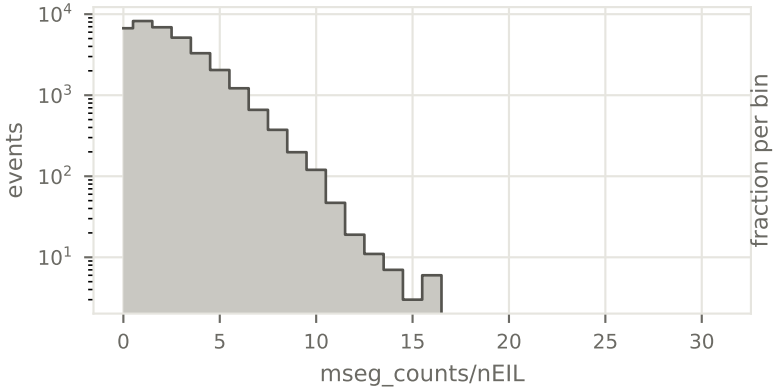


2b · combined signal vs background · normalized

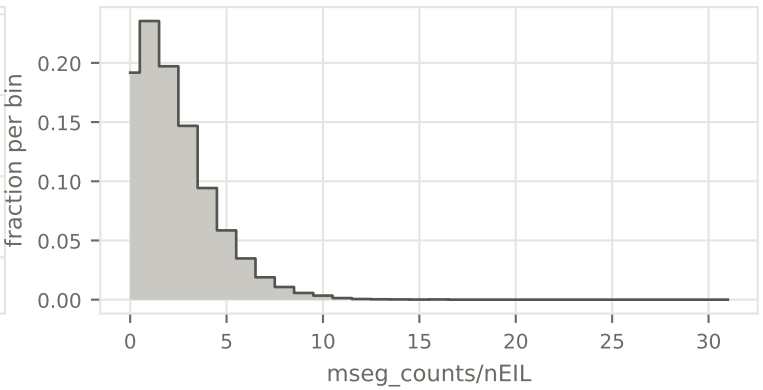


2 · combined signal
background n 34,970
combined, pooled D 0.080 n 9,996
equal weight per mass D 0.093

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

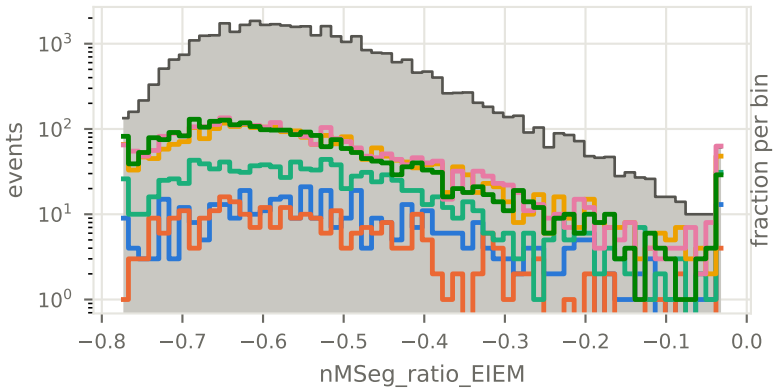
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

nMSeg_ratio_EIEM

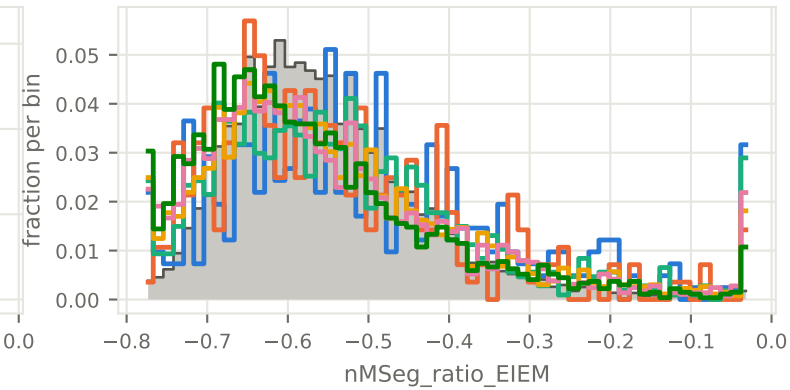
region: ENDCAP · rank 81 / 98 · in: NN1 endcap

D_equal 0.086 (noise 0.022) D_pooled 0.092 min 0.078 (mH400_mS100) max 0.177 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

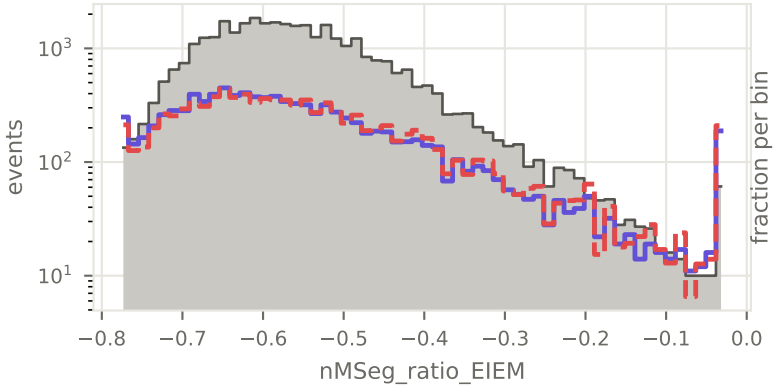


1b · each mass point vs background · normalized

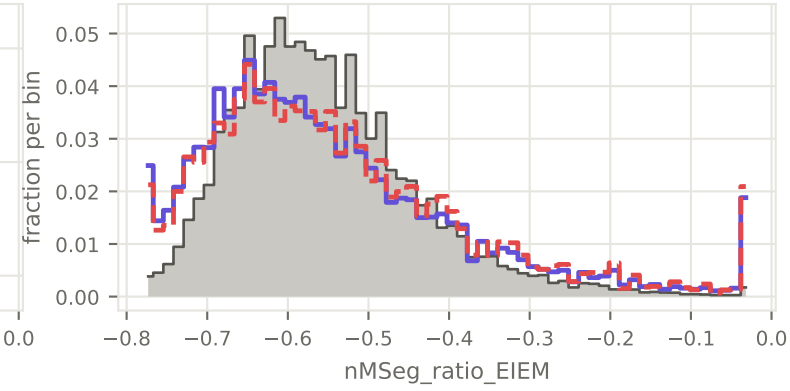


1 · each mass point
background n 34,970
mH125_mS15 D 0.177 n 411
mH125_mS55 D 0.084 n 281
mH200_mS50 D 0.107 n 1,071
mH400_mS100 D 0.078 n 2,648
mH600_mS150 D 0.100 n 2,882
mH1000_mS275 D 0.142 n 2,703

2a · combined signal vs background · raw

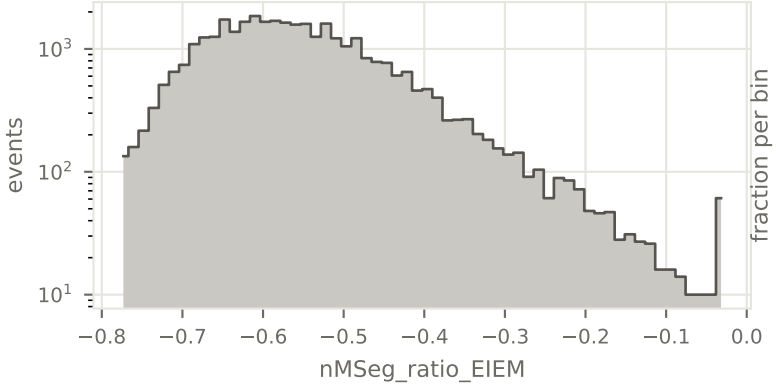


2b · combined signal vs background · normalized

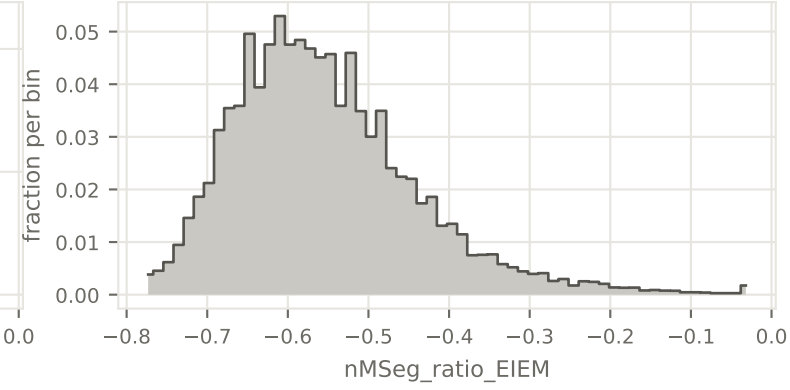


2 · combined signal
background n 34,970
combined, pooled D 0.092 n 9,996
equal weight per mass D 0.086

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

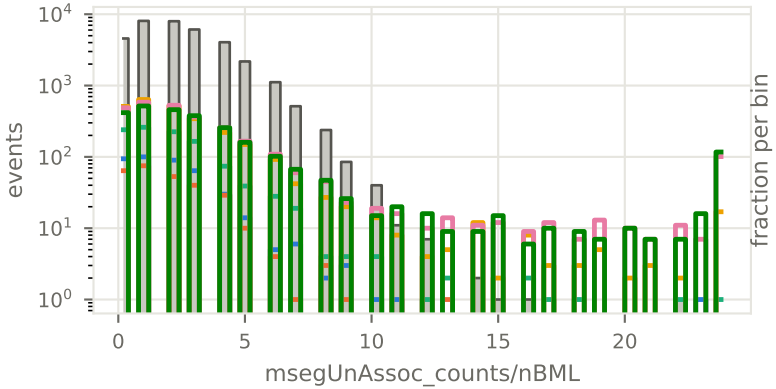
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msegUnAssoc_counts/nBML

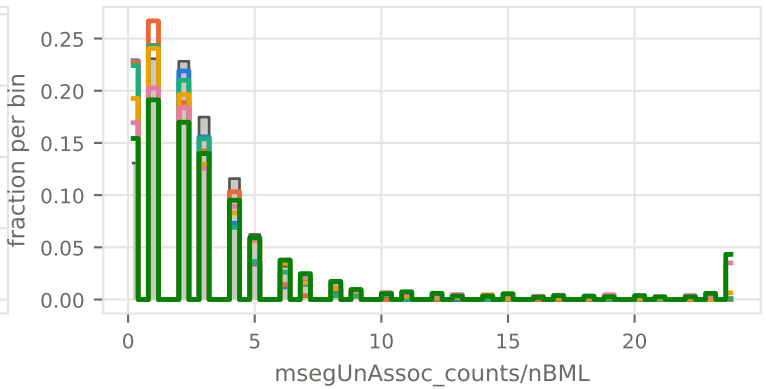
region: ENDCAP · rank 82 / 98 · in: NN1 barrel

D_equal 0.069 (noise 0.022) D_pooled 0.080 min 0.072 (mH400_mS100) max 0.133 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

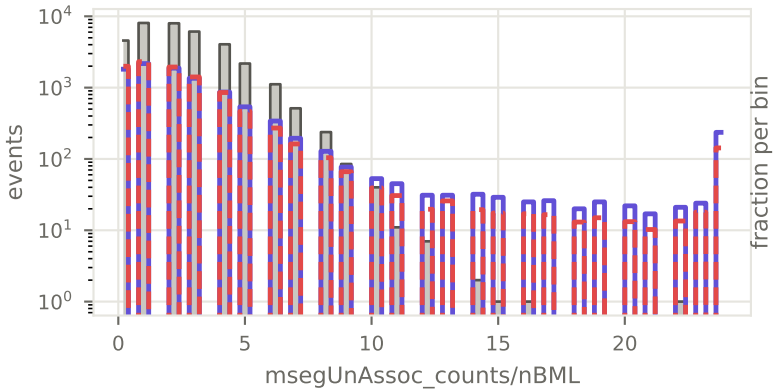


1b · each mass point vs background · normalized

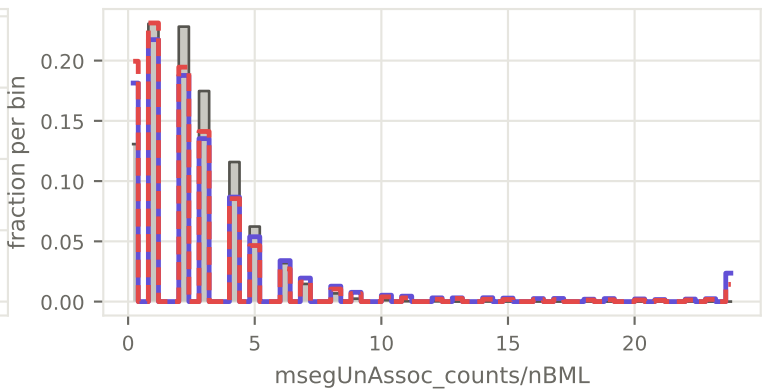


1 · each mass point
- background n 34,970
- mH125_mS15 D 0.111 n 411
- mH125_mS55 D 0.133 n 281
- mH200_mS50 D 0.106 n 1,071
- mH400_mS100 D 0.072 n 2,648
- mH600_mS150 D 0.115 n 2,882
- mH1000_mS275 D 0.133 n 2,703

2a · combined signal vs background · raw

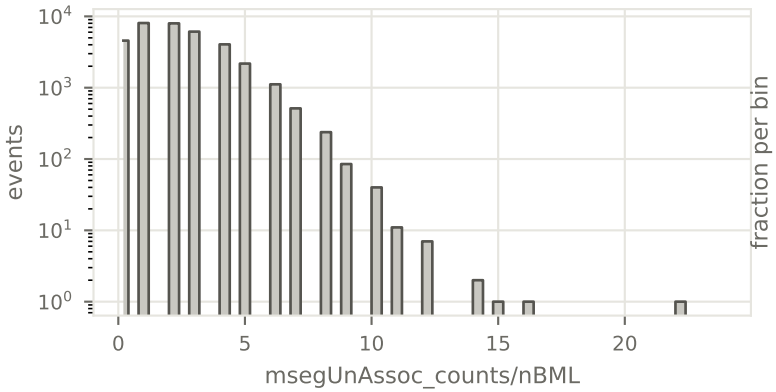


2b · combined signal vs background · normalized

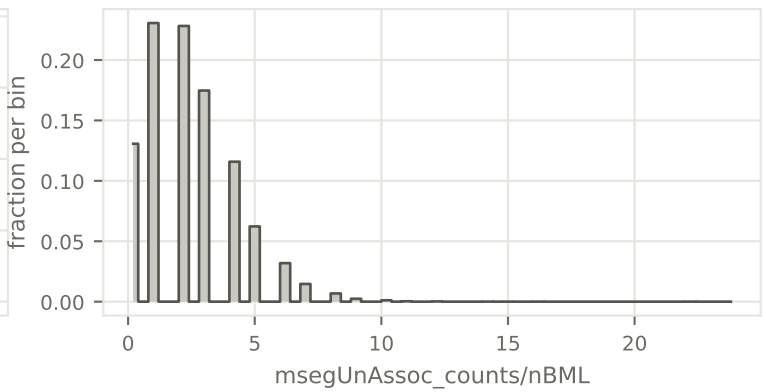


2 · combined signal
- background n 34,970
- combined, pooled D 0.080 n 9,996
- equal weight per mass D 0.069

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
- background n 34,970

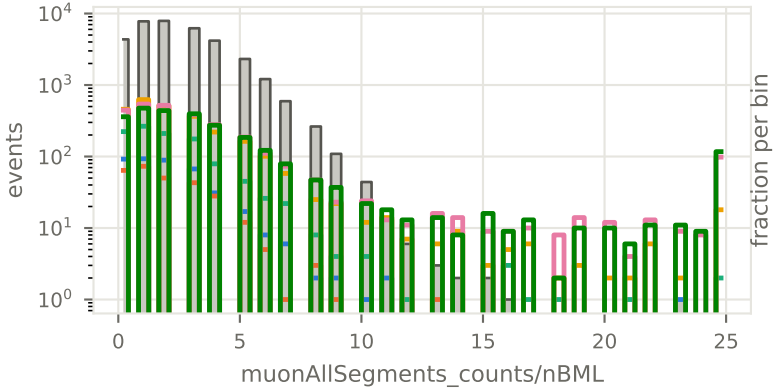
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

muonAllSegments_counts/nBML

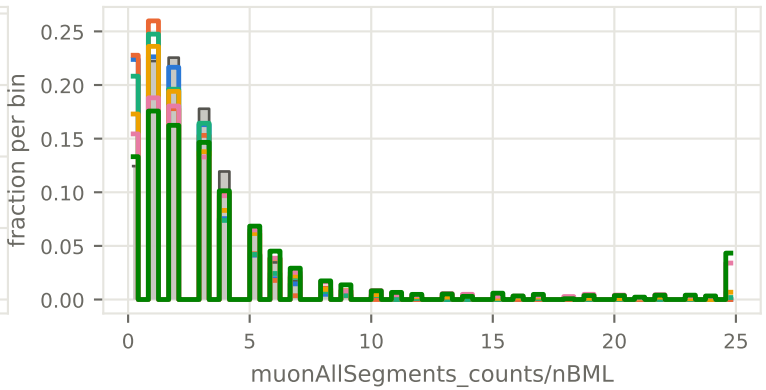
region: ENDCAP · rank 83 / 98

D_equal 0.062 (noise 0.022) D_pooled 0.087 min 0.062 (mH400_mS100) max 0.150 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

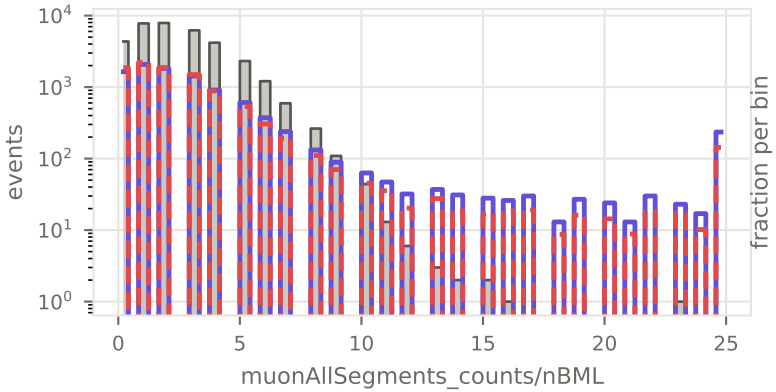


1b · each mass point vs background · normalized

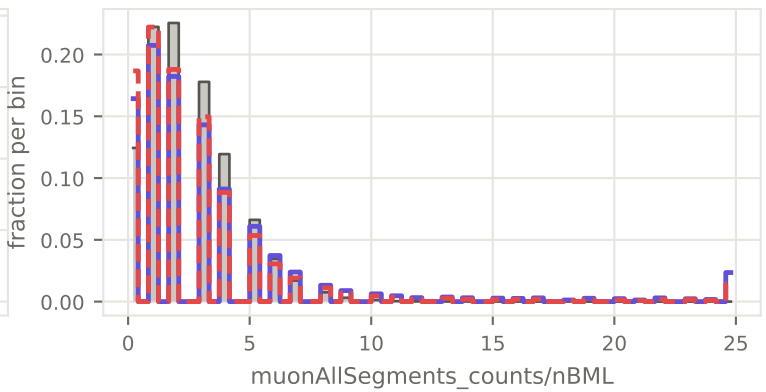


1 · each mass point
background n 34,970
mH125_mS15 D 0.103 n 411
mH125_mS55 D 0.141 n 281
mH200_mS50 D 0.109 n 1,071
mH400_mS100 D 0.062 n 2,648
mH600_mS150 D 0.118 n 2,882
mH1000_mS275 D 0.150 n 2,703

2a · combined signal vs background · raw

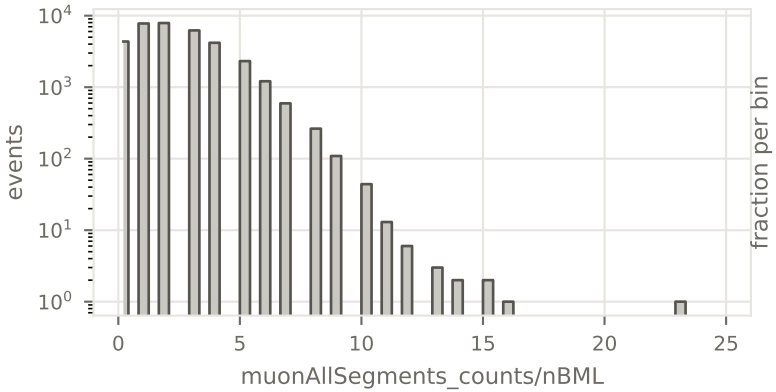


2b · combined signal vs background · normalized

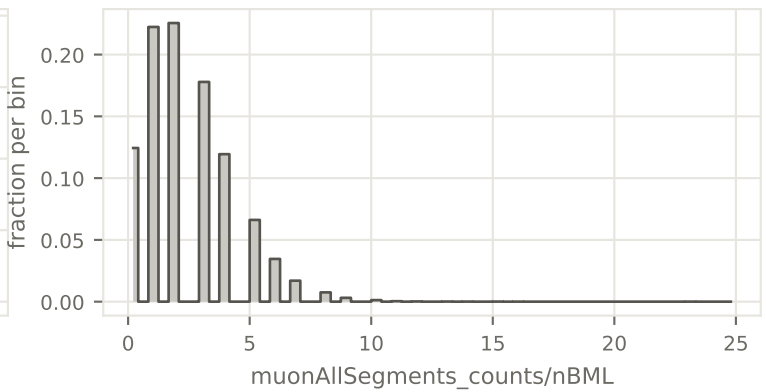


2 · combined signal
background n 34,970
combined, pooled D 0.087 n 9,996
equal weight per mass D 0.062

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

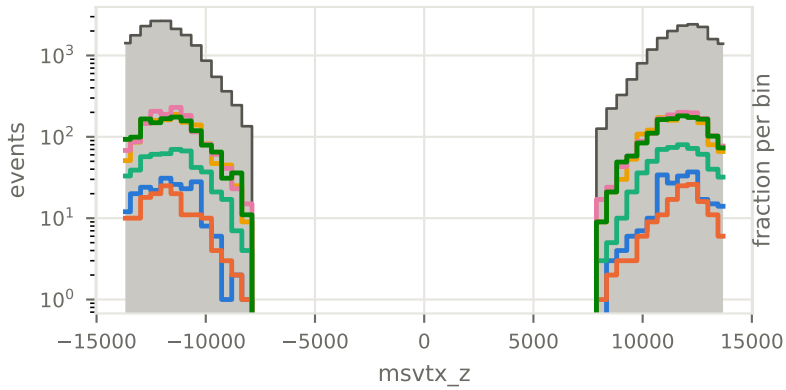
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_z

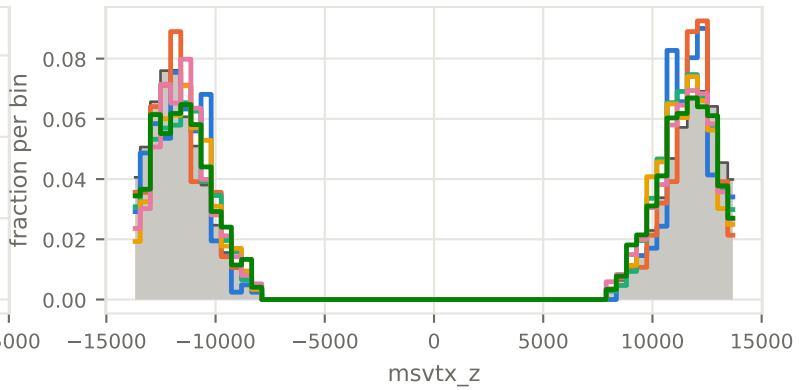
region: ENDCAP · rank 84 / 98

D_equal 0.057 (noise 0.022) D_pooled 0.066 min 0.039 (mH125_mS55) max 0.081 (mH200_mS50) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

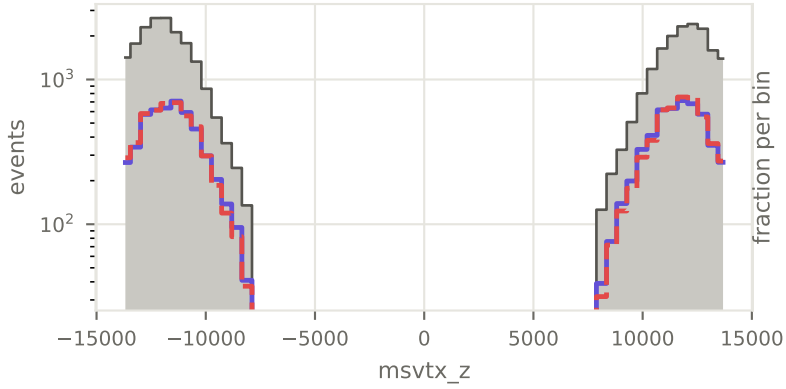


1b · each mass point vs background · normalized

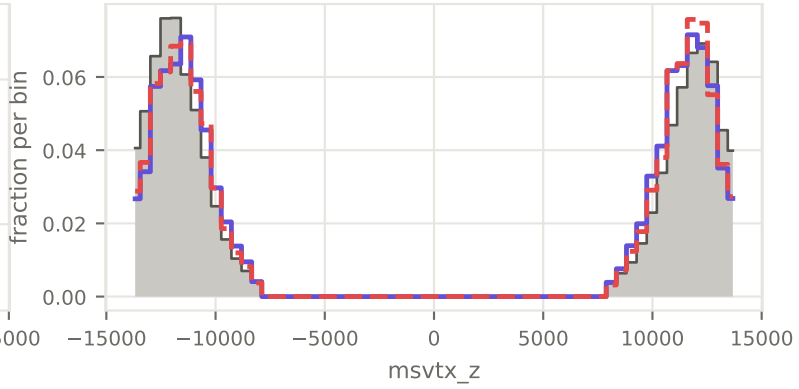


1 · each mass point
background n 34,970
mH125_mS15 D 0.050 n 411
mH125_mS55 D 0.039 n 281
mH200_mS50 D 0.081 n 1,071
mH400_mS100 D 0.077 n 2,648
mH600_mS150 D 0.069 n 2,882
mH1000_mS275 D 0.062 n 2,703

2a · combined signal vs background · raw

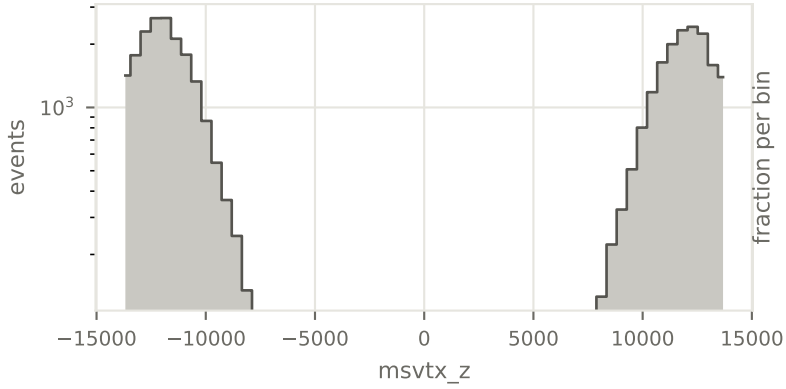


2b · combined signal vs background · normalized

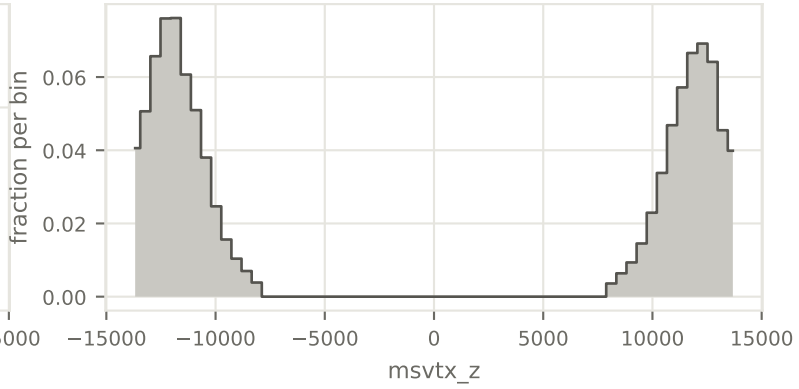


2 · combined signal
background n 34,970
combined, pooled D 0.066 n 9,996
equal weight per mass D 0.057

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

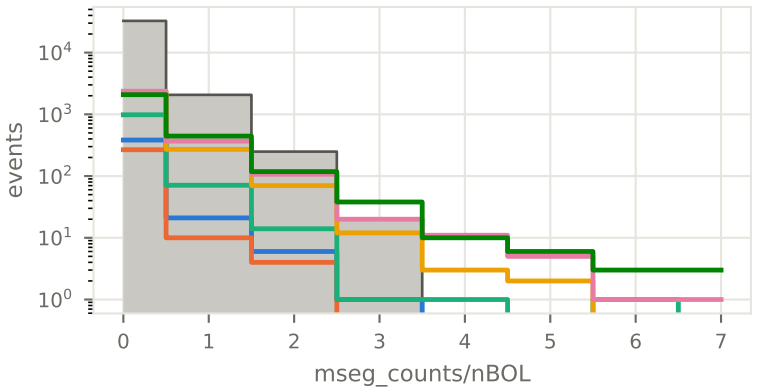
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_counts/nBOL

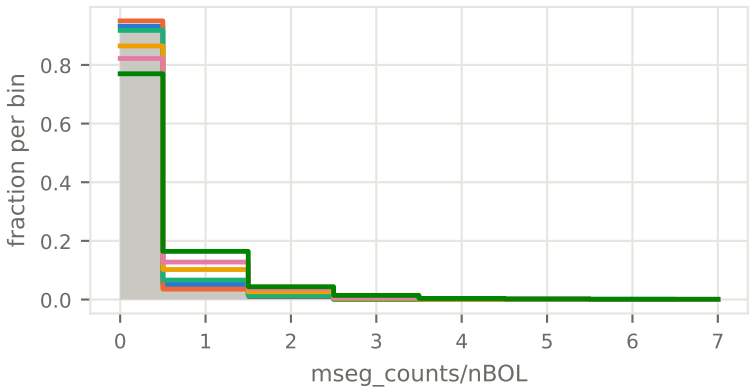
region: ENDCAP · rank 85 / 98

D_equal 0.057 (noise 0.022) D_pooled 0.096 min 0.009 (mH125_mS15) max 0.163 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

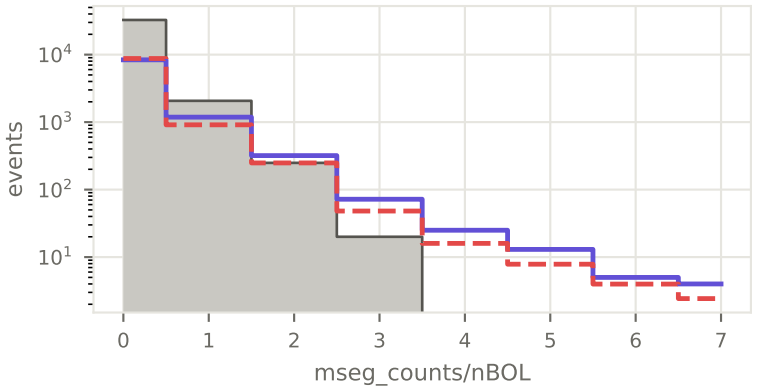


1b · each mass point vs background · normalized

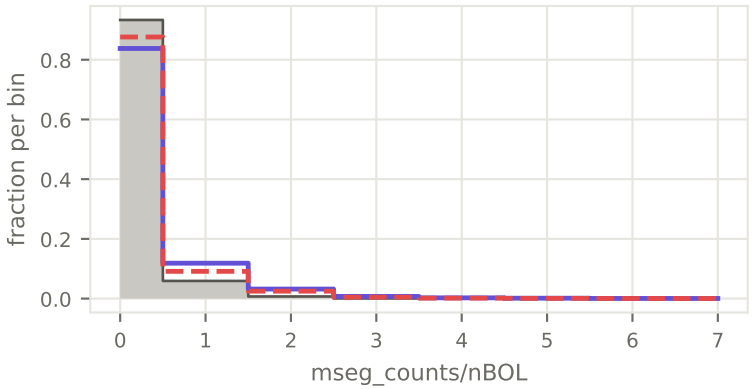


1 · each mass point
background n 34,970
mH125_mS15 D 0.009 n 411
mH125_mS55 D 0.017 n 281
mH200_mS50 D 0.015 n 1,071
mH400_mS100 D 0.068 n 2,648
mH600_mS150 D 0.111 n 2,882
mH1000_mS275 D 0.163 n 2,703

2a · combined signal vs background · raw

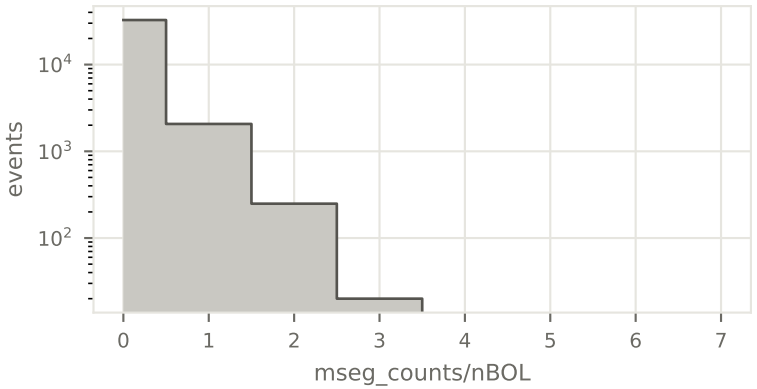


2b · combined signal vs background · normalized

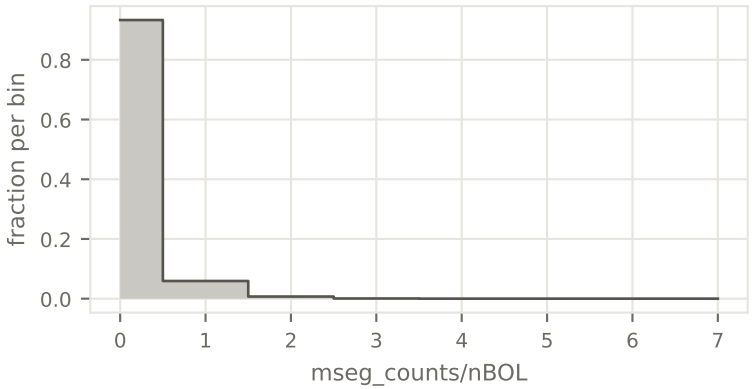


2 · combined signal
background n 34,970
combined, pooled D 0.096 n 9,996
equal weight per mass D 0.057

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

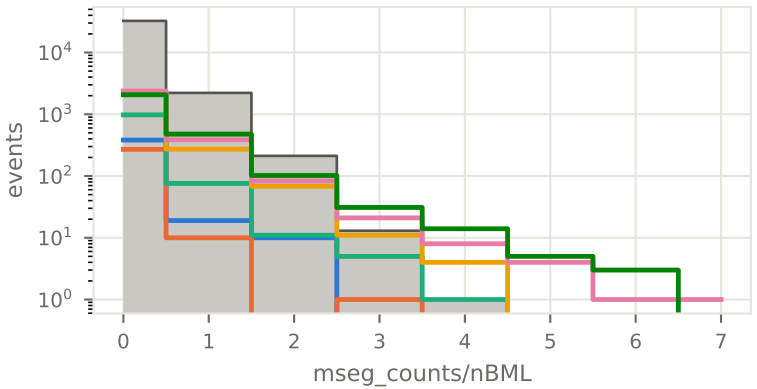
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_counts/nBML

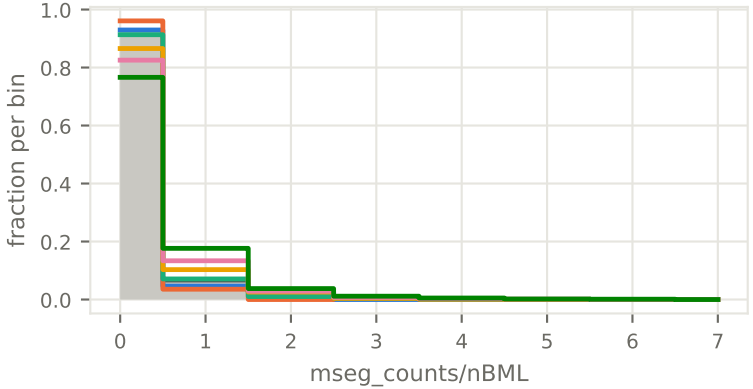
region: ENDCAP · rank 86 / 98

D_equal 0.053 (noise 0.022) D_pooled 0.093 min 0.017 (mH200_mS50) max 0.164 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

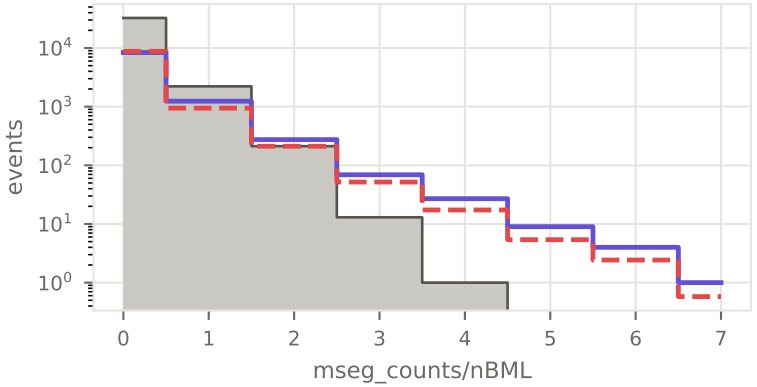


1b · each mass point vs background · normalized

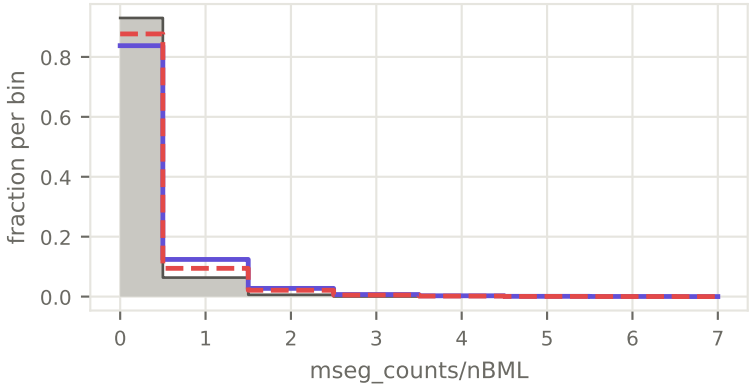


1 · each mass point
background n 34,970
mH125_mS15 D 0.018 n 411
mH125_mS55 D 0.031 n 281
mH200_mS50 D 0.017 n 1,071
mH400_mS100 D 0.065 n 2,648
mH600_mS150 D 0.105 n 2,882
mH1000_mS275 D 0.164 n 2,703

2a · combined signal vs background · raw

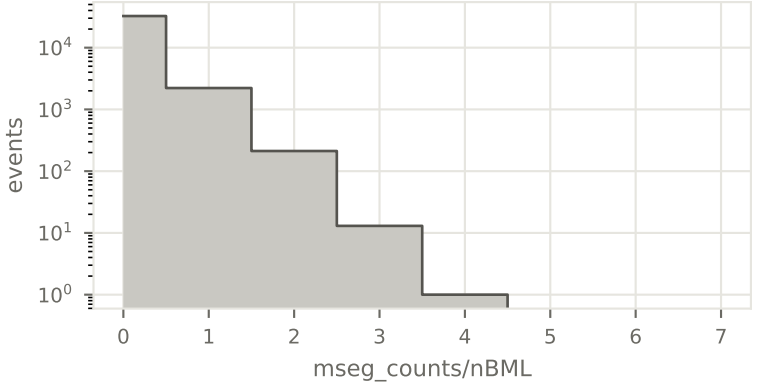


2b · combined signal vs background · normalized

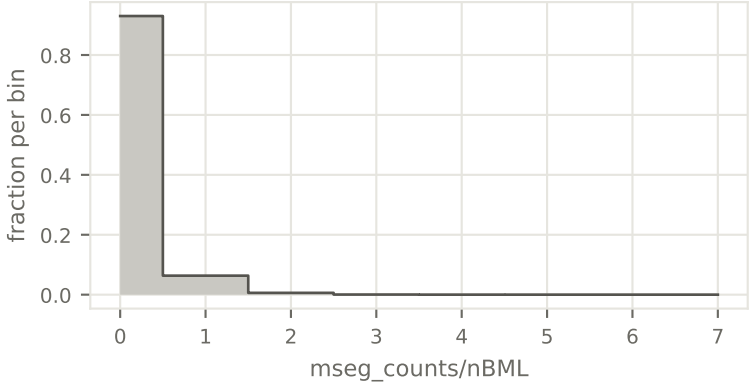


2 · combined signal
background n 34,970
combined, pooled D 0.093 n 9,996
equal weight per mass D 0.053

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

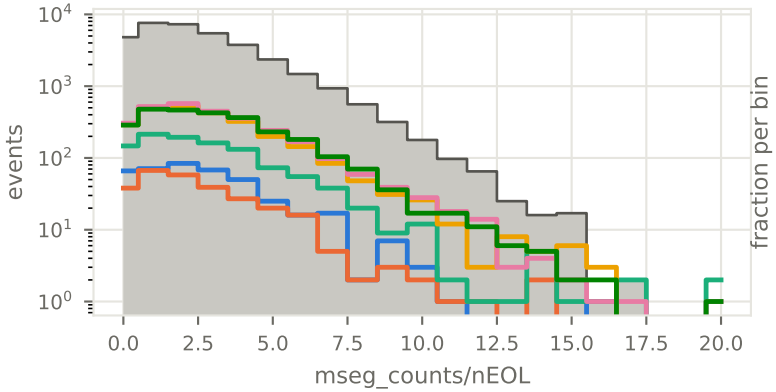
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_counts/nEOL

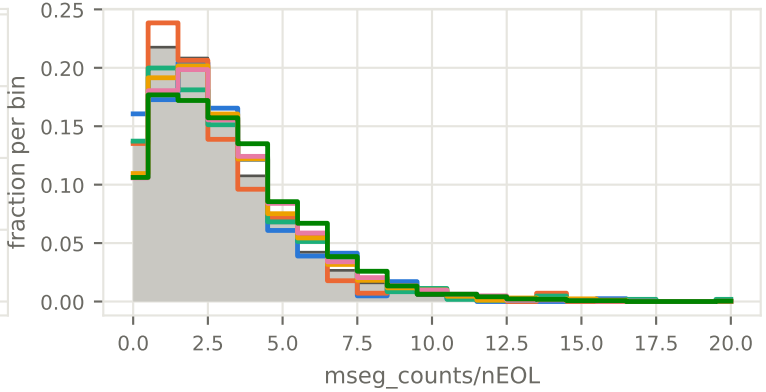
region: ENDCAP · rank 87 / 98

D_equal 0.052 (noise 0.022) D_pooled 0.074 min 0.018 (mH125_mS55) max 0.109 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

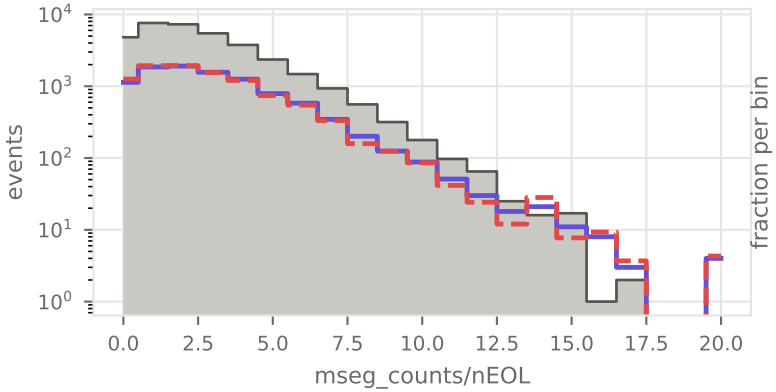


1b · each mass point vs background · normalized

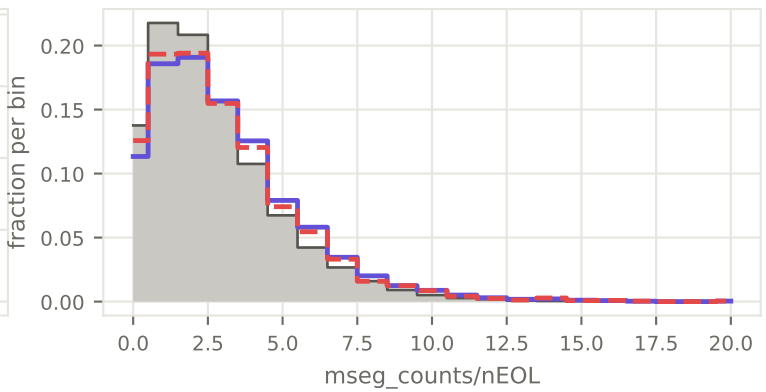


1 · each mass point
background n 34,970
mH125_mS15 D 0.026 n 411
mH125_mS55 D 0.018 n 281
mH200_mS50 D 0.050 n 1,071
mH400_mS100 D 0.061 n 2,648
mH600_mS150 D 0.079 n 2,882
mH1000_mS275 D 0.109 n 2,703

2a · combined signal vs background · raw

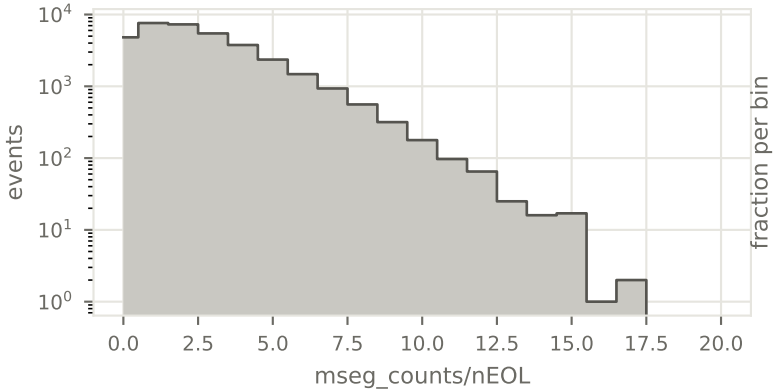


2b · combined signal vs background · normalized

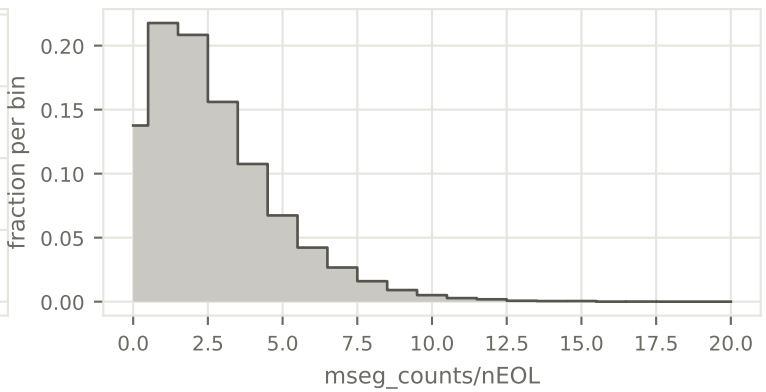


2 · combined signal
background n 34,970
combined, pooled D 0.074 n 9,996
equal weight per mass D 0.052

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

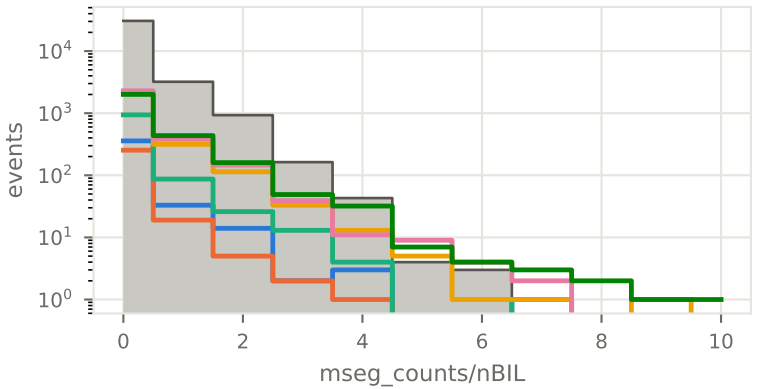
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_counts/nBIL

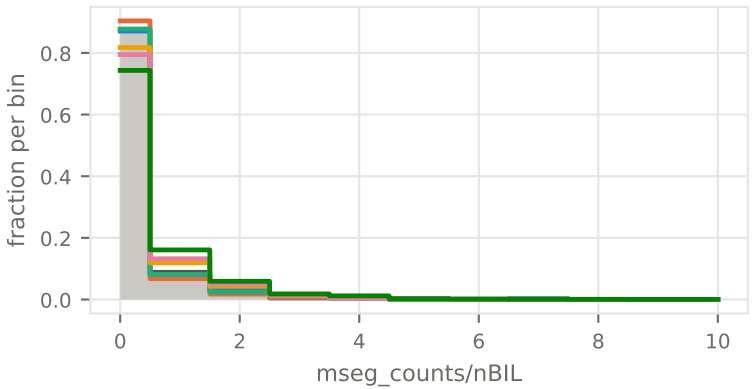
region: ENDCAP · rank 88 / 98

D_equal 0.041 (noise 0.022) D_pooled 0.074 min 0.011 (mH200_mS50) max 0.132 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

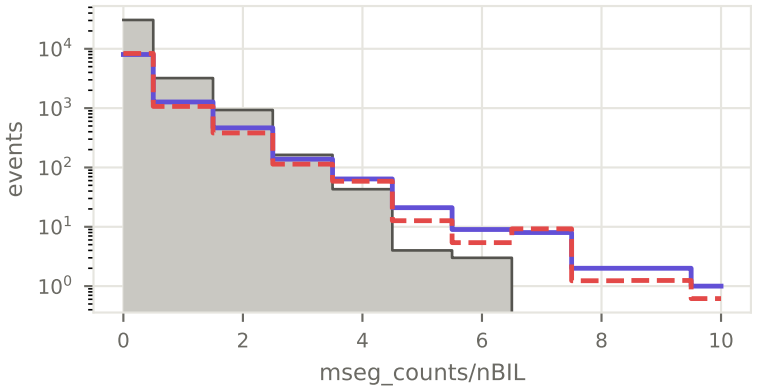


1b · each mass point vs background · normalized

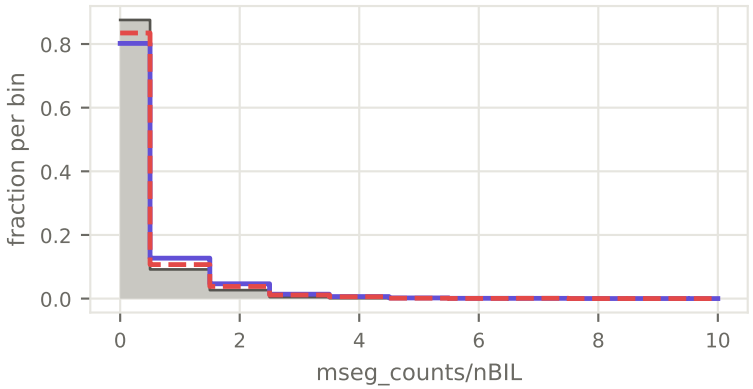


1 · each mass point
background n 34,970
mH125_mS15 D 0.016 n 411
mH125_mS55 D 0.028 n 281
mH200_mS50 D 0.011 n 1,071
mH400_mS100 D 0.058 n 2,648
mH600_mS150 D 0.081 n 2,882
mH1000_mS275 D 0.132 n 2,703

2a · combined signal vs background · raw

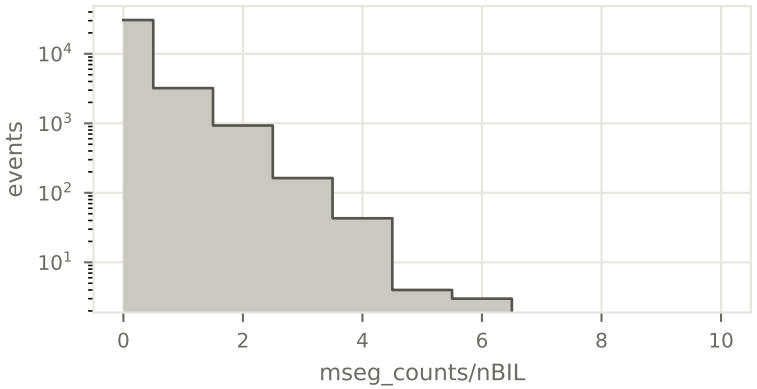


2b · combined signal vs background · normalized

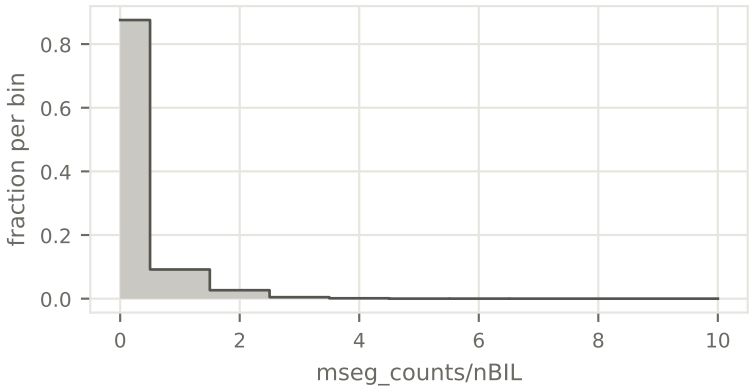


2 · combined signal
background n 34,970
combined, pooled D 0.074 n 9,996
equal weight per mass D 0.041

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

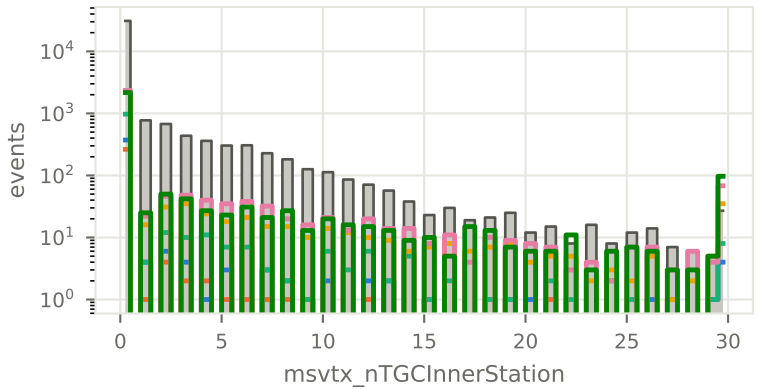
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCInnerStation

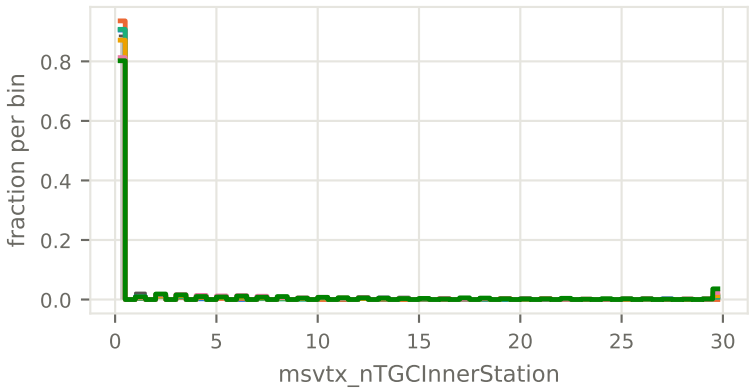
region: ENDCAP · rank 89 / 98

D_equal 0.039 (noise 0.022) D_pooled 0.062 min 0.020 (mH125_mS15) max 0.097 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

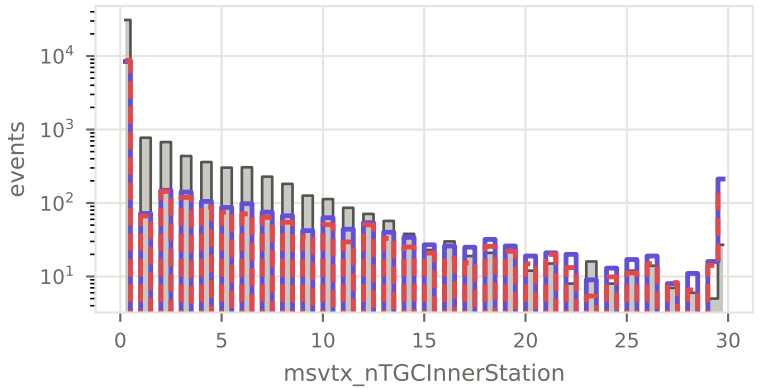


1b · each mass point vs background · normalized

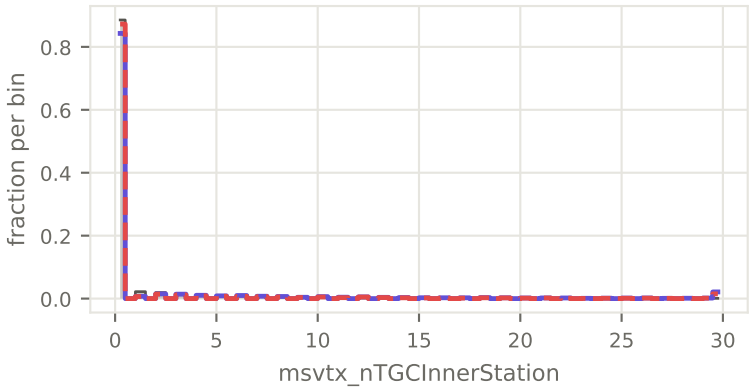


1 · each mass point
background n 34,970
mH125_mS15 D 0.020 n 411
mH125_mS55 D 0.050 n 281
mH200_mS50 D 0.022 n 1,071
mH400_mS100 D 0.042 n 2,648
mH600_mS150 D 0.091 n 2,882
mH1000_mS275 D 0.097 n 2,703

2a · combined signal vs background · raw

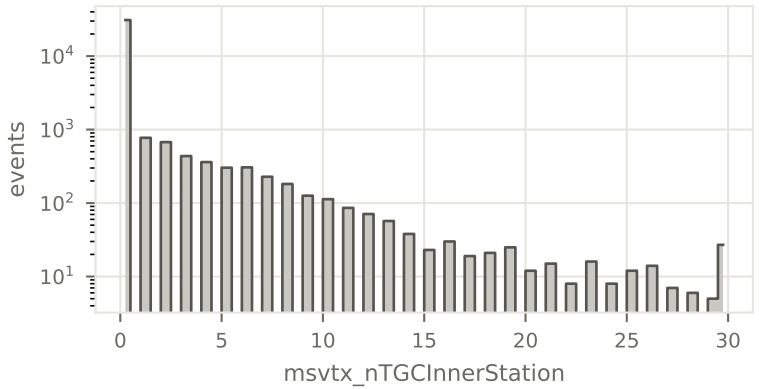


2b · combined signal vs background · normalized

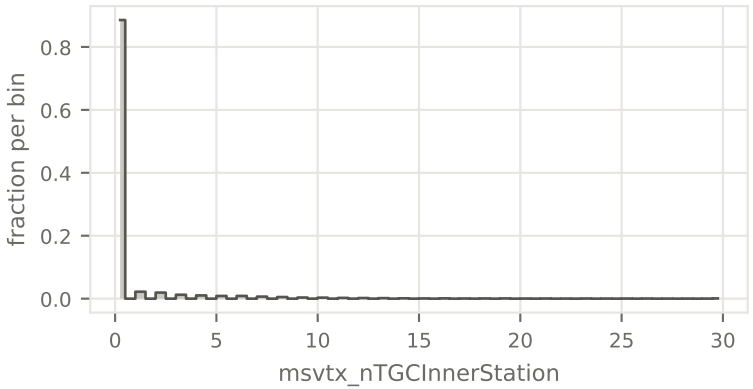


2 · combined signal
background n 34,970
combined, pooled D 0.062 n 9,996
equal weight per mass D 0.039

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

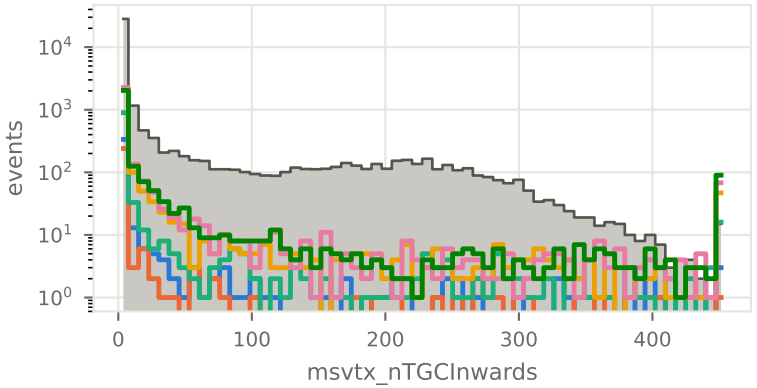
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTGCInwards

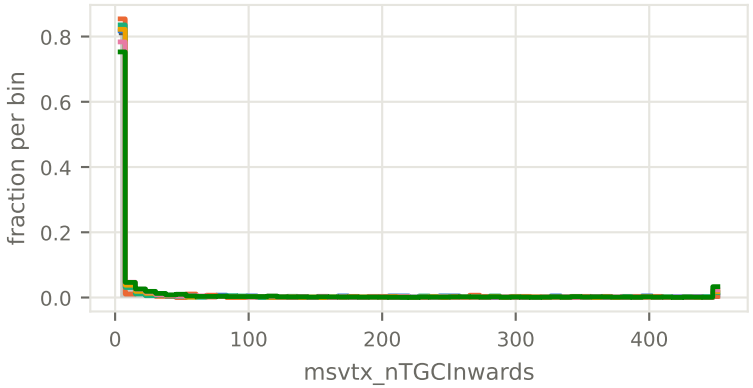
region: ENDCAP · rank 90 / 98

D_equal 0.038 (noise 0.022) D_pooled 0.036 min 0.038 (mH600_mS150) max 0.097 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

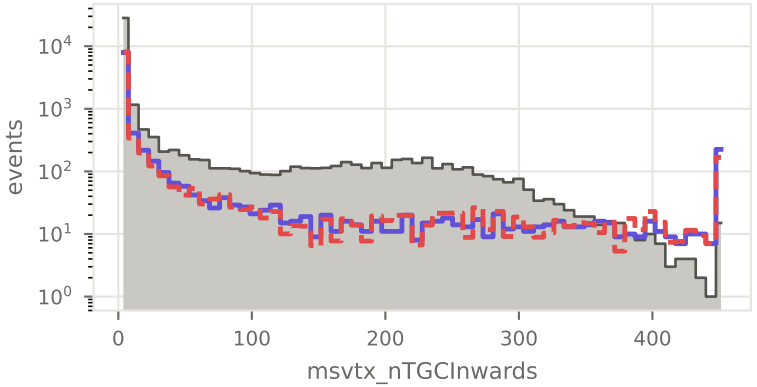


1b · each mass point vs background · normalized

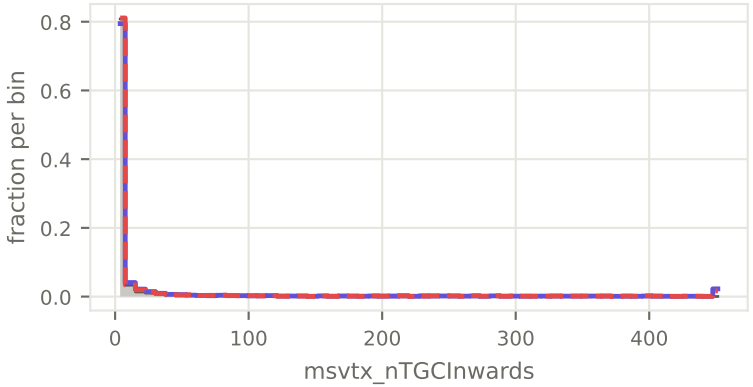


1 · each mass point
background n 34,970
mH125_mS15 D 0.053 n 411
mH125_mS55 D 0.097 n 281
mH200_mS50 D 0.069 n 1,071
mH400_mS100 D 0.052 n 2,648
mH600_mS150 D 0.038 n 2,882
mH1000_mS275 D 0.057 n 2,703

2a · combined signal vs background · raw

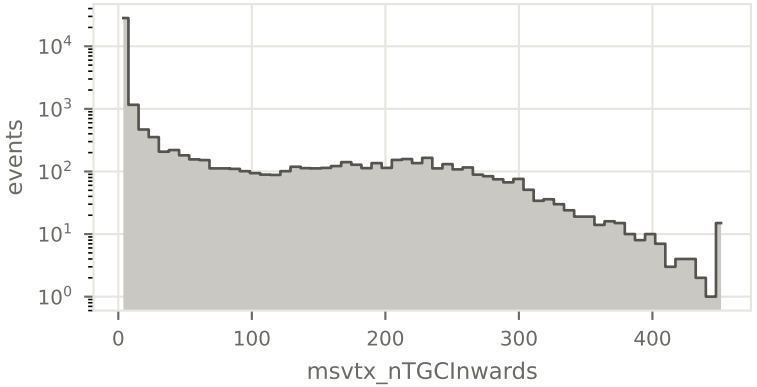


2b · combined signal vs background · normalized

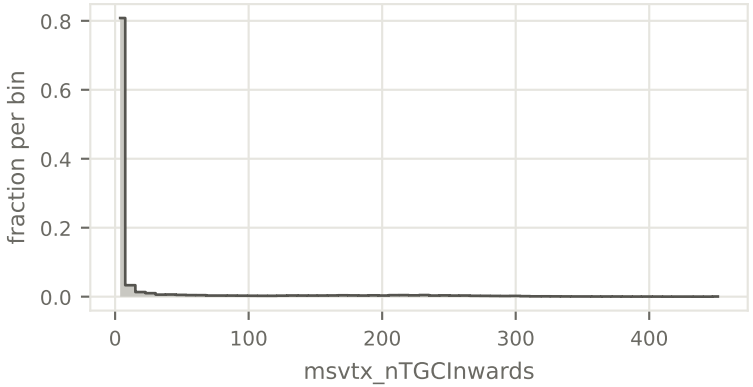


2 · combined signal
background n 34,970
combined, pooled D 0.036 n 9,996
equal weight per mass D 0.038

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

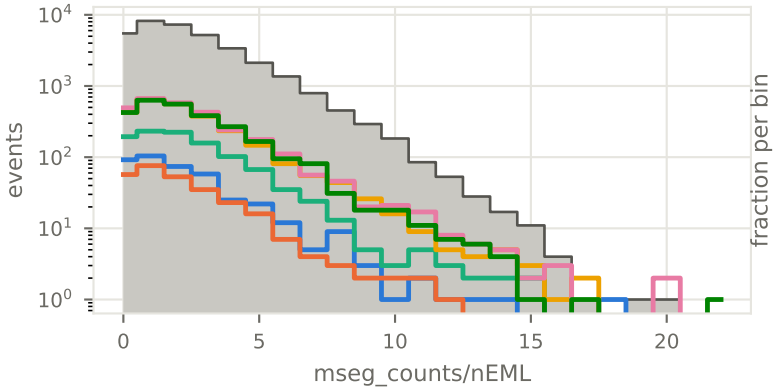
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_counts/nEML

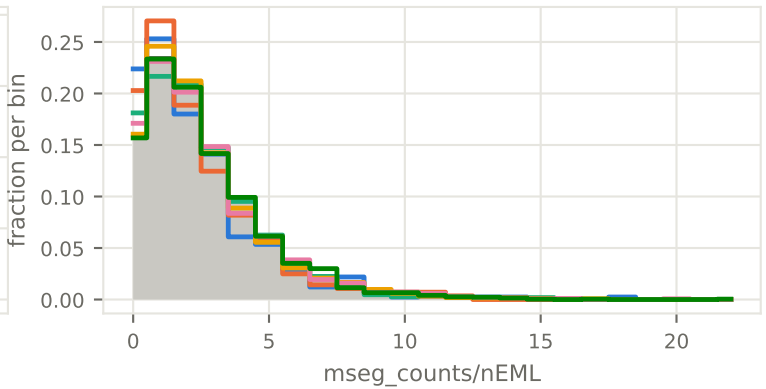
region: ENDCAP · rank 91 / 98

D_equal 0.033 (noise 0.022) D_pooled 0.013 min 0.011 (mH1000_mS275) max 0.085 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

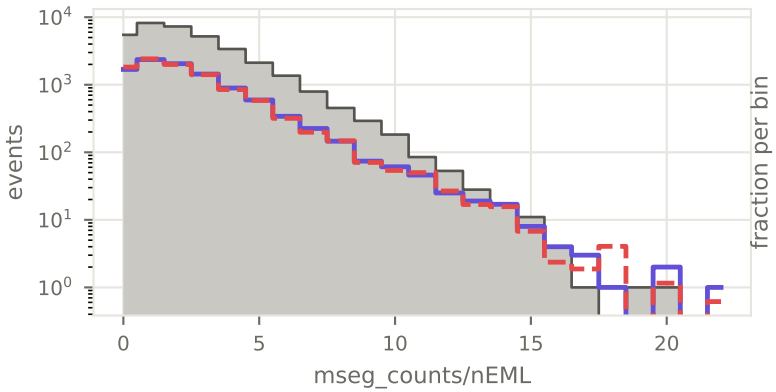


1b · each mass point vs background · normalized

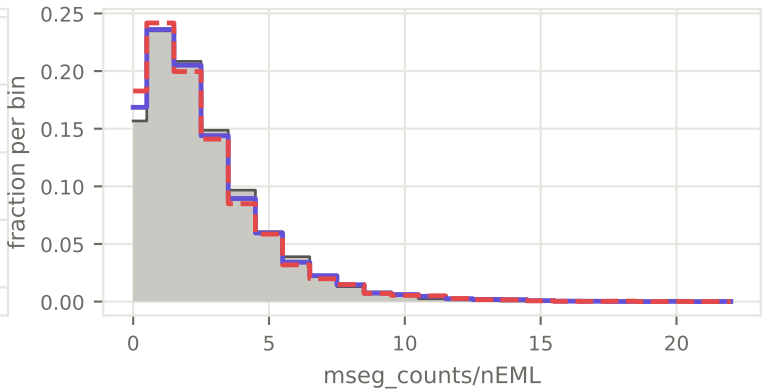


1 · each mass point
background n 34,970
mH125_mS15 D 0.085 n 411
mH125_mS55 D 0.082 n 281
mH200_mS50 D 0.024 n 1,071
mH400_mS100 D 0.018 n 2,648
mH600_mS150 D 0.014 n 2,882
mH1000_mS275 D 0.011 n 2,703

2a · combined signal vs background · raw

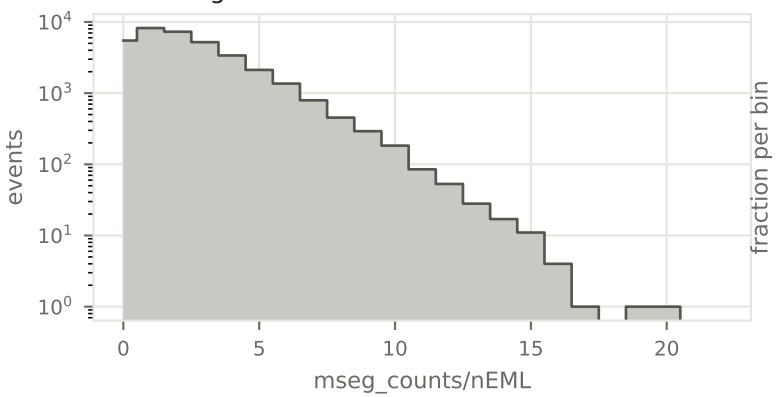


2b · combined signal vs background · normalized

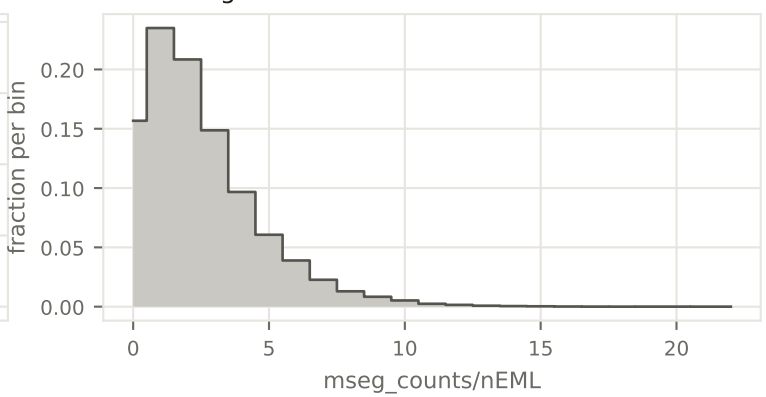


2 · combined signal
background n 34,970
combined, pooled D 0.013 n 9,996
equal weight per mass D 0.033

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

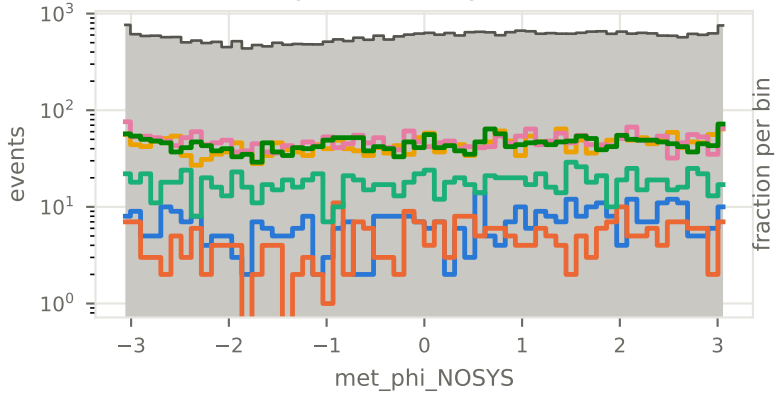
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

met_phi_NOSYS

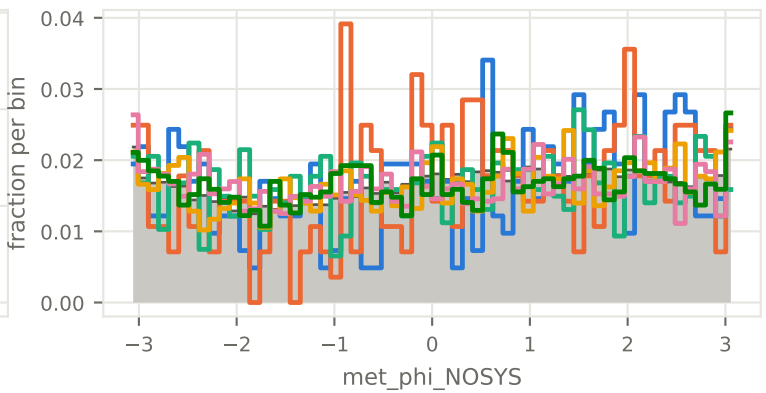
region: ENDCAP · rank 92 / 98

D_equal 0.013 (noise 0.022) D_pooled 0.018 min 0.018 (mH400_mS100) max 0.073 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

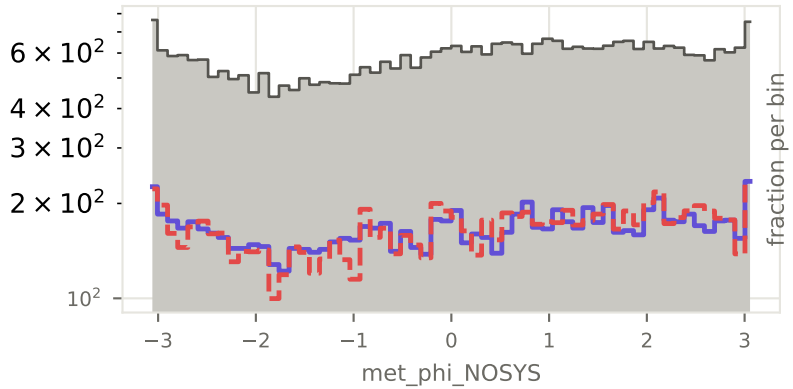


1b · each mass point vs background · normalized

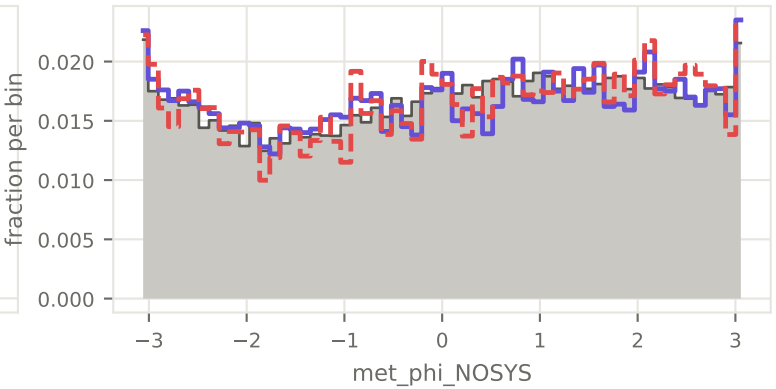


1 · each mass point
background n 34,970
mH125_mS15 D 0.063 n 411
mH125_mS55 D 0.073 n 281
mH200_mS50 D 0.027 n 1,071
mH400_mS100 D 0.018 n 2,648
mH600_mS150 D 0.037 n 2,882
mH1000_mS275 D 0.026 n 2,703

2a · combined signal vs background · raw

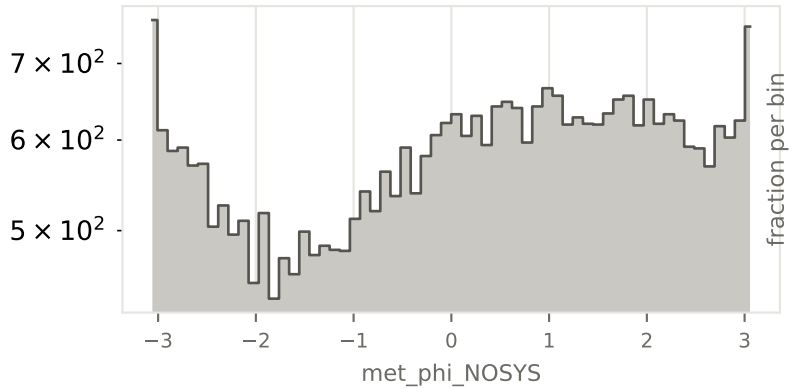


2b · combined signal vs background · normalized

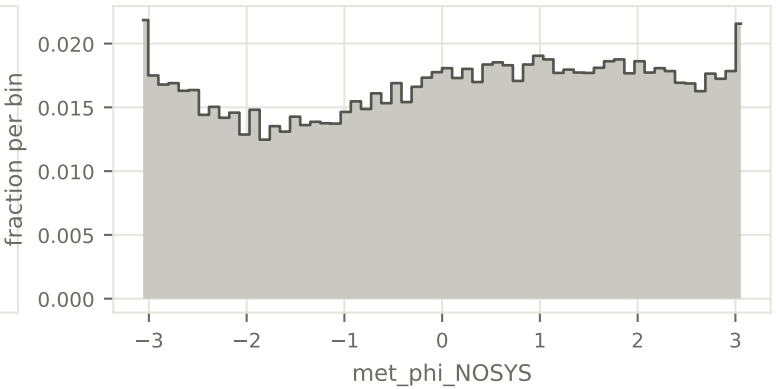


2 · combined signal
background n 34,970
combined, pooled D 0.018 n 9,996
equal weight per mass D 0.013

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

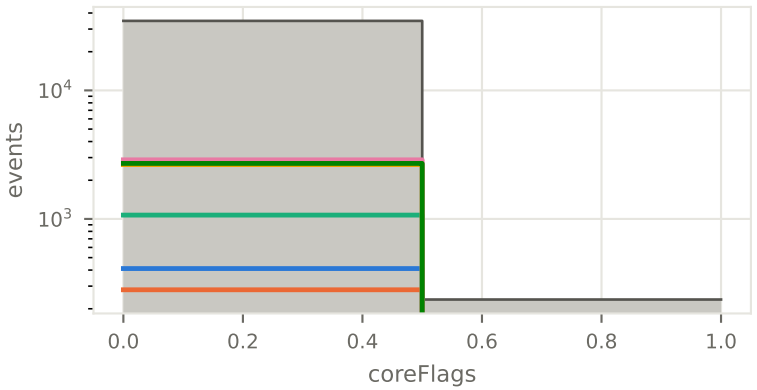
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

coreFlags

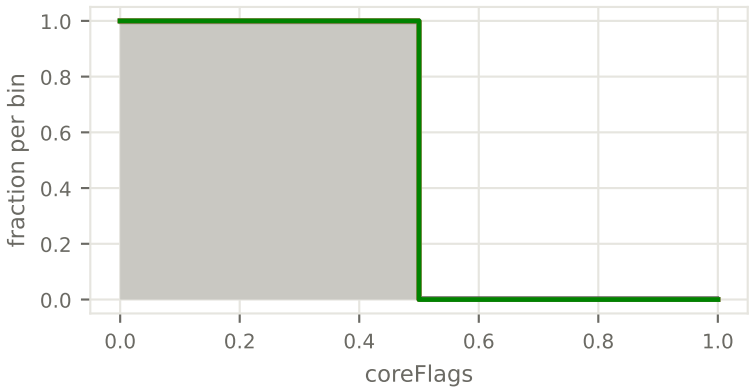
region: ENDCAP · rank 93 / 98

D_equal 0.007 (noise 0.022) D_pooled 0.007 min 0.007 (mH1000_mS275) max 0.007 (mH400_mS100) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

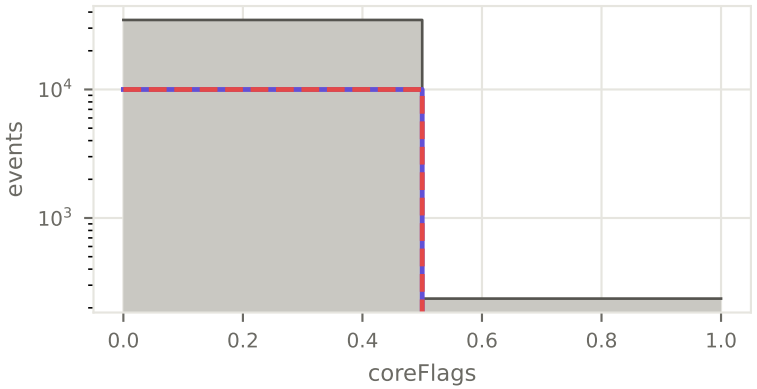


1b · each mass point vs background · normalized

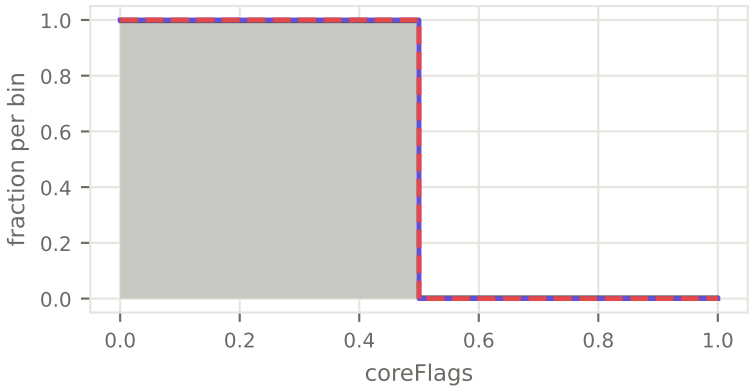


1 · each mass point
background n 34,970
mH125_mS15 D 0.007 n 411
mH125_mS55 D 0.007 n 281
mH200_mS50 D 0.007 n 1,071
mH400_mS100 D 0.007 n 2,648
mH600_mS150 D 0.007 n 2,882
mH1000_mS275 D 0.007 n 2,703

2a · combined signal vs background · raw

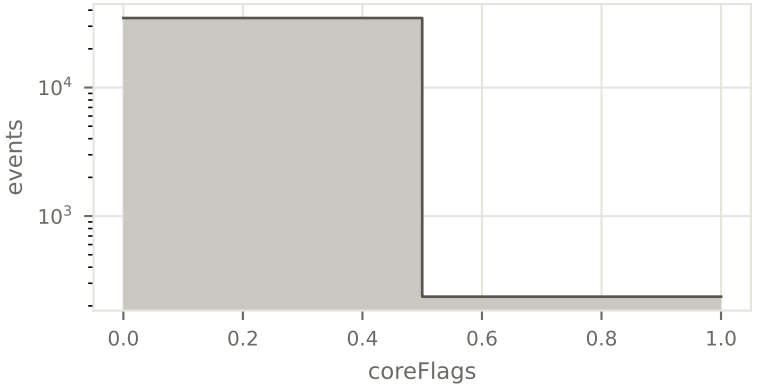


2b · combined signal vs background · normalized

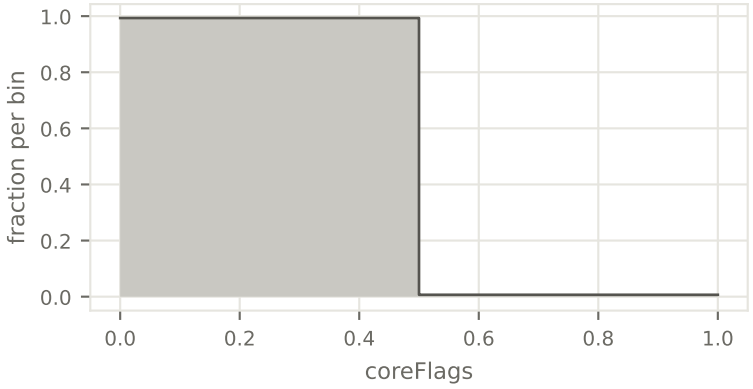


2 · combined signal
background n 34,970
combined, pooled D 0.007 n 9,996
equal weight per mass D 0.007

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed